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SECTION 01 11 00

SUMMARY OF WORK
08/15, CHG 2: 08/21

PART 1 GENERAL

1.1 FAR 52.211-10, Commencement, Prosecution and Completion of Work

- a. The Contractor is required to (i) commence work under this contract within 5 calendar days after the date the Contractor receives the notice to proceed, (ii) prosecute the work diligently, and (iii) complete the entire work ready for use not later than ~~852760~~ calendar days after the date the Contractor receives the notice to proceed. The time stated for completion must include the final cleanup of the premises. (FAR 52.212-3)
- b. All work shall be in accordance with the drawings and specifications or instructions attached hereto and made a part thereof, or to be furnished hereafter by the Contracting Officer and subject, in every detail, to his supervision, direction, and instructions.
- c. The Contractor must furnish all labor, materials, equipment, and services for the following work as described in the construction documents.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Excavation Permits; G

1.3 WORK COVERED BY CONTRACT DOCUMENTS

1.3.1 Project Description

The work includes replacement of deteriorated Runway 03-21 lighting, full depth reconstruction of shoulder pavement for Runway 03-21, reconstruction of the southern 1,400 feet of Runway 03-21, replacement of runway end lighting and approach lighting in the area of the runway pavement replacement, replacement of visual NAVAID infrastructure for Runway 03-21, replacement of signage for Runway 03-21, replacement of Building 2067 with a new Airfield Lighting Vault, replacement of Airfield Lighting Controls (in the ATCT, Base Operations Office, and AFLV), Runway 03-21 approach lighting, airfield lighting for Runway 15-33, North Ramp lighting, and incidental related work.

This work includes the development of an approved concrete pavement mix design as detailed in Section 32 13 14.13 CONCRETE PAVING FOR AIRFIELDS AND OTHER HEAVY DUTY PAVEMENTS. This process requires a minimum of 135 calendar days to complete if progressed without interruption or

re-testing. Section 01 30 00 ADMINISTRATIVE REQUIREMENTS paragraph 1.13 requires submittal of the Contractor's preliminary schedule including detailed description of proposed schedule for development of the approved concrete mix design. Note that a six-month interim CPARS rating will be issued based on the Contractor's success in delivering an approved mix design against the approved preliminary schedule.

1.3.2 Location

The work is located at the Wheeler-Sack Army Airfield, located at the Fort Drum Military Installation, in Jefferson County, NY. The site of the work is on a U.S. military reservation and all rules and regulations issued by the Commanding Officer covering general safety, security, and sanitary requirements, etc., must be observed by the Contractor. The exact location is as depicted on the contract plans.

1.4 OCCUPANCY OF PREMISES

ATC, base operations building, and the existing AFLV building(s) will be occupied during performance of work under this Contract. Occupancy notifications will be posted in a prominent location in the work area.

Before work is started, arrange with the Contracting Officer a sequence of procedure, means of access, space for storage of materials and equipment, and use of approaches, corridors, and stairways.

All airfield pavements, runways, taxiways, and aprons must be open and operational during Phase 1 of construction. Temporary closures of airfield pavements are permitted only when specified work mandates closure of a runway, taxiways, or apron and is coordinated by the contractor in advance with the installation through the COR.

1.5 DFARS 236.204, Disclosure of the Magnitude of Construction Projects

This project is estimated to be between \$25,000,000 to \$100,000,000.

1.6 EXISTING WORK

In addition to FAR 52.236-9 Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements:

- a. Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.
- b. Repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.
- c. The responsibility for damage to the permanent work must be as set forth in the article of the contract clause entitled "PERMITS AND RESPONSIBILITIES". However, if in the judgment of the Contracting Officer, the permanent work performed by the Contractor is damaged by flood, earthquake, hurricane, severe coastal storm or tornado, which damage is not due to the failure of the Contractor to take reasonable precautions or to exercise sound engineering and construction practices in the conduct of the work, the Contractor must make the repairs as ordered by the Contracting Officer and full compensation

for repairs will be made at the applicable Contract unit or lump-sum prices as fixed and established in the Contract. If, in the opinion of the Contracting Officer, there are no Contract unit or lump sum prices applicable to the work, an equitable adjustment, pursuant to Contract Clause entitled CHANGES, will be made as full compensation for the repairs of that part of the permanent work for which there are not applicable Contract unit or lump-sum prices. Except as herein provided, damage to all work, utilities, materials, equipment, and plant, including temporary construction and utilities, pavements, and other property along the routes used by the Contractor's pipelines and land vehicles, must be repaired to the satisfaction of the Contracting Officer, the State of New York, and the utilities companies, at the Contractor's expense regardless of the cause of such damage.

1.7 LOCATION OF UNDERGROUND UTILITIES

Obtain [excavation permits](#) prior to start of excavation, and comply with Installation requirements for locating and marking underground utilities. Contact local utility locating service a minimum of 10 business days prior to excavating, to mark utilities, and within sufficient time required if work occurs on a Monday or after a Holiday. Verify horizontal and vertical location of all existing utilities indicated on contract drawings or marked in the field, within area of work, prior to completing any excavation work.

Contractor must comply with the Garrison Policy Memorandum 23-11, Excavation Permits (20 Jul 23) attached following this specification section.

Identify and mark all other utilities not managed and located by the local utility companies. Scan the construction site with Ground Penetrating Radar (GPR), electromagnetic, or sonic equipment, and mark the surface of the ground or paved surface where existing underground utilities or utilities are discovered, whether marked or unmarked. Verify the elevations of existing piping, utilities, and any type of underground or encased obstruction not indicated, or specified to be removed, that is indicated or discovered during scanning, in locations to be traversed by piping, ducts, and other work to be conducted or installed. Verify elevations before installing new work closer than nearest manhole or other structure at which an adjustment in grade can be made.

1.7.1 Notification Prior to Excavation

Notify the Contracting Officer at least 15 days prior to starting excavation work.

1.8 RECYCLED MATERIAL AND EQUIPMENT

~~Items designated by the Contracting Officer to be recycled remain the property of the Government. Metal materials for recycling shall remain the property of the government.~~ The ~~government~~ contractor must will provide, and pickup roll off dumpsters, for contractor to segregate and itemize items for metal recycling.

1.9 SPARE PARTS

Contractor to provide and transfer to the Government (prior to final acceptance of completed work) no less than 10% of all lighting fixtures specified on this project for future use as spares (by type and size) and

Pre-Award Re-Phasing

17 April 2026

in all applicable lens colors.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --



DEPARTMENT OF THE ARMY
US ARMY INSTALLATION MANAGEMENT COMMAND
HEADQUARTERS, UNITED STATES ARMY GARRISON, FORT DRUM
1780 RESTORE HOPE DRIVE
FORT DRUM, NEW YORK 13602-4200

AMIM-DRG-ZA

20 July 2023

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Garrison Policy 23-11, Excavation Permits

1. References:

- a. 16 NYCRR (New York State Rules and Regulations), part 753, 4 January 2012.
- b. Army Regulation (AR) 380-49, Industrial Security Program, 20 March 2013.

2. Supersession: This memorandum supersedes and replaces Garrison Policy 21-11, Excavation Permits, 23 July 2021.

3. Purpose: Establish local policies and procedures regulating and requiring the use of excavation (digging) permits on Fort Drum.

4. Concept: Every excavation on the installation has impacts to safety, environment, and operations. New York State law mandates coordination with subsurface utilities prior to start of excavation. Fort Drum extends the requirements of the New York State law to cover the utilities owned and operated by the installation and all the land owned by the Department of the Army and managed by the Fort Drum Garrison.

5. Applicability: This policy applies to all commands, contractors, tenant organizations, and all directorates located on or working within the Fort Drum training areas, and installation boundary.

6. General: The primary purpose of this policy is to establish a uniform set of rules for excavation permits. A utility strike affects life, health, safety, security, and preparedness. The excavation permitting process is the only authorized method for assessing risks. Failure to comply with the requirements of this policy will invoke an immediate 'stop work' order until compliance is verified.

a. 16 NYCRR (New York Codes, Rules and Regulations) Part 753, applies to all excavations within the boundaries of Fort Drum. This policy contains requirements in addition to, but not in conflict with, 16 NYCRR Part 753..

b. Several agencies on Fort Drum are involved in approving an excavation permit application. No excavation can begin on Fort Drum without an approved excavation permit, except for an emergency permit as described in paragraph 9e. For projects with

a government employee as the Project Manager (PM), the PM or designated Construction Control Representative (Con Rep) will issue the completed permit to the contractor.

7. An excavation is any intrusion into the surface of the earth. This includes, but is not limited to, the placement of tent pins, excavation for utility lines, and excavation for foundation elements for outdoor recreational devices, antennas, utilities, and buildings. Hand excavation with hand tools and vacuum excavation that is no more than 12 inches into the ground is allowed without a permit. Unpermitted hand and vacuum excavation shall stop if a utility line is uncovered.

a. Within the cantonment areas, driving stakes, pins, or rods are considered to be excavation and requires an excavation permit.

b. In the range area, an excavation permit is required for any intrusion into the surface of the earth at any fixed training facility (range proper) and within 50 feet from the edges of any range road. In the training area an excavation permit for areas that are more than 50 feet from the edges of the road and outside a fixed training facility (range proper) is not required.

8. The Fort Drum Excavation Permit Process is automated through the Web-based Excavation Permit Application (WEPA) at <https://mapsrus.drum.army.mil/wepa.html>. This application directs the collection of information and routes permit requests to approving officials. This policy formalizes the requirements and provides guidance on the data to be collected. WEPA is served through the Fort Drum Intranet and is not available on the Internet. Access is limited to Common Access Card (CAC) holders with a Fort Drum installation campus area network account. Project Managers may recommend contractors for CAC and network access in accordance with the requirements of AR 380-49 (Industrial Security Program). Refer to Garrison Policy Memorandum 21-20, Common Access Card Credentialing for Eligible Department of Defense (DoD) and Other Federal Contractors. For contracts valued at less than \$250K, government employees may administratively enter the permit request on behalf of contractors but shall not be named as the requestor on the permit. Any CAC holder with Fort Drum computer network access rights may make a permit request through the website.

9. The following requirements apply to every excavation permit requested on Fort Drum:

a. The UDig NY Program must be contacted in accordance with 16 NYCRR Part 753 before applying for a Fort Drum excavation permit. The ticket number provided by the UDig NY Program is required by WEPA before an excavation permit request may be submitted. For commercial telephone companies, cable TV providers and the Development Authority of the North Country (DANC), UDig NY will contact those entities, mark the construction site, and locate utilities. To create a new ticket call 1-800-

962-7962 or 811 or access through the internet at <https://www.udignyny.org/>. The requestor shall maintain a copy of official communications from the UDig NY Program pertaining to each excavation request.

b. For entities other than those identified in paragraph 9.a, and as stated in 16 NYCRR Part 753, the excavator is responsible for outlining the anticipated excavation area with white markings prior to submitting an excavation permit application. Government utility locators will then locate utilities within the excavation area and up to 15 feet beyond the excavation area. The area drawn by the requestor in its permit request shows the general location of excavation but is not the official excavation area. Requests to locate utilities and mark areas beyond 15' of the excavation limits will be disapproved. The official excavation area will be identified in the approved permit.

c. Fort Drum personnel that approve excavation permits have ten (10) business days from the date of receipt of the request to approve each permit. Excavation is not authorized until after the permit has been issued.

d. Any disapproval of a permit application stops the permit process and requires submission of a new permit application.

e. An emergency excavation permit is available only to the Public Works Operations and Maintenance Division, Public Works Environmental Division and Network Enterprise Center (NEC). An emergency is defined as a break in utility service to a facility on the installation where immediate excavation is required to respond to and mitigate a spill and/or effect repairs. Supervisors of these organizations have unilateral approval authority for excavations outside the normal duty day. Permit application and approval may be retroactive. Verbal approvals shall be noted in subsequent approval documentation.

f. Approved excavation permits are emailed to the requestor by the government PM, or other Government Official if no government PM has been assigned. Excavation permits are considered issued when all required approvals are obtained, and the signed permit is sent to the requestor via e-mail.

g. Excavation permits are valid for starting excavation for thirty (30) calendar days from the date of issuance. The excavation permit remains valid until the work is completed or until work ceases on the site for 30 consecutive calendar days, whichever occurs first. If additional work is required in an area where the permit has expired or work has previously been completed, then another permit shall be requested unless the requestor has maintained utility markings and is granted written authorization by the PM or government designated representative overseeing the project, and it is documented with a new Certification of Verification of Utility Markings (Enclosure 1).

h. Requestors shall hard stake utility markings in all unpaved areas and shall survey and maintain all utility markings in paved and unpaved areas prior to beginning construction activities. Contractors are required to maintain all utility markings through completion of the project. It is the contractor's responsibility to replace all utility markings that are damaged, removed, or destroyed within the limits of the excavation permit boundary. Requestors that fail to maintain utility markings during the period of construction will have their excavation permit revoked, will be required to stop work, and will have to request and receive a new excavation permit prior to resuming work. Government costs for remarking utilities resulting from the contractor's failure to maintain utility markings, expiration of the permit or permit revocation will be charged to the contractor in the form of a contract modification crediting those costs back to the government.

i. Prior to a requestor starting excavation work, the requestor shall schedule a site meeting with the government's Con Rep, or designated representative, as applicable. At the meeting the requestor shall inspect and certify to the government that all utilities for the excavation permit are properly marked. The requestor shall complete and sign the Fort Drum Excavation Permit – Certification of Verification of Utility Markings (Enclosure 1) and forward it to the government representative that witnessed the inspection; if acceptable, the government representative will sign the certification. The completed certification and approved excavation permit will then be issued by the government.

j. The approved excavation permit shall be carried by the requestor/excavator at the dig site and shall be presented when requested by any government representative. Failure to provide an approved excavation permit is a life, health, safety violation and requires the excavator to immediately cease any mechanical excavation until the contractor can demonstrate approval for the excavation.

k. If the proposed work lines are within five (5) feet of utility markings, then the excavator is expected to hand dig or vacuum excavate to verify depth and position of the indicated utilities. Failure to locate the utilities within five (5) feet of either side of where they are indicated shall halt further excavation until the area is cleared by the owner of the utility.

l. The requestor is liable for repairs to any utilities damaged by its work, regardless of the condition or accuracy of above ground markings.

10. WEPA is not a project approval process. An excavation permit shall not be requested unless the project has been approved by the appropriate authority. Housing residents must coordinate with Fort Drum Mountain Community Homes (FDMCH). Soldiers will coordinate with their chain of command. Any contractor questions regarding excavations shall be coordinated through the PM and Contracting Officer

AMIM-DRG-ZA

SUBJECT: Garrison Policy Memorandum 23-11, Excavation Permits

Representative (COR) before applying for an excavation permit. For contracted work, the government contract number shall be provided.

11. The point of contact for this policy is Mr. Richard Bator, Chief, Engineering Plans and Services Division, at (315) 772-7911, or richard.j.bator.civ@army.mil.

2 Encls

1. Fort Drum WEPA Instructions
2. Excavation Permit Application Checklist

MATTHEW R. MYER
Colonel, IN
Garrison Commander

DISTRIBUTION: A

FORT DRUM NY

Web Excavation Permit Application (WEPA) Instructions



© Public Works Fort Drum NY
4205 Po Valley Rd
Fort Drum, NY 13602
Fort Drum IGI&S Team
usarmyftdrumusagigis@army.mil

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Web Excavation Permit Application (WEPA)

WEPA (Web Excavation Permit Application) is an online tool used to place a request for a dig permit on Fort Drum. This tool streamlines the dig permit requesting process and allows for faster approval.

What to Know Before Submission

Prior to requesting a permit via WEPA there are a few things to know and remember. A copy of the final permit must be on site and ready to be presented on demand by any government representative.

UDIG-NY/811

Before requesting a dig permit through WEPA you must contact UDIG-NY/811 in accordance with New York State law. They can be reached at 1-800-272-4480 or online at <https://udigny.org/>

A New York permit id number must be entered before WEPA will allow you to proceed.

Ground Markings

The requestor/contractor will be responsible for marking the area of excavation in white prior to the Fort Drum locator team arrival and be available or onsite during the Fort Drum locating and marking process. The requestor/contractor will also be responsible for maintaining the markings throughout the excavation project.

Timeline

A WEPA request may take up to **10 business days** to process. The final permit will be valid for 30 calendar days. Within that time, you must start the excavation. If additional days are required, you must request a new permit.

Government Contracts

If your permit request is for a government contract, you must submit your contract number and project manager's information on the WEPA form.

WEB EXCAVATION PERMIT APPLICATION (WEPA)

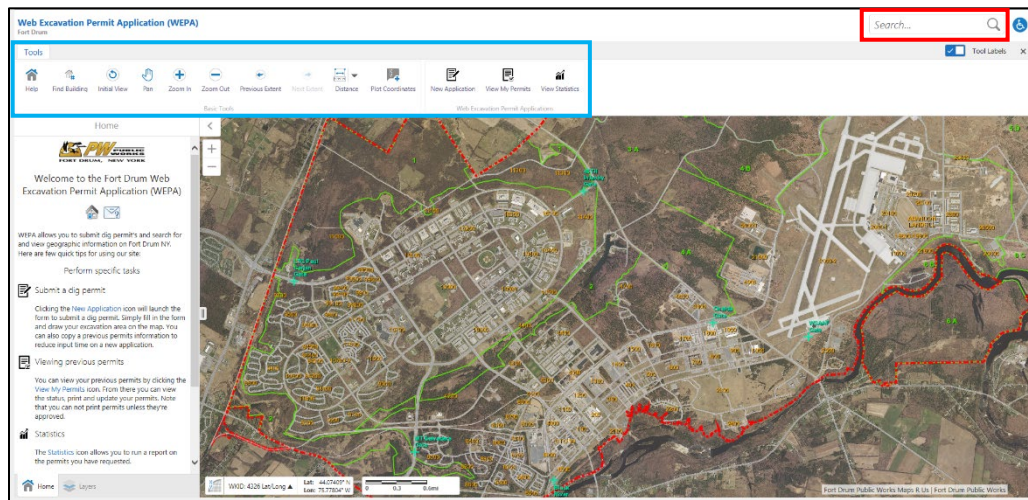
Before getting started

While working through this guide you will find various colored text and boxes to better describe the areas of the form. For example, the following text will be colored the same as the boxes on the image below to easier find the area of the image being described.

Global Search Bar

Toolbar

How to Request a Permit in WEPA

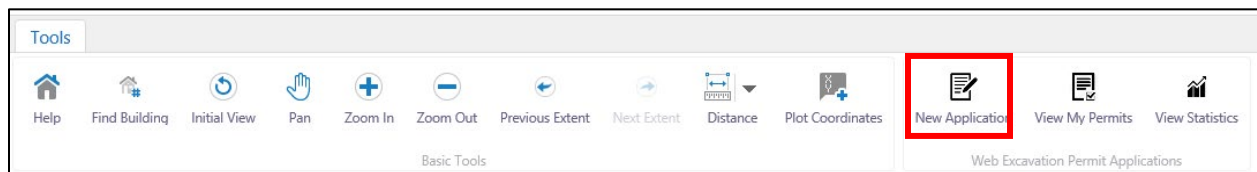


Note: The Web Excavation Permit Application (WEPA) is only available from within the Fort Drum intranet to CAC enrolled users and cannot be reached from outside the installation.

The web address is <https://mapsrus.drum.army.mil/wepa>

To Submit a New Application

Under the “Web Excavation Permit Applications” portion of the “Tools” tab click the “New Application” icon.



Next, please read the customer agreement and check the box if you agree. Click “Continue” to proceed.

Start Permit Request

Your application may take up to 10 business days to process.

The permit is valid for 30 calendar days. Within that time, you must start the excavation. If additional days are required before excavation start, you must request a new permit.

I Agree that:

1. The requestor/contractor will mark area of excavation using white markings prior to Fort Drum inspection team arrival.
2. The requestor/contractor will maintain all markings throughout the excavation project.
3. The party responsible for excavation will be present or available during the Fort Drum inspection and marking process.
4. All requestor and excavating parties will comply with New York State and Federal laws and regulations.

It is your responsibility to check the status of the permit at this website. A complimentary email will be sent once permit is issued.

☐ I Understand

[Close](#) [Continue](#)

New Application

What would you like to do?

☒ New Permit (blank)

☐ New Permit (copy a previous permit)

[Close](#) [Back](#) [Continue](#)

The next step is to select whether this is a “New Permit (Blank)” or “New Permit (Copy a previous permit)”.

The copy-a-previous-permit function will allow you to copy information such as your requestor and excavator information, excavation details, NY Permit ID and excavation boundary to a new permit. You can do this by typing in a **previous permit ID** or clicking the **drop down box** to display your last five permits. This is very handy for re-submitting a previous permit that may have been denied, or needs to be re-submitted after the 30 day timeline. It is also useful in copying the requestor and excavator information so that you don’t have to retype your details each time.

Next, please input your New York permit ID.

Dig Safely New York Ticket ID

Provide Dig Safely New York Ticket ID. The ticket ID is provided by the State of New York and is required prior to completing this application. For more information visit www.digsafelynewyork.com

Ticket: *

[Dig Safely New York website](#)

[Close](#) [Back](#) [Continue](#)

New Application

What would you like to do?

☐ New Permit (blank)

☒ New Permit (copy a previous permit)

Select Previous Permit ID

Permit ID: *

☒ Copy NY Dig Safely ID

☒ Copy Requestor Information

☒ Copy Excavator Information

☒ Copy Excavation Details

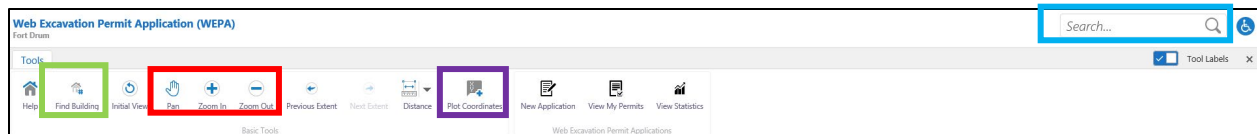
☒ Copy Excavation boundary (Shape geometry)

[Close](#) [Back](#) [Continue](#)

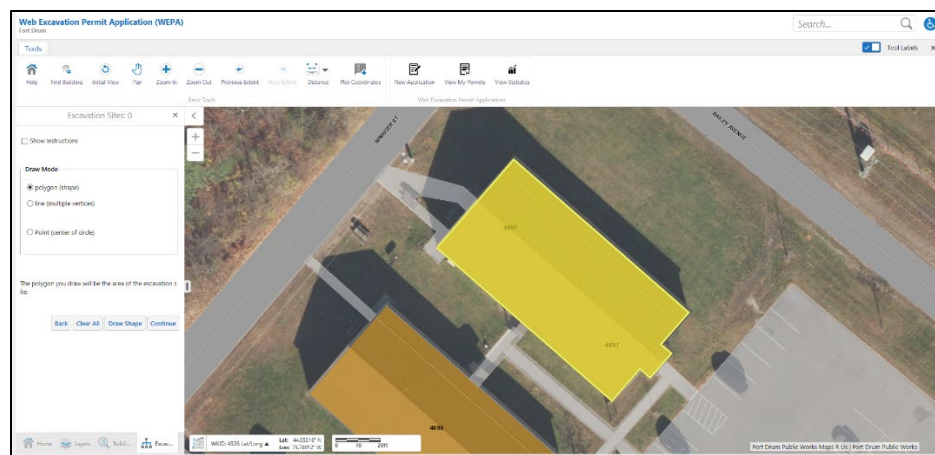
Moving on, please select your permit type.

- Standard- Most permits requires this routing
- Satellite Dish- Special reduced routing for housing
- Emergency- Only to be used during emergency situations and by maintenance shops and Environmental

The next step will be to draw your excavation area on the map. You can use the toolbar tools such as “Pan”, “Zoom In” and “Zoom Out” or the mouse wheel and left mouse click to navigate the map. If you are searching for a building, road, range or training area then you can use the “Find Building” tool, the “Global search bar” or “Plot Coordinates”. For more information on finding locations on the map see the [Searching for Locations](#) section.



Drawing your excavation area-



To create the permit area you may draw a polygon, line or point. Lines and points will require a buffer distance in feet to draw. Typically, the polygon tool will suffice for most inputs.

Select the drawing mode you would like to use and click “Draw Shape”. Draw a polygon on the map that represents the area boundary of the excavation site. Single click for each corner of the polygon and double-click to stop drawing.

Multiple excavation sites may be drawing on the map, allowing you to apply for multiple permits at once. Each shape drawn on the map will be a submitted as a separate permit.

When you have drawn your excavation site click “Draw Shape” again if you are submitting multiple excavation sites or “Continue” if you are done.

Next, please fill out the requestor and excavator information forms. The requestor is the person asking for the excavation permit. **NOTE:** The person transcribing the request is **NOT** the requestor unless they are the party responsible for the excavation.

Your Information (Requestor)

Provide your contact information as the requestor. The primary communication channel that will be used is your email. Other communication channels may be used to determine excavation eligibility.

Personal	Address
First Name: * Clarence	Street Address: 4897 Nininger st
Last Name: * Dugger	City: Fort Drum
Email: * clarence.b.dugger.civ@mail.mil	State: NY
Confirm Email: * 	Zipcode: 13602
Phone: * 315-772-0438	Country: United States of America
Title: GIS ENG Tech	

Company:

Office Name:

[Close](#) [Back](#) [Continue](#)

Excavator Information

Provide contact information for the party conducting the excavation even if it is a third-party.

☐ Same as Requestor

Personal	Address
First Name: * Clarence	Address: 4897 Nininger st
Last Name: * Dugger	City: Fort Drum
Email: * clarence.b.dugger.civ@mail.mil	State: NY
Phone: * 315-772-0438	Zipcode: 13602
Title: GIS ENG Tech	Country: United States of America

Company:

Office Name:

[Close](#) [Back](#) [Continue](#)

Next, provide the excavation specifics. Remember, if your dig permit is associated with a contract you must put the contract number and Project Manager's contact information.

Excavation Specifications

Provide as much detail as possible for the planned excavation to avoid delays during the approval process. (May need to scroll down to see buttons)

Purpose for excavation:

Tool: *
Select tools that will be used

Start Date: *

Completion Date: *

Depth of excavation (ft): *

Fire Alarm/Building Impact: *

Traffic Impact: *

Blasting: *

MCA Project: *

Comments:

*Required

[Back](#) [Continue](#) [Close](#)

Finally, review your application summary, to ensure complete and accurate information. Click "Submit Application" if information is correct otherwise click "back" to make any changes.

Application Summary

Ensure all the information is correct before submitting this application. Electronic documents may be attached to this application after submittal.

Dig Safely ID: 12345-123-123

Permit Type: Standard (Standard permit)

Requestor

FirstName: Clarence

LastName: Dugger

Email: clarence.b.dugger.civ@mail.mil

EmailConfirm: clarence.b.dugger.civ@mail.mil

Phone: 315-772-0438

Title: GIS ENG Tech

StreetAddress: 4897 Nininger st

City: Fort Drum

State: NY

Zipcode: 13602

Country: United States of America

Company:

OfficeName:

Excavator

FirstName: Clarence

LastName: Dugger

Email: clarence.b.dugger.civ@mail.mil

EmailConfirm: clarence.b.dugger.civ@mail.mil

Phone: 315-772-0438

Title: GIS ENG Tech

StreetAddress: 4897 Nininger st

City: Fort Drum

State: NY

Zipcode: 13602

Country: United States of America

Company:

OfficeName:

Permit Information

Purpose: test

Tool: Machine Auger

DateStart: 10/19/2017 10:06:28 AM

DateComplete: 11/18/2017 10:06:28 AM

Depth: 2

FireAlarm: No

Traffic: No

After your application is submitted you will be asked if you would like to submit attachments, this option is available as an edit to your submission. You can submit CAD, PDF and JPG files not to exceed 10MB. Attachments should only be included if necessary in aiding the permit approval process. Click “Ok” to submit Attachments or “Cancel” to skip. Please see the [Managing Attachments](#) section for how to submit an Attachment, after submitting an application.

Attach Document

Would you like to add document attachments to this permit

OK

Cancel

Viewing and Editing Previous Permits

Viewing Previous Permits

To view previously submitted permits click the “**View My Permits**” icon.

Tools

Help

Find Building

Initial View

Pan

Zoom In

Zoom Out

Previous Extent

Next Extent

Distance

Plot Coordinates

New Application

View My Permits

View Statistics

Basic Tools

Web Excavation Permit Applications

The “Search for Permit” form will appear with three choices to search by. You can search by “My Applications”, “Permit ID” and “Requestor Name”.

6

ENCLOSURE 1

Searching by Permit ID and Requestor Name, you simply select the appropriate criteria and type in the permit id or requestor name and click “Search”. Selecting my applications will show your requested permits.

Once the search has been applied you will see the “Permits” box.

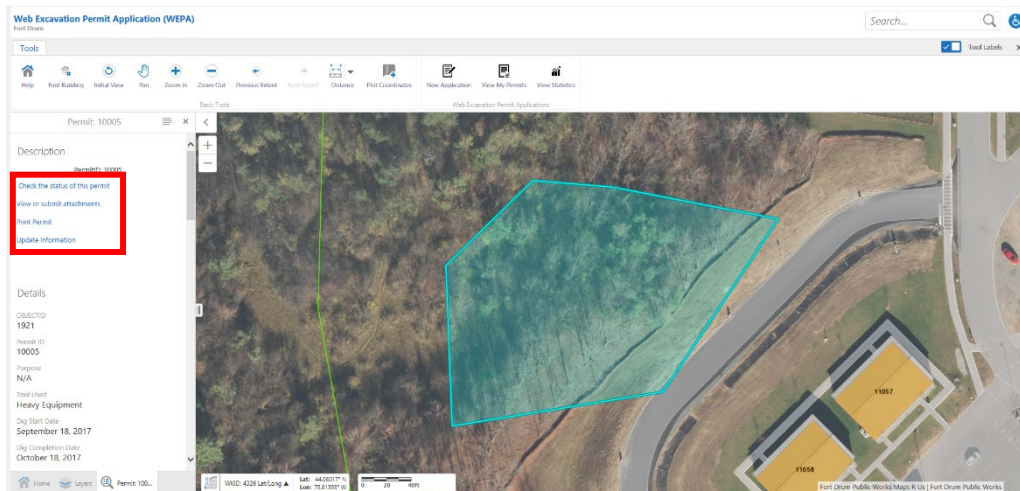
Permits				
Permits are listed based on your search results. When the "Status" of the request is "Approved" you will be allowed to print the permit. Printed permit must be available on site during excavation.				
Requests				
Permit ID	Status	Request Date	Contract Number	Project Number
10000	Approved	2017/10/04	-- (undefined) --	-- (undefined) --
10001	Pending	2017/10/04	-- (undefined) --	-- (undefined) --
10002	Pending	2017/10/04	-- (undefined) --	-- (undefined) --
10003	Pending	2017/10/04	-- (undefined) --	-- (undefined) --

[Print Approved Permit](#) [Select Permit](#) [Status](#) [Sort](#) [Close](#)

From here you can see the permit id, status, request date and other information.

If the permit is pending you can highlight the permit and select “Status” to view which disciplines have approved your permit.

To view the permit area on the map and see the details, select a permit from the list and click “Select Permit”. The map will zoom to and highlight the permit area of the selected permit id.

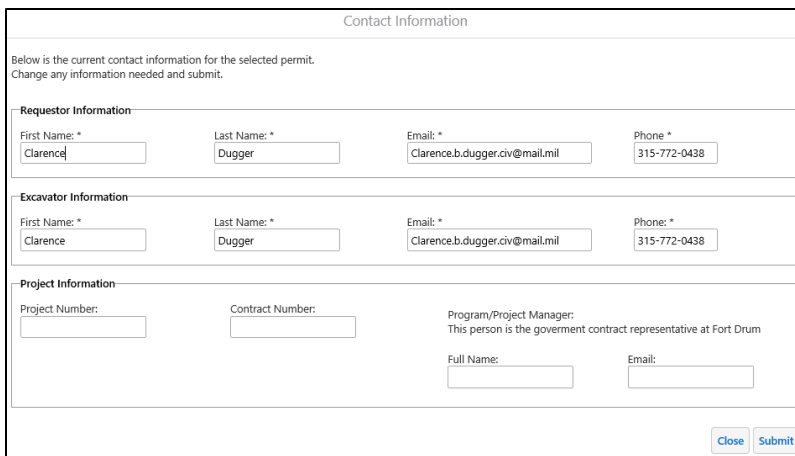


On the left hand side of the screen under “Description” you can view the details for the selected permit. A few options available at this point can be found as **hyperlinks (blue text)**. From here you can also “view the status”, “view or submit attachments”, “print permit” (pending or approved) and “Update Information”. If your permit is not approved yet, you can update the information on the permit. Please see the next section for updating previous permits.

Update Previous Permit

Once a permit is submitted and awaiting approval you can still update the permit information. Updating the excavation boundary is not allowed, you must submit a new application if this is the case.

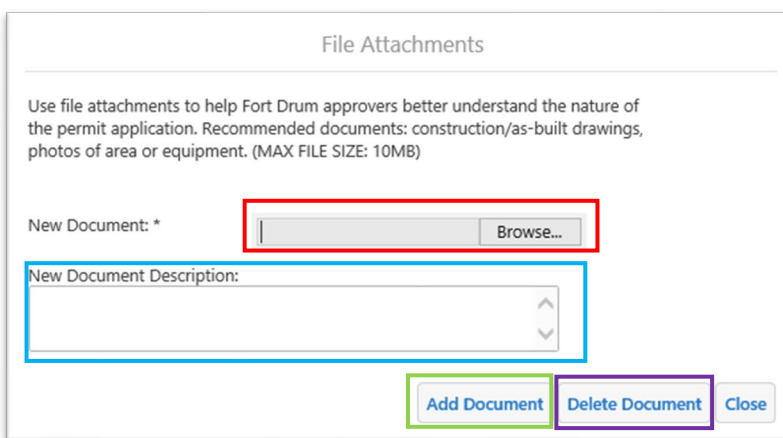
The “Update Information” link allows you to update only the contact information, such as the requestor, excavator or project information.



The "Contact Information" form is divided into three sections: Requestor Information, Excavator Information, and Project Information. Each section contains fields for First Name, Last Name, Email, and Phone. The Requestor and Excavator sections are identical, with the same values entered: First Name: Clarence, Last Name: Dugger, Email: Clarence.b.dugger.civ@mail.mil, and Phone: 315-772-0438. The Project Information section includes fields for Project Number, Contract Number, Program/Project Manager (with a note: "This person is the government contract representative at Fort Drum"), Full Name, and Email. At the bottom right, there are "Close" and "Submit" buttons.

Managing Attachments

After submitting the application select the permit as described in the [Viewing Previous Permits](#) section and click “View or Submit Attachments”.



The "File Attachments" form provides instructions: "Use file attachments to help Fort Drum approvers better understand the nature of the permit application. Recommended documents: construction/as-built drawings, photos of area or equipment. (MAX FILE SIZE: 10MB)". It features a "New Document: *" field with a "Browse..." button, a "New Document Description:" text area, and three buttons at the bottom: "Add Document" (highlighted with a green box), "Delete Document" (highlighted with a purple box), and "Close".

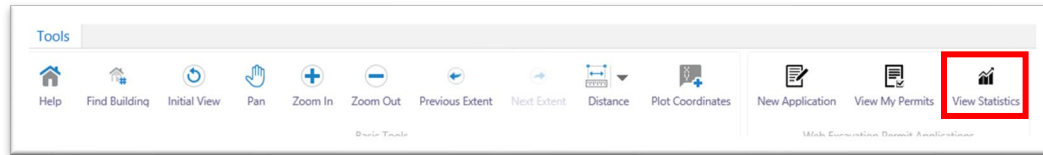
The File Attachments form will launch. To add a new attachment click “Browse” and navigate to the file you would like to attach, highlight the file then click “Open”. Additionally, you can enter a “Document Description”. After you are done attaching the document and entering a description then click “Add Document” to finalize it. Note: files are limited to 10MB in size and the format must be PDF, Photo files (JPG, BMP, PNG etc.) and CAD files (DGN, DWG etc.).

To delete an attachment you will follow the same workflow for submitting except you will highlight the attachment and click “Delete Document”.

To view an attachment select the permit for viewing as described in the [Viewing Previous Permits](#) section and then clicking “View or Submit Attachments”. A hyperlink (blue text) of all submitted attachments will be available. Click the hyperlink of the respective permit you wish to view and it will be opened in a new browser window.

View Statistics

The “View Statistics” workflow in WEPA allows you to run a usage report for a given time period.



To generate a usage report click the “View Statistics” icon. Select a “time period” and then a “Report Type”. Finally, click “Generate Report”.

A screenshot of the 'Statistics' dialog box in WEPA. The dialog has a title bar with a close button. It contains a 'Filter' section with the text 'Show statistics on WEPA usage based on period of time.' Below this is a 'Past: *' label followed by a dropdown menu showing 'Select past time period', which is highlighted with a red box. The 'Report Type' section has two radio buttons: 'Adobe (.pdf)' (selected) and 'Microsoft Excel (.xls)', with this section highlighted by a blue box. At the bottom, there is a 'Report for: *' label followed by a dropdown menu showing 'Requestor'. Below the dropdown are two buttons: 'Generate Report' (highlighted with a green box) and 'Cancel'.

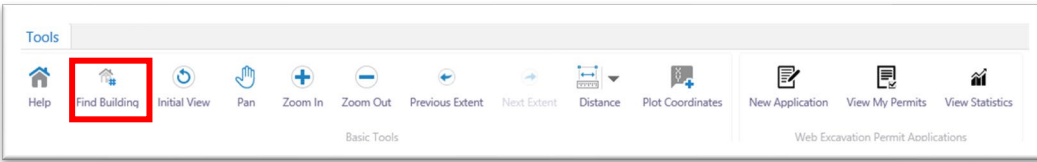
After the report is generated you will see a box stating that your report is ready. Click the hyperlink (Blue text) to open your report in a new window. The report will give you statistics for the amount of permits requested, approved and disapproved, the amount of each type of permit applied for, how many are currently pending and what type of permits the pending ones are.

Searching for Locations

If you are not too familiar with the installation, tools are available to find a particular building, range, training area or a specific area with coordinates to aid you in finding your excavation area on the map.

Finding a Building

To find a building on the installation simply launch the “Find Building” icon.



A form will launch on the left hand side of the screen called “Search Buildings by ID”. Start typing the building number you are looking for, as you type the box will drop down with a filtered list of available numbers that meet your search criteria. Select a number from the list and click “**Search**”. The map will select and zoom to the building.

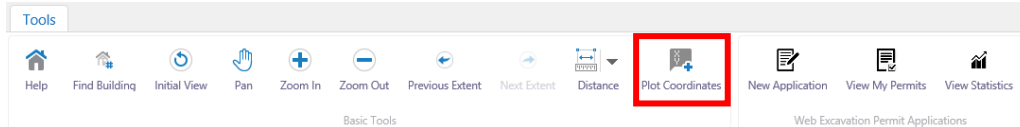
Finding a Range, Training Area or Road

To find a Range, Training Area or Road on the installation you can use the “Global Search” box in the upper right hand corner of the screen.

Simply type in the road name, training area number or range number and a “Search Results” box will appear on the left hand side of screen. Click the corresponding search term from the list of results and the map will zoom it.

Finding a Location by Coordinates

The “**Plot Coordinates**” tool allows you to input coordinates to display on the map by entering them or clicking a specific location on the map.



After clicking the “Plot Coordinates” icon the plot coordinates window will open on the left hand side of the screen. You can change the “**Coordinate System**” under the drop down (available coordinate systems include MGRS, Lat/Long, and NYS Plane). Coordinates can now be typed in to the box and click “**Add**” to plot on the map. Alternatively, you can click the map to plot coordinates. Click the “**Add Another Coordinate**” button at the bottom to add a new input box if they have all been removed. Note: if you type in your own coordinate you must pan or zoom in and out to refresh the map for the coordinate to appear.

 A screenshot of a mobile application window titled 'Plot Coordinates'. At the top, there is a dropdown menu labeled 'Coordinate System:' with 'WKID: 4326 Lat/Long' selected. Below this is a list of coordinates. The first entry, labeled '1', shows 'Lat 44.06121° N' and 'Lon 75.70142° W'. The second entry, labeled '2', has empty input fields for 'Lat' and 'Lon'. To the right of the second entry are 'Add' and 'Cancel' buttons. At the bottom of the window is a button labeled '+ Add Another Coordinate'.

Questions about your permit application or need immediate assistance?

Customer/Agency	Point of Contact	Email	Phone (315)
Family, Morale, Welfare and Recreation (FMWR)	Kim Irey (primary)	kimberley.a.irey.naf@mail.mil	772-3096
Troop Construction	David Huston (primary)	david.m.huston.civ@army.mil	772-5093
Troop Construction (Cantonment)	Division Engineer (primary)		774-2706
Troop Construction (Cantonment)	Rick Medeiros	ricardo.c.medeiros.civ@army.mil	772-1467
Troop Exercises (Range)	Bill Mandigo (primary)	william.b.mandigo4.civ@army.mil	772-5595
Troop Exercises (Range)	Jim P. Moore	james.p.moore.civ@army.mil	772-5413
Fort Drum Mountain Community Homes Resident	Shanna Turpin (primary)	shanna.turpin@fdmch.com	955-6803
Fort Drum Mountain Community Homes Resident	Ed Siebels	ed.siebels@fdmch.com	955-6695
Fort Drum Mountain Community Homes Maintenance Management	Alyson Ranger	alyson.ranger.fdmch.com	955-6823
Lendlease Construction	Randy Lajuett	randy.lajuett@lendlease.com	955-6294
Directorate of Public Works (DPW) Construction Branch	Michael Jones	michael.j.jones6.civ@army.mil	772-8810
Directorate of Public Works (DPW) Engineering	Shane Berghorn	shane.k.berghorn.civ@army.mil	772-0441
Directorate of Public Works (DPW) Master Planning	David Huston	david.m.huston.civ@army.mil	772-5093

Directorate of Public Works (DPW) O&M	Eugene Spencer	Eugene.w.spencer2.civ@army.mil	772-4947
NEC	Lena McConahay	lena.r.mcconahay.civ@army.mil	774-6400
NEC	Lyle Decillis (John)	Lyle.j.decillis.civ@army.mil	772-4000
NEC	Jack Hatcher	Jack.l.hatcher.civ@army.mil	772-9365
MEDAC	Steve Woodard	steven.e.woodard4.civ@army.mil	772-9336
Army Corp of Engineers	Scott Britton (primary)	scott.a.britton.civ@usace.army.mil	774-5200
Army Corp of Engineers	Andrew Mitchell	andrew.j.mitchell@usace.army.mil	772-2041
AAFES	Thea Sarver	sarvert@aafes.com	773-0061
Unable to reach agency representative and need immediate assistance	Cheryl Steiner (primary)	cheryl.l.steiner.civ@army.mil	772-2893
Unable to reach agency representative and need immediate assistance	Larry Ryan	Larry.f.ryan.civ@army.mil	772-5808
Unable to reach agency representative and need immediate assistance	R. Justin Bator, PE	richard.j.bator.civ@army.mil	772-7911

Excavation Permit Application Checklist

Fort Drum Public Works Engineering Plans & Services Division

This checklist does not replace, amend, or subjugate in any manner, Fort Drum's Policy, Procedures, or Authorities for issuance of an Excavation Permit. The intent of this checklist is provide supplemental guidance for submitting applications for Excavation Permits.

Excavation permits are requested through Fort Drum's Web Excavation Permit Application (WEPA) at the link below:

<https://mapsrus.drum.army.mil/wepa>

- ☐ Read the Web Excavation Permit Application (WEPA) Instructions (Attached) and is available on WEPA
- ☐ After receipt of the Notice to Proceed (NTP), mark the area of excavation with white markings.
- ☐ Contact UDig NY at 1-800-962-7962, 811 or online at <https://www.udignyn.org> A Dig Safely New York permit ID number is required before WEPA will allow you to proceed with applying for a Fort Drum Excavation Permit.
- ☐ Complete the online Excavation Permit Application on WEPA. When completing the online application, ensure ALL Contact Information fields are filled in; to include the Project Information Section circled in red.

The screenshot shows the 'Contact Information' form in the WEPA system. The form is titled 'Contact Information' and includes a sub-header: 'Below is the current contact information for the selected permit. Change any information needed and submit.' The form is divided into three main sections: 'Requestor Information', 'Excavator Information', and 'Project Information'. The 'Requestor Information' section contains fields for First Name, Last Name, Email, and Phone. The 'Excavator Information' section contains fields for First Name, Last Name, Email, and Phone. The 'Project Information' section, which is circled in red, contains fields for Project Number, Contract Number, Program/Project Manager (with a note: 'This person is the government contract representative at Fort Drum'), Full Name, and Email. At the bottom right of the form are 'Close' and 'Submit' buttons.

- ☐ When all reviewing disciplines have "approved" the excavation permit, the permit status will change from Pending to Approved. This does not mean that the permit is officially issued. The Status means that the excavation permit is approved to be issued to the contractor. The permit is not issued until signed by a government representative.

Excavation Permit – Certification of Verification of Utility Markings

Fort Drum Public Works
Engineering, Plans & Services Division
Fort Drum, New York 13601

Work Order No: Click here to enter text.

Title: Click here to enter text.

Contractor: Click here to enter text.

Contract No: Click here to enter text. Task Order No: Click here to enter text.

This is to certify that I, Click here to enter text., as an official representative of Click here to enter text., obtained from Fort Drum's Engineering, Plans & Services (EP&S) Division Real Property Branch GIS Section utility mapping for each underground utility and verified onsite on Click here to enter text. that all underground utilities within the area defined under Fort Drum Excavation Permit No. Click here to enter text., dated Click here to enter text., have been marked. Utility markings are consistent with the EP&S Real Property GIS mapping.

Certified by:

Click here to enter text.

Print Name
Contractor/Excavator Duly Authorized Representative

Signature Date

Witnessed by:

Click here to enter text.

Print Name
Government Representative

Signature Date

SECTION 01 14 00

WORK RESTRICTIONS
11/22, CHG 1: 02/23

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

List of Contact Personnel

1.2 SPECIAL SCHEDULING REQUIREMENTS

- a. Have materials, equipment, and personnel required to perform the work at the site prior to the commencement of the work.
- b. The Wheeler-Sack Army Airfield will remain in operation during the entire construction period. The Contractor must conduct his operations so as to cause the least possible interference with normal operations of the activity. Contractor shall coordinate the sequencing order of work with Contracting officer, Fort Drum DPW and Fort Drum Airfield manager before beginning the work and submit for Government review. Government review will be completed within 14 days after the submission.
- c. Permission to interrupt any Activity roads, airfield pavements (including taxiways, runways, and aprons) or utility service must be requested in writing a minimum of 15 calendar days prior to the desired date of interruption.
- d. Communication outages require a 30 day notice prior to the desired interruption.
- e. The work under this contract requires special attention to the scheduling and conduct of the work in connection with existing operations. Identify on the construction schedule each factor which constitutes a potential interruption to operations.

The following conditions apply:

- (1) Closure of airfield pavements.
- (2) Utility Outages (lasting longer than 4-hours).
- (3) Outages of Airfield Lighting Systems - making connections to existing lighting circuits not being replaced under this project.
- (4) Outages of Airfield Lighting Controls.

1.3 CONTRACTOR ACCESS AND USE OF PREMISES

1.3.1 Activity Regulations

Ensure that Contractor personnel employed on the Activity become familiar with and obey Activity regulations including safety, fire, traffic and security regulations. Keep within the limits of the work and avenues of ingress and egress. Wear appropriate personal protective equipment (PPE) in designated areas. Do not enter any restricted areas unless required to do so and until cleared for such entry. Ensure all Contractor equipment, include delivery vehicles, are clearly identified with their company name.

1.3.1.1 Subcontractors and Personnel Contacts

Provide a [list of contact personnel](#) of the Contractor and subcontractors including addresses and telephone numbers for use in the event of an emergency. As changes occur and additional information becomes available, correct and change the information contained in previous lists.

1.3.1.2 Site Security And Vehicle Identification

- a. The Contractor is responsible for the security of the areas within the contract limits. When the Government takes possession of certain areas, the Contractor is responsible for the areas remaining under Contractor control.
- b. The Contract must maintain continual base security with personnel security guard and temporary fences. Wherever there are fence disruptions due to the construction of this project, the Contractor must continually monitor and maintain the fence.
- c. The Contractor is responsible for furnishing an identification required by Fort Drum to each employee. The Contractor must provide an updated list of all employees working on the site. This list must be provided on a monthly basis or when requested by the Contracting Office throughout the duration of this contract. Refer to paragraph 1.3.1.6 Security Badges.
- d. The Contractor is responsible for requiring each employee engaged on the work to display such identification as may be approved and directed by the Contracting Officer. All prescribed identification must immediately be delivered to the Contracting Officer for cancellation upon release of any employees. When required by the Contracting Officer, the Contractor must obtain and submit fingerprints of all persons employed or to be employed on the project. Refer to Paragraph 1.3.1.4 DoD ID Credential.
- e. The Contractor and subcontractors must register all company and private vehicles that will be used in the execution of this contract with the Installation Provost Marshall's office prior to start of work by the Contractor/subcontractor. Entry to the installation requires the following prior coordination with the Contracting Officer or designated representative; current vehicle registration, proof of insurance, valid driver's license for the vehicle driver, and procure identification for other personnel. Contractors/Subcontractors and their employees requiring access to the installation will be required to comply with the installation access control policy/procedures. The government will not be responsible for damages due to delay/stoppages caused by failure to comply.

- f. All vehicles and personnel are subject to search and seizure of contraband and/or unauthorized government property. Contractor vehicles (Contractor-owned and personal), contractor personnel, and their personal property must be subject to searches upon entering or leaving the installation. The search and seizure provisions of AR 190-22 must apply to contractor personnel entering or leaving Fort Drum or activities/installations in the Fort Drum AR 5-9 area of responsibility which require access by Contractor personnel in performance of this contract. Refer to Paragraph 1.3.1.5 Search and Control.

1.3.1.3 Gate Access Forms for entering the Fort Drum Installation

Gate Access is coordinated with the Contracting Officer. Required forms for access will be provided after award.

- a. Contractors fill out and emails the forms to the COR a minimum of 10 days prior to their date access is required.
- b. On the date of arrival, contractors must inprocess at the Ft Drum Inprocessing Center adjacent to the Main Gate (Rt 11/Interstate 781). All individuals are required to have a valid photo Id, and if driving, proof of insurance and registration, a DD Form 3150. A Fort Drum Resident Office representative will sponsor the Contractor. **It is highly recommended that the Contractor contact the COR to coordinate the time they plan to arrive to minimize wait time at the Inprocessing Center.

1.3.1.4 DoD ID Credential

- a. Each employee who requires access to Fort Drum to perform work under any contract, at any tier, must obtain a DoD identification credential to obtain access to Fort Drum. A DoD identification credential is required if the employee requires access to Fort Drum.
- b. The Contractor shall provide information as required by the Fort Drum Chief, Physical Security to enable the Government to conduct a criminal history background check (CHBC) on Contractor employees who require access to Fort Drum. If the CHBC is not adverse, a 30 day temporary access pass will be prepared by the Fort Drum Physical Security Division. The Contractor employee will make an appointment for a Local Access Badge (LAB) if access is needed longer than 30 days. A LAB will be issued for a period of no longer than 365 days. If the CHBC is adverse, the employee will be denied access. If access is denied for any reason, the Contractor's employee may request a waiver from the installation and will not be allowed access until the waiver is requested and approved. Waiver denials may be appealed through the Directorate of Emergency Services. The final determination to allow entry and the issuance of a DoD credential rests with the Garrison Commander or designee, the Director, Emergency Services.
- c. If an employee no longer needs a DoD identification credential for any reason (e.g. quits his/her job or no longer performs under the contract), the Contractor is required to collect the DoD identification credential from the employee and surrender it to the Fort Drum Physical Security Division within four (4) calendar days of such change. If the DoD identification credential cannot be returned within the required time frame for any reason, the Contractor shall

immediately notify the Physical Security Division verbally, followed up in writing the next work day. Email notification is authorized. An employee's inability to obtain entrance to a Government installation because he/she does not have the required identification badge or visitor's pass shall not excuse timely performance of the requirements of this contract. DoD identification credentials are currently issued at the Fort Drum Physical Security Division located in the Fort Drum MP Station. The Government may change the location at which DoD identification credential are issued or returned, with or without advance notice to the contractor.

- d. A DoD identification credential will not be reproduced or copied by the contractor, its subcontractors, or their employees. If an employee's DoD identification credential is lost, stolen, or reproduced, the Contractor shall verbally report the loss, theft, or reproduction to the Fort Drum Physical Security Division on the day such loss, theft, or reproduction is discovered, followed by a written report of the circumstances to the Fort Drum Physical Security Division within two (2) calendar days.
- e. Each Contractor employee shall wear the DoD identification credential while performing work under the contract. The DoD identification credential shall be worn on the upper front of the outer garment unless precluded by OSHA regulation(s). The DoD identification credential will not be used for access to any Government installation except for performance of work under the contract for which is was issued.
- f. The Contractor shall, upon expiration or termination of the contract, collect all DoD identification credentials and turn them into the Fort Drum Physical Security Division.

1.3.1.5 Search And Control

Fort Drum is a controlled access post. All vehicles and personnel are subject to search and seizure of contraband and/or unauthorized Government property. Contractor vehicles (Contractor owned and personal), Contractor employees, and their personal property shall be subject to searches and seizure of contraband and/or unauthorized Government property upon entering or leaving the Installation. The search and seizure provisions of AR 190-16 shall apply to Contractor personnel entering or leaving Fort

1.3.1.6 Security Badges

Contractor personnel shall wear a distinctive Contractor furnished security (ID) badge readily identifying the individual as a Contractor employee. The badge shall be worn in a conspicuous place at all times when performing work under the contract. A listing of issued identification badges shall be furnished to the Contracting Officer prior to contract performance date and updated quarterly to reflect Contractor and Subcontractor personnel changes. Badges shall be worn and displayed on outer clothing and shall contain at a minimum, the following information:

- Photograph of Employee
- Title of Contract
- Name of Worker
- Contractor's Name
- Contractor's Phone Number

1.3.1.7 No Smoking Policy

Smoking is prohibited within and outside of all buildings on installation, except in designated smoking areas. This applies to existing buildings, buildings under construction and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The Contracting Officer will identify designated smoking areas.

1.3.2 Working Hours

Regular working hours will consist of an 8 hour workday, within the period of 7 a.m. and 5 p.m., Monday through Friday, excluding Federal Government holidays.

1.3.3 Work Outside Regular Hours

Work outside regular working hours requires Contracting Officer approval. Make application 15 calendar days prior to such work to allow arrangements to be made by the Government for inspecting the work in progress, giving the specific dates, hours, location, type of work to be performed, contract number and project title. Based on the justification provided, the Contracting Officer may approve work outside regular hours. During periods of darkness, the different parts of the work must be lighted in a manner approved by the Contracting Officer. Make utility cutovers after normal working hours or on Saturdays, Sundays, and Government holidays unless directed otherwise. Contractor must plan for interruptions to work due to airfield operations for a maximum of 24 business days throughout the contract period of performance (may or may not be consecutive days).

1.3.4 Camera Usage and Restriction

Contractor to be briefed on camera usage with Contracting Officer.

1.3.5 Occupied Facilities and Active Operation

The Contractor shall be working around existing facilities and active aircraft operation. These buildings will be active and utilized for the entire period of this contract. The Contractor shall maintain all utilities and systems operational at all times except during outages as approved by Contracting Officer.

The Contractor will be working in and around existing buildings which are occupied. Do not enter the buildings without prior approval of the Contracting Officer.

The existing buildings and their contents must be kept secure at all times. Provide temporary closures as required to maintain security as directed by the Contracting Officer.

Provide dust covers or protective enclosures to protect existing work that remains, and Government material located in existing buildings during the construction period.

Relocate movable furniture approximately 6 feet away from the Contractor's working area as required to perform the work, protect the furniture, and replace the furniture in its original location upon completion of the work. Leave attached equipment in place, and protect it against damage, or temporarily disconnect, relocate, protect, and reinstall it at the

completion of the work.

Contractor to provide 72 hours advance notification for access to Base Operations building. Contractor to notify Bill Munger or Ed Chiasson, WSAAF Base Operations, for access accordingly. Contractor is required to be escorted by Base Operations personnel while working in the building, and must follow direction of the Base Operations personnel.

Contractor to provide 1 week advance notification for access to existing AFLV building. Contractor to notify Robert Chenoweth for access accordingly. Contractor is required to be escorted by WSAAF personnel while working in the existing AFLV, and must follow direction of WSAAF personnel.

Contractor to provide 48 hours advance notification if access is needed to the TDAU room in the weather station. Contractor to notify Todd Shippee or Scott Shattuck for access accordingly. Contractor is required to be escorted by WSAAF personnel while working in the TDAU room, and must follow direction of WSAAF personnel.

Contractor to host a scheduling meeting with Bill Munger to coordinate access and coverage for any work occurring in the ATCT. Provide at least 7 days advance notification. Contractor will be escorted by WSAAF personnel during any work within the ATCT.

Contractor to coordinate with Maurice Mitchell for access to NEC Switch Node 7, a minimum of 7 days in advance. Contractor will be escorted by NEC personnel during any work within Switch Node 7.

COR will provide contact info for all contacts to the Contractor after NTP.

1.3.6 Special Safety Precautions For Airfield Work

When work is ordered the following additional requirements shall apply (Note: it is imperative that these requirements be rigidly adhered to and complete cooperation be given to the Airfield Traffic Controllers in order to ensure the safety of all personnel on or near the airfield.)

All work on the Airfield must be coordinated with the Airfield Manager.

Notify the Airfield Manager through the Contracting Officer's representative 30 days in advance of beginning the work on the Airfield.

- a. All areas undergoing repair shall be barricaded against traffic during the time of construction. Barricades shall be of sufficient weight to prevent them from becoming airborne. Barricades shall be placed across pavement width on both ends of the taxiway, runways, and areas under repair or as directed by the Contracting Officer. ~~Barricades shall be the collapsible type with flashing safety lights and alternating color bands of orange and white stripes.~~ Barricades shall be constructed to be clearly visible for at least 100 yards. Acceptability of barricades with regard to visibility and type of construction shall be determined by the Public Works Construction Representative with approval of the Airfield Facilities Manager.
- b. All motor vehicles and equipment used by the Contractor shall have flashing lights functioning at all times when on the airfield.
- c. The Contractor shall maintain constant radio contact with the Airfield

Control Tower and shall observe proper radio procedures, as required by the Contracting Officer, when operating within radio controlled areas of the airfield. The contractor shall provide a radio for itself, as well as its traffic manager's use. Radios shall operate on tower ground frequency 121.9 MHz, or as Directed by the Airfield Traffic Control Tower.

- d. The Contractor shall be prepared to respond immediately to any directives from the Airfield Control Tower. The Air Traffic Controller has the authority to direct the Contractor to remove personnel, vehicles, and equipment from the work area. If directed by the Air Traffic Controller, the Contractor shall remove all personnel, vehicles, and equipment to a point beyond 500 feet from the centerline of the runway or taxiway with a period of time not to exceed five minutes and shall remain beyond that limit until such time as clearance to return is given by the Control Tower. The Contractor shall be familiar with and shall adhere to light signals from the Control Tower.
- e. Debris and waste materials resulting from construction shall be removed daily. Debris deposited on any active portion of the airfield shall be removed continuously during the course of work. Runways, taxiways, and aprons shall be swept clean of all debris prior to being turned over to the Government for use by aircraft.
- f. The Contractor shall attend a coordination/safety meeting at the airfield operations center, building 2065, prior to the start of any work. This meeting shall be conducted by airfield personnel. The meeting will explain airfield safety requirements. Also, work to be done by the Contractor will be coordinated so as not to interfere with ongoing scheduled aviation activity.

1.3.7 Utility Cutovers and Interruptions

- a. Make utility cutovers and interruptions after normal working hours or on Saturdays, Sundays, and Government holidays. Conform to procedures required in paragraph WORK OUTSIDE REGULAR HOURS. All outage work must be performed continuously, tested and approved by the Contracting Officer at no additional cost to the Government.
- b. Ensure that new utility lines are complete, except for the connection, before interrupting existing service.
- c. Interruption to water, sanitary sewer, storm sewer, telephone service, electric service, air conditioning, heating, fire alarm, and compressed air are considered utility cutovers pursuant to the paragraph WORK OUTSIDE REGULAR HOURS. Such interruptions are further limited to 4 hours. This time limit includes time for deactivation and reactivation.
- d. Operation of Garrison Utilities: The Contractor must not operate nor disturb the setting of control devices in the station utilities system, including water, sewer and, electrical, services. The Government will operate the control devices as required for normal conduct of the work. The Contractor must notify the Contracting Officer giving reasonable advance notice when such operation is required.
- e. All utility outages shall be scheduled by the Contractor and approved

by the Contracting Officer in coordination with Installation requirements. No outage will be approved which will adversely affect the current operation or mission performance. Outages shall only be approved to perform tie-ins of new or temporary utilities to existing lines. No electric or gas outages will be approved during the heating season. "Live"/"hot" utility connections shall be used whenever possible as to minimize end user down time. Contractor shall submit a written work plan and have it approved by the Contracting Officer 30 days prior to planned outage to ensure proper union and Installation notifications. The Contractor shall also be responsible for any repairs or start-up procedures in the affected facilities caused by the outages. The Contractor shall coordinate with the Contracting Officer and representatives of the Installation regarding the work that the Contractor must accomplish in various locations to re-establish the utilities to proper working conditions. The request for the approval of a utility outage shall include, as minimum, the following: description of impacted utility components; time and duration of the outage; Installation areas and systems affected; proof that all preparatory work is complete; proof that all necessary materials, equipment and manpower are in place; utility lines have been verified; and a contingency plan is in place.

- f. The Contractor shall have on-site all materials, equipment, manpower, etc. to continuously perform and complete all work during the approved duration of the outage unless another schedule is specifically approved otherwise by the Contracting Officer. All utilities and systems shall be fully tested and operational prior to the end of the approved outage. Unscheduled outages shall be repaired immediately. Repairs and corrective actions shall proceed continuously in a diligent manner until all services and utilities are restored to their original condition. All work shall be performed to the satisfaction of the Contracting Officer at no additional cost to the Government.

1.4 EXCAVATION PERMITS

- a. The COR shall coordinate with the Contractor to issue necessary excavation and utility clearances. The Contractor shall request such clearances in writing to the COR not less than 10 calendar days prior to the date in which commencement of actual digging is anticipated (Exhibit D, Data Item #D014). The Contractor shall utilize a Web based excavation permit system by inputting data into a government system located in several government facilities. The Contractor will mark/stake the construction area immediately prior to requesting the excavation permit. A date for issuance of the excavation permit shall then be scheduled and the Contractor notified. Fort Drum will locate and mark government owned utilities. It shall be contractor's responsibility to have privately owned utilities located and marked through DIG SAFELY NY (UFPO). The Government shall provide written certification that the Fort Drum owned utilities have been located and marked prior to approval of the Permit. It shall be the Contractor's responsibility to protect all utilities that lie in areas where excavation is required, by hand digging or other measures as necessary. All costs for repair and temporary utilities required due to disruption of service shall be borne by the Contractor.
- b. The Contractor is responsible for maintaining the utility markings after the excavation permit is issued.

See attached Excavation Permit Policy.

1.5 ENDANGERED SPECIES RESTRICTIONS

The implementing instrument for endangered species act consultation requirements for the federally-endangered indiana bat and the federally-threatened northern long-eared bat on Fort Drum military installation is included as an attachment for reference on work restrictions specifically for tree removal activities.

1.6 REQUIRED TRAINING

All Contractor personnel must attend Airfield Ground Safety training before mobilizing to the airfield jobsite. Contractor may also be required to go through the process of obtaining a flight line driver's license if the Contractor requires access outside of the proposed construction site flight line side. Coordinate with Fort Drum Airfield Manager and COR for the required trainings.

1.7 AIRFIELD DRIVER TRAINING

Any Contractor employee (including subcontractors) that will operate a motor vehicle on the airfield must take airfield driver training. Material suppliers (i.e., concrete/asphalt delivery drivers) do not have to have the driver training if they are escorted onto and off from the airfield by a Contractor employee who has taken the airfield driver training. If not escorted, then the material supplier drivers must also take the airfield driver training.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

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**IMPLEMENTING INSTRUMENT FOR ENDANGERED SPECIES ACT
CONSULTATION REQUIREMENTS FOR THE FEDERALLY-
ENDANGERED INDIANA BAT AND THE FEDERALLY-THREATENED
NORTHERN LONG-EARED BAT ON
FORT DRUM MILITARY INSTALLATION**

JANUARY 2018

**Prepared By:
U.S. Army Garrison Fort Drum
Fish & Wildlife Management Program
Environmental Division, Directorate of Public Works**

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HOW TO USE THIS DOCUMENT:

There is duplication throughout this document. This is by design so the user may review specific requirements and measures without referring to other sections of the document that do not apply to their organization. Please identify and apply appropriate measures/requirements pertaining to your specific activities. While all attempts have been made to include all applicable requirements, certain scenarios may not be covered within this document. If there are any questions, please contact Fort Drum's Endangered Species Biologist at 315-772-4999 prior to implementing any action.

MANDATORY TRAINING REQUIREMENT:

The DPW - Environmental Division will provide mandatory training related to this document for all project personnel that are responsible for associated actions conducted on Fort Drum. Personnel responsible for these types of actions should immediately be identified to the Environmental Division. Please contact 772-6108 or 772-4999 to schedule training.

ANNUAL REPORTING REQUIREMENTS:

Information on construction and vegetation management activities (e.g., maintaining right-of-ways for roads, railroad tracks, and utility lines; clearing vegetation from trails and fencelines; mowing open areas; removing individual trees due to hazards or aesthetics, roots growing into underground pipes or branches growing into utility lines; pesticide/herbicide application, etc.) must be tracked and submitted quarterly (March 31, June 30, September 30, December 31) to the Fort Drum Fish and Wildlife Management Program. Mowing of landscaped yards does not need to be reported.

Additionally, electronic maps of the project limits must be submitted to Fort Drum's Fish and Wildlife Management Program in the form of an appropriately formatted and scaled GIS shapefile or DGN prior to the project (as accurate as possible). A final electronic GIS shapefile or DGN "As Built" must be submitted after the completion of the project (accurately surveyed). If only individual trees greater than 3 inches (diameter breast height) are to be removed, then a list of the tree count, diameters, and species must be submitted via the Record of Environmental Consideration (REC) process prior to the action. DPW-Environmental can assist with this requirement as requested.

PLANS AND REGULATIONS:

Any changes to *Fort Drum Regulation 350-6 Assignment and Operational Use of Local Training Areas*; *Fort Drum Regulation 420-7 Environmental Regulation*; *Fort Drum's Integrated Pest Management Plan*; *Fort Drum's Integrated Wildland Fire Management Plan*; *Fort Drum's Utility Design Standards*; and/or any other Plan, Regulation or document referencing these requirements must be coordinated with Fort Drum's Fish and Wildlife Management Program (772-4999) since some Conservation Measures are based on these regulations.

WHAT TO DO IF YOU FIND BATS:

All bats found must be immediately reported to Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636). Do not attempt to handle any live bats, regardless of

condition. If a live bat is found in a building please contact 772-2072 (Pest Control) or 772-4999 (Fort Drum Fish and Wildlife Management) to have it humanely removed.

CONSTRUCTION:

1. If any bats are discovered during the demolition or repair of buildings (to include work such as complete and partial building demo, removal/replacement of roofs, siding, etc.), all work must cease and Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636) must be immediately contacted. If the building has pre-existing known bat colonies, then Fort Drum's Fish and Wildlife Management Program must be contacted before work is to occur. (Currently, only buildings within Fort Drum's historic LeRay Area and the 2500 area are known to contain bat colonies).

2. A Bat Conservation Area (BCA) has been established and will protect Indiana and northern long-eared bat roosting and foraging areas from permanent development within the Cantonment Area and Training Areas 3A and 4A. No permanent facility is to be constructed within the BCA, with the exception of some small facilities (e.g. cabins, picnic shelters, parking lots, a campground, etc.) that may impact no more than 8 ac (3 ha). These actions must be coordinated with Fort Drum's Fish and Wildlife Management Program. The BCA GIS layer is available from the Public Works GIS offices (772-5709 or 772-1502).

3. No new trails may be constructed within the Bat Conservation Area (BCA) without prior coordination with Fort Drum's Fish and Wildlife Management Program (772-4999) and consultation with the US Fish and Wildlife Service.

4. Felling of trees greater than 3 inches (diameter breast height) must occur between October 16- April 15.

5. For bona fide military training emergencies/ immediate construction needs occurring north and east of US Military Highway in the Training Area, no more than 10 forested ac (4 ha; with no more than 5 ac (2.02 ha) per project) may be removed between August 16 - October 15. Prior to tree clearing for training emergencies in August- October, project sites must be surveyed for potential northern long-eared bat habitat. Coordinate with Fort Drum's Fish and Wildlife Management Program (772-4999) prior to initiating this type of project.

6. All roost trees used by female bats, including roosts identified in the future, will be protected from construction for the lifespan of the roost tree. Additionally, a buffer will be placed around all of these roosts to protect the roost from disturbance and to maintain a semblance of a natural environment for Indiana and northern long-eared bats. The size and shape of a buffer will be determined on a case by case basis by Fort Drum's Fish and Wildlife Management Program in consultation with the US Fish and Wildlife Service. Coordinate with Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636) to determine roost locations.

7. Clearing of natural vegetation (e.g., shrubs and trees) less than 3 inches (diameter breast height) must typically occur between August 1 - April 15 to minimize the impact to migratory birds and to maintain foraging areas for bats. The project proponent must ensure coordination with the Fish and Wildlife Management Program, via the REC process or direct contact, prior to clearing of any natural vegetation less than 3 inches (diameter breast height).

8. If any vegetation or tree clearing is required for a given construction project, flagging or signs must be used to demarcate construction limits prior to any construction activities. Flagging must be removed upon completion of the project.

9. New buildings should be appropriately designed and constructed so cracks and crevices are not created, vents are screened, etc. Properly constructed buildings will discourage bats from roosting in buildings, thus minimizing human/bat conflicts in occupied dwellings.

10. Project planning and design should attempt to minimize building footprints by combining infrastructure (i.e. roads, utility lines, etc.) for multiple buildings or by constructing multi-story versus multiple or expanded single story buildings whenever possible.

11. Light minimization measures are to be incorporated for all exterior lighting that may include but are not limited to full cutoffs, reflectors, shields, downward angling of lights, and/or reducing correlated color temperature of fixtures. See Appendix 2 for detailed requirements that must be followed to the maximum extent possible. Deviations from these requirements must be coordinated with Fort Drum's Fish and Wildlife Management Program (772-4999).

12. No night lighting for construction projects is authorized without prior coordination and authorization from the project manager and Fort Drum's Fish and Wildlife Management Program.

13. All construction projects over an acre are required to prepare a sediment and erosion control plan or a storm water pollution prevention plan (SWPPP), which details all erosion and sediment control practices and, when necessary, post-construction storm water management practices.

14. Stormwater management plans should be reviewed with the objective of moving towards integrated infrastructure to reduce the number or completely eliminate the need for stormwater retention ponds and the excessive land use required.

15. All personnel responsible for construction activities will be informed about the need to follow design plans, stay within flagging, minimize impacts to wildlife, and other environmental concerns. This may be done through training, Environmental Protection Plans and/or other contract language.

16. The project proponent must ensure coordination with the Forest Management Program and the Fish and Wildlife Management Program prior to felling trees greater than 3 inches (diameter breast height).

17. If wetland mitigation is required, only areas (both on- and off-post) that have no or minimal (e.g., a few isolated trees within a landscape of open grass or shrubland) tree removal will be recommended for wetland mitigation sites. The exception to this would be restoring or creating forested wetlands.

MECHANICAL VEGETATION REMOVAL AND MANAGEMENT:

1. A Bat Conservation Area (BCA) has been established and will protect Indiana and northern long-eared bat roosting and foraging areas from permanent development within the Cantonment Area and Training Areas 3A and 4A. No permanent facility is to be constructed within the BCA, with the exception of some small facilities (e.g. cabins, picnic shelters, parking lots, a campground, etc.) that may impact no more than 8 ac (3 ha). No more than 50 ac per year (and no more than 25 ac in a contiguous block) will be mechanically managed within the BCA annually. These actions must be coordinated with Fort Drum's Fish and Wildlife Management Program. The BCA GIS layer is available from the Public Works GIS offices (772-5709 or 772-1502).
2. All roost trees used by female bats, including roosts identified in the future, will be protected from construction for the lifespan of the roost tree. Additionally, a buffer will be placed around all of these roosts to protect the roost from disturbance and to maintain a semblance of a natural environment for Indiana and northern long-eared bats. The size and shape of a buffer will be determined on a case by case basis by Fort Drum's Fish and Wildlife Management Program in consultation with the US Fish and Wildlife Service. Coordinate with Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636) to determine roost locations.
3. Felling of trees greater than 3 inches (diameter breast height) must occur between October 16 - April 15, unless a high risk hazard tree has been identified by the Forest Management Program. All potential hazard trees must be assessed by the Forest Management Program (772-5614 or 772-3170). If a hazard tree is determined to be of high risk, is greater than 3 inches (diameter breast height), and needs to be removed between April 16 – October 15, the Fort Drum Fish and Wildlife Management Program must be notified in advance so they may also assess the tree. If appropriate, an emergence survey will be conducted and if no bats are observed, then the hazard tree must be promptly removed.
4. The project proponent must ensure coordination with the Forest Management Program and the Fish and Wildlife Management Program prior to felling trees greater than 3 inches (diameter breast height).
5. Trees that are considered aesthetically unappealing for landscaping purposes may be removed during the tree clearing window (October 16 - April 15) if it is not a known roost tree. These trees are typically trees that have structural defects, do not leaf-out completely, and/or are dead. Because they do not pose an immediate risk to harming people or damaging property, trees that need to be removed for aesthetic reasons are not the same as hazard trees. See # 3 above for a description of hazard tree management.
6. Clearing of natural vegetation (e.g., shrubs and trees) less than 3 inches (diameter breast height) must typically occur between August 1 - April 15 to minimize the impact to migratory birds and to maintain foraging areas for bats. The project proponent must ensure coordination with the Fish and Wildlife Management Program, via the REC process or direct contact, prior to clearing of any natural vegetation less than 3 inches (diameter breast height). Mowing of landscaped yards, which are generally defined as those areas that are adjacent to buildings and/or contain grasses less than 4 inches tall, are not considered for this requirement.

7. Mowing/ vegetation removal will not occur within 100 ft of known roost trees to avoid disturbing roosting bats and maintaining cover around the roosts.

8. If soils are impacted by vegetation clearing, degraded areas should be repaired via actions that may include grading, compacting, seeding, and application of fertilizer, lime, and mulch. This minimizes erosion run-off into waterways, and thus protects water quality and associated invertebrates, which may be prey for Indiana bats.

9. Vegetation management activities should typically avoid delineated water bodies/wetlands. Although there is no formal buffer requirement around wetlands, a 20-30 ft (6-9 m) buffer is typically maintained around identified wetlands as a best management practice. This leads to less impacts to water quality and protects water sources for Indiana bats.

10. No more than 50 ac per year (and no more than 25 ac in a contiguous block) will be mechanically removed within the BCA annually.

LAND CONVERSION (NOT CONSTRUCTION):

1. Bat Conservation Area. Approximately 2,200 ac (890 ha) have been set aside for Indiana bats. This BCA will also provide the same protections to northern long-eared bats. Land conversion will not occur within the BCA without additional consultation with the USFWS.

2. Roost Tree Protection. No female roost trees, including roosts identified in the future, will be felled for the lifespan of the roost, unless there is a human health and safety concern. This includes roost trees in and outside of the BCA.

3. Roost Tree Avoidance. Land conversion activities will not occur within 0.75 mi (1.2 km) of known maternity roost trees located outside the BCA without further consultation with the USFWS. An exception to this requirement would be the forested areas in and immediately adjacent to Wheeler Sack Army Airfield (WSAAF). In order to meet federal regulations for air safety, some of these areas may be converted from forest to grassland for ease of maintenance and to minimize catastrophic bird-aircraft strikes. These areas were originally clearcut in 2005 and contain trees primarily less than 4 inches dbh. They have now regrown to heights that are once again becoming a safety concern. Some areas will be maintained as forest, but will be clearcut approximately every 5-10 years to keep them at the appropriate height. Other areas will be completely converted to grass.

4. No more than a total of 50 ac/year in each category (100 ac total for military training and wildlife habitat benefits) of land conversion will occur in forested areas with > 3 inches dbh trees. This will help to ensure large areas within a contiguous area will not be removed, minimizing the potential to remove a large percentage of unknown roost trees.

5. Time of Year Restriction. A time of year restriction for clearing trees (> 3 inches DBH) has been established to protect roosting bats during non-hibernation seasons. Felling of trees must take place between October 16 - April 15 while most Indiana or northern long-eared bats are not on Fort Drum.

6. No cutting of trees will occur within or along the bed or bank of streams protected under Article 15 of the New York State Environmental Conservation Law unless required to meet specific management goals and only after obtaining a permit from NYSDEC.

7. A minimum of 70 sq ft of residual basal area, all snags, and all live trees > 16 inches DBH that have noticeable cracks, crevices, or exfoliating bark will be retained around all perennial streams and open waterbodies (2 ac or greater in size) on Fort Drum. A perennial stream is defined as having flowing water year-round during a typical year. The water table is located above the stream bed for most of the year. Groundwater is the primary source of water for stream flow. Runoff from rainfall is a supplemental source of water for stream flow. If land conversion treatments are needed that do not meet this conservation measure and that do not have a "no effect" determination, then individual consultation will be required with the USFWS. This buffer protects water quality and provides foraging habitat for Indiana and northern long-eared bats. Indiana bats are known to utilize riparian corridors that have suitable vegetative cover for foraging and for roosting in nearby trees.

8. For annual reporting purposes, the proponent of the land conversion activities will provide shapefiles of converted areas and vegetative cover types pre- and post-conversion (within a scaled map to Fort Drum's Fish and Wildlife Management Program. This information will be used to describe the vegetative cover types and habitat modification on Fort Drum and will be reported annually to the USFWS.

PESTICIDE/HERBICIDE APPLICATION:

1. Only pesticides registered by the Environmental Protection Agency and New York State may be applied, and only in accordance with their label.

2. Aerial application of pesticides in the Bat Conservation Area (BCA) is prohibited. The BCA GIS layer or map is available from the Public Works GIS offices (772-5709 or 772-1502).

3. Other pesticide application within the BCA will be limited to 50 ac per year (no more than 25 ac in a contiguous block) for tow-behind power blowers and 25 ac per year for spot/ground application.

4. Tow-behind power blowers will not be utilized until after August 15 in all forested areas to allow bat pups to reach volancy and exit an area if disturbed by this activity. Deviations from this conservation measure will require further consultation with the USFWS.

5. Pesticides applied from tow-behind power blowers will use drift control additives and will be applied using low pressure to reduce drift and potential swirling motion from the blower. All efforts will be made to not spray more than 10 feet above ground level.

6. Whenever possible, herbicides that have low toxicity to mammals will be utilized with the tow-behind power blowers. Herbicides that may be somewhat toxic to mammals will be mixed and applied at a rate that should minimize any potential exposure concerns.

7. Aerial application of pesticides outside the BCA must occur between the hours of sunrise and 1 hour before sunset.

8. Application of pesticides from ground-mounted vehicles (trucks, tractors, atvs, etc.) that spray chemicals directly onto the ground and do not result in broad dispersal will be conducted at least 100 ft (30 m) away from known roost trees (including roosts identified in the future) and 250 ft (76 m) from known primary roosts. Pesticides applied from ground mounted vehicles will use drift control additives and droplet sizes appropriate for reducing drift. Coordinate with Fort Drum's Fish and Wildlife Management Program to determine known roost locations. Location-specific applications (i.e., hatchet injections of trees, individual application to specific plants) may be used within 100-250 ft (30-76 m) of known roosts.

9. Application of pesticides that result in broad dispersal (e.g., tow behind power blowers) will be conducted at least 250 ft (76 m) away from known roost trees (including roosts identified in the future) and 500 ft (152 m) from known primary roosts. Pesticides will be applied between sunrise and one hour before sunset.

10. Pesticides will not be applied outdoors when the wind speed exceeds 10 mi/hr for all ground applications except power mist blowers. Pesticides applied via power mist blower will only be applied with wind speeds 8 mi/hr or less. This is to reduce the risk of pesticide drift, which could impact water quality or non-target areas. Care will be taken to make sure that any spray drift is kept away from non-target areas and individuals.

11. Areas where herbicide will be applied aerially must be delineated by painted boundaries or other markers on the ground.

12. Aerial application will utilize helicopters and employ large droplet technology through special nozzles on drop tubes to ensure good precision and accuracy in application and that the herbicide stays on target.

13. Pesticides applied aerially will only be applied with wind speeds 8 mi/hr or less.

14. Pesticides will not be applied to any protected wetlands, streams, or other waters of NY State without obtaining the appropriate permits. Pesticides will not be applied to any created wetland mitigation sites without coordination with Fort Drum's Wetland Management office. Only herbicide approved for aquatic areas will be used in any other wetland areas that may be over-sprayed.

15. If a bat colony is found roosting in a building, then insecticides are to be used sparingly and no foggers are to be used.

16. For each pesticide application, Applicators will utilize the Pesticide Usage Database that is available through the Natural Resources Branch or the Pest Control office to report applicable information about: who applied the pesticide; what and how much was applied; where it was applied; why it was applied; and when it was applied. Maps of the treated areas may also be required for annual reporting purposes to the USFWS. For pesticides applied indoors or immediately along the exterior of the building, no map would be required.

SILVICULTURE AND FOREST MANAGEMENT:

1. A Bat Conservation Area (BCA) has been established and will protect Indiana and northern long-eared bat roosting and foraging areas from permanent development and habitat

loss within the Cantonment Area and Training Areas 3A and 4A. No timber harvests will occur within the BCA until an appropriate management plan has been developed and consulted on with the US Fish and Wildlife Service. If there is a bona fide need for timber harvesting in the BCA, consultation will have to be reinitiated. The BCA GIS layer is available from the Public Works GIS offices (772-5709 or 772-1502).

2. All roost trees used by female bats, including roosts identified in the future, will be protected for the lifespan of the roost tree. Additionally, a buffer will be placed around all of these roosts to protect the roost from disturbance and to maintain a semblance of a natural environment for Indiana and northern long-eared bats. The size and shape of a buffer will be determined on a case by case basis by Fort Drum's Fish and Wildlife Management Program in consultation with the US Fish and Wildlife Service. Coordinate with Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636) to determine roost locations.

3. Clearcutting and overstory tree removal cannot occur within 0.75 mi (1.2 km) of known maternity roost trees located outside the BCA without further consultation with the US Fish and Wildlife Service. An exception to this requirement is a small number of small forested patches (ranging from ~5-15 acres) that will be clearcut at or near WSAAF to meet federal regulations for air safety. The majority of these patches contain trees primarily less than 4 inches dbh. They will be maintained as forest, but will be clearcut approximately every 5-10 years to keep them at the appropriate height. Selective thinning may not occur within one tree height of the known roost tree to minimize the risk of accidentally felling a known maternity roost during the non-hibernation season. Tree height is based on the average height of the stand (~80 ft (24 m)) surrounding the roost tree. For selective thinning harvests within 0.75 mi (1.2 km) of a known maternity roost, all snags and live trees >16 inches (diameter breast height) that have noticeable cracks, crevices, or exfoliating bark are to be retained. Further consultation will be needed with the USFWS for timber harvests that do not follow this conservation measure.

4. Clearing of natural vegetation (e.g., shrubs and trees) less than 3 inches (diameter breast height) should typically occur between August 1 - April 15 to minimize the impact to migratory birds and to maintain foraging areas for bats. The project proponent must ensure coordination with the Fish and Wildlife Management Program, via the REC process or direct contact, prior to clearing any natural vegetation less than 3 inches (diameter breast height)

5. Felling of trees greater than 3 inches (diameter breast height) must occur between October 16 - April 15, unless there a high risk hazard tree has been identified by the Forest Management Program. If a hazard tree is determined to be of high risk, is greater than 3 inches (diameter breast height), and needs to be removed between April 16 – October 15, the Fort Drum Fish and Wildlife Management Program must be notified in advance so they may also assess the tree. If appropriate, an emergence survey will be conducted and if no bats are observed, then the hazard tree must be promptly removed.

6. Felling of standing trees for firewood harvest must occur between October 16 - April 15. The known primary Indiana bat roosting areas (those areas behind Guthrie Clinic and Cool Road) have been made off limits to firewood cutting during April 16 – October 15. Firewood may be removed between October 16 – April 15. Although firewood harvest only removes trees that are lying on the ground, this restriction will help avoid any associated noise or disturbance in the roosting areas from chainsaws and/or tractors used in the harvest of the wood.

7. All snags will be left in silvicultural treatments unless there is a safety concern for the contractor or the military units training in the stands (e.g., maneuver corridors), or unless the treatment is a salvage harvest or clearcut. Snags should be distributed and retained throughout the landscape.

8. No trees are to be cut within or along the bed or bank of streams protected under Article 15 of the New York State Environmental Conservation Law unless required to meet specific management goals and only after obtaining a permit from New York State Department of Environmental Conservation (NYSDEC).

9. Vegetation management activities should typically avoid delineated water bodies/wetlands. Although there is no formal buffer requirement around wetlands, a 20-30 ft (6-9 m) buffer is typically maintained around identified wetlands as a best management practice. This leads to less impacts to water quality and protects water sources for Indiana bats.

10. A minimum of 70 sq ft of residual basal area, all snags, and all live trees greater than 16 inches (diameter breast height) that have noticeable cracks, crevices, or exfoliating bark must be retained around all perennial streams and open waterbodies (2 ac or greater in size) on Fort Drum. A perennial stream is defined as having flowing water year-round during a typical year.

11. Whenever possible, new log landings should be constructed at least 200 ft (61 m) from water bodies and wetlands.

12. Spill kits and oil absorbent mats must be present on log landings in case of fuel, lubricant or hydraulic fluid spills or leaks.

13. If soils are impacted by vegetation clearing, degraded areas should be repaired via actions that may include grading, compacting, seeding, and application of fertilizer, lime, and mulch.

14. Where possible, skid trail grade needs to be maintained at less than 15%. Where higher grade is unavoidable, the grade should be broken, drainage structures should be installed, and soil stabilization practices should be used where needed to minimize runoff and erosion.

15. Debarking and other damage to residual trees should be minimized wherever possible.

16. Stream crossings should only be used when absolutely necessary.

17. Streams should be crossed by the most direct route.

18. Ruts will be filled in, and water bars and erosion barriers should be installed to prevent or minimize erosion and sedimentation from roads, skid trails and log landings.

19. Erosion control measures need to be inspected within 24 hours after a rain event and checked once per week. Erosion controls should be maintained or removed as needed.

20. No machinery is to be operated in streams protected under Article 15 of the New York State Environmental Conservation Law without first obtaining a permit from NYSDEC.

21. During hardwood removals, dead or dying oak trees that may have been typically removed from the stand should be left in the targeted units. This is limited to areas that receive large amounts of sunlight during the day (e.g. the edge of the stand, near an opening within the stand, etc.) to provide roost trees for Indiana bats and other wildlife.

22. Whenever possible, a percentage of suitable live trees (i.e., trees that look as if they have the potential to develop into future snags) should be retained, so cavities appropriate for wildlife may develop and for future snag recruitment. Suitable trees should be long lived hardwoods >15 inches (diameter breast height) and have the greatest potential to develop cavities. In wetland areas 10 ac (4 ha) or larger with open water and shorelines greater than 30 m apart, 20 suitable trees should be left for every 50 ac (20 ha) harvested within 0.5 mi (0.8 km) of wetlands.

23. When possible, unique forest openings (e.g. patch cuts of aspen varying from 1-10 ac in size removed from the stand) should be provided.

VERTEBRATE PEST CONTROL:

1. All bats found must be immediately reported to Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636).

2. If any colonies of bats are found in structures and there is a requirement to remove/exclude them, any actions should only be done through a systematic process and after coordination with Fort Drum's Fish and Wildlife Management.

3. No lethal control methods are permitted for bats unless there is a suspected human health risk for exposure to rabies or other disease. If individual bats are in buildings and there is no evidence of maternity use, then all efforts must be made to safely capture and release individual bats. Or, bats may be excluded by establishing one-way valves over the roost's exit (if feasible). Exclusions may only be done from August - early May.

4. Sealing cracks and crevices in buildings must also be done during the late fall, winter, or early spring.

5. No adhesive traps used for rodents or insects are to be placed in such a manner that they could capture bats—glue traps are not to be placed in any crawl space or attic compartment within buildings or in areas where bats are known to occur.

MILITARY TRAINING:

1. A Bat Conservation Area (BCA) has been established and will protect Indiana and northern long-eared bat roosting and foraging areas from permanent development and habitat loss within the Cantonment Area and Training Areas 3A and 4A. No permanent facility is to be constructed within the BCA. Temporary facilities for training purposes may be constructed within the BCA with prior coordination with Fort Drum's Fish and Wildlife Management Program (772-4999). The BCA GIS layer is available from the Public Works GIS offices (772-5709 or 772-1502).

2. No tree felling will occur in the BCA for training purposes. If tree felling must occur outside the BCA, prior coordination and approval by Fort Drum's Forest Management and Fish and Wildlife Management Programs must occur. Additionally, tree felling will be in accordance with Construction and Vegetation Removal Requirements above.

3. No new trails may be constructed within the Bat Conservation Area (BCA) without prior coordination with Fort Drum's Fish and Wildlife Program (772-4999) and consultation with the US Fish and Wildlife Service.

4. Felling of trees greater than 3 inches (diameter breast height) must occur between October 16 - April 15, unless there is a bona fide training emergency or a high risk hazard tree has been identified by the Forest Management Program.

5. For bona fide military training emergencies/ immediate construction needs occurring north and east of US Military Highway in the Training Area, no more than 10 forested ac (4 ha; with no more than 5 ac (2.02 ha) per project) may be removed between August 16 - October 15. Prior to tree clearing for training emergencies in August- October, project sites must be surveyed for potential northern long-eared bat habitat. Coordinate with Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636) prior to initiating this type of project.

6. All roost trees used by female bats, including roosts identified in the future, will be protected for the lifespan of the roost tree. Additionally, a buffer will be placed around all of these roosts to protect the roost from disturbance and to maintain a semblance of a natural environment for Indiana and northern long-eared bats. The size and shape of a buffer will be determined on a case by case basis by Fort Drum's Fish and Wildlife Management Program in consultation with the US Fish and Wildlife Service. Coordinate with Fort Drum's Fish and Wildlife Management Program (772-4999) to determine roost locations.

7. All trees that are believed to be hazards need to be assessed by the Forest Management Program (772-5614 or 772-3170). They will determine whether or not the trees are truly a hazard, and it will then be determined when the trees need to be removed. If it is determined to be of high risk, is greater than 4 inches (diameter breast height), and needs to be removed between April 16- October 15, the Fort Drum Fish and Wildlife Management Program must be notified in advance so they may also assess the tree. If appropriate, an emergence survey will be conducted and if no bats are observed, then the hazard tree must be promptly removed.

8. No Category 1 smoke operation will be conducted within 1,000 m of the installation boundary, public roads, Cantonment Area, ammunition supply point or WSAAF in accordance with *Fort Drum Regulation 350-4 Range Regulation* and *Fort Drum Regulation 350-6 Assignment and Operational Use of Local Training Areas (LTAs)*.

9. In the Training Area, Category 1 smoke and obscurants can only be used greater than 330 ft (100 m) from known Indiana or northern long-eared bat maternity roost areas (including roosts identified in the future) between April 16 – October 15. Coordinate with Fort Drum's Fish and Wildlife Management Program (772-4999) to determine roost locations. Training missions will be aware of maternity areas via the NEPA process and will be directed to avoid these areas. The use of Category 1 smoke and obscurants must also be rotated among training areas to minimize impacts to any one area.

10. The use of Category 2 smoke, to include colored smoke, smoke grenades, and smoke pots (aka, pyrotechnics) may be used in the Training Areas at any time within 1,000 m of the installation boundary, but will not be used within (330 ft) 100 m of any known Indiana or northern long-eared bat roost areas between April 16 - October 15.

11. Category 2 smoke may not be used within 100 m of any forested areas within the LTAs between April 16 - October 15. Approval from Range Control and NEPA review (via the REC process) is required prior to any use of Category 2 smoke, and these reviews will help ensure that Category 2 smoke use is in accordance with this conservation measure.

12. Category 2 smoke may be periodically used at three mobile MOUTs within the LTAs during April 15 - October 15. Only infrequent use of colored smoke is expected to be used in the mobile MOUTs. With the exception of the Category 2 colored smoke used at the mobile MOUTS, no other smoke or obscurant may be used in the BCA.

13. Vehicular traffic is restricted to open grassy areas and within easy access of the road in the Local Training Areas in accordance with *Fort Drum Regulation 350-6 Assignment and Operational Use of Local Training Areas*. Vehicles are not permitted to cross streams, ditches, wetlands, or dense vegetation in order to reach grassy areas without prior NEPA review (via the REC process).

14. Petroleum, Oil, and Lubricant operations are prohibited in the Local Training Areas in accordance with *Fort Drum Regulation 350-6 Assignment and Operational Use of Local Training Areas*.

15. Fort Drum must abide by the most up-to-date Fort Drum Integrated Wildland Fire Management Plan, unless training under special circumstances that are approved by the commander. Military activities that may spark fires are not to be conducted during moderate to high danger ratings in order to prevent unintentional wildfires.

RECREATION:

1. The current skeet range is located adjacent to the Bat Conservation Area (BCA). Shooting at this facility sends fields of fire over known fall, summer, and assumed spring foraging locations for Indiana bats. From April 15 - October 15 the skeet range must not be operated between 1 hour before sunset to 30 minutes after sunrise to prevent the accidental shooting of an Indiana or northern long-eared bat. The skeet range may be operated during daylight hours from 30 minutes after sunrise to 1 hour before sunset.

APPENDIX 1: LANGUAGE FOR CONTRACTS, SOWs, RFPs, EPPs, etc:

1. The federally-endangered Indiana bat and the federally-threatened northern long eared bat are present on Fort Drum and must be considered in all actions. Actions on Fort Drum, must be in accordance with requirements negotiated with the US Fish & Wildlife Service under Section 7 of the Endangered Species Act. Projects involving ANY habitat modification (e.g., vegetation removal) must be addressed by Fort Drum's Fish and Wildlife Management Program. In general, no trees greater than 3 inches (diameter breast height) may be cut from 16 April - 15 October to avoid any direct adverse effects to these bats. Clearing of natural vegetation (e.g., shrubs and trees) less than 3 inches (diameter breast height) must typically occur between August 1 - April 15 to minimize the impact to migratory birds and to maintain foraging areas for bats. The contractor (and all subcontractors) shall be responsible for coordination with Fort Drum's Fish & Wildlife Management Program to ensure compliance with the Endangered Species Act and all other applicable laws and regulations pertaining to wildlife. Overall the contractor (and all subcontractors) shall minimize interference with, disturbance to, and damage to plants, fish, wildlife, and their habitats on and adjacent to the project area.
2. Flagging or signs must be used to demarcate construction limits of a given project prior to any construction activity, vegetation, and/or tree clearing. Clearing must only occur within the flagged area, and flagging must be removed upon completion of the project.
3. If any bats are discovered during the demolition or repair of buildings (to include work such as complete and partial building demolition, removal/replacement of roofs, siding, etc.), all work must cease and Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636) must be immediately contacted. If the building has pre-existing known bat colonies, then Fort Drum's Fish and Wildlife Management Program must be contacted before demolition is to occur. (At this time, the only structures known to contain bat colonies are within the historic LeRay Area and the 2500 area).
4. All bats found must be immediately reported to Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636). Do not attempt to handle any live bats, regardless of condition. If a live bat is found in a building please contact 772-2072 or 772-4999.
5. Light minimization measures are to be incorporated for all exterior lighting that may include but are not limited to full cutoffs, reflectors, shields, downward angling of lights and/or reduction of correlated color temperatures to reduce blue-rich, white light. See Appendix 2 for detailed requirements that must be followed to the maximum extent possible.
6. No night lighting for construction projects is authorized without prior coordination and authorization from the project manager and Fort Drum's Fish and Wildlife Management Program.
7. Project planning and design should attempt to minimize building footprints by combining infrastructure (i.e. roads, utility lines, etc.) for multiple buildings or by constructing multi-story versus multiple or expanded single story buildings whenever possible.

Appendix 2. Outdoor Lighting Minimization Measures.

Purpose

The purpose of the Fort Drum Outdoor Lighting Guidelines is to regulate outdoor lighting in order to reduce or prevent light pollution. This means to the extent reasonably possible the reduction or prevention of glare and light trespass, the conservation of energy, and promotion of safety and security. These Guidelines will ensure appropriate outdoor lighting in compliance with the Endangered Species Act and in accordance with the Fort Drum's Army Strategic Plan for Sustainability.

Definitions

- a. Fixture Height: height of the fixture shall be the vertical distance from the ground directly below the centerline of the fixture to the lowest direct light emitting part of the fixture.
- b. Foot-candles: a unit of illumination of a surface that is equal to one lumen per square foot. For the purposes of these regulations, foot-candles shall be measured at a height of 3 ft. above finished grade.
- c. Fully Shielded Light: light fixtures shielded or constructed so that no light rays are directly emitted by the installed fixture at angles above the horizontal plane as certified by a photometric test report. The fixture must also be properly installed to effectively down direct light in order to conform with the definition.
- d. Light Trespass: the shining of light produced by a light fixture beyond the boundaries of the property on which it is located.
- e. Lumen: the unit of luminous flux, the total amount of light falling uniformly on or passing through an area of 1 square foot, each of which is 1 foot from a 1-candela source, yielding an illuminance of 1 foot candle at that distance (the output of lamps and bulbs is customarily measured in lumens, a common 100 watt incandescent light bulb, for example, having an output less than 1,800 lumens).
- f. Point Light Source: the exact place from which illumination is produced (i.e., a light bulb filament or discharge capsule).
- g. Sag-lens or Drop-lens: A clear or prismatic refracting lens that extends below the lowest opaque portion of a light fixture.

Applicability

All outdoor lighting fixtures installed, retro-fitted, or replaced on Fort Drum property shall comply with these regulations. These regulations do not apply to interior lighting.

Exemptions

The following are exempt from the provisions of these guidelines:

- a. Traffic control signals and devices.
- b. Temporary emergency lighting (i.e., fire, police, repair workers).
- c. Moving vehicle lights.
- d. Navigation lights (i.e., airports, heliports, radio/television towers).
- e. Seasonal decorations with individual lights in place no longer than 60 days.
- f. Lighting for flags. Efforts should be made in these areas to minimize sky glow and light trespass whenever feasible.
- g. Sports field outdoor lighting (i.e. ball fields, football, soccer, ice rink, etc.). Sports outdoor lighting is to be turned off when a sporting event is not occurring.
- h. Other special situations for temporary or periodic events (i.e. fairs, festivals, carnivals, night-time construction).
- i. Security lights of any wattage that are controlled by a motion-sensor switch and which do not remain on longer than 10 minutes after activation.
- j. Access points, Army Supply points, or other high security areas subject to AR 190-11 or TM-8-583-2. Efforts should be made in these areas to minimize sky glow and light trespass whenever feasible.

Additional exemptions may be provided after coordination with Fort Drum's Fish and Wildlife Management Program.

General Standards

All building exterior lighting and site lighting shall be at a minimum in accordance with these requirements and/or the most recent Fort Drum Utility Design Standards. A Professional Engineer must review any lighting plan in and ensure it is sound and meets minimization requirements. The following general standards shall apply to all outdoor lighting installed, retrofitted, or replaced on Fort Drum, which is not exempted above :

- a. Outdoor lighting must be hooded, fully shielded (i.e. full cutoff fixtures), and/or aimed downward. Outdoor lighting used to illuminate parking spaces, driveways, maneuvering areas, or buildings shall conform to the definition for "fully shielded light fixtures" and be designed, arranged and screened so that the point light source shall not be visible from adjoining lots (i.e. woodlands) or streets.

- b. The intensity of light within a site shall not exceed two (2) footcandles at any property line, edge of pavement, or road. There shall be no or minimal measureable light output behind the light pole.
- c. The hood or shield must mask the direct horizontal surface of the light source. The light must be aimed to insure that the illumination is only pointing downward onto the ground surface, with no escaping light permitted to contribute to sky glow by shining upward into the sky.
- d. Any bright light shining onto adjacent properties (i.e. woodlands) or streets which would result in a nuisance glare or a disabling glare shall not be permitted. Light trespass beyond property boundaries or above the horizontal plane shall be considered non-compliant.
- e. Existing fixtures may be adapted to comply with these guidelines by adding a properly designed hood or shield, or by pointing any upward-mounted, shielded fixture downward onto the ground surface.
- f. All outdoor lighting fixtures shall be designed, installed, located and maintained such that nuisance glare onto adjacent properties (i.e. woodlands) or streets shall be minimized and all direct illumination kept within the boundaries of a building's property.
- g. Accent lighting shall be directed downward onto the building or object and not toward the sky or onto adjacent properties (i.e. woodlands). Direct light emissions shall not be visible above the roof line or beyond the building edge.
- h. Spotighting on landscaping and foliage shall be limited to 150 watts (2220 lumens output) and lighting is to be angled downwards. The lamp shall be fully shielded and not create disabling or nuisance glare.
- i. No sag-lens or drop-lens are to be used.
- j. LED light fixtures shall be utilized to the maximum extent practicable and in accordance with Fort Drum's latest Utility Standards. These fixtures shall be long life, coupled with high efficient drivers. LED lights shall incorporate measures to reduce blue-rich white light output. These measures will be coordinated with Fort Drum's electric shop and Natural Resources Branch prior to implementation, but could include things such as targeting wavelengths between 550-650 nm, and/or reducing the correlated color temperature of the fixture to 3000K or lower. A map is available that designates areas in which utilizing lower correlated color temperatures is required. This can be viewed through Fort Drum's Fish and Wildlife Management Program and/or the electric shop.

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS
05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 52.2 (2017) Method of Testing General
Ventilation Air-Cleaning Devices for
Removal Efficiency by Particle Size

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.34 (2021) Protection of the Public on or
Adjacent to Construction Sites

ANSI/ASSP A10.44 (2020) Control of Energy Sources
(Lockout/Tagout) for Construction and
Demolition Operations

ASTM INTERNATIONAL (ASTM)

ASTM D6245 (2012) Using Indoor Carbon Dioxide
Concentrations to Evaluate Indoor Air
Quality and Ventilation

ASTM D6345 (2010) Standard Guide for Selection of
Methods for Active, Integrative Sampling
of Volatile Organic Compounds in Air

ASTM F855 (2020) Standard Specifications for
Temporary Protective Grounds to Be Used on
De-energized Electric Power Lines and
Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048 (2016) Guide for Protective Grounding of
Power Lines

IEEE C2 (2023) National Electrical Safety Code

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1 (2014; R 2019) American National Standard
for Industrial Head Protection

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 51B	(2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2023) National Electrical Code
NFPA 70E	(2024) Standard for Electrical Safety in the Workplace
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations
NFPA 306	(2024) Standard for the Control of Gas Hazards on Vessels

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

ANSI/SMACNA 008	(2007) IAQ Guidelines for Occupied Buildings Under Construction, 2nd Edition
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U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements
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U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 20	Standards for Protection Against Radiation
29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1926	Safety and Health Regulations for Construction
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further clarification is needed, contact the Contracting Officer.

1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health

(SOH) program according to the Contract, EM 385-1-1, applicable federal, state, and local requirements.

1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1.

1.2.1.2 Level Two SSHO

A designated employee with Level Two SSHO responsibility that meets and follows the requirements of EM 385-1-1. Level Two SSHOs cannot be assigned to projects that have a residual Risk Assessment Code (RAC) of high or extremely high.

1.2.1.3 Level Three SSHO

A designated Qualified Person or Competent Person with SOH responsibility that meets and follows the requirements of EM 385-1-1. Level 3 SSHOs cannot be assigned to projects that have a residual RAC of high or extremely high.

1.2.1.4 Alternate SSHO

An employee that meets the definition of the contract-required level SSHO, but is not the primary SSHO.

1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

1.2.3 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractors for their actions or liability.

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

Final IAQ Management Plan

Indoor Air Quality (IAQ) Management Plan; G

SD-06 Test Reports

Accident Reports; G

LHE Inspection Reports

Monthly Exposure Reports

SD-07 Certificates

Activity Hazard Analysis (AHA); G

Certificate of Compliance

Hot Work Permit

License Certificates

Portable Gauge Operations Planning Worksheet; G

Radiography Operation Planning Work Sheet; G

Standard Lift Plan; G

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by the Occupational Safety and Health Administration (OSHA) www.osha.gov, the Centers for Disease Control and Prevention (CDC) www.cdc.gov, and as required by federal, state and local requirements.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report by the fifth of each month. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the progress payment.

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE [EM 385-1-1](#), and the following federal, state, and local laws, ordinances, criteria, rules and regulations at the date of the Solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Site Safety and Health Officer (SSHO)

1.7.1.1 Qualifications of SSHO

All SSHOs will have met the training, experience requirements identified in the EM 385-1-1 and this Contract. SSHO's must have also completed the 40-hour HAZWOPER training prior to the Contractor beginning work in any known ~~/suspected~~ PFAS contaminated areas, and have five (5) years of experience on construction projects as on on-site Safety and Health Officer.

1.7.1.2 Duties of SSHO

All SSHOs will carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1. All SSHOs will be designated on an ENG Form 6282, provided by the Contracting Officer. Superintendent, QC Manager, and SSHO are subject to dismissal if their required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

The SSHO may not serve as the Quality Control Manager, Superintendent, Project Manager, QC Specialist, or any other key personnel position required to be provided by the Contractor for this project under the terms of this contract.

1.7.1.3 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, or foremen must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.7.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1 are met for the project. The Prime Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. If the required SSHO has to temporarily (that is, up to 24 hours / 1 day) leave the site of work due to unforeseen or emergency situations, a Level One, Two, or Three SSHO may be used in the interim and must be on the site of work at all times when work is being performed.

If the SSHO must be off-site for a period longer than 24 hours / 1 day, a qualified alternate that meets the contract requirements must be onsite.

- a. Prime contractor must ensure all SSHOs will:

- (1) Are designated on an ENG Form 6282.
- (2) Meet minimum training and experience requirements identified in EM 385-1-1.
- (3) Execute roles and responsibilities identified in EM 385-1-1.

See related requirements for SSHO's in Sectopm 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS, Paragraph 1.5.1, for working in Known/~~Suspected~~ Areas of PFAS Contamination on this project.

1.7.3 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements

Provide a minimum of one Level One SSHO that meets the requirements of EM 385-1-1 for this project.

1.7.4 Competent Person for Confined Space Entry

Provide a CP for Confined Space Entry who meets the requirements of EM 385-1-1 and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with EM 385-1-1.

Since this work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with NFPA 306, 29 CFR 1915, "Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment," or as applicable in 29 CFR 1910 for general industry, 29 CFR 1926 for construction.

1.7.5 Preconstruction Conference

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. The creation of the APP and Schedule will be created after being given Notice to Proceed.
- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 Accident Prevention Plan (APP)

Submit the Accident Prevention Plan (APP) for review and acceptance by the Government at least 10 calendar days prior to the date of the preconstruction conference for acceptance, after being given Notice to Proceed. A competent person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of [EM 385-1-1](#), ENG Form 6293, and herein. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and occupational health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with the APP and ENG Form 6293 Accident Prevention Plan Worksheet. The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer within 30 calendar days of Contract award and not less than 10 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e., imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by [ANSI/ASSP A10.34](#)), and the environment.

1.8.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in [EM 385-1-1](#), including the following:

- a. Names and qualifications (resumes including education, training,

experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.

- b. As a minimum, designate and submit qualifications of Competent Persons (CP) for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance. Designate and submit qualifications for additional CPs as applicable to the work performed under this Contract.

1.8.3 Plans

Provide plans in the APP in accordance with the requirements outlined in [EM 385-1-1](#), including the following:

1.8.3.1 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section [02 41 00](#) DEMOLITION and referenced sources.

1.9 [ACTIVITY HAZARD ANALYSIS \(AHA\)](#)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with [EM 385-1-1](#) or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAS must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOW must review the AHA for that work and sign a signature log specifically

maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.11 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. The Government has no responsibility to provide emergency medical treatment.

1.12 NOTIFICATIONS AND REPORTS

1.12.1 Accident Notification

Notify the Contracting Officer in accordance with the EM 385-1-1 Accident Reporting Timeline.

Table Accident Reporting Required Timeline		
Accident Type	Notify KO or COR	Complete Final Accident Report on ENG 3394 and provide to KO or COR
Fatality, in-patient hospitalization, amputation, eye loss, or property damage over \$600,000.	Immediately, no later than (NLT) 8 Hours	Within 7 Days
All other accidents and near misses	Immediately, no later than (NLT) 24 Hours	Within 7 Days

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.12.2 Accident Reports

- a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. All accidents are reportable, regardless of whether or not it is recordable. Complete the applicable USACE Accident Report, ENG Form 3394, and provide the

report to the Contracting Officer within 7 calendar days of the accident. The Contracting Officer will provide copies of any required or special forms. All accidents are reportable, regardless of whether or not it is recordable.

- b. Near Misses: For Army projects, report all "Near Misses" to the COR, using local accident reporting procedures, within 24 hours. The COR will provide the Contractor the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.

1.12.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a Certificate of Compliance for LHE entering an activity under this Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1 and using Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

1.13 HOT WORK PERMIT

1.13.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e., welding or cutting) or operating other flame-producing/spark producing devices, from the installation Fire Prevention Office, Building No. T-801, telephone no. 315-772-6338. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. Contractors are required to meet all criteria before a permit is issued. Provide at least two 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of 1 hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency installation fire department phone number. Report any fire, no matter how small, to the responsible installation fire department immediately.

1.13.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "Hot Work" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres. Contractor is required to obtain installation-specific confined space permit required by Fort Drum prior to

commencing work in manholes, handholes, and any other confined spaces on the project.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in [EM 385-1-1](#).

1.14 RADIATION SAFETY REQUIREMENTS

Submit [License Certificates](#), employee training records, and Leak Test Reports for radiation materials and equipment to the Contracting Officer and Radiation Safety Office (RSO) for all specialized and licensed material and equipment proposed for use on the construction project (excludes portable machine sources of ionizing radiation including moisture density and X-Ray Fluorescence (XRF)). Maintain on-site records whenever licensed radiological materials or ionizing equipment are on Government property.

Protect workers from radiation exposure in accordance with [10 CFR 20](#), ensuring any personnel exposures are maintained As Low As Reasonably Achievable.

1.14.1 Radiography Operation Planning Work Sheet

Submit a Gamma and X-Ray [Radiography Operation Planning Work Sheet](#) to Contracting Officer 14 days prior to commencement of operations involving radioactive materials or radiation generating devices. For portable machine sources of ionizing radiation, including moisture density and XRF, use and submit the [Portable Gauge Operations Planning Worksheet](#) instead. The Contracting Officer will review the submitted worksheet and provide questions and comments.

Contractors must use primary dosimeters process by a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

1.14.2 Site Access and Security

Coordinate site access and security requirements with the Contracting Officer for all radiological materials and equipment containing ionizing radiation that are proposed for use on a government facility. For gamma radiography materials and equipment, a Government escort is required for any travels on the Installation. The Government authorized representative will meet the Contractor at a designated location outside the Installation, ensure safety of the materials being transported, and will escort the Contractor for gamma sources onto the Installation, to the job site, and off the Installation. For portable machine sources of ionizing radiation, including moisture density and XRF, the Government authorized representative will meet the Contractor at the job site.

Provide a copy of all calibration records, and utilization records for radiological operations performed on the site.

1.14.3 Loss or Release and Unplanned Personnel Exposure

Loss or release of radioactive materials, and unplanned personnel exposures must be reported immediately to the Contracting Officer, RSO,

and Base Security Department Emergency Number.

1.14.4 Site Demarcation and Barricade

Properly demark and barricade an area surrounding radiological operations to preclude personnel entrance, in accordance with EM 385-1-1, Nuclear Regulatory Commission, and Applicable State regulations and license requirements, and in accordance with requirements established in the accepted Radiography Operation Planning Work Sheet.

Do not close or obstruct streets, walks, and other facilities occupied and used by the Government without written permission from the Contracting Officer.

1.14.5 Security of Material and Equipment

Properly secure the radiological material and ionizing radiation equipment at all times, including keeping the devices in a properly marked and locked container, and secondarily locking the container to a secure point in the Contractor's vehicle or other approved storage location during transportation and while not in use. While in use, maintain a continuous visual observation on the radiological material and ionizing radiation equipment. In instances where radiography is scheduled near or adjacent to buildings or areas having limited access or one-way doors, make no assumptions as to building occupancy. Where necessary, the Contracting Officer will direct the Contractor to conduct an actual building entry, search, and alert. Where removal of personnel from such a building cannot be accomplished and it is otherwise safe to proceed with the radiography, position a fully instructed employee inside the building or area to prevent exiting while external radiographic operations are in process.

1.14.6 Transportation of Material

Comply with 49 CFR 173 for Transportation of Regulated Amounts of Radioactive Material. Notify Local Fire authorities and the site Radiation Safety Officer (RSO) of any Radioactive Material use.

1.14.7 Schedule for Exposure or Unshielding

Actual exposure of the radiographic film or unshielding the source must not be initiated until after 5 p.m. on weekdays.

1.14.8 Transmitter Requirements

Adhere to the base policy concerning the use of transmitters, such as radios and cell phones. Obey Emissions control (EMCON) restrictions.

1.15 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with EM 385-1-1, 29 CFR 1926, 29 CFR 1910, and Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used. Contractor must apply for and obtain installation specific confined space entry permit prior to starting any work in confined spaces.

1.15.1 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue

from a confined space.

1.16 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Submit an IAQ Management Plan within 15 calendar days after notice to proceed and not less than 10 calendar days before the preconstruction conference. Revise and resubmit Plan as required by the Contracting Officer. Make copies of the final plan available to all workers on site. Include provisions in the Plan to meet the requirements specified below and to ensure safe, healthy air for construction workers and building occupants. Submit [Final IAQ Management Plan](#) for inclusion in the Sustainability eNotebook, in accordance with Section [01 33 29](#) SUSTAINABILITY REQUIREMENTS AND REPORTING.

1.16.1 Requirements During Construction

Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with [ASTM D6245](#). Provide for evaluation of volatile organic compounds (VOCs) in indoor air in accordance with [ASTM D6345](#). Use filters with a Minimum Efficiency Reporting Value (MERV) of 8 in permanently installed air handlers during construction.

1.16.1.1 Control Measures

Meet or exceed the requirements of [ANSI/SMACNA 008](#) to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

- a. HVAC protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.
- b. Source control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building. Designate receiving/storage areas for incoming material that minimize IAQ impacts.
- c. Pathway interruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.
- d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.
- e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

1.16.1.2 Moisture Contamination

- a. Remove accumulated water and keep work dry.
- b. Use dehumidification to remove moist, humid air from a work area.

- c. Do not use combustion heaters or generators inside the building.
- d. Protect porous materials from exposure to moisture.
- e. Remove and replace items which remain damp for more than a few hours.

1.16.2 Requirements After Construction

After construction ends and prior to occupancy, conduct a building flush-out or test the indoor air contaminant levels. Flush-out must be a minimum 2 weeks with MERV-13 filtration media as determined by ASHRAE 52.2 at 100 percent outside air. Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Indoor Air. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

1.17 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

- a. Head Protection that meets ANSI/ISEA Z89.1
- b. Long Pants
- c. Appropriate Safety Footwear
- d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e., 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during demolition, repair, renovation, or construction operations. Stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification.

3.1.4 FAA Requirements

Reference Section 01 35 13 SPECIAL PROJECT PROCEDURES, paragraph 3.1.2 for contractor's FAA Form 7460 requirements. Contractor to notify WSAAF base operations upon submission of FAA Forms 7460-1 and 2.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages at least 14 days in advance. Apply for communications outages at least 30 days in advance. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1. In accordance with EM 385-1-1, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of Hazardous

Energy Control Program (HECP) and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Installation representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1, 29 CFR 1910, 29 CFR 1915, ANSI/ASSP A10.44, NFPA 70E.

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, each lock and

tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with EM 385-1-1.

3.5.1 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 and 29 CFR 1926.

3.5.1.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.1.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers

such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1.

3.6 EQUIPMENT

3.6.1 Use of Explosives

Explosives are strictly prohibited on the project site, WSAAF, and the Fort Drum installation.

3.7 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1.

3.7.1 Electrical Work

As described in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety footwear, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.

3.7.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State, Local requirements applicable to where work is being performed.

3.7.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.7.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

3.7.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

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SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

11/20, CHG 2: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C511 (2017; R 2021) Reduced-Pressure Principle Backflow Prevention Assembly

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD (2009; Rev 2012) Manual on Uniform Traffic Control Devices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section **01 33 00** SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Site Plan; G

Traffic Control Plan; G

Haul Road Plan; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

SD-06 Test Reports

Backflow Preventer Tests

SD-07 Certificates

Backflow Tester Certification

Backflow Preventers Certificate of Full Approval

1.3 CONNECTION WITH WORK OF OTHER CONTRACTS

During the period of this contract, other contracts may be in force for the construction of other features of work on or adjacent to the site of work being accomplished under this contract. The Contractor must arrange his plant and must schedule and perform the work as to cooperate with all other contractors and Government agencies. It is the Contractor's responsibility to know the extent of the limits of his contract. No direct or extra compensation will be allowed because of the cooperation required.

- a. At all points of connection with work of other contracts, the Contractor must have weekly coordination meetings until all connections have been completed with the adjoining contractor(s) to insure proper and timely connections. Contractor must notify COR's of any respective coordination meetings with contractors of other projects on the installation at least 48 hours in advance.
- b. Where the work under this contract is completed before that of the adjoining contractor, the Contractor must terminate his work in an approved manner ready for future connection by the adjoining contractor. Pipes and conduits must be closed with suitable caps or plugs that will prevent entry of dirt or debris, but that are readily removable when final connections are made. For underground lines that are back-filled, approved type markers that extend above the ground surface must be provided to facilitate future location of the lines by the adjoining contract.
- c. Where the work of the adjoining contractor is already in place, the Contractor must perform all work required to effect the necessary connection, including locations of underground lines, removing of caps, providing necessary adapters or joining pieces, and all related incidental work for necessary for a proper, secure connection. The Contractor is required to perform tests to verify secure connections are achieved at standard testing parameters, no leakage is observed, and all deficiencies are corrected at no additional cost to the Government.

1.4 Coordination of Trades

The contract drawings are in part diagrammatic and show the general arrangement. The Contractor must have a competent engineer on the project site to coordinate all fieldwork and shop drawings of the various trades prior to installation and/or submission of field or shop drawings for approval. The Contractor must allot spaces to the various trades prior to installation of the work. In spaces where all the various installations cannot be accommodated, the Contractor cannotify the Contracting Officer and must submit alternate solutions at no additional cost to the Government. The decision of the Contracting Officer will be final.

1.5 Layout of Work

The contractor must lay out his work from the dimensions indicated on the approved drawings and is responsible for all measurements in connection therewith.

1.6 CONSTRUCTION SITE PLAN

Prior to the start of work, submit for Government approval a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.7 BACKFLOW PREVENTERS CERTIFICATE

1.7.1 Backflow Tester Certificate

Prior to testing, submit to the Contracting Officer certification issued by the State or local regulatory agency attesting that the [backflow tester](#) has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with a company participating in other phases of this Contract.

1.7.2 Backflow Prevention Training Certificate

Submit a certificate recognized by the State or local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current.

1.8 DOD CONDITION OF READINESS (COR)

DOD will set the Condition of Readiness (COR) based on the weather forecast for sustained winds 50 knots ([58 mph](#)) or greater. Contact the Contracting Officer for the current COR setting.

Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, [EM 385-1-1](#) Section 01 Emergency Planning and the instructions below.

Unless otherwise directed by the Contracting Officer, comply with:

- a. Condition FOUR (Sustained winds of [58 mph](#) or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than [3.3 feet](#) high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness. Contact Contracting Officer for weather and COR updates and completion of required actions.

- b. Condition THREE (Sustained winds of 58 mph or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general base areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.
- c. Condition TWO (Sustained winds of 58 mph or greater expected within 24 hours): Secure the jobsite, and leave Government premises.
- d. Condition ONE. (Sustained winds of 58 mph or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

1.9 CYBERSECURITY DURING CONSTRUCTION

{For Reference Only: This subpart (and its subparts) relates to AC-18, SA-3, CCI-00258.} Meet the following requirements throughout the construction process.

1.9.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, contractor computers must meet the following requirements:

1.9.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.9.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

1.9.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.9.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor

owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.9.2 Temporary IP Networks

Temporary contractor-installed IP networks are to be provided by the Contractor for Contractor use during construction. Contractor to procure a hot spot for Contractor field office use if necessary. When used, temporary contractor-installed IP networks must meet the following requirements:

1.9.2.1 Contractor Provided Wi-Fi Network

Provide standalone Wi-Fi network for authorized Government personnel. Network access must be available throughout the jobsite and for the entire Contract period. The Government will use the Wi-Fi service for coordination in the field and related activities. Do not attach or connect the temporary Wi-Fi network to Government networks or systems unrelated to construction activities.

Password information must adhere to requirements outlined in Paragraph PASSWORDS AND PASSPHRASES. Provide password and Wi-Fi network name (SSID) to the Contracting Officer for Government personnel access. Wi-Fi performance must be capable of handling bandwidth to support Government and Contractor activities, including drawing and model access onsite. Provide a point of contact for reporting Wi-Fi outages.

1.9.2.2 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited. Contractor access to existing Government furnished networks will not be provided.

1.9.3 Government Access to Network

Government personnel, as defined, prescribed, and identified by the Contracting Officer, must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification.

1.9.4 Temporary Wireless IP Networks

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.9.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.9.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to the commencement of work activities, provide a clear weatherproof covered bulletin board not less than 36 by 48 inches in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, Safety and Health Information as required by EM 385-1-1 Section 01 and other information approved by the Contracting Officer. Coordinate requirements herein with 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, and in location as approved by the Contracting Officer.

2.1.2 Project Identification Signs

The requirements for the signs, their content, and location are as specified in Section 01 58 00 PROJECT IDENTIFICATION. Erect signs within 15 days after receipt of the notice to proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

2.1.3 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1 Section 04. Attach signs to the perimeter fencing every 150 feet warning the public of the presence of construction hazards. Signs must require unauthorized persons to keep out of the construction site. Correct the data required by safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Transportation Facilities

- a. Railroads: CSX serves the locality of the proposed work. Railhead is located approximately 15 miles from the project site. The Contractor must make all arrangements at his expense for the use of sidings necessary for the delivery of materials, equipment, supplies, and other facilities required for completion of the work. The Contractor's use of sidings must be arranged so as not to interrupt or delay the operation of the Military reservation.
- b. Highways and Roads: Routes US 11 and NYS Routes 26 & 3 serve the

locality of the proposed work. Roads within the military reservation proposed to be used by the Contractor, must be subject to prior approval of the Post authorities and such roads, if used, must be maintained throughout construction and must be restored to as good condition as existed prior to their use.

The Contractor may make use of old NYS Routes 29 and 37, but usage by other vehicles may not be interrupted without prior approval from the Contracting Officer and in compliance with paragraph 19(b). The Contractor must also construct such temporary haul roads and bridges as may be necessary for the conduct of his work. Such temporary construction must be restored to its original condition. All costs for the use of existing transportation facilities, for the construction of temporary facilities, and for maintenance, repair, removal and restoration must be borne by the Contractor.

2.2.2 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this Contract in accordance with EM 385-1-1 Section 04. Construct with suitable grades and widths; avoid sharp curves, blind corners, and dangerous cross traffic. Submit haul road plan for approval. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. The method of dust control, although optional, must be adequate to ensure safe operation at all times. Location, grade, width, and alignment of construction and haul roads are subject to approval by the Contracting Officer.

2.2.3 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Barricades are required whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night at all times, even if construction is not ongoing.

2.3 FENCING

Provide fencing along the construction site and at all open excavations and tunnels to control access by unauthorized personnel. Safety fencing must be highly visible to be seen by pedestrians and vehicular traffic. All fencing must meet the requirements of EM 385-1-1. Remove the fence upon completion and acceptance of the work.

Security of staging area is contractor's means and methods with respect to type of fencing to be installed. Note that chain link fencing is not permitted on the airfield, but orange safety fencing is required for safety of excavations in accordance with EM 385-1-1.

2.3.1 Post-Driven Chain Link Fencing

Temporary post-driven fencing must be galvanized chain link fencing 6 feet high supported by an tightly secured to galvanized steel posts driven below grade. Fence posts must be located on minimum 10 foot centers. Posts may be set in various surfaces such as sand, soil, asphalt or concrete as necessary. Chain link fencing must remain rigid and taut with

a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Completely remove fencing and posts at the completion of construction and restore surfaces disturbed or damaged to its original condition. Locate and identify underground utilities prior to setting fence posts. Equip fence with a lockable gate. Gate must remain locked when construction personnel are not present.

2.4 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1 Section 11, NFPA 241 and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

2.5 BACKFLOW PREVENTERS

Certificate of Full Approval from FCCCHR List, University of Southern California, attesting that the design, size and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval is not acceptable.

Reduced pressure principle type conforming to the applicable requirements AWWA C511. Provide backflow preventers complete with 150 pound flanged cast iron, bronze, mounted gate valve and strainer, 304 stainless steel or bronze, internal parts. The particular make, model/design, and size of backflow preventers to be installed must be included in the latest edition of the List of Approved Backflow Prevention Assemblies issued by the FCCCHR List and be accompanied by a Certificate of Full Approval from FCCCHR List. After installation conduct Backflow Preventer Tests and provide test reports verifying that the installation meets the FCCCHR Manual Standards.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

Construction Contract employees must park privately owned vehicles in an area designated by the Contracting Officer. Employee parking must not interfere with existing and established parking requirements of the Government installation.

3.2 Temporary Bulletin Board

Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, as approved by the Contracting Officer.

3.3 AVAILABILITY AND USE OF UTILITY SERVICES

3.3.1 Temporary Utilities

Connection to existing utilities (other than potable water and electric) is not permitted and will not be available within the Project Site. The Government will permit the Contractor to tie into existing sanitary, but if the Contractor does not want to connect to it, then the Contractor will be responsible for providing their own temporary sanitary line and connections. Provide temporary utilities required for construction. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards.

3.3.2 Payment for Utility Services

- a. The Government will make all reasonably required utilities available from existing outlets and supplies, as specified in the Contract. Unless otherwise provided in the Contract, the amount of each utility service consumed will be charged to or paid at prevailing rates charged to the Government or, where the utility is produced by the Government, at reasonable rates determined by the Contracting Officer. Carefully conserve utilities furnished without charge.
- b. Reasonable amounts of the following utilities will be made available without charge.

Utility Services
Electricity
Potable Water

- c. The point at which the Government will deliver such utilities or services and the quantity available must be coordinated with the Contracting Officer. Pay all costs incurred in connecting, converting, and transferring the utilities to the work. Make connections, including providing backflow-preventing devices on connections to domestic water lines; providing meters; and providing transformers; and make disconnections.
- d. The Contractor must provide their own utilities, unless otherwise specified above.

3.3.3 Meters and Temporary Connections

Provide and maintain necessary temporary connections, distribution lines, meters, and meter bases required to measure the amount of each utility used for the purpose of determining charges. Notify the Contracting Officer, in writing, 5 working days before final electrical connection is desired so that a utilities contract can be established. The Government will provide a meter and make the final hot connection after inspection and approval of the Contractor's temporary wiring installation. Do not make the final electrical connection.

3.3.4 Final Meter Reading

Before completion of the work and final acceptance of the work by the Government, notify the Contracting Officer, in writing, 5 working days before termination is desired. Before completion, the Government will take final meter readings. The Contractor, at no additional cost to the Government, will disconnect, and remove all the temporary distribution lines, meter bases, and associated appurtenances. Pay all outstanding utility bills, for utilities not provided by the Government, before final acceptance of the work by the Government.

3.3.5 Sanitation

Provide and maintain within the construction, laydown, and staging areas minimum field-type sanitary facilities in accordance with [EM 385-1-1](#) Section 02. Locate the facilities behind the construction fence or out of the public view. Clean units and empty wastes at least once a week or

more frequently into a municipal, district, or station sanitary sewage system, or remove waste to a commercial facility. Obtain approval from the system owner prior to discharge into a municipal, district, or commercial sanitary sewer system. Penalties or fines associated with improper discharge will be the responsibility of the Contractor. Sanitary wastes from the Contractor's sanitary facilities will not be discharged or disposed of into the installation's sanitary sewer system. Comply with the Contractor's selected sanitary waste disposal facility policies, procedures, and regulations. Maintain these conveniences at all times. Include provisions for pest control and elimination of odors. Government toilet facilities will not be available to Contractor's personnel.

3.3.6 Telephone

Make arrangements and pay all costs for telephone facilities desired.

3.3.7 Obstruction Lighting of Cranes

Provide a minimum of 2 aviation red or high intensity white obstruction lights on temporary structures (including cranes) over **100 feet** above ground level. Light construction and installation must comply with FAA AC 70/7460-1. Lights must be operational during periods of reduced visibility, darkness, and as directed by the Contracting Officer.

Cranes under **100 feet** in height must be lowered during non-working hours, and booms must be flagged at all times.

3.3.8 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

3.4 STATION OPERATION AFFECT ON CONTRACTOR OPERATIONS

The WSAAF will remain operational during construction of this project. Airfield operations to support mission and training must take priority over construction activities. Comply with all directions from the WSAAF Base Operations and the ATCT.

3.5 TRAFFIC PROVISIONS

3.5.1 Maintenance of Traffic

- a. Conduct operations in a manner that will not close a thoroughfare or interfere with traffic on railways or highways except with written permission of the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide a **Traffic Control Plan** for Government approval detailing the proposed controls to traffic movement for approval. The plan must be in accordance with State and local regulations and the **MUTCD**, Part VI. Make all notifications and obtain all permits required for modification to traffic movements outside Station's jurisdiction. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the highway authority have been met.
- b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Comply with authorities requirements having jurisdiction of the access routes

to be traversed, and obtain approval from the Contracting Officer prior to starting any activity that will obstruct traffic.

- c. Provide, erect, and maintain, at Contractor's expense, lights, barriers, signals, passageways, detours, and other items, that may be required by the Life Safety Signage, overhead protection authority having jurisdiction.
- d. Provide cones, signs, barricades, lights, or other traffic control devices and personnel required to control traffic. Do not use foil-backed material for temporary pavement marking because of its potential to conduct electricity during accidents involving downed power lines.
- e. Comply with all applicable NYSDOT requirements.

3.5.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Provide self-illuminated (lighted) barricades during hours of darkness. Brightly-colored (orange) vests are required for all personnel working in roadways. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of damage to roads caused by construction operations.

3.5.3 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer. Coordinate dust control methods with 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.

3.6 REDUCED PRESSURE BACKFLOW PREVENTERS

Provide an approved reduced pressure backflow prevention assembly at each location where the Contractor taps into the Government potable water supply. Submit to the Fort Drum Environmental Compliance Manager the backflow preventer certification and DOH 1013 Form filled out completely for review and approval. Fort Drum Environmental Compliance is the only approving authority for backflow preventers.

Perform backflow preventer tests using test equipment, procedures, and certification forms conforming to those outlined in the latest edition of the Manual of Cross-Connection Control published by the FCCCHR Manual. Test and tag each reduced pressure backflow preventer upon initial installation (prior to continued water use) and monthly thereafter. Tag must contain the following information: make, model, serial number, dates of tests, results, maintenance performed, and signature of tester. Record test results on certification forms conforming to requirements cited earlier in this paragraph.

3.6.1 Hydrant Use Requirements

Contractor to obtain hydrant use permit from installation fire department.

3.7 CONTRACTOR'S TEMPORARY FACILITIES

Contractor is responsible for security of their property and connecting, providing, and maintaining all required utilities to the Contractor's temporary facilities. Provide adequate outside security lighting at the temporary facilities. Trailers must be anchored to resist high winds and meet applicable state or local standards for anchoring mobile trailers. Coordinate anchoring with [EM 385-1-1](#) Section 04. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply: Contractor's trailer must have sufficient space for all contractor personnel requiring office space.

3.7.1 Administrative Field Offices

Provide and maintain administrative field office facilities within the construction area at the designated site. Government office and warehouse facilities will not be available to the Contractor's personnel.

3.7.2 Storage Area

Construct a temporary [6 foot](#) high chain link fence around trailers and materials. Include plastic strip inserts, colored brown, so that visibility through the fence is obstructed. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on the current day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

Keep fencing in a state of good repair and proper alignment. If the Contractor elects to traverse grassed or unpaved areas which are not established roadways with construction equipment or other vehicles, cover the grassed or unpaved areas with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways; gravel gradation must be at the Contractor's discretion. Mow and maintain grass located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, structures, under trailers, and in areas not accessible to mowers must be edged or trimmed neatly.

3.7.3 Supplemental Storage Area

Upon request, and pending availability, the Contracting Officer will designate another or supplemental area for the use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the installation boundaries. Maintain the area in a clean and orderly fashion and secured if needed to protect supplies and equipment. Utilities will not be provided to this area by the Government.

3.7.4 Appearance of Trailers

- a. Trailers must be roadworthy and comply with all appropriate state and local vehicle requirements. Trailers which are rusted, have peeling paint or are otherwise in need of repair will not be allowed on Installation property. Trailers must present a clean and neat exterior appearance and be in a state of good repair.
- b. Maintain the temporary facilities. Failure to do so will be sufficient reason to require their removal at the Contractor's expense.

3.7.5 Maintenance of Storage Area

Keep fencing in a state of good repair and proper alignment. Grassed or unpaved areas, which are not established roadways, and will be traversed with construction equipment or other vehicles, will be covered with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways, should the Contractor elect to traverse them with construction equipment or other vehicles. Remove snow and mow and maintain grass located within and outside the boundaries of the storage area. The Contractor is responsible for snow removal, mowing, and maintenance of vegetation a minimum of 20 feet outside the perimeter fencing of the storage area for the duration of the project. Grass and vegetation along fences under trailers, and in areas not accessible to mowers will be edged or trimmed neatly.

3.7.6 Maintenance of Construction Areas

Contractor is responsible for snow removal in all areas required to perform work through the duration of the project. Coordinate snow removal with Fort Drum Public Works Roads & Grounds and WSAAF Base Operations for locations where snow removed is to be stockpiled at WSAAF. Contractor to transport all removed snow to either Mountain Ramp or Guard Ramp, depending on which area is closest to area snow is being removed from. Contractor to coordinate with installation DPW through COR for blowing snow from these two dump locations. Contractor is responsible for mowing within the limits of construction through the duration of the project (all areas required to perform work and all areas confined by construction fencing). Complete mowing and vegetation management within the limits of construction consistent with government mowing outside the limits of construction. The frequency of government mowing is weather and seasonally dependent. Continually during every workday and at the end of each workday inspect construction areas for construction debris, trash or other materials that are a Foreign Object Debris (FOD) hazard to aircraft. Remove and dispose of FOD materials immediately from the construction area. Maintain construction debris, excavated soils, and stockpiles in conditions that prevent FOD hazards to aircraft, to include wind-borne dust and debris, though duration of the project. The prevailing winds are generally from the southwest, west, and northwest at WSAAF.

3.7.7 Security Provisions of Storage Area

The Contractor is responsible for the security of its equipment, materials, supplies, vehicles, and/or temporary office trailers. Under no circumstances shall the Government be responsible for any loss, theft, or damage to any contractor or subcontractor, at any tier, to its equipment, materials, supplies, vehicles, and/or temporary office trailers. Security lighting must be approved by Fort Drum and WSAAF Base Operations prior to

installation.

3.7.8 Safety Systems

Protect the integrity of all installed safety systems or personnel safety devices. Obtain prior approval from the Contracting Officer if entrance into systems serving safety devices is required. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish Contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.7.9 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the building are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the building from damage.

3.7.9.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.8 GOVERNMENT FIELD OFFICE

3.8.1 Resident Engineer's Office

Provide the Government Resident Engineer and installation construction representative with a keyed, locking office, minimum 800-900 square feet in floor area (20 foot by 40 to 45 foot double-wide office trailer), located where directed and providing space heat, air conditioning unit, electric light and power, and toilet facilities consisting of a lavatory and with connections to water. Provide a minimum of 6 sets of keys. Provide one sanitary facility in accordance with EM 385-1-1, exclusively for government use. The field office must have a functional bathroom with running water and flush toilet internal to the structure. A portable restroom will not be accepted. The contractor shall be responsible for all costs for maintenance of, and disposal of sanitary wastes from, the Government's sanitary facility. Provide a mail slot in the door or a lockable mail box mounted on the surface of the door. Include a 5 CF refrigerator, 3 by 6 foot plastic folding table, fire extinguisher, first aid kit, coffee pot, 1000W microwave, high speed internet service, 4 by 8 foot plan table with plan rack, two 3 by 4 by 1 foot bookshelves, two 3 by 5 foot whiteboards with dry erase markers and erasers, minimum four, computer work spaces (with a standard size office desk and ergonomic task chair, garbage can, and two minimum 24 inch flat screen monitors), a conference room (with large conference table, minimum 10 ergonomic conference chairs, and minimum 60" flat screen TV with HDMI hookup), a lockable, legal sized, four-drawer file cabinet. High speed internet must be supplied to each work station and the conference room (no less than 1.0 gbps download/ upload speed). Contractor shall pay all costs associated

with this service including ISP fees. Contractor shall provide and install all required equipment (i.e., Switch, Hub, Ethernet NIC Cards, modems, etc.) for use with Government furnished field office computers). No separate payment shall be made for providing the items and all costs in connection therewith will be considered the obligation of the Contractor. Provide an ACE-IT approved copier/ scanner/ printer/ fax with necessary tele/data connection. Hardware and cables to be furnished and installed so that Government work stations can utilize printer on a private network in field office. Janitorial services must be provided (minimum of two weekly visits, including mopping of the floor, dusting, and garbage removal). Drinking water must be provided (5 gallon hot and cold water cooler, replenish throughout contract as needed). Provide soap, paper towels, toilet paper, printing paper, ink/toner, and replenish throughout contract as necessary. Note: Computer hardware and software used in the field office will be Government furnished. At completion of the project, the office and all furnishings will remain the property of the Contractor and be removed from the site. Utilities must be connected and disconnected in accordance with local codes and to the satisfaction of the Contracting Officer. Compliance with safety and appearance requirements for temporary facilities stated in previous paragraphs is required. Government Resident Engineer and installation construction representative's field office trailer is to be separate from the Contractor's field office trailer. Submit proposed office plan to the Government for approval.

3.8.2 Trailer-Type Mobile Office

The option is available to, furnish and maintain a trailer-type mobile office acceptable to the Contracting Officer to meet the requirements of the minimum facilities specified above. Securely anchor the trailer to the ground at all four corners to guard against movement during high winds. Coordinate requirements for proper anchoring with EM 385-1-1 Section 04.

3.9 PLANT COMMUNICATIONS

Whenever the individual elements of the plant are located so that operation by normal voice between these elements is not satisfactory, install a satisfactory means of communication, such as telephone or other suitable devices and make available for use by Government personnel.

3.10 TEMPORARY PROJECT SAFETY FENCING

As soon as practicable, but not later than 15 days after the date established for commencement of work, furnish and erect temporary project safety fencing at the work site. Maintain the safety fencing during the life of the Contract and, upon completion and acceptance of the work, remove from the work site.

3.11 DUMPSTERS

~~Equip~~ Provide and equip dumpsters with a secure cover and paint the standard installation color. Keep dumpsters covered, except when being loaded with trash and debris. Locate dumpsters within the secured staging area out of public view. Empty site dumpsters at least once a week, or as needed to keep the site free of debris and trash. ~~Fort Drum will provide and pickup metal recycling dumpsters for the Contractor's use.~~ Trash, debris, recycling, or other materials disposed in the Government-provided metal recycling dumpsters must be removed by the Contractor. ~~at its own~~

~~expense, prior to Fort Drum picking up the dumpsters.~~ If necessary, provide 55 gallon trash containers painted the darker installation color to collect debris in the construction site area. For large demolitions, large dumpsters without lids are acceptable, but must not have debris higher than the sides before emptying.

3.12 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store all salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

The contractor shall continually inspect construction site activities and at the end of the workday. Remove and dispose of foreign object debris (FOD) immediately when discovered. Inspect airfield pavements to include, but not limited to runways, taxiways, tow-lanes, aprons, and ramps adjacent to construction sites for the presence of FOD that may have been transported outside the limits of construction. Immediately remove and dispose of any FOD observed on airfield pavements, regardless of if transported from the construction area or not. Request approval from the ATCT for inspection of active airfield pavements for the presence of FOD.

3.13 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and all other temporary products from the site. After removal of trailers, materials, and equipment from within the fenced area, remove the fence. Restore areas used during the performance of the Contract to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary.

-- End of Section --

SECTION 01 57 19

TEMPORARY ENVIRONMENTAL CONTROLS
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NEW YORK STATE DEPARTMENT OF TRANSPORTATION MATERIALS BUREAU
(NYSDOT)

6 NYCRR Parts 360-366 and 369 Standards for Solid Waste Management

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1053	Respirable Crystalline Silica
29 CFR 1910.1200	Hazard Communication
29 CFR 1926.1153	Respirable Crystalline Silica
40 CFR 50	National Primary and Secondary Ambient Air Quality Standards
40 CFR 60	Standards of Performance for New Stationary Sources
40 CFR 63	National Emission Standards for Hazardous Air Pollutants for Source Categories
40 CFR 64	Compliance Assurance Monitoring
40 CFR 82	Protection of Stratospheric Ozone
40 CFR 112	Oil Pollution Prevention
40 CFR 241	Guidelines for Disposal of Solid Waste
40 CFR 243	Guidelines for the Storage and Collection of Residential, Commercial, and Institutional Solid Waste
40 CFR 258	Subtitle D Landfill Requirements
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 261.7	Residues of Hazardous Waste in Empty Containers
40 CFR 262.11	Hazardous Waste Determination and

Recordkeeping

40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 273	Standards for Universal Waste Management
40 CFR 273.2	Standards for Universal Waste Management - Batteries
40 CFR 273.3	Standards for Universal Waste Management - Pesticides
40 CFR 273.4	Standards for Universal Waste Management - Mercury Containing Equipment
40 CFR 273.5	Standards for Universal Waste Management - Lamps
40 CFR 273.6	Applicability - Aerosol Cans
40 CFR 279	Standards for the Management of Used Oil
40 CFR 300	National Oil and Hazardous Substances Pollution Contingency Plan
40 CFR 300.125	National Oil and Hazardous Substances Pollution Contingency Plan - Notification and Communications
40 CFR 355	Emergency Planning and Notification
40 CFR 403	General Pretreatment Regulations for Existing and New Sources of Pollution
49 CFR 171	General Information, Regulations, and Definitions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 173	Shippers - General Requirements for Shipments and Packagings

1.2 DEFINITIONS

1.2.1 Class I and II Ozone Depleting Substance (ODS)

Class I ODS is defined in Section 602(a) of The Clean Air Act. A list of Class I ODS can be found on the EPA website at the following weblink.
<https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances>.

Class II ODS is defined in Section 602(s) of The Clean Air Act. A list of Class II ODS can be found on the EPA website at the following weblink.
<https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances>.

1.2.2 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e., methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

1.2.3 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.4 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.5 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.6 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with 40 CFR 261. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with 40 CFR 261.

1.2.7 Hazardous Materials

Hazardous material is any material that: Is defined in 49 CFR 171, listed in 49 CFR 172, regulated as a hazardous material in accordance with 49 CFR 173; or requires a Safety Data Sheet (SDS) in accordance with 29 CFR 1910.1200; or during end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste as defined by 40 CFR 261 Subparts A, B, C, or D. Designation of a material by this definition, when separately regulated or controlled by other sections or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this section for "control" purposes. Such material includes ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs).

1.2.8 Hazardous Waste

Hazardous Waste is any material that meets the definition of a solid waste and exhibits a hazardous characteristic (ignitability, corrosivity, reactivity, or toxicity) as specified in 40 CFR 261, Subpart C, or contains a listed hazardous waste as identified in 40 CFR 261, Subpart D, or meets a state, local, or host nation definition of a hazardous waste.

1.2.9 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, discharge into defined drainage areas, or discharge into the "waters of the United States" must occur. Comply with federal, state, and local laws and regulations.

1.2.10 Municipal Separate Storm Sewer System (MS4) Permit

MS4 permits are those held by municipalities or installations to obtain NPDES permit coverage for their stormwater discharges.

1.2.11 National Pollutant Discharge Elimination System (NPDES)

The NPDES permit program controls water pollution by regulating point sources that discharge pollutants into waters of the United States.

1.2.12 Oily Waste

Oily waste are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with other state and local requirements.

This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. These materials may be land filled provided that: It is not prohibited in other state regulations or local ordinances; the amount generated is "de minimus" (a small amount); it is the result of minor leaks or spills resulting from normal process operations; and free-flowing oil has been removed to the practicable extent possible. Large quantities of this material, generated as a result of a major spill or in lieu of proper maintenance of the processing equipment, are a solid waste. As a solid waste, perform a hazardous waste determination prior to disposal. As this can be an expensive process, it is recommended that this type of waste be minimized through good housekeeping practices and employee education.

1.2.13 Pesticide

Pesticide is any substance or mixture of substances intended for preventing, destroying, repelling, or mitigating any pest, or intended for use as a plant regulator, defoliant or desiccant.

1.2.14 Pesticide Treatment Plan

A plan for the prevention, monitoring, and control to eliminate pest

infestation.

1.2.15 Pests

Pests are arthropods, birds, rodents, nematodes, fungi, bacteria, viruses, algae, snails, marine borers, snakes, weeds and other organisms (except for human or animal disease-causing organisms) that adversely affect readiness, military operations, or the well-being of personnel and animals; attack or damage real property, supplies, equipment, or vegetation; or are otherwise undesirable.

1.2.16 Project Pesticide Coordinator

The Project Pesticide Coordinator (PPC) is an individual who resides at a Civil Works Project office and who is responsible overseeing of pesticide application on project grounds.

1.2.17 Regulated Waste

Regulated waste are solid wastes that have specific additional federal, state, or local controls for handling, storage, or disposal.

1.2.18 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.19 Solid Waste

Solid waste is a solid, liquid, semi-solid or contained gaseous waste. A solid waste can be a hazardous waste, non-hazardous waste, or non-Resource Conservation and Recovery Act (RCRA) regulated waste. Types of solid waste typically generated at construction sites may include:

1.2.19.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 2.5-inch particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may be reinforced with or contain ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.19.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.19.3 Material Not Regulated As Solid Waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as

amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.19.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with 40 CFR 261.

1.2.19.5 Recyclables

Recyclables are materials, equipment and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing and mirrors that are recovered and sold as recyclable, wiring, insulated/non-insulated copper wire cable, wire rope, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of lead contaminated or lead based paint contaminated may not be included as recyclable if sold to a scrap metal company. Paint cans that meet the definition of empty containers in accordance with 40 CFR 261.7 may be included as recyclable if sold to a scrap metal company.

1.2.19.6 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

1.2.19.7 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.19.8 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.20 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, or creeks meeting the definition of "waters of the United States". Surface discharges from construction sites are discrete, identifiable sources and require a permit from the governing agency. Comply with federal, state, and local laws and regulations.

1.2.21 Wastewater

Wastewater is the used water and solids that flow through a sanitary sewer to a treatment plant.

1.2.21.1 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.22 Waters of the United States

Waters of the United States means Federally jurisdictional waters, including wetlands, that are subject to regulation under Section 404 of the Clean Water Act or navigable waters, as defined under the Rivers and Harbors Act.

1.2.23 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.2.24 Universal Waste

The universal waste regulations streamline collection requirements for certain hazardous wastes in the following categories: batteries, pesticides, mercury-containing equipment (for example, thermostats), and lamps (for example, fluorescent bulbs). The rule is designed to reduce hazardous waste in the municipal solid waste (MSW) stream by making it easier for universal waste handlers to collect these items and send them for recycling or proper disposal. These regulations can be found at [40 CFR 273](#).

1.2.25 Location Specific Universal Waste

Any of the following dangerous waste that are subject to the universal waste requirements of [6 NYCRR Parts 360-366 and 369](#): Batteries as described in [6 NYCRR Parts 360-366 and 369](#); Lamps as described in [6 NYCRR Parts 360-366 and 369](#); Mercury-containing equipment as described in [6 NYCRR Parts 360-366 and 369](#).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section [01 33 00](#) SUBMITTAL PROCEDURES:

[SD-01 Preconstruction Submittals](#)

[Environmental Manager Qualifications; G](#)

[Employee Training Records; G](#)

Environmental Protection Plan; G

Dirt and Dust Control Plan; G

Stormwater Pollution Prevention Plan (SWPPP); G

Contractor and Subcontractor Certifications and Contractor's
Qualified Inspector(s) Certifications

Spill Prevention Control And Countermeasure (SPCC) Plan; G

Batch Plant Stormwater Pollution Prevention Plan; G

SD-06 Test Reports

Monthly Solid Waste Disposal Report; G

Inspection Reports

SD-07 Certificates

Employee Training Records; G

Erosion and Sediment Control Inspector Qualifications

SD-11 Closeout Submittals

Stormwater Pollution Prevention Plan Compliance Notebook; G

Stormwater Notice of Termination (for NPDES coverage under the
general permit for construction activities); G

As-Build Record Site and Topographic Survey

Waste Determination Documentation; G

Project Solid Waste Disposal Documentation Report; G

Sales Documentation; G

1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with federal, state, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with Applicable Environmental Laws may be required. Analytical work must be performed by qualified laboratories; and where required by law, the laboratories must be certified.

Provide a qualified stormwater inspector in accordance with the requirements stated in the most recent version of the NYSDEC Construction Stormwater Inspection Manual for SWPPP inspections, as well as the requirements listed in GP-0-25-001, throughout the project.

1.4.1 Conformance with the Environmental Management System

Perform work under this contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the contract.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.5 SPECIAL ENVIRONMENTAL REQUIREMENTS

Comply with the special environmental requirements listed here when completing construction activities within known ~~/suspected areas of~~ PFAS contamination known areas:

- a. Stormwater Pollution Prevention Plan (SWPPP), Notice of Intent (NOI), and Notice of Termination.

1.5.1 Surveillance of Work in Known ~~/Suspected Areas~~ of PFAS Contamination

The Contractor's SHSO must be on-site at the work area(s) when construction activities are performed in areas of known ~~/suspected~~ PFAS contamination that result in ground disturbance, or when work occurs in areas where ground disturbance has already occurred and known ~~/suspected~~ PFAS contaminated soil is exposed. The Contractor's SSHO is responsible for ensuring all contractor and subcontractor employees working within known ~~/suspected~~ areas of PFAS contamination are performing work in compliance with the contract requirements and the Contractor's approved Accident Prevention and Environmental Protection Plans.

1.5.2 Management of Excavated Soils from Known ~~/Suspected~~ Areas of PFAS Contamination

~~Excavated soils in known/suspected areas of PFAS shall be used to backfill excavations only in the area where the soil was excavated. To the extent feasible, excess excavated soils shall be spread on-site to grade and~~

~~feathered to match existing grades within the defined know/suspected area of PFAS contamination where it was excavated. Excessive spreading of soils that impede stormwater drainage/ runoff from airfield pavements or existing drainage features will not be permitted. Where excessive soil is spread, remove and regrade the area as directed by the COR to the satisfaction of the Government. Removal and grading of areas of excessive soil spreading will be at no additional cost to the Government. Stabilize soils spread to prevent wind and stormwater runoff migration of soils from these areas until the areas are restored with topsoil and hydroseed. Excess soil that cannot be spread within the area of excavation shall be stockpiled within the area it was excavated. The contractor is responsible for the management of the stockpiled soil to prevent migration of potential PFAS-contaminated soil from the stockpile. Do not comingle soils from areas outside of the known/suspected areas of contamination with soils excavated from these areas. See Ft Drum SOP, Decision Tree, and Department of War Detailed Analysis of Options for the Destruction and Disposal of Covered Materials Containing Per- and Polyfluoroalkyl Substances attachments following 02 61 13 EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL.~~

1.5.3 Sampling and Analysis of Soil from Known/~~Suspected~~ Areas of PFAS

~~Upon completion of all work that will disturb soils in an area of known/suspected contamination and all soils in that area are stockpiled, coordinate with Fort Drum Environmental for sampling and analysis of stockpiled soils. Fort Drum Environmental will complete the sampling and laboratory analysis at no cost to the contractor. Stockpiles of comingled soils will not be sampled by Fort Drum Environmental for laboratory analysis. Fort Drum Environmental requires 7 business days advance notification prior to sampling the stockpiles. Fort Drum will submit samples for laboratory analysis with a standard turn-around time of three weeks. If the contractor requires expedited laboratory analysis results, all costs associated with such expedited analysis will be at the contractor's expense. See Ft Drum SOP, Decision Tree, and Department of War Detailed Analysis of Options for the Destruction and Disposal of Covered Materials Containing Per- and Polyfluoroalkyl Substances attachments following 02 61 13 EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL.~~

1.5.4 Management and Disposal of ~~Known/Suspected~~ PFAS Contaminated Soil

~~See Ft Drum SOP, Decision Tree, and Department of War Detailed Analysis of Options for the Destruction and Disposal of Covered Materials Containing Per- and Polyfluoroalkyl Substances attachments following 02 61 13 EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL. Based on laboratory results of soil samples from the stockpiled soil, Fort Drum Environmental will determine if the soil meets NYSDEC regulations as material meeting the requirements as clean for unrestricted use, or soil that does not meet the regulations for clean unrestricted use. Fort Drum Environmental is the only Authority that is permitted to make this determination.~~

1.5.4.1 Clean Unrestricted Use Soil

Clean unrestricted use soils may be used by the Contractor for on-site backfill/ fill within the limits of construction. Any excess soil will remain the property of the government, and the contractor shall be responsible for loading, transporting, and unloading of soils from the site to a Site to be Determined on or near WSAF, as directed by the Contracting Officer.

1.5.4.2 Soils not Classified as Clean for Unrestricted Use

Soils that are not classified as clean for unrestricted use shall remain the property of the Contractor. The Contractor, at no additional cost to the Government, is responsible for all costs to legally dispose of the soil off from Government property.

1.5.4.3 Comingled Soils

Comingled soils shall remain the property of the Contractor. The Contractor will be responsible for all costs to legal disposal of comingled soils, to include all laboratory sampling and analysis required by the Contractor's designated disposal facility.

1.5.4.4 Transportation Manifest

The Contractor shall prepare, if required by Fort Drum Environmental, transportation manifests to haul material to its selected disposal facility. The manifests shall be reviewed by and signed by the Fort Drum Environmental manager prior to transport. Copies of signed manifests from the Contractor's selected disposal facility shall be provided to Fort Drum Environment NLT 2 days after disposal at the facility, demonstrating legal disposal off/ from Government property.

1.5.4.5 Weight Tickets

The Contractor shall provide weight tickets from the disposal facility for both manifested and non-manifested loads transported off from Government property. Weight ticket shall be used to demonstrate legal disposal off from Government property when transportation manifests are not required.

1.6 QUALITY ASSURANCE

1.6.1 Preconstruction Survey and Protection of Features

This paragraph supplements the Contract Clause PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS. Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and the Contracting Officer will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

1.6.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with federal, state and local regulations. In cases where the Government will also provide public notification (such as stormwater permitting), coordinate with the Contracting Officer. Submit copies of regulatory notifications to the Contracting Officer at least 30 days prior to commencement of work

activities. Typically, regulatory notifications must be provided for the following (this listing is not all-inclusive): demolition, renovation, NPDES defined site work, construction, removal or use of a permitted air emissions source, and remediation of controlled substances (asbestos, hazardous waste, lead paint).

1.6.3 Environmental Brief

Attend an environmental brief to be included in the preconstruction meeting. Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the installation; and types and quantities of wastes/wastewater that may be generated during the Contract. Discuss the results of the Preconstruction Survey at this time.

Prior to initiating any work on site, meet with the Contracting Officer and installation Environmental Office to discuss the proposed Environmental Protection Plan (EPP) or equipment local requirement. Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural and cultural resources, required reports, required permits, permit requirements (such as mitigation measures), and other measures to be taken.

1.6.4 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for coordinating contractor compliance with federal, state, local, and installation requirements. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB transformers). This can be a collateral position; however, the person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in 40 CFR requirements in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance and close-out. Submit [Environmental Manager Qualifications](#) to the Contracting Officer.

1.6.5 Employee Training Records

Prepare and maintain [Employee Training Records](#) throughout the term of the contract meeting applicable 40 CFR requirements. Provide Employee Training Records in the Environmental Records Binder. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with federal, state and local regulatory requirements for RCRA Large Quantity Generator. Provide a Position Description for each employee, by subcontractor, based on the Davis-Bacon Wage Rate designation or other equivalent method, evaluating the employee's association with hazardous and regulated wastes. This Position Description will include training requirements as defined in [40 CFR 265](#) for a Large Quantity Generator facility. Submit these Assembled Employee Training Records to

the Contracting Officer at the conclusion of the project, unless otherwise directed.

Train personnel to meet EPA requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of the United States, and endangered species and their habitat that are known to be in the area. Provide copy of the [Erosion and Sediment Control Inspector](#) Qualifications as defined by EPA.

1.6.6 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with federal, state or local environmental laws or regulations, permits, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. FAR 52.242-14 Suspension of Work provides that a suspension, delay, or interruption of work due to the fault or negligence of the Contractor allows for no adjustments to the contract for time extensions or equitable adjustments. In addition to a suspension of work, the Contracting Officer may use additional authorities under the contract or law.

1.7 COMPLIANCE WITH ENDANGERED SPECIES ACT AT FT. DRUM

Comply with the requirements listed below:

- (1) The federally-endangered Indiana bat and the federally-threatened northern long eared bat are present on Fort Drum and must be considered in all actions. Actions on Fort Drum, must be in accordance with requirements negotiated with the US Fish & Wildlife Service under Section 7 of the Endangered Species Act. Projects involving ANY habitat modification (e.g., vegetation removal) must be addressed by Fort Drum's Fish and Wildlife Management Program. In general, no trees greater than 3 inches (diameter breast height) may be cut from 16 April - 15 October to avoid any direct adverse effects to these bats. Clearing of natural vegetation (e.g., shrubs and trees) less than 3 inches (diameter breast height) should typically occur between August 1 - April 15 to minimize the impact to migratory birds and to maintain foraging areas for bats. The contractor (and all subcontractors) must be responsible for coordination with Fort Drum's Fish & Wildlife Management Program to ensure compliance with the Endangered Species Act and all other applicable laws and regulations pertaining to wildlife. Overall the contractor (and all subcontractors) must minimize interference with, disturbance to, and damage to plants, fish, wildlife, and their habitats on and adjacent to the project area.
- (2) Flagging or signs must be used to demarcate construction limits of a

given project prior to any construction activity, vegetation, and/or tree clearing for a given project. Clearing must only occur within the flagged area, and flagging must be removed upon completion of the project.

- (3) If any bats are discovered during the demolition or repair all work must cease and Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636) must be immediately contacted. If the building has pre-existing known bat colonies, then Fort Drum's Fish and Wildlife Management Program must be contacted before demolition is to occur. (At this time, the only structures known to contain bat colonies are within the historic LeRay Area.).
- (4) All bats found must be immediately reported to Fort Drum's Fish and Wildlife Management Program (772-4999 or 772-9636). Do not attempt to handle any live bats, regardless of condition. If a live bat is found in a building please contact 772-2072 or 772-4999.
- (5) Light minimization measures are to be incorporated for all exterior lighting that may include but are not limited to full cutoffs, reflectors, shields, downward angling of lights and/or reduction of correlated color temperatures to reduce blue-rich, white light. See paragraph 1.55 for detailed requirements that must be followed to the maximum extent possible.
- (6) No night lighting for construction projects is authorized without prior coordination and authorization from the project manager and Fort Drum's Fish and Wildlife Management Program.
- (7) Project planning and design should attempt to minimize building footprints by combining infrastructure (i.e. roads, utility lines, etc.) for multiple buildings or by constructing multi-story versus multiple or expanded single story buildings whenever possible.

1.8 ENVIRONMENTAL PROTECTION PLAN

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction. Incorporate construction related objectives and targets from the installation's EMS into the EPP. Include in the EPP measures for protecting natural and cultural resources, required reports, and other measures to be taken. Meet with the Contracting Officer or Contracting Officer Representative to discuss the EPP and develop a mutual understanding relative to the details for environmental protection including measures for protecting natural resources, required reports, and other measures to be taken. Submit the EPP within 30 calendar days after NTP and not less than 10 days before the preconstruction conference. Revise the EPP throughout the project to include any reporting requirements, changes in site conditions, or contract modifications that change the project scope of work in a way that could have an environmental impact. No requirement in this section will relieve the Contractor of any applicable federal, state, and local environmental protection laws and regulations. During Construction, identify, implement, and submit for approval any additional requirements to be included in the EPP. Maintain the current version onsite.

The EPP includes, but is not limited to, the following elements:

1.8.1 General Overview and Purpose

1.8.1.1 Descriptions

A brief description of each specific plan required by environmental permit or elsewhere in this Contract such as stormwater pollution prevention plan, spill control plan, solid waste management plan, wastewater management plan, air pollution control plan, contaminant prevention plan, Non-Hazardous Solid Waste Disposal Plan, and borrowing material plan.

1.8.1.2 Duties

The duties and level of authority assigned to the person(s) on the job site who oversee environmental compliance, such as who is responsible for adherence to the EPP, who is responsible for spill cleanup and training personnel on spill response procedures, who is responsible for manifesting hazardous waste to be removed from the site (if applicable), and who is responsible for training the Contractor's environmental protection personnel.

1.8.1.3 Procedures

A copy of any standard or project-specific operating procedures that will be used to effectively manage and protect the environment on the project site.

1.8.1.4 Communications

Communication and training procedures that will be used to convey environmental management requirements to Contractor employees and subcontractors.

1.8.1.5 Contact Information

Emergency contact information contact information (office phone number, cell phone number, and e-mail address).

1.8.2 General Site Information

1.8.2.1 Drawings

Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, jurisdictional wetlands, material storage areas, structures, sanitary facilities, storm drains and conveyances, and stockpiles of excess soil.

1.8.2.2 Work Area

Work area plan showing the proposed activity in each portion of the area and identify the areas of limited use or nonuse. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site, and a traffic control plan.

Show where any fuels, hazardous substances, solvents, or lubricants will be stored. Provide a spill plan to address any releases of those materials.

1.8.2.3 Documentation

A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

1.8.3 Management of Natural Resources

- a. Land resources
- b. Tree protection
- c. Replacement of damaged landscape features
- d. Temporary construction
- e. Stream crossings
- f. Fish and wildlife resources
- g. Wetland areas

1.8.4 Protection of Historical and Archaeological Resources

- a. Objectives
- b. Methods

1.8.5 Stormwater Management and Control

- a. Ground cover
- b. Erodible soils
- c. Temporary measures
 - (1) Structural Practices
 - (2) Temporary and permanent stabilization
- d. Effective selection, implementation and maintenance of Best Management Practices (BMPs).
- e. Stormwater Pollution Prevention Plan (SWPPP).

1.8.6 Protection of the Environment from Waste Derived from Contractor Operations

Control and disposal of solid and sanitary waste.

As a minimum, include the following:

- a. Procedures to ensure a written waste determination is made for appropriate wastes that are to be generated
- b. Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory

certifications

- c. Management procedures for storage, labeling, transportation, and disposal of waste (treatment of waste is not allowed unless specifically noted)
- d. Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar
- e. Used oil management procedures in accordance with 40 CFR 279; Hazardous waste minimization procedures

1.8.7 Prevention of Releases to the Environment

Procedures to prevent releases to the environment

Notifications in the event of a release to the environment

1.8.8 Regulatory Notification and Permits

List what notifications and permit applications must be made. Some permits require up to 180 days to obtain. Demonstrate that those permits have been obtained or applied for by including copies of applicable environmental permits. The EPP will not be approved until the permits have been obtained.

1.8.9 Clean Air Act Compliance

1.8.9.1 Haul Route

Submit truck and material haul routes along with a [Dirt and Dust Control Plan](#) for controlling dirt, debris, and dust on Installation roadways. As a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways.

1.8.9.2 Pollution Generating Equipment

Identify air pollution generating equipment or processes that may require federal, state, or local permits under the Clean Air Act. Determine requirements based on any current installation permits and the impacts of the project. Provide a list of all fixed or mobile equipment, machinery or operations that could generate air emissions during the project to the Installation Environmental Office (Air Program Manager). Ensure required permits are obtained prior to installing and operating applicable equipment/processes.

1.8.9.3 Stationary Internal Combustion Engines

Identify portable and stationary internal combustion engines that will be supplied, used or serviced. Comply with 40 CFR 60 Subpart IIII, 40 CFR 60 Subpart JJJJ, 40 CFR 63 Subpart ZZZZ, and local regulations as applicable. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and EPA emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation and delineate between maintenance/testing, emergency, and non-emergency operation.

1.8.9.4 Refrigerants

Identify management practices to ensure that heating, ventilation, and air conditioning (HVAC) work involving refrigerants complies with 40 CFR 82 requirements. Technicians must be certified, maintain copies of certification on site, use certified equipment and log work that requires the addition or removal of refrigerant. Any refrigerant reclaimed is the property of the Government, coordinate with the Installation Environmental Office to determine the appropriate turn in location.

1.8.9.5 Air Pollution-engineering Processes

Identify planned air pollution-generating processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used.

1.8.9.6 Compliant Materials

Provide the Government a list of SDSs for all hazardous materials proposed for use on site. Materials must be compliant with all Clean Air Act regulations for emissions including solvent and volatile organic compound contents, and applicable National Emission Standards for Hazardous Air Pollutants requirements. The Government may alter or limit use of specific materials as needed to meet installation permit requirements for emissions.

1.9 OUTDOOR LIGHTING MINIMIZATION

The purpose of the Fort Drum Outdoor Lighting Guidelines is to regulate outdoor lighting in order to reduce or prevent light pollution. This means to the extent reasonably possible the reduction or prevention of glare and light trespass, the conservation of energy, and promotion of safety and security. These Guidelines will ensure appropriate outdoor lighting in compliance with the Endangered Species Act and in accordance with the Fort Drum's Army Strategic Plan for Sustainability.

Definitions:

- a. Fixture Height: height of the fixture must be the vertical distance from the ground directly below the centerline of the fixture to the lowest direct light emitting part of the fixture.
- b. Foot-candles: a unit of illumination of a surface that is equal to one lumen per square foot. For the purposes of these regulations, foot-candles must be measured at a height of 3 ft. above finished grade.
- c. Fully Shielded Light: light fixtures shielded or constructed so that no light rays are directly emitted by the installed fixture at angles above the horizontal plane as certified by a photometric test report. The fixture must also be properly installed to effectively down direct light in order to conform to the definition.
- d. Light Trespass: the shining of light produced by a light fixture beyond the boundaries of the property on which it is located.
- e. Lumen: the unit of luminous flux, the total amount of light falling uniformly on or passing through an area of 1 square foot, each of

which is 1 foot from a 1-candela source, yielding an illuminance of 1 foot candle at that distance (the output of lamps and bulbs is customarily measured in lumens, a common 100 watt incandescent light bulb, for example, having an output less than 1,800 lumens).

- f. Point Light Source: the exact place from which illumination is produced (i.e., a light bulb filament or discharge capsule).
- g. Sag-lens or Drop-lens: A clear or prismatic refracting lens that extends below the lowest opaque portion of a light fixture.

Applicability:

All outdoor lighting fixtures installed, retro-fitted, or replaced on Fort Drum property must comply with these regulations. These regulations do not apply to interior lighting.

Exemptions

The following are exempt from the provisions of these guidelines:

- a. Traffic control signals and devices.
- b. Temporary emergency lighting (i.e. fire, police, repair workers).
- c. Moving vehicle lights.
- d. Navigation lights (i.e. airports, heliports, radio/television towers).
- e. Seasonal decorations with individual lights in place no longer than 60 days.
- f. Lighting for flags. Efforts should be made in these areas to minimize sky glow and light trespass whenever feasible.
- g. Sports field outdoor lighting (i.e. ball fields, football, soccer, ice rink, etc.). Sports outdoor lighting is to be turned off when a sporting event is not occurring.
- h. Other special situations for temporary or periodic events (i.e. fairs, festivals, carnivals, night-time construction).
- i. Security lights of any wattage that are controlled by a motion- sensor switch and which do not remain on longer than 10 minutes after activation.
- j. Access points, Army Supply points, or other high security areas subject to AR 190-11 or TM-8-583-2. Efforts should be made in these areas to minimize sky glow and light trespass whenever feasible.

Additional exemptions may be provided after coordination with Fort Drum's Fish and Wildlife Management Program.

General Standards:

The following general standards must apply to all outdoor lighting installed, retrofitted, or replaced on Fort Drum, which is not exempted above:

- a. Outdoor lighting must be hooded, fully shielded (i.e. full cutoff fixtures), and/or aimed downward. Outdoor lighting used to illuminate parking spaces, driveways, maneuvering areas, or buildings must conform to the definition for "fully shielded light fixtures" and be designed, arranged and screened so that the point light source must not be visible from adjoining lots (i.e. woodlands) or streets.
- b. The intensity of light within a site must not exceed two (2) foot-candles at any property line, edge of pavement, or road.
- c. The hood or shield must mask the direct horizontal surface of the light source. The light must be aimed to insure that the illumination is only pointing downward onto the ground surface, with no escaping light permitted to contribute to sky glow by shining upward into the sky.
- d. Any bright light shining onto adjacent properties (i.e. woodlands) or streets which would result in a nuisance glare or a disabling glare must not be permitted. Light trespass beyond property boundaries or above the horizontal plane must be considered non-compliant.
- e. Existing fixtures may be adapted to comply with these guidelines by adding a properly designed hood or shield, or by pointing any upward-mounted, shielded fixture downward onto the ground surface.
- f. All outdoor lighting fixtures must be designed, installed, located and maintained such that nuisance glare onto adjacent properties (i.e. woodlands) or streets must be minimized and all direct illumination kept within the boundaries of a building's property.
- g. Accent lighting must be directed downward onto the building or object and not toward the sky or onto adjacent properties (i.e. woodlands). Direct light emissions must not be visible above the roof line or beyond the building edge.
- h. Spotlighting on landscaping and foliage must be limited to 150 watts (2220 lumens output) and lighting is to be angled downwards. The lamp must be fully shielded and not create disabling or nuisance glare.
- i. No sag-lens or drop-lens must be used.
- j. LED light fixtures must be utilized to the maximum extent practicable. These fixtures must be long life, coupled with high efficient drivers. LED lights must incorporate measures to reduce blue-rich white light output. These measures will be coordinated with Fort Drum's electric shop and Natural Resources Branch prior to implementation, but could include things such as targeting wavelengths between 550-650 nm, and/or reducing the correlated color temperature of the fixture to 3000K or lower. A map is available that designates areas in which utilizing lower correlated color temperatures is required. This can be viewed through Fort Drum's Fish and Wildlife Management Program, the Master Planning Division, and/or the electric shop.

1.10 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7 Permits and Responsibilities. Notify the Government of all equipment that may require permits or special

approvals that the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7 Permits and Responsibilities.

a. The following permits have been obtained by the Government:

(1) NYSDEC Stormwater Permit for Construction Activity (GP-0-25-001)

1.11 ENVIRONMENTAL RECORDS BINDER

Maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

1.12 PESTICIDE DELIVERY, STORAGE, AND HANDLING

Application of Pesticides by the Contractor is prohibited on Fort Drum and WSAAF.

1.13 SOLID WASTE MANAGEMENT PERMIT

Provide the Contracting Officer with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location state and local Solid Waste Management Permit or license showing such agency's approval of the disposal plan before transporting wastes off Government property.

1.13.1 Monthly Solid Waste Disposal Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

1.14 FACILITY HAZARDOUS WASTE GENERATOR STATUS

Ft. Drum is designated as a Large Quantity Generator. Meet the regulatory requirements of this generator designation for any work conducted within the boundaries of this Installation. Comply with provisions of federal, state, and local regulatory requirements applicable to this generator status regarding training and storage, handling, and disposal of construction derived wastes.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 PROTECTION OF NATURAL RESOURCES

Minimize interference with, disturbance to, and damage to fish, wildlife, and plants, including their habitats. Prior to the commencement of activities, consult with the Installation Environmental Office as applicable, regarding rare species or sensitive habitats that need to be

protected. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's responsibility.

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Office or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified. Refer to Article 1.7 within this Specification for Compliance with Endangered Species Act at Fort Drum.

3.1.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified and permitted.

3.1.2 Vegetation

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor is responsible for any resultant damage.

Protect existing trees that are to remain to ensure they are not injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. Coordinate with the Contracting Officer and Installation Environmental Office to determine appropriate action for trees and other landscape features scarred or damaged by equipment operations.

3.2 STORMWATER

Do not discharge stormwater from construction sites to the sanitary sewer. If the water is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization in advance from the Installation Environmental Office for any release of contaminated water.

3.2.1 [Stormwater Notice of Termination](#) for Construction Activities

Submit a Notice of Termination to the Contracting Officer for approval once construction is complete and final stabilization has been achieved on all portions of the site for which the permittee is responsible. Once approved, submit the Notice of Termination to the appropriate state or federal agency. Prepare [as-build record site and topographic survey](#) information required by the permitting agency for certification of the stormwater management system, and provide to the Contracting Officer.

3.2.2 [Stormwater Pollution Prevention Plan Compliance Notebook](#)

Create and maintain a three ring binder of documents that demonstrate compliance with the Construction General Permit. Include a copy of the permit Notice of Intent, proof of permit fee payment, SWPPP and SWPPP update amendments, inspection reports and related corrective action records, copies of correspondence with the the New York State Permitting

Agency, and a copy of the permit Notice of Termination in the binder. At project completion, the notebook becomes property of the Government. Provide the compliance notebook to the Contracting Officer.

3.2.3 Batch Plant Stormwater Pollution Prevention Plan

Contractor is responsible for the sizing and layout of the Contractor's onsite batch plant area, and therefore, responsible for preparing a stormwater prevention plan that complies with the New York State Department of Environmental Conservation (NYSDEC). This plan must be submitted to the Fort Drum Environmental Department for approval before implementation. The plan will consist of, at a minimum;

- a. Drawing plan showing location of all erosion, sediment control, and temporary stormwater practices, in accordance with NYSDEC requirements for stormwater management from construction activities.
- b. Calculations showing that practices have been sized to handle the rainfall required by the NYSDEC. Once Fort Drum has approved of the plan, it must be included in the onsite SWPPP as an addendum.

Contractor is then responsible for implementing all erosion, sediment control, and stormwater practices and maintaining them throughout construction.

3.2.4 Inspection Reports

Submit "Inspection Reports" to the Contracting Officer in accordance with the State of New York Construction General Permit.

3.2.5 Erosion and Sediment Control Measures

Provide erosion and sediment control measures in accordance with state and local laws and regulations. Preserve vegetation to the maximum extent practicable.

Erosion control inspection reports may be compiled as part of a stormwater pollution prevention plan inspection reports.

3.2.5.1 Erosion Control

Prevent erosion by mulching, Compost Blankets, Geotextiles. Stabilize slopes by sodding, seeding, or such combination of these methods necessary for effective erosion control. Use of hay bales is prohibited.

Provide seeding in accordance with Section 32 92 19 SEEDING.

3.2.5.2 Sediment Control Practices

Implement sediment control practices to divert flows from exposed soils, temporarily store flows, or otherwise limit runoff and the discharge of pollutants from exposed areas of the site. Implement sediment control practices prior to soil disturbance and prior to creating areas with concentrated flow, during the construction process to minimize erosion and sediment laden runoff. Include the following devices: silt fence, storm drain inlet protection, and rock check dams. Location and details of installation and construction are indicated on the drawings.

3.2.6 Work Area Limits

Mark the areas that need not be disturbed under this Contract prior to commencing construction activities. Mark or fence isolated areas within the general work area that are not to be disturbed. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, all markers must be visible in the dark. Personnel must be knowledgeable of the purpose for marking and protecting particular objects.

3.2.7 Contractor Facilities and Work Areas

Place field offices, staging areas, stockpile storage, and temporary buildings in areas designated on the drawings or as directed by the Contracting Officer. Move or relocate the Contractor facilities only when approved by the Government. Provide erosion and sediment controls for onsite borrow and spoil areas to prevent sediment from entering nearby waters. Control temporary excavation and embankments for plant or work areas to protect adjacent areas.

3.2.8 Municipal Separate Storm Sewer System (MS4) Management

Comply with the Installation's MS4 permit requirements.

3.3 SURFACE AND GROUNDWATER

3.3.1 Cofferdams, Diversions, and Dewatering

Construction operations for dewatering, must be constantly controlled to maintain compliance with existing state water quality standards and designated uses of the surface water body. Comply with the State of New York water quality standards. Do not discharge excavation ground water to the sanitary sewer, storm drains, or to surface waters without prior specific authorization in writing from the Installation Environmental Office or Contracting Officer. Discharge of hazardous substances will not be permitted under any circumstances. Use sediment control BMPs to prevent construction site runoff from directly entering any storm drain or surface waters.

If the construction dewatering is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization for any contaminated groundwater release in advance from the Installation Environmental Officer and the federal or state authority, as applicable. Discharge of hazardous substances will not be permitted under any circumstances.

3.4 PROTECTION OF CULTURAL RESOURCES

3.4.1 Archaeological Resources

If, during excavation or other construction activities, any previously unidentified or unanticipated historical, archaeological, and cultural resources are discovered or found, activities that may damage or alter such resources will be suspended. Resources covered by this paragraph include, but are not limited to: any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rock or coral alignments, pavings, wall, or other constructed features; and any indication of agricultural or other human activities. Upon such discovery

or find, immediately notify the Contracting Officer so that the appropriate authorities may be notified and a determination made as to their significance and what, if any, special disposition of the finds should be made. Cease all activities that may result in impact to or the destruction of these resources. Secure the area and prevent employees or other persons from trespassing on, removing, or otherwise disturbing such resources. The Government retains ownership and control over archaeological resources.

- a. Cultural resources on Federal property are protected and managed by the Archaeological Resources Protection Act of 1979 and other applicable laws. The Contractor shall exercise care so as not to disturb or damage artifacts or fossils (should any be uncovered) during the excavation operations. Should the Contractor or any parties operating or associated with the performance of this contract discover evidence of possible scientific, prehistoric, historic or archaeological finds within the work limit lines or adjacent to work area shall immediately cease work at that location and notify the Contracting Officer. The Contractor shall provide the Contracting Officer with all information as to the specific location and nature of the findings. The Contractor shall cooperate fully with the Contracting Officer, except that all notifications by the Contractor shall be to the Contracting Officer and that all directions to the Contractor will be from the Contracting Officer. Where appropriate by reason of discovery, the Contracting Officer may order delays in time of performance or changes in the work or both. If such delays or changes are ordered, an equitable adjustment will be made in the contract in accordance with the applicable clauses of the contract.
- b. Protected properties on Fort Drum, marked with "Off-Limits" signs, shall not be entered.

3.5 AIR RESOURCES

Equipment operation, activities, or processes will be in accordance with 40 CFR 64 and state air emission and performance laws and standards.

3.5.1 Preconstruction Air Permits

Notify the Air Program Manager, through the Contracting Officer, at least 6 months prior to bringing equipment, assembled or unassembled, onto the Installation, so that air permits can be secured. Necessary permitting time must be considered in regard to construction activities. Clean Air Act (CAA) permits must be obtained prior to bringing equipment, assembled or unassembled, onto the Installation. Contractor to obtain any required air permits.

3.5.2 Burning

Burning is prohibited on the Government premises.

3.5.3 Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer. Intentional venting of refrigerants (including most Non-ODS substitute refrigerants) is prohibited per 40 CFR 82.

3.5.4 EPA Certification Requirements

Heating and air conditioning technicians must be certified through an EPA-approved program. Maintain copies of certifications at the employees' places of business; technicians must carry certification wallet cards, as provided by environmental law.

3.5.5 Dust Control

Keep dust down at all times, including during nonworking periods. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster. Since these products contain Crystalline Silica, comply with the applicable OSHA standard, 29 CFR 1910.1053 or 29 CFR 1926.1153 for controlling exposure to Crystalline Silica Dust.

3.5.5.1 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed 40 CFR 50, state, and local air pollution standards or that would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators, or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs. Comply with state and local visibility regulations.

3.5.5.2 Abrasive Blasting

Blasting operations are prohibited.

3.5.6 Odors

Control odors from construction activities. The odors must be in compliance with state regulations and local ordinances and may not constitute a health hazard.

3.6 WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the EPP. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

3.6.1 Salvage, Reuse and Recycle

Identify anticipated materials and waste for salvage, reuse, and recycling. Describe actions to promote material reuse, resale or recycling. To the extent practicable, all scrap metal must be sent for reuse or recycling and will not be disposed of in a landfill.

Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the state or local permit (cover) or license for recycling.

3.6.2 Nonhazardous Solid Waste Diversion Report

Maintain an inventory of nonhazardous solid waste diversion and disposal of construction and demolition debris. Submit a report to the Contracting Officer on the first working day after each fiscal year quarter, starting the first quarter that nonhazardous solid waste has been generated. Include the following in the report:

Construction and Demolition (C&D) Debris Disposed	cubic yards, as appropriate
C&D Debris Recycled	cubic yards, as appropriate
C&D Debris Composted	cubic yards, as appropriate
Total C&D Debris Generated	cubic yards, as appropriate
Waste Sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount)	cubic yards, as appropriate

3.7 WASTE MANAGEMENT AND DISPOSAL

3.7.1 Waste Determination Documentation

Complete a Waste Determination form (provided at the pre-construction conference) for Contractor-derived wastes to be generated. All potentially hazardous solid waste streams that are not subject to a specific exclusion or exemption from the hazardous waste regulations (e.g., scrap metal, domestic sewage) or subject to special rules, (lead-acid batteries and precious metals) must be characterized in accordance with the requirements of 40 CFR 262.11 or corresponding applicable state or local regulations. Base waste determination on user knowledge of the processes and materials used, and analytical data when necessary. Consult with the Installation environmental staff for guidance on specific requirements. Attach support documentation to the Waste Determination form. As a minimum, provide a Waste Determination form for the following waste (this listing is not exhaustive): oil- and latex-based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

3.7.2 Solid Waste Management

3.7.2.1 Project Solid Waste Disposal Documentation Report

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation must include the receiver's tax identification number and business, EPA or state registration number, along with the receiver's delivery and business addresses and telephone numbers. For each solid waste retained for the Contractor's own use, submit the information previously described in this paragraph on the solid waste disposal report. Prices paid or received do not have to be reported to the Contracting Officer unless required by other provisions or specifications of this Contract or public law.

3.7.2.2 Control and Management of Solid Wastes

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste. Transport solid waste off Government property and dispose of it in compliance with 40 CFR 260, state, and local requirements for solid waste disposal. A Subtitle D RCRA permitted landfill is the minimum acceptable offsite solid waste disposal option. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate. Solid waste disposal offsite must comply with most stringent local, state, and federal requirements, including 40 CFR 241, 40 CFR 243, and 40 CFR 258.

Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with 49 CFR 173.

3.7.3 Control and Management of Hazardous Waste

3.7.3.1 Universal Waste Management

Manage the following categories of universal waste in accordance with federal, state, and local requirements and installation instructions:

- a. Batteries as described in 40 CFR 273.2
- b. Lamps as described in 40 CFR 273.5
- c. Mercury-containing equipment as described in 40 CFR 273.4
- d. Aerosol cans as described in 40 CFR 273.6
- e. Pesticides as described in 40 CFR 273.3

Mercury is prohibited in the construction of this facility, unless specified otherwise, and with the exception of mercury vapor lamps and fluorescent lamps. Dumping of mercury-containing materials and devices such as mercury vapor lamps, fluorescent lamps, and mercury switches, in rubbish containers is prohibited. Remove without breaking, pack to

prevent breakage, and transport out of the activity in an unbroken condition for disposal as directed.

3.7.3.2 Electronics End-of-Life Management

Recycle or dispose of electronics waste, including, but not limited to, used electronic devices such computers, monitors, hard-copy devices, televisions, mobile devices, in accordance with 40 CFR 260-262, state, and local requirements, and installation instructions.

3.7.4 Releases/Spills of Oil and Hazardous Substances

3.7.4.1 Response and Notifications

Exercise due diligence to prevent, contain, and respond to spills of sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with 40 CFR 300. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the Installation Fire Department, the Installation Command Duty Officer, the Installation Environmental Office, the Contracting Officer and the state or local authority.

Submit verbal and written notifications as required by the federal (40 CFR 300.125 and 40 CFR 355), state, local regulations and instructions. Provide copies of the written notification and documentation that a verbal notification was made within 20 days. Spill response must be in accordance with 40 CFR 300 and applicable state and local regulations. Contain and clean up these spills without cost to the Government.

3.7.4.2 Clean Up

Clean up hazardous and non-hazardous waste spills. Reimburse the Government for costs incurred including sample analysis materials, clothing, equipment, and labor if the Government will initiate its own spill cleanup procedures, for Contractor- responsible spills, when: Spill cleanup procedures have not begun within one hour of spill discovery/occurrence; or, in the Government's judgment, spill cleanup is inadequate and the spill remains a threat to human health or the environment.

3.7.5 Wastewater

3.7.5.1 Disposal of Wastewater

Disposal of wastewater must be as specified below.

3.7.5.1.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged prior to being treated to remove pollutants. Dispose of the construction- related waste water off-Government property in accordance with 40 CFR 403, state, regional, and local laws and regulations.

3.7.5.1.2 Surface Discharge

For discharge of ground water, obtain a state or federal permit specific for pumping and discharging ground water prior to surface discharging. Surface discharge in accordance with the requirements of the NPDES or state STORMWATER DISCHARGES FROM CONSTRUCTION SITES permit.

3.7.5.1.3 Land Application

Water generated from the flushing of lines after disinfection or disinfection in conjunction with hydrostatic testing must be discharged into the sanitary sewer with prior approval and notification to the Wastewater Treatment Plant's Operator. Disinfection wastewater shall not be discharge to the ground. Coordinate with Fort Drum Utilities Branch for the disposal location into the Fort Drum Sanitary Sewer System. Provide Fort Drum Environmental three (3) calendar days (not including weekends and holidays) advance notification prior to disposal of disinfection wastewater into the Fort Drum sewer system. Fort Drum Environmental must provide notification to the Watertown, NY wastewater treatment plant.

3.8 HAZARDOUS MATERIAL MANAGEMENT

Include hazardous material control procedures in the Safety Plan, in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. Do not bring hazardous material onto Government property that does not directly relate to requirements for the performance of this contract. Submit an SDS and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the installation. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste, in accordance with 40 CFR 261 and state and installation requirements.

3.9 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds. Consult with the U.S. Department of Agriculture jurisdictional office for additional cleaning requirements.

3.10 MILITARY MUNITIONS

In the event military munitions, as defined in 40 CFR 260, are discovered or uncovered, immediately stop work in that area and immediately inform the Contracting Officer.

3.11 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and

cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with EPA 40 CFR 112, and other federal, state, regional, and local laws and regulations. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or navigable waters of the United States. Describe in the EPP (see paragraph ENVIRONMENTAL PROTECTION PLAN) how POL tanks and containers must be stored, managed, and inspected and what protections must be provided. Storage of fuel on the project site must be in accordance with EPA, state, and local laws and regulations and paragraph OIL STORAGE INCLUDING FUEL TANKS. Storage of fuel on the project site may be permitted, but only outside of airfield fencing limits and the well field protection zone (inside the Contractor's staging/laydown area). Contractor must comply with NYSDEC requirements for secondary containment and location and methods for storage must also be approved by Ft Drum Environmental. Contractor is solely responsible for any environmental cleanup requirements relating to use and storage of petroleum products on the project site at no additional cost to the Government.

3.11.1 Used Oil Management

Manage used oil generated on site in accordance with 40 CFR 279. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE DISPOSAL.

3.11.2 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 5 inches freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 1320 gallons will be used onsite (only containers with a capacity of 55 gallons or greater are counted), provide and implement a Spill Prevention Control and Countermeasure (SPCC) plan meeting the requirements of 40 CFR 112. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Monitor and remove any rainwater that accumulates in open containment dikes or berms. Inspect the accumulated rainwater prior to draining from a containment dike to the environment, to determine there is no oil sheen present.

3.12 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

If petroleum-contaminated soil, or suspected hazardous waste is found

during construction that was not identified in the Contract documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.13 SOUND INTRUSION

Make the maximum use of low-noise emission products, as certified by the EPA. Blasting or use of explosives are not permitted.

Keep construction activities under surveillance and control to minimize environment damage by noise. Comply with the provisions of the State of New York Department of Environmental Conservation (DEC) rules.

3.14 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause: "Cleaning Up". Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform with surrounding contours.

-- End of Section --

Fort Drum

Record of Environmental Consideration (REC)

Action Name/Project Number: WHEELER-SACK ARMY AIRFIELD RUNWAY 03-21 AIRFIELD LIGHTING REPAIR PROJECT
Date and Duration of Proposed Action: From: 10/1/2025 To: 12/31/2027
Proponent: (unit/directorate, address, bldg. #, POC, phone, email)
Engineering - 4205 Po Valley Road; Fort Drum, NY 13602
POC: Ryan Churchill, Phone:315-608-1119
Email:ryan.g.churchill2.civ@army.mil

(Include Location (8 digit coordinate and TA) , Radial distance in meters from center grid, Level of Activity(i.e., number of personnel and equipment) Type of activity (i.e., POL operations, mess/laundry/bath, demolition, real property actions, grading, borrow pit, construction or renovation actions, excavation, fielding/testing actions, etc.) ALL SIGNATURE BLOCKS SHOULD BE ON FIRST PAGE: COUTINUE ON REVERSE

PROJECT # EL-33441-2J

Training Area WSAAF

GRID LOCATION: 31978698

DESCRIPTION OF ACTIVITY:

This project will repair the Runway 03-21 edge and in-pavement lighting fixtures, visual NAVAIDs, and infrastructure to include the repair by replacement of the Airfield Lighting Vault. A related and required project element includes Runway 03 end pavement repair and shoulder pavement repairs.

Note - Design Analysis without attachments has been provided for reference along with overall plan. Entire design package is very large and can be sent by DoD Safe if necessary.

Signature of this document is solely a determination of compliance with the provisions of the National Environmental Policy Act (NEPA). This does not preclude, nor constitute a determination of preclusion from, any additional requirements pursuant to submittals or permits ensuring compliance with other applicable federal, state, or local laws, regulations, requirements, and/or guidelines.

Signed Churchill, Ryan G CIV USARMY ID-READINESS (USA)

Proponent

Date: 6/12/2025

APPROVAL

See Requirements page for list of actions needed for Compliance with Environmental Laws and Post Policies.

Guidance is available to download, and provide to MICC, project personnel, contractors, and operators with the REC documents.

Signed Jason Wagner

PW - Environmental Division Use Only

NEPA

Date: 6/12/2025

*It has been determined that the action:

☐ a. Is adequately covered in existing EA ☐ EIS ☐

☒ b. Is categorically excluded under the provisions of 32 CFR 651, Appendix B, Section II (g) (1)

LOG# 25.0841.00

Approved 6/12/2025

Received 6/12/2025

40 CFR 93.153 para c2(iv) exempt

Fort Drum Form 541

1 Nov 96

IAW: 32 CFR 651, AR 200.2, FD Reg 350-4

Additional Requirements/Guidance for Proponent Record of Environmental Consideration (REC)

--- Follow established procedures for the collection, transportation, disposal, and reporting of Non-Hazardous Solid Waste, Salvageable, and/or Recyclable materials. POC is Walker Heap, (315) 783-5810.

--- IAW EPA & NYS laws and Fort Drum policies, Best Management Practices are required as needed. Contact Mike Reome for guidance, (315) 772-9471.

--- Time of Year Restrictions apply. Prior site-specific coordination and approval is required if vegetation removal is part of this request. Vegetation can only be removed after Aug 1 or before April 15th. POC is Chris Dobony, (315) 778-6348.

--- Time of Year Restrictions apply. Prior site-specific coordination and approval is required if tree removal is part of this request. Trees greater than 3" DBH can only be removed after Oct 15 or before April 15th. POC is Chris Dobony, (315) 778-6348.

--- This project area is currently in the CERCLA RI phase for the reason of having exceeded DoD screening level criteria. Any excavated soil must remain at the project site. If the soil cannot be reused at the site, contact the IRP Manager and/or Solid Waste Program Manager for excess soil protocol in an area with known or suspected PFAS contamination.

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

02/19, CHG 3: 11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 273 Standards for Universal Waste Management

49 CFR 173 Shippers - General Requirements for Shipments and Packagings

49 CFR 178 Specifications for Packagings

1.2 DEFINITIONS

1.2.1 Co-mingle

The practice of placing unrelated materials together in a single container, usually for benefits of convenience and speed.

1.2.2 Construction Waste

Waste generated by construction activities, such as scrap materials, damaged or spoiled materials, temporary and expendable construction materials, and other waste generated by the workforce during construction activities.

1.2.3 Demolition Debris/Waste

Waste generated from demolition activities, including minor incidental demolition waste materials generated as a result of Intentional dismantling of all or portions of a building, to include clearing of building contents that have been destroyed or damaged.

1.2.4 Disposal

Depositing waste in a solid waste disposal facility, usually a managed landfill or incinerator, regulated in the US under the Resource Conservation and Recovery Act (RCRA).

1.2.5 Diversion

The practice of diverting waste from disposal in a landfill or incinerator, by means of eliminating or minimizing waste, or reuse of materials.

1.2.6 Final Construction Waste Diversion Report

A written assertion by a material recovery facility operator identifying

constituent materials diverted from disposal, usually including summary tabulations of materials, weight in short-ton.

1.2.7 Recycling

The series of activities, including collection, separation, and processing, by which products or other materials are diverted from the solid waste stream for use in the form of raw materials in the manufacture of new products sold or distributed in commerce, or the reuse of such materials as substitutes for goods made of virgin materials, other than fuel.

1.2.8 Reuse

The use of a product or materials again for the same purpose, in its original form or with little enhancement or change.

1.2.9 Salvage

Usable, salable items derived from buildings undergoing demolition or deconstruction, parts from vehicles, machinery, other equipment, or other components.

1.2.10 Source Separation

The practice of administering and implementing a management strategy to identify and segregate unrelated waste at the first opportunity.

1.3 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)

Divert a minimum of 60 percent by weight of the project construction waste and demolition debris/waste from the landfill or incinerator. Follow applicable industry standards in the management of waste. Apply sound environmental principles in the management of waste. (1) Practice efficient waste management when sizing, cutting, and installing products and materials and (2) use all reasonable means to divert construction waste and demolition debris/waste from landfills and incinerators and to facilitate the recycling or reuse of excess construction materials.

1.4 CONSTRUCTION WASTE MANAGEMENT

Implement a Construction Waste Management Program for the project. Take a pro-active, responsible role in the management of construction construction waste, recycling process, disposal of demolition debris/waste, and require all subcontractors, vendors, and suppliers to participate in the Construction Waste Management Program. Establish a process for clear tracking, and documentation of construction waste and demolition debris/waste.

1.4.1 Implementation of Construction Waste Management Program

Develop and document how the Construction Waste Management Program will be implemented in a Construction Waste Management Plan. Submit a Construction Waste Management Plan to the Contracting Officer for approval. Construction waste and demolition debris/waste materials include un-used construction materials not incorporated in the final work, as well as demolition debris/waste materials from demolition activities or deconstruction activities. In the management of waste, consider the availability of viable markets, the condition of materials, the ability to

provide material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates.

1.4.2 Oversight

The Quality Control Manager, as specified in Section 01 45 00 QUALITY CONTROL, is responsible for overseeing and documenting results from executing the Construction Waste Management Plan for the project.

1.4.3 Special Programs

Implement special programs involving rebates or similar incentives related to recycling of construction waste and demolition debris/waste materials. Retain revenue or savings from salvaged or recycling, unless otherwise directed. Ensure firms and facilities used for recycling, reuse, and disposal are permitted for the intended use to the extent required by federal, state, and local regulations.

1.4.4 Special Instructions

Provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the projects. Designation of single source separating or commingling will be clearly marked on the containers.

1.4.5 Waste Streams

Delineate waste streams and characterization, including estimated material types and quantities of waste, in the Construction Waste Management Plan. Manage all waste streams associated with the project. Typical waste streams are listed below. Include additional waste streams not listed:

- a. Land Clearing Debris
- b. Asphalt
- c. Masonry and CMU
- d. Concrete
- e. Metals (Includes, but is not limited to, banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, bronze.)
- f. Wood (nails and staples allowed)
- g. Glass
- h. Paper
- i. Plastics (PET, HDPE, PVC, LDPE, PP, PS, Other)
- j. Gypsum
- k. Non-hazardous paint and paint cans
- l. Carpet

- m. Ceiling Tiles
- n. Insulation
- o. Beverage Containers

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Waste Management Plan; G

SD-06 Test Reports

Quarterly Reports

Annual Report

SD-11 Closeout Submittals

Final Construction Waste Diversion Report

1.6 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Construction Waste Management Plan and to develop a mutual understanding relative to the management of the Construction Waste Management Program and how waste diversion requirements will be met.

The requirements of this meeting may be fulfilled during the coordination and mutual Understanding meeting outlined in Section 01 45 00 QUALITY CONTROL. At a minimum, discuss and document waste management goals at following meetings:

- a. Preconstruction meeting.
- b. Regular Quality Control meetings.
- c. Work safety meeting (if applicable).

1.7 CONSTRUCTION WASTE MANAGEMENT PLAN

Submit Construction Waste Management Plan within 15 calendar days after notice to proceed. Revise and resubmit Construction Waste Management Plan until it receives final approval from the Contracting Officer, in order for construction to begin. Execute demolition or deconstruction activities in accordance with Section 02 41 00 DEMOLITION. Manage demolition debris/waste or deconstruction materials in accordance with the approved construction waste management plan.

An approved Construction Waste Management Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Ensure all subcontractors receive a copy of the approved Construction Waste Management Plan. The plan demonstrates how to meet the project waste diversion requirement. Also, include the following in the plan:

- a. Identify the names of individuals responsible for waste management and waste management tracking, along with roles and responsibilities on the project.
- b. Actions that will be taken to reduce solid waste generation, including coordination with subcontractors to ensure awareness and participation.
- c. Description of the regular meetings to be held to address waste management.
- d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of materials.
- e. Name of landfill and incinerator to be used.
- f. Identification of local and regional re-use programs, including non-profit organizations such as schools, local housing agencies, and organization that accept used materials such as material exchange networks and resale stores. Include the name, location, phone number for each re-use facility identified, and provide a copy of the permit or license for each facility.
- g. List of specific materials, by type and quantity, that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, address, and phone number.
- h. Identification of materials that cannot be recycled or reused with an explanation or justification, to be approved by the Contracting Officer.
- i. Description of the means by which materials identified in item (g) above will be protected from contamination.
- j. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).
- k. Copy of training plan for subcontractors and other services to prevent contamination by co-mingling materials identified for diversion and waste materials.
- l. Facilities or subcontractors offering construction waste transport on-site or off-site must ensure that proper shipping orders, bill of lading, manifests, or other shipping documents containing waste diversion information meet requirements of 40 CFR 273 Universal Waste Management, 49 CFR 173 Shippers - General Requirements for Shipments and Packagings, and 49 CFR 178 Specifications for Packaging. Individuals signing manifests or other shipping documents should meet

the minimum training requirements.

- m. List each supplier who deliver construction materials, in bulk, or package products in returnable containers or returnable packaging, or have take-back programs. List each program and the applicable material to actively monitor and track to assist in meeting waste diversion requirements on the project.

Distribute copies of the waste management plan to each subcontractor, [Quality Control Manager](#) , and the Contracting Officer.

1.8 RECORDS (DOCUMENTATION)

1.8.1 General

Maintain records to document the types and quantities of waste generated and diverted through re-use, recycling and sale to third parties; through disposal to a landfill or incinerator facility. Provide explanations for materials not recycled, reused or sold. Collect and retain manifests, weight tickets, sales receipts, and invoices specifically identifying diverted project waste materials or disposed materials.

1.8.2 Accumulated

Maintain a running record of materials generated and diverted from landfill disposal, including accumulated diversion rates for the project. Make records available to the Contracting Officer during construction or incidental demolition activities. Provide a copy of the diversion records to the Contracting Officer upon completion of the construction, incidental demolitions or minor deconstruction activities.

1.9 REPORTS

1.9.1 General

Maintain current construction waste diversion information on site for periodic inspection by the Contracting Officer. Include in the quarterly reports, annual reports and final reports: the project name, contract information, information for waste generated, diverted and disposed of for the current reporting period and show cumulative totals for the project. Reports must identify quantities of waste by type and disposal method. Also include in each report, supporting documentation to include manifests, weigh tickets, receipts, and invoices specifically identifying the project and waste material type and weighted sum.

1.9.2 Quarterly Reporting

Provide cumulative reports at the end of each quarter (December, March, June, and September, corresponding with the federal fiscal year for reporting purposes). Submit [quarterly reports](#) not later than 15 calendar days after the preceding quarter has ended. Submit Quarterly Reports to the Department of Public Works (DPW) Solid Waste Manager..

1.9.3 Annual Reporting

Provide a cumulative construction waste diversion report annually. Submit [annual report](#) not later than 30 calendar days after the preceding fourth quarter has ended. Provide copy of annual construction waste diversion

report to the Department of Public Works (DPW) Solid Waste Manager.

1.10 FINAL CONSTRUCTION WASTE DIVERSION REPORT

A Final Construction Waste Diversion Report is required at the end of the project. Provide [Final Construction Waste Diversion Report](#) 60 days prior to the Beneficial Occupancy Date (BOD). The final Construction Waste Diversion Report must be included in the Sustainability eNotebook in accordance with Section [01 33 29](#) SUSTAINABILITY REQUIREMENTS AND REPORTING.

1.11 COLLECTION

Collect, store, protect, and handle reusable and recyclable materials at the site in a manner which prevents contamination, and provides protection from the elements to preserve their usefulness and monetary value. Provide receptacles and storage areas designated specifically for recyclable and reusable materials and label them clearly and appropriately to prevent contamination from other waste materials. Keep receptacles or storage areas neat and clean.

~~Installation will provide Contractor with a roll-off dumpster for metal waste to be recycled. Contractor to provide 72 hours advance notification to Installation for pickup of full and delivery of new/empty dumpsters. Contractor to only send metal waste to recycling center.~~ Contractor is responsible for off-site disposal of [recycling and](#) all other waste.

Train subcontractors and other service providers to either separate waste streams or use the co-mingling method as described in the Construction Waste Management Plan. Handle hazardous materials in accordance with applicable regulations and coordinate with Section [01 57 19](#) TEMPORARY ENVIRONMENTAL CONTROLS and Section [02 61 13](#) Excavation and Handling of Contaminated Materials. Separate materials by one of the following methods described herein:

1.11.1 Source Separation Method

Separate waste products and materials that are recyclable from trash and sort as described below into appropriately marked separate containers and then transport to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the category types as defined in the Construction Waste Management Plan.

1.11.2 Co-Mingled Method

Place waste products and recyclable materials into a single container and then transport to an authorized recycling facility, which meets all applicable requirements to accept and dispose of recyclable materials in accordance with all applicable local, state and federal regulations. The Co-mingled materials must be sorted and processed in accordance with the approved Construction Waste Management Plan.

1.11.3 Other Methods

Other methods proposed by the Contractor may be used when approved by the Contracting Officer.

1.12 DISPOSAL

Control accumulation of waste materials and trash. Recycle or dispose of collected materials off-site at intervals approved by the Contracting Officer and in compliance with waste management procedures as described in the waste management plan. Except as otherwise specified in other sections of the specifications, dispose of in accordance with the following:

1.12.1 Reuse

Give first consideration to reusing construction and demolition materials as a disposition strategy. Recover for reuse materials, products, and components as described in the approved Construction Waste Management Plan. Coordinate with the Contracting Officer to identify onsite reuse opportunities or material sales or donation available through Government resale or donation programs. Sale of recovered materials is not allowed on the Installation. Consider the use of surplus industrial supply broker services, who match entities with reusable or repurpose industrial materials with entities with need of such materials.

1.12.2 Recycle

Recycle non-hazardous construction and demolition/debris materials that are not suitable for reuse. Track rejection of contaminated recyclable materials by the recycling facility. Rejected recyclables materials will not be counted as a percentage of diversion calculation. Recycle all fluorescent lamps, HID lamps, mercury (Hg) -containing thermostats and ampoules, and PCBs-containing ballasts and electrical components as directed by the Contracting Officer. Do not crush lamps on site as this creates a hazardous waste stream with additional handling requirements.

1.12.3 Waste

Dispose by landfill or incineration only those waste materials with no practical use, economic benefit, or recycling opportunity.

1.13 INSTALLATION SPECIFIC CONSTRUCTION & DEMOLITION DEBRIS

1.13.1 Unsalvageable Waste

All unsalvageable solid waste must be disposed of off the installation at a federal or state permitted facility post in accordance with local, state and federal laws and regulations. In case of a conflict among these laws and regulations, the most stringent law or regulation shall apply. Some of the project sites may be adjacent to wetlands. Wetlands will be identified by the Government prior to giving notice to proceed for any Task Order requiring work near these areas. Contractor shall not place or store material in any area designated as a wetland. Contractor is responsible for protecting and barricading any dumpsters to prevent unwarranted items being dumped.

The contractor shall retain all weight tickets for solid waste disposed of off the installation (Exhibit A, Data Item #A012).

The weight of all unsalvageable solid waste disposed of off the installation shall be reported to the COR, who will forward the information to Fort Drum DPW-Environmental Division in pounds or tons. Location of disposal shall be reported to the COR for inclusion in 360

Disposal Reporting.1.13.2 Salvageable Waste, Recycle, and Re-use

Definition: Any material recovered from construction, renovation, and demolition projects that are not disposed of at a permitted landfill. Salvage materials include but are not limited to metals, wood, plastics, glass, paper, or cardboard.

Contractor's salvageable materials become the property of the Contractor upon removal from the job site.

All salvageable material (including recycle) shall be removed from the site by the Contractor and disposed of off post in accordance with local, state and federal laws and regulations. In case of a conflict among these laws and regulations, the most stringent law or regulation shall apply. Some of the project sites may be adjacent to wetlands. Wetlands will be identified by the Government prior to giving notice to proceed for any Task Order requiring work near these areas. Contractor shall not place or store material in any area designated as a wetland. Contractor is responsible for protecting and barricading any dumpsters to prevent unwarranted items being dumped.

Clean, uncontaminated spoils include but are not limited to, concrete, asphalt (not to exceed ½ cubic yard in size), brick, masonry, stone, aggregate, soils, chipped brush and stumps. These materials are generally utilized in borrow pit rehab, mitigation projects, or re-used on-site as fill or grade materials. Access to borrow pits is controlled and monitored by DPW Master Planning.

Weights for salvage, recycle and re-use materials will be reported to the Fort Drum DPW-Environmental Division monthly on the first of the month. The Contractor may weigh the materials and retain the weight tickets, or if weighing/weight tickets are not available the contractor may estimate the weight. Weights shall be reported in pounds or tons.

1.13.3 Hazardous Waste

Contractor is responsible for removal of all Hazardous waste from the Contract site in accordance with local, state, and federal laws and regulations. In case of a conflict among these laws and regulations, the most stringent law or regulation shall apply. Some of the project sites may be adjacent to wetlands.

Shipping documents (e.g. manifests, bills of lading, land ban forms, waste profiles) for transportation of hazardous materials/wastes off site shall be submitted for approval to the Hazardous Waste Manager, Environmental Division, Public Works at least three days prior to shipment of the material. These documents shall be completed in accordance with local, state and federal laws and regulations. In case of a conflict among these laws and regulations, the most stringent law or regulation shall apply.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used. -- End of Section --

SECTION 02 41 00

DEMOLITION
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

(2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

1.2.1.2 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.2 Demolition

Prepare a [Demolition Plan](#) and submit proposed salvage, demolition, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress, a disconnection schedule of utility services, and airfield lighting, a detailed description of methods and equipment to be used for each operation and of the sequence of operations. Identify components and materials to be salvaged for reuse or recycling with reference to paragraph Existing Facilities to be Removed. Append tracking forms for all removed materials indicating type, quantities, condition, destination, and end use. Coordinate with Waste Management Plan in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Provide procedures for safe conduct of the work in accordance with EM 385-1-1. Plan must be approved by Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition until authorization is received from the Contracting Officer. The work of this section is to be performed in a manner that maximizes the value derived from the salvage and recycling of materials. Remove rubbish and debris from the project site; do not allow accumulations on airfield pavements. The work includes demolition, salvage of identified items and materials, and removal of resulting

rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with EM 385-1-1, Section 23, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract. Do not overload pavements to remain. Provide new supports and reinforcement for existing construction weakened by demolition, deconstruction, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove snow, dust, dirt, and debris from work areas daily.

1.3.2 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition and deconstruction operations. Prior to start of work, utilities serving each area of alteration or removal will be shut off by the Government and disconnected and sealed by the Contractor.

1.3.3 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted.

1.5 AVAILABILITY OF WORK AREAS

Areas in which the work is to be accomplished will be available in accordance with the Section 01 14 00 WORK RESTRICTIONS.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL

PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G

Existing Conditions

SD-07 Certificates

Notification; G

1.7 QUALITY ASSURANCE

Submit timely **notification** of demolition to the Contracting Officer, in writing 10 working days, prior to the commencement of work. Comply with federal, state, and local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the requirements in accordance with Sections **01 14 00** WORK RESTRICTIONS, **01 35 13** SPECIAL PROJECT PROCEDURES, and **01 57 19** TEMPORARY ENVIRONMENTAL CONTROLS. Use of explosives will not be permitted.

1.7.1 Dust and Debris Control

Prevent the spread of dust and debris on airfield pavements and avoid the creation of a nuisance or hazard in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution. Sweep pavements as often as necessary to control the spread of debris that may result in foreign object damage potential to aircraft.

1.8 PROTECTION

1.8.1 Traffic Control Signs

a. Where aircraft safety is endangered in the area of removal work, use traffic barricades with flashing lights. Anchor barricades in a manner to prevent displacement by wind, jet or prop blast. Notify the Contracting Officer prior to beginning such work.

1.8.2 Protection of Personnel

Before, during and after the demolition work continuously evaluate the condition of the site specific features being demolished and take immediate action to protect all personnel working in and around the project site.

1.9 FOREIGN OBJECT DAMAGE (FOD)

Aircraft and aircraft engines are subject to FOD from debris and waste material lying on airfield pavements. Remove all such materials that may appear on operational aircraft pavements due to the Contractor's operations. If necessary, the Contracting Officer may require the Contractor to install a temporary barricade at the Contractor's expense to control the spread of FOD potential debris. Provide a barricade consisting of a fence covered with a fabric designed to stop the spread of debris. Anchor the fence and fabric to prevent displacement by winds or jet/prop blasts. Remove barricade when no longer required.

1.10 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

1.11 EXISTING CONDITIONS

Before beginning any demolition or deconstruction work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer or the Contracting Officer's Representative showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer or the Contracting Officer's Representative.

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

Inspect and evaluate existing structures onsite for reuse. Disassemble existing construction scheduled to be removed for reuse. Dismantled and removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for reuse, remanufacture, recycling, or other disposal, as specified. Designate materials for reuse onsite whenever possible.

3.1.1 Utilities and Related Equipment

3.1.1.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition or deconstruction work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.1.2 Disconnecting Existing Utilities

Remove existing utilities, as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting

Officer prior to further work in that area. Remove meters and related equipment and deliver to a location in accordance with instructions of the Contracting Officer.

3.1.2 Paving and Slabs

Demolition of concrete pavement on airfields and other heavy duty pavements will follow methods in Section 32 13 14.13 CONCRETE PAVING FOR AIRFIELDS AND OTHER HEAVY DUTY PAVEMENTS. Provide neat sawcuts at limits of pavement removal as indicated. Remove demolished pavements and slabs from the installation at no additional cost to the Government.

3.1.3 Airfield Lighting

Remove existing airfield lighting as indicated and terminate in a manner satisfactory to the Contracting Officer. Remove edge dispose of off station.

3.2 CONCURRENT EARTH-MOVING OPERATIONS

Do not begin excavation, filling, and other earth-moving operations that are sequential to demolition work in areas occupied by structures to be demolished until all demolition in the area has been completed and debris removed. Fill holes and other hazardous openings.

3.3 DISPOSITION OF MATERIAL

3.3.1 Title to Materials

Except for salvaged items specified in related Sections, and for materials or equipment scheduled for salvage, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition and deconstruction, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

3.3.2 Reuse of Materials and Equipment

Remove and store materials and equipment listed in the Demolition Plan indicated to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture re-use of materials in the diversion calculations for the project.

3.3.3 Salvaged Materials and Equipment

Remove materials and equipment that are listed in the Demolition Plan indicated and specified to be removed by the Contractor and that are to remain the property of the Government, and deliver to a storage site, as directed by the Contracting Officer.

- a. Salvage items and material to the maximum extent possible.

- b. Store all materials salvaged for the Contractor as approved by the Contracting Officer and remove from Government property before completion of the contract. Coordinate the salvaged materials with tracking requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture salvaged materials in the diversion calculations for the project.
- c. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers. Deliver the following items reserved as property of the Government to the areas designated.
- d. Remove the following items reserved as property of the using service prior to commencement of work under this contract.
- ~~e. All electrical conductors removed as part of this project are property of the government. Transport recyclable electrical conductors to the staging area metal recycling dumpster provided by the government~~

3.4 CLEANUP

Remove debris and rubbish from project site and similar excavations. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

3.5 DISPOSAL OF REMOVED MATERIALS

3.5.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable federal, state and local regulations.

3.5.2 Burning on Government Property

Burning of materials removed from demolished and deconstructed structures will not be permitted on Government property.

3.6 REUSE OF SALVAGED ITEMS

Salvaged items shall not be reused on this project. Materials, equipment, and the like specifically specified to be removed and reinstalled shall not be considered "Salvaged." Replace specified items to be remove and reinstalled that are damaged by the contractor during removal or storage of items prior to reinstallation.

-- End of Section --

SECTION 02 61 13

EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL

02/10, CHG 1: 02/21

PART 1 GENERAL

1.1 DEFINITIONS

1.1.1 Contaminated Soil

Soil determined through laboratory testing to exceed the NYSDEC Part 360 classification for "unrestricted use".

1.1.2 Unrestricted Use Soil

Soil determined through laboratory testing to meet the NYSDEC Part 360 classification for "unrestricted use".

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

(2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926

Safety and Health Regulations for Construction

40 CFR 302

Designation, Reportable Quantities, and Notification

1.3 DESCRIPTION OF WORK

The work consists of excavation and temporary storage of known ~~or suspected~~ areas of PFAS contamination as indicated on the drawings. Submit a **Work Plan** as specified below. Notify the Contracting Officer within 24 hours, and before excavation, if contaminated soil is discovered that has not been previously identified or if other discrepancies between data provided and actual field conditions are discovered. Backfill material is not available onsite. Ground water was not encountered in any borings drilled a depth of 14 feet below existing surface conditions. Comply with requirements in Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.

1.3.1 Scheduling

Notify the Contracting Officer 14 calendar days prior to the start of excavation of contaminated material. The Contractor is responsible for contacting regulatory agencies in accordance with the applicable reporting requirements. Coordinate waste collection and disposal in advance with the Fort Drum Environmental Office.

1.3.2 Work Plan

Submit a Work Plan at least 30 calendar days prior to beginning construction activities. Do not perform work at the site, with the exception of site inspections and surveys, until the Work Plan is approved. Allow 30 calendar days in the schedule for the Government's review. No adjustment for time or money will be made if resubmittals of the Work Plan are required due to deficiencies in the plan. At a minimum, include the following in the Work Plan:

- a. Schedule of activities.
- b. Method of excavation and equipment to be used.
- c. Shoring or side-wall slopes proposed.
- d. Storage methods and locations for solid contaminated material.
- e. Spill contingency plan.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Work Plan; G

SD-02 Shop Drawings

Surveys

SD-06 Test Reports

Compaction

1.5 REGULATORY REQUIREMENTS

1.5.1 Permits and Licenses

Obtain required federal, state, and local permits for excavation and storage of contaminated material. Obtain permits at no additional cost to the Government.

1.5.2 Air Emissions

Monitor and control air emissions in accordance with Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.

PART 2 PRODUCTS

2.1 SPILL RESPONSE MATERIALS

Provide appropriate spill response materials including, but not limited to the following: containers, adsorbents, shovels, and personal protective equipment. Make spill response materials available at all times when contaminated materials/wastes are being handled or transported. Provide spill response materials that are compatible with the type of materials and contaminants being handled.

2.2 BACKFILL

Soil excavated from known/~~suspected~~ areas of PFAS contamination may be used as backfill with the area that is excavated. Manage excess soils excavated with areas of known/~~suspected~~ PFAS contamination in accordance with Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.

PART 3 EXECUTION

3.1 WORK WITHIN AREAS OF KNOWN ~~OR-SUSPECTED~~ PFAS CONTAMINATION

All work within the areas of known ~~or-suspected~~ PFAS contaminations must be completed in accordance with the following related sections and plans, as applicable:

- a. 01 14 00 WORK RESTRICTIONS
- b. 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS
- c. 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS
- d. 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
- e. Contractor's approved Environmental Health and Safety Plan (EHSP)
- f. Contractor's approved work plan

3.2 SURVEYS

Perform surveys immediately prior to and after excavation of contaminated material to determine the volume of contaminated material removed. Determine the volume of excess soil excavated and stockpiled in areas of known/~~suspected~~ areas of PFAS contamination prior to spreading on site. Feather edges to blend/match existing grades. Determine if excess soil after spreading onsite will remain that will require government sampling and analysis. Following removal and legal disposal of excess stockpiled soil and prior to placement of topsoil, complete a survey of the limits of known/~~suspected~~ PFAS contaminated soil spread onsite at 25-foot intervals and at a 50-foot grid interval through the entire areas where excess soils were spread. Provide survey data to the Government for review and approval. No additional costs or schedule extensions will be considered by the government for regrading or removal excessively spread material that impedes stormwater runoff from airfield pavements or within existing stormwater drainage features. The limits of each area where known/~~suspected~~ PFAS contaminated soil is spread on site shall be shown on the As-built Condition and Topographic Survey.

3.3 EXISTING STRUCTURES AND UTILITIES

Do not perform excavation until site utilities have been field located. Take the necessary precautions to ensure no damage occurs to existing structures and utilities. Repair damage to existing structures and utilities resulting from the Contractor's operations at no additional cost to the Government. Do not disturb utilities encountered that were not previously shown or otherwise located without approval from the Contracting Officer.

3.4 CONTAMINATED MATERIAL REMOVAL

3.4.1 Excavation

Excavate areas of contamination to the depth and extent required to complete the work included in the project unless directed by the Contracting Officer. Perform excavation in a manner that will eliminate the potential for contaminated material to be mixed with uncontaminated material. Soils excavated outside the limits of known/suspected PFAS contamination as shown on the drawings must not be commingled with soils excavated within the limits on known/suspected PFAS contamination, or vice versa. Soils excavated within as an area of known/suspected PFAS contamination will not be use, transported, or stockpiled anywhere other than the area from which it was excavation. A trained Environmental, Health, and Safety (EH&S) Officer must be on site at all times during construction activities near or within the potential areas of concern. Adhere to all applicable federal, state and local rules and regulations while performing work within or near the potential areas of concern to include wetting of excavation areas prior to and throughout excavation activities and use of the appropriate personal protective equipment.

3.4.2 Shoring

If workers must enter the excavation, evaluate, shore, slope or brace it as required by EM 385-1-1 and 29 CFR 1926 section 650.

3.5 CONTAMINATED MATERIAL STORAGE

Place material in temporary storage immediately after excavation. The following paragraphs describe acceptable methods of material storage.

3.5.1 Stockpiles

Construct temporary stockpiles to isolate stored excess contaminated material from the environment. Temporarily stockpile all excavated material located within the potential areas of concern in one location near the original source of the material. The height of the stockpile must not penetrate above the obstruction elevations indicated on the drawings. All contaminated material must be stockpiled in a maximum of one location for each designated construction free zone indicated on the drawings. Protect stockpiles from potential contaminate migration in accordance with all applicable federal, state and local rules and regulations.

3.6 SAMPLING

3.6.1 Sampling of Stored Material

The government will complete sampling and laboratory analysis of stockpiled excess soil to determine handling and disposal requirements.

Laboratory analytical reports will be provided by the Government to the Contractor. Sampling and Analysis will be completed in accordance with the provisions of Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS. The Government will not complete sampling and laboratory analysis for the Contractor's own use or for compliance with its EH&S Plan. Contractor required sampling and laboratory analysis, if any, will be at the contractor's own experience and at no additional cost to the Government.

3.7 SPILLS

In the event of a spill or release of a hazardous substance (as designated in 40 CFR 302), pollutant, contaminant, or oil (as governed by the Oil Pollution Act (OPA), 33 U.S.C. 2701 et seq.), notify the Contracting Officer immediately. If the spill exceeds the reporting threshold, follow the pre-established procedures as described in the Base Wide Contingency Plan for immediate reporting and containment. Take immediate containment actions to minimize the effect of any spill or leak. Perform cleanup in accordance with applicable federal, state, and local regulations. As directed by the Contracting Officer, perform additional sampling and testing to verify spills have been cleaned up. Perform spill cleanup and testing at no additional cost to the Government.

3.8 BACKFILL AND COMPACTION

Complete backfill and compaction in accordance with Section 31 00 00 EARTHWORK.

3.9 SOIL STABILIZATION

Contractor must provide temporary soil stabilization measures to prevent wind-borne or stormwater runoff migration of exposed soils within known ~~/suspected~~ areas of PFAS contamination through construction until final restoration and vegetative growth is established in accordance with Section 32 92 19 SEEDING. Contractor selected method(s) of soil stabilization must not generate or pose a FOD hazard to aircraft. Immediately repair areas of soil stabilization that pose a FOD hazard to aircraft or when wind or stormwater runoff shows migration/erosion of exposed soils.

3.10 DISPOSAL REQUIREMENTS

Complete offsite disposal of known ~~/suspected~~ PFAS contaminated soil and provide documentation of disposal in accordance with Section 01 57 19 Temporary Environmental Controls.

-- End of Section --

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Directorate of Public Works – Environmental Division

Management Policy for PFAS Contaminated Soils Originating on Fort Drum

EXTERNAL DISTRIBUTION

Current as of 23-Jan-2026

Fort Drum PFAS Contaminated Soils Management Policy

A. Background Information

1. A PFAS Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) Remedial Investigation (RI) is currently underway at Fort Drum to determine the extent of PFAS contamination in soil, groundwater, and surface water at a minimum of 27 locations (“Areas of Interest or “AOIs”) across the installation.
2. The RI is a multi-year effort through 2027 at minimum.
3. In intent of this policy is to avoid creating additional PFAS hazardous waste sites on Fort Drum when relocating contaminated soils from a construction site to an alternative Fort Drum location, including the Land Clearing Debris (LCD) Landfill.

B. Protocol:

Until contamination has been thoroughly delineated, Environmental Division will utilize all existing analytical results, groundwater flow models, and project characteristics to make soil management determinations on a case-by-case basis according to the following and associated decision tree (*Appendix A*):

1. During the project citing and planning process, Directorate of Public Works (DPW) EP&S staff should avoid construction in areas of known PFAS contamination to the greatest extent possible in accordance with Installation Management Command (IMCOM) Policy Memorandum – *Timely Identification and Mitigation of Contaminated Soil at Proposed Construction Sites on Installation Management Command Installations* (NOV 25).
2. For projects that are known or suspected of having PFAS contamination and avoiding construction in the area is not possible, contract Statements of Work shall specify that contractors are required to maintain all soil within the construction footprint and proximal to the original source of the material. Excavated soil should first be used as backfill and then spread/feathered when no additional backfill material is required near the source of the material. Contractors are not authorized to haul excess soil off the project site. Given the

variability of contamination across the installation, consult with Fort Drum DPW Environmental Division to ensure proper placement of excess soil.

3. If the contractor is unable to use excess soils within the construction footprint but review of historic analytical sampling results performed by the Environmental Division shows favorable results (i.e., no exceedances compared to Office of the Secretary of Defense (OSD) 2023 Risk Screening Levels for Soil (RSLs)¹) then excess soil can be directly hauled to the designated location at the Fort Drum Land Clearing Debris (LCD) Landfill (*Appendix B*). No temporary stockpiling on site or composite sampling required. Access to the LCD Landfill will be coordinated through DPW. DPW staff will control access to the LCD Landfill and will ensure that no soil relocated to the designated area (surrounded by silt fence and posted with off-limits signage) is used for future projects. Contract Statements of Work shall require/direct:

- a) The placement of excess spoils at the LCD Landfill after the Contractor has exhausted all means to manage excess soils on site. Every effort should be made to keep as much soil on site as possible to minimize the amount of soil that is hauled to the LCD Landfill.
- b) The Contractor to report the total cubic yards to the Project COR after soils have been relocated to the LCD Landfill. The report will include marked up maps showing excavation location(s) and a log detailing dates hauled, and volumes moved (*Appendix C*) to accurately and thoroughly document how much soil has been relocated from a given project area to the LCD Landfill.

4. If the contractor is unable to use excess soils within the construction footprint and review of historic analytical sampling results performed by the Environmental Division show exceedances compared to OSD 2023 Risk Screening Levels¹ then:

- a) If the project footprint accommodates the temporary stockpiling of excess soils, the contractor will temporarily stockpile excess soils onsite. All excess soils from a given project area will be consolidated and stockpiled in one location as proximal to the original source of the material as possible. A Construction Representative (CON REP) from Fort Drum DPW EP&S will provide oversight when temporary stockpiles are developed to ensure proper placement within the project footprint/area. Fort Drum will complete all

required PFAS sampling²/testing to determine management strategy for all soils that will require removal from the site. A minimum of seven business days of advanced notification will be required to complete soil sampling. Analytical analysis costs will be at the contractor's expense.

- i. If results indicate that PFAS contamination exceeds OSD 2023 Risk Screening Levels¹ the contractor will be required to dispose of excess soils IAW all applicable legal guidelines and OSD policies. Fort Drum will review all disposal documentation prior to removal. Disposal of contaminated soils will be at the contractor's expense.
 - ii. If results indicate that PFAS contamination is below OSD 2023 Risk Screening Levels¹, excavated soils will be hauled to a designated location at the Fort Drum LCD Landfill. Hauling of excess soils will be at the contractor's expense and will be reported per the guidance in *Section 3(b)* above.
- b) If the project footprint cannot accommodate the temporary stockpiling of excess soils, the contractor will haul and temporarily stockpile excess soils at a designated location outside of the project footprint, on hardstand and surrounded by silt fence, identified by EP&S Division in coordination with Environmental Division. All excess soils from a given project area will be consolidated and stockpiled in one location. Fort Drum will complete all required PFAS sampling²/testing to determine management strategy for all soils that will require removal from the site. A minimum of seven business days of advanced notification will be required to complete soil sampling. Analytical analysis costs will be at the contractor's expense.
- i. If results indicate that PFAS contamination exceeds OSD 2023 Risk Screening Levels¹, the contractor will be required to dispose of excess soils IAW all applicable legal guidelines and OSD policies. Fort Drum will review all disposal documentation prior to removal. Disposal of contaminated soils will be at the contractor's expense.
 - ii. If results indicate that PFAS contamination is below OSD 2023 Risk Screening Levels¹, excavated soils will be hauled to a designated location at the Fort Drum LCD Landfill. Hauling of excess soils will be at the

contractor's expense and will be reported per the guidance in *Section 3(b)* above.

Director
DPW

DRAFT

Footnotes

1. OSD 2023 Residential Risk Screening Levels (RSL) for Soil*

Chemical	RSL (mg/kg)
PFOS	0.013
PFOA	0.019

*Detections of target PFAS constituents greater than OSD 2023 RSLs would trigger a CERCLA investigation.

2. Soil Sampling Methodology

- a. PFAS are considered Semi-Volatile Organic Compounds (SVOCs) therefore the number of samples will be based on cubic yards of “spoils”.
- b. Table for determination of the number of composite samples to pull from the stockpile

Cubic Yards (yd ³) of “Spoils”	Number of Composite Samples
0-300	1
300-1,000	2
>1,000	1 additional composite for each additional 1,000 yd ³ of “spoils”

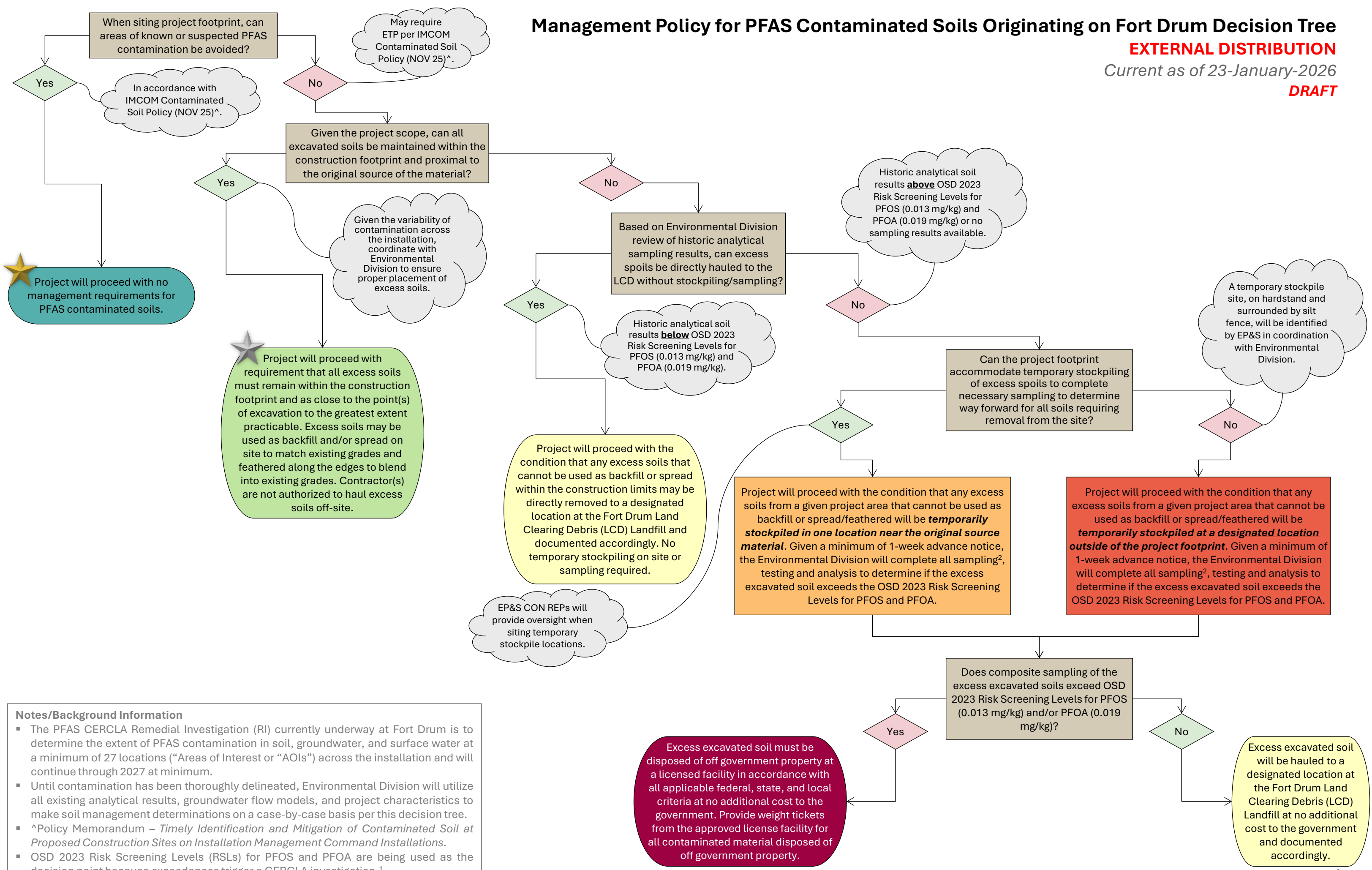
- c. Each composite sample for analysis is created from 3-5 discrete samples from representative locations of the “spoils”.

Management Policy for PFAS Contaminated Soils Originating on Fort Drum Decision Tree

EXTERNAL DISTRIBUTION

Current as of 23-January-2026

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Notes/Background Information

- The PFAS CERCLA Remedial Investigation (RI) currently underway at Fort Drum is to determine the extent of PFAS contamination in soil, groundwater, and surface water at a minimum of 27 locations (“Areas of Interest or “AOIs”) across the installation and will continue through 2027 at minimum.
- Until contamination has been thoroughly delineated, Environmental Division will utilize all existing analytical results, groundwater flow models, and project characteristics to make soil management determinations on a case-by-case basis per this decision tree.
- ^Policy Memorandum – *Timely Identification and Mitigation of Contaminated Soil at Proposed Construction Sites on Installation Management Command Installations.*
- OSD 2023 Risk Screening Levels (RSLs) for PFOS and PFOA are being used as the decision point because exceedances trigger a CERCLA investigation.¹

1. PFAS Contaminated Soil Guidance Values

Organization	Guidance/Screening Value	PFOA (ppm or mg/kg)	PFOS (ppm or mg/kg)
NYSDEC	Unrestricted Site Use	0.00066	0.00088
NYSDEC	Protection of Groundwater	0.00080	0.00100
NYSDEC	Ecological Screening Value	--	0.00870
NYSDEC	Residential Use	0.00660	0.00880
OSD	Residential RSL	0.01900	0.01300
NYSDEC	Restricted Residential	0.03300	0.04400
NYSDEC	Commercial	0.50000	0.44000
NYSDEC	Ecological Screening Value	0.57000	--
NYSDEC	Industrial	0.60000	0.44000

- NYSDEC Guidance and Screening Values from CP-51, *Soil Cleanup Guidance* (June 2024 Proposed Revisions).
- OSD Residential Risk Screening Levels (RSLs) from OSD Memorandum: *Investigating Per- and Polyfluoroalkyl Substances within the Department of Defense Cleanup Program*, August 2023. Detections of target PFAS constituents greater than the 2023 RSLs have and would currently trigger a CERCLA investigation.

2. Soil Sampling Methodology

- a) PFAS are considered SVOCs therefore the number of samples will be based on cubic yards (yd³) of “spoils”.
- b) Table for determination of number of composite samples to pull from the stockpile:

Cubic Yards (yd³) of “Spoils”	Number of Composite Samples
0-300	1
300-1,000	2
>1,000	1 additional composite sample for each additional 1,000 yd³ of “spoils”

- c) Each composite sample for analysis is created from 3-5 discrete samples from representative locations of the “spoils”.

Department of War Detailed Analysis of Options for the Destruction and Disposal of Covered Materials Containing Per- and Polyfluoroalkyl Substances



February 2026

Office of the Assistant Secretary of War
for Energy, Installations, and Environment

CLEARED
For Open Publication

Mar 03, 2026

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I. INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) are a family of thousands of chemicals that have unique physical and chemical properties that impart oil, water, stain, and soil repellency; chemical and thermal stability; and friction reduction to a range of products. Since the 1950s, many products commonly used by consumers and industry have been manufactured with or from PFAS. These products have application in several industries, including the aerospace, semiconductor, medical, automotive, construction, electronics, firefighting, and aviation industries, as well as in consumer applications such as carpets, clothing, furniture, outdoor equipment, and food packaging¹.

Due to their widespread historical use, PFAS are found worldwide in soil, sediments, surface water, groundwater, rainwater, and air. PFAS are persistent chemicals that break down very slowly over time and may be linked to harmful health effects in humans and animals.² The most studied PFAS chemicals in the environment include perfluorooctane sulfonic acid (PFOS) and perfluorooctanoic acid (PFOA), both of which are no longer manufactured in the U.S. PFOS production was voluntarily phased out between 2000 and 2002 and PFOA production phase out began in 2006.³

Since the 1970s, the Department of War (DoW) has used Aqueous Film-Forming Foam (AFFF) to rapidly suppress fires that involve jet fuel. AFFF quickly extinguishes petroleum-based fires, thus minimizing loss of life and protecting equipment and facilities. AFFF formulations contained various PFAS over the years, including PFOS and PFOA. Over the past few years, the Department has undertaken a significant initiative to develop and demonstrate PFAS-free alternatives for AFFF, referred to as fluorine-free firefighting agents (F3). The Department published a new Military Performance-based specification (MILSPEC) for fluorine free foam (F3) on January 6, 2023.⁴ Six F3 agents have completed the military specification qualification process and are available for use. Additional F3 products are undergoing qualification testing to meet DoW MILSPEC standards and the Department continues funding for research, demonstration, and validation projects aimed at improving the firefighting performance of F3 agents. The Department continues to transition to the use of these fluorine-free alternative products on its installations. The Military Departments are also evaluating available technologies, in addition to alternative foams, to replace current AFFF systems in facilities.

¹ ITRC. 2023. PFAS Uses and Products. September 2023. Available online: <https://pfas-1.itrcweb.org/2-5-pfas-uses/>. Accessed September 2025.

² EPA. 2021. PFAS Strategic Roadmap: EPA's Commitments to Action 2021-2024. EPA-100-K-21-002. October 2021. Available online: <https://www.epa.gov/system/files/documents/2021-10/pfas-roadmap_final-508.pdf>. Accessed December 16, 2022.

³ US EPA. 2016. Factsheet PFOA & PFOS Drinking Water Health Advisories. EPA 800-F-16-003.

⁴ MIL-PRF-32725 "FIRE EXTINGUISHING AGENT, FLUORINE-FREE FOAM (F3) LIQUID CONCENTRATE, FOR LAND-BASED, FRESH WATER APPLICATIONS", available online: <https://media.defense.gov/2023/Jan/12/2003144157/-1/-1/1/MILITARY-SPECIFICATION-FOR-FIRE-EXTINGUISHING-AGENT-FLUORINE-FREE-FOAM-F3-LIQUID-CONCENTRATE-FOR-LAND-BASED-FRESH-WATER-APPLICATIONS.PDF>.

DoW is required by section 322 of the National Defense Authorization Act (NDAA) for Fiscal Year (FY) 2020 to cease the use of AFFF on military installations (i.e., land-based applications) and inherent in this process is the removal of existing AFFF. The total volume of AFFF that requires disposal is estimated to be over two million gallons⁵. In addition, firefighting equipment (e.g., firetrucks) and facilities with AFFF systems (e.g., airplane hangars) will need to be rinsed before the new F3 firefighting agent is installed, and this PFAS-containing rinsate will also require disposal. DoW estimates that there will be an additional 1.5 million gallons of rinsate requiring management. Because Congress has dictated timelines for replacement of AFFF, there is a large volume of AFFF and rinsate across all of DoW that requires management. Per section 322, as of October 2023, DoW has prohibited new procurement of AFFF for use at its installations. Use of existing AFFF at military installations will cease by October 2026.⁶

In addition to DoW's transition from the use of AFFF at its military installations, quantities of PFAS-containing materials are generated from DoW's nationwide cleanup program, emergency use of AFFF, and spills of AFFF. These combined quantities exceed DoW's long-term storage capabilities, so DoW requires a comprehensive destruction and disposal strategy to address covered DoW PFAS-containing materials being sent to an offsite commercial destruction or disposal facility.

II. PRIOR EPA AND DOW PFAS DESTRUCTION AND DISPOSAL INTERIM GUIDANCE DOCUMENTS

In December 2020, the U.S. Environmental Protection Agency (EPA) published its *Interim Guidance on the Destruction and Disposal of Perfluoroalkyl and Polyfluoroalkyl Substance and Materials Containing Perfluoroalkyl and Polyfluoroalkyl Substances*, December 18, 2020. This EPA Interim Guidance evaluated destruction and disposal technologies that were currently available for the management of PFAS, including permitted deep well injection (also known as Class I wells), permitted hazardous waste landfills (also known as Resource Conservation and Recovery Act (RCRA) Subtitle C landfills), solid waste landfills (also known as RCRA Subtitle D landfills), hazardous waste combustors (including incinerators), and other thermal treatment.

The DoW PFAS Task Force considered this EPA Interim Guidance when developing the *DoW Interim Guidance on Destruction or Disposal of Materials Containing PFAS Substances in the U.S.*, July 11, 2023. This July 2023 DoW interim guidance was developed to comply with section 343 of the FY 2022 NDAA and identified the following six commercially available alternatives to destroy or dispose of DoW PFAS-containing materials, in order of consideration:

⁵ Assistant Secretary of War for Energy, Installations, & Environment. 2023. *Interim Guidance on Destruction or Disposal of Materials Containing Per- and Polyfluoroalkyl Substances in the United States*. July 11, 2023. *Hereafter referred to as DoW's 2023 interim guidance*.

⁶ Section 322 of the NDAA for FY 2020 allows a maximum of two, one-year waivers if certain conditions are met and the Secretary of Defense certifies to Congress that the waiver (i.e., extension) is necessary. On August 2, 2024, the Secretary of Defense provided Congress with the first certification of the waiver. On July 31, 2025, the Secretary of Defense provided Congress with the second certification of the waiver. The prohibition of use of AFFF on military installations is now extended to October 1, 2026.

1. Carbon reactivation units with environmental permits (for used granular activated carbon (GAC) only)
2. Hazardous waste landfills with environmental permits
3. Solid waste landfills that have composite liners, and gas and leachate collection and treatment systems with environmental permits
4. Underground injection control wells with environmental permits
5. Other PFAS destruction technologies with environmental permits or accepted by environmental regulators
6. Hazardous waste incinerators (HWIs) with environmental permits

On July 14, 2023, DoW issued *Guidance on Incineration of Materials Containing Per- and Polyfluoroalkyl Substances*, prohibiting the use of incineration as a matter of policy until further notice from Office of the Assistant Secretary of War for Energy, Installations and Environment (OASW(EI&E)).

In April 2024, the EPA published *Interim Guidance on the Destruction and Disposal of Perfluoroalkyl and Polyfluoroalkyl Substance and Materials Containing Perfluoroalkyl and Polyfluoroalkyl Substances – Version 2 (2024)*, which is an update to their December 2020 guidance.⁷ The update included new destruction and disposal technology testing results available after the 2020 guidance was published. The 2024 guidance encourages the use of destruction and disposal options that have a lower potential for releasing PFAS into the environment. Technologies EPA evaluated include: permitted class 1 non-hazardous industrial or hazardous waste injection wells; permitted hazardous waste landfills; municipal solid waste landfills with composite liners and leachate and gas collection treatment; GAC reactivation units with thermal oxidizers; and permitted hazardous waste combustors that operate under certain conditions. Additionally, the updated guidance included a technology evaluation framework to help managers of PFAS-containing materials evaluate emerging technologies and inform decisions about destruction and disposal of PFAS-containing materials.

III. UPDATED 2025 DOW PFAS DESTRUCTION AND DISPOSAL GUIDANCE AND ANALYSIS

DoW is now issuing updated interim guidance based upon this analysis to help DoW make informed decisions in the evaluation of existing destruction and disposal options for covered DoW PFAS-containing materials. This 2025 analysis takes into account new PFAS disposal and destruction information, including the EPA's 2024 updated guidance, and

⁷“Interim Guidance on the Destruction and Disposal of Perfluoroalkyl and Polyfluoroalkyl Substances and Materials Containing Perfluoroalkyl and Polyfluoroalkyl Substances—Version 2 (2024).” U.S. Environmental Protection Agency, 8 Apr. 2024. *Referred to as “EPA Interim PFAS Disposal Guidance (Apr. 2024)” in later footnotes.* Available online: <https://www.epa.gov/system/files/documents/2024-04/2024-interim-guidance-on-pfas-destruction-and-disposal.pdf>

incorporates and updates the July 2023 DoW PFAS destruction and disposal guidance and analysis.

This analysis was used in the development of DoW's 2025 interim guidance and considers options for DoW's PFAS-containing materials being sent to a commercial facility for disposal or destruction, unless the materials are from household or consumer products, or the material qualifies as a hazardous waste.⁸

This document analyzes PFAS-containing waste management options to identify options that minimize PFAS emissions to the environment and potential human health exposures. This analysis identifies the following waste management options in no specific order:

1. Carbon Reactivation Units with environmental permits (for use with GAC only).
2. Hazardous Waste Landfills with environmental permits.
3. Solid Waste Landfills that have composite liners and leachate collection and treatment systems, with environmental permits (not for AFFF concentrate).
4. HWIs with environmental permits that meet specific temperature requirements.
5. Underground injection control wells with environmental permits (for liquids only).
6. Thermal desorption units that use off-gas collection and thermal oxidation with environmental permits (for soils).
7. Other PFAS destruction technologies with Environmental permits or regulator-acceptance.

IV. EVALUATION OF AVAILABLE OFF-SITE COMMERCIAL DESTRUCTION & DISPOSAL OPTIONS

A. Screening Criteria for Options

In choosing among disposal and destruction options, one of the key factors for DoW's analysis is the additional oversight and controls provided via environmental permitting requirements at existing storage, disposal and destruction facilities. DoW recognizes the statutory authority and responsibility of EPA and State environmental regulatory agencies to regulate the disposal and destruction of wastes that may threaten human health or the environment, and to issue environmental permits that protect human health and the environment.

⁸The analysis does not apply to PFAS cleanup destruction technologies as remedial alternatives are evaluated and selected with environmental regulator involvement and in accordance with the federal cleanup law, the Comprehensive Environmental Response, Compensation, and Liability Act. Any waste streams generated from DoW PFAS cleanup actions, however, that are sent to commercial disposal or destruction facilities (e.g., GAC) used to treat drinking water in a cleanup that must now be regenerated or disposed) are covered by this analysis. Additionally, materials classified as hazardous wastes, even if mixed with PFAS, are not covered by this analysis as hazardous waste regulations (e.g., land disposal restrictions) identify the appropriate destruction or disposal options. Per the 2025 DoW guidance, DoW Components shall apply the criteria in this guidance (e.g, minimum temperature required for hazardous waste incinerators) for the specific destruction or disposal option utilized for the hazardous waste, in addition to those required by RCRA regulations.

All options discussed in this section are governed by that environmental oversight provided by EPA and State regulators.⁹

Consistent with EPA's 2020 and 2024 interim guidance, DoW has tried to consider waste management options for AFFF and other PFAS-containing waste to develop alternatives that are as protective as possible of human health and the environment. DoW therefore considered the following criteria:

- Currently available waste management options for the disposal of large quantities of AFFF and PFAS-containing material;
- Identified options in DoW's 2023 interim guidance or EPA's 2024 interim guidance;
- Options with protections in place to minimize emissions or releases;
- Environmentally permitted regulator approved options (where permits are not required); and
- The type and composition of the materials to be destroyed or disposed, including the relative concentrations of PFAS.

While cost was not a screening factor for this analysis, the costs and availability of identified options will need to be evaluated by the DoW Components for disposal or destruction of installation-specific PFAS-containing materials.

The analysis that follows carefully evaluated the most effective means for destroying or disposing of PFAS-containing materials and wastes, and prioritized use of destruction and disposal technologies that have a lower potential for PFAS release to the environment to help support protection of human health and the environment. One important factor DoW considered was the additional oversight and requirements that are associated with a facility obtaining an environmental permit. DoW also had to consider that most of the PFAS-containing materials that DoW generates are from its AFFF transition, AFFF spill responses, or its cleanup program where known PFAS exposures have occurred and are being mitigated and otherwise addressed. Direct human exposures (e.g., PFAS in drinking water that is being addressed under DoW's cleanup program which generates materials for destruction and disposal) had to be balanced against the potential environmental releases from destruction and disposal facilities. Even if releases occur in very small quantities at destruction and disposal facilities, they do not necessarily present any exposure risks to specific populations in the surrounding community.¹⁰

⁹Department of War (DoW). 2025. Updated Interim Guidance on Destruction or Disposal of Materials Containing Per- and Polyfluoroalkyl Substances in the United States. February 20, 2026.

¹⁰See EPA 2024 Guidance at page 1 which states, "While the purpose of an effective destruction and disposal technology is to prevent or minimize environmental releases, it is also important to distinguish between a potential environmental release and a direct human exposure. A PFAS release does not inherently imply direct and immediate human exposure and a release does not necessarily present an unacceptable risk to specific populations."

B. Options Analysis

1. Carbon Reactivation Units with environmental permits (for use with granular activated carbon only)

Because DoW, and others, have widely used GAC to remove PFAS from drinking water and groundwater, DoW evaluated carbon reactivation units. While carbon reactivation units “use high temperatures to thermally desorb contaminants from GAC, which allows for the carbon to be used again,”¹¹ they are not defined as “incinerators” and instead are a form of recycling/preserving virgin materials. While there are about seventeen commercial carbon reactivation units across the country, currently only four “operate under RCRA permits and applicable air permits” which “provide additional regulatory oversight and include operating requirements and emission limitations to safely and effectively treat the hazardous contaminants.”¹² Due to these additional safeguards, RCRA-permitted carbon reactivation units “may operate under conditions more conducive to destroying PFAS and controlling related [products of incomplete combustion].”¹³ Additionally, GAC reactivation is economically favored over replacement with virgin carbon.¹⁴ As identified in DoW’s 2023 guidance and this analysis, carbon reactivation units are currently available, receive environmental permits, are an identified option in DoW’s and EPA’s guidance, and have protections in place to minimize emissions or releases. Therefore, DoW has identified RCRA and CAA-permitted carbon reactivation units as an available destruction option to address PFAS-containing GAC.

2. Hazardous Waste Landfills with environmental permits

In the EPA 2024 Guidance, EPA recognized hazardous waste landfills with an environmental permit as one of three technologies that have a lower potential for environmental release of PFAS compared to other technologies in the same category, and are viewed as one of the more protective technologies.¹⁵ Hazardous waste landfills “have the most stringent environmental controls in place for minimizing environmental releases and migration of some PFAS from disposed waste.”¹⁶ Because they typically do not accept biodegradable wastes and are not biologically active, they do not generally need to be equipped to collect and control landfill gas (unlike solid waste landfills). According to EPA, “permitted hazardous waste landfills have the most protective landfill engineering controls and practices for the containment of PFAS waste and would be more effective at minimizing PFAS release into the environment than other landfill types.”¹⁷ Engineering controls and containment practices for hazardous waste landfills include double liner systems with leachate collection and leak detection and extensive record keeping. As identified in our 2023 guidance and this analysis, hazardous waste landfills

¹¹EPA Interim PFAS Disposal Guidance (Apr. 2024). “Interim Guidance on the Destruction and Disposal of Perfluoroalkyl and Polyfluoroalkyl Substances and Materials Containing Perfluoroalkyl and Polyfluoroalkyl Substances,” page 48.

¹²EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 48.

¹³EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 52.

¹⁴EPA Interim PFAS Disposal Guidance (Apr. 2024). Page D-2.

¹⁵EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 2.

¹⁶EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 12.

¹⁷EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 2.

are currently available, receive environmental permits, are an identified option in DoW's and EPA's guidance, and have protections in place to minimize emissions or releases.

The DoW Components will need to consider the type of PFAS-containing materials when considering the use of hazardous waste landfills. For example, liquids must be solidified to remove any free liquids before disposal in a landfill, which may increase the volume significantly (e.g., threefold).¹⁸

3. Solid Waste Landfills that have composite liners and leachate collection and treatment systems, with environmental permits. (Not an option for AFFF concentrate)

The EPA 2020 and 2024 guidances identify a variety of solid waste landfills: municipal solid waste, ash monofill, industrial, and construction and demolition landfills.¹⁹ Because environmental controls can vary at landfills, EPA evaluated the viability of landfilling as a means of containing PFAS. In DoW's 2023 guidance, DoW identified solid waste landfills with composite liners and gas and leachate collection and management in place as available disposal option that maximizes reduction of PFAS releases or emissions to the environment and human health exposures. EPA's 2024 guidance, however, raised additional concerns with potential releases of PFAS from solid waste landfills, even with these controls.²⁰ As an additional precaution, any solid waste landfill DoW uses for PFAS-containing materials must have a composite liner, gas and leachate collection and management systems, and an environmental permit. In addition, based on the type of media, chemical concentrations, and volume of media to be disposed, DoW Components must consider the availability of technologies that may contain PFAS as described in Table 3-4 of the EPA 2024 Guidance.²¹ The DoW Components are directed to use other options for the disposal or destruction of AFFF concentrate (i.e., unmixed liquid concentrate) waste and other liquid PFAS-containing materials.

4. Hazardous waste incinerators with environmental permits that meet specific temperature requirements

DoW also considered thermal treatment technologies, recognizing that these options have data gaps in the understanding of their ability to control the migration of all PFAS compounds into the environment. Thermal treatment technologies include a wide-variety of technologies and controls, including hazardous waste combustors (e.g., incinerators, cement kilns, lightweight aggregate kilns), as well as other thermal treatment (e.g., carbon reactivation units, sewage sludge incinerators, municipal waste combustors, thermal oxidizers).²² In the EPA 2024 Guidance, EPA recognized hazardous waste combustors with an environmental permit as one of three technologies that have a lower potential for environmental release of PFAS compared to other technologies in the same category, and are viewed as one of the more protective

¹⁸Geoengineer. Available online: <https://www.geoengineer.org/education/web-class-projects/cee-549-geoenvironmental-engineering-winter-2013/assignments/stabilization-solidification> Accessed March 16, 2025

¹⁹EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 12 and 13.

²⁰EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 71-76.

²¹EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 77-79.

²²EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 46-51.

technologies.²³ EPA acknowledged that HWIs are designed to optimize temperatures, residence times, turbulence, and other parameters to ensure compliance with organic destruction removal efficiencies.²⁴ EPA, notwithstanding its acknowledgment of uncertainties with PFAS thermal treatment technologies, recognized that the subset of permitted hazardous waste combustors “may operate under conditions more conducive to destroying PFAS and controlling related [products of incomplete combustion] PICs relative to thermal treatment units that do not have both RCRA and [Clean Air Act] CAA permits.”²⁵ Additionally, EPA emphasized that all HWIs, cement kilns, and lightweight aggregate kilns are subject to RCRA and CAA permitting requirements and include operating requirements and emissions limitations to safely and effectively treat regulated hazardous contaminants and may adequately destroy PFAS and minimize PICs (emissions).²⁶ These operating requirements include emission limitations for metals, dioxins/furans, particulate matter, hydrogen chloride and chlorine gas, and carbon monoxide or hydrocarbons, as well as limits on minimum organic destruction and removal efficiency (DRE) (e.g. 99.99 percent for organic chemicals).²⁷ After considering the latest studies and additional information presented in the next section V of this guidance on implementation of section 330 of the FY 2020 NDAA, DoW has identified RCRA and CAA-permitted HWIs with operating temperatures greater than 1100°C in the afterburner/secondary combustion chamber as an available destruction option if the commercial incinerator has conducted a “PFAS Emissions Field Test” following EPA’s 2024 Appendix A recommendations. As identified in DoW’s 2023 guidance and this analysis, HWIs are currently available, receive environmental permits, are an identified option in DoW’s and EPA’s guidance, and have protections in place to minimize emissions or releases.

5. Underground injection control wells with environmental permits (for liquids only)

In the EPA 2024 Guidance, EPA recognized underground injection control, also known as deep well injection, with an environmental permit as one of three technologies that have a lower potential for environmental release of PFAS compared to other technologies in the same category, and are viewed as one of the more protective technologies.²⁸ EPA acknowledged underground injection has the capability to control migration of PFAS to the environment, but “the limited number of these wells currently receiving PFAS and accepting off-site waste, the well location, and waste transportation logistics may significantly limit the type and quantity of PFAS-containing fluids appropriate for underground injection control.”²⁹ Because of the limited availability of underground injection control locations, use for only liquid materials, and the volume of disposal required for DoW PFAS-containing materials, DoW believes this disposal option will rarely be an available option for DoW. DoW, however, has identified underground injection control as a disposal option. As identified in DoW’s 2023 guidance and this analysis,

²³EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 2.

²⁴EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 47.

²⁵EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 46.

²⁶EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 53.

²⁷EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 46.

²⁸EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 2.

²⁹EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 92.

underground injection control wells are currently available, receive environmental permits, are an identified option in DoW's and EPA's guidance, and have protections in place to minimize emissions or releases.

6. Thermal desorption units that use off-gas collection and thermal oxidation with environmental permits (for soils)

DoW mentioned permitted thermal desorption units in its 2023 guidance, as an example of an alternative to HWIs for PFAS-impacted soils. EPA's 2024 Guidance also acknowledge that "though the efficacy of thermal and catalytic oxidizers in destroying PFAS is currently unknown, a properly optimized thermal oxidizer can readily achieve a DRE of 99.99% for volatile organic carbons"³⁰. Thermal treatment technologies that utilize a desorption stage have been demonstrated to remove organic contaminants from soil, sludge, or sediment by heating them to evaporate the contaminants from the solid material and then collect the vapor in a vapor treatment system.³¹ Some thermal desorption units are configured with thermal oxidizers to treat process gases and destroy volatile organic compounds and organic hazardous air pollutants, with some thermal oxidizers being able to also accept liquid feed streams³². In the EPA 2024 Guidance, EPA recognized that "Each thermal oxidizer is designed to optimize temperatures, residence times, and turbulence to address the composition of the particular feed stream at that unit and meet the requirements of a regulation or permit. Depending on the type of thermal oxidizer (direct-fired, regenerative, recuperative, or flameless), the operating temperature ranges from 760°C to 1,200°C (1,400°F to 2,190°F). The residence time of thermal oxidizers ranges from 0.5 to 2 seconds, depending on site-specific criteria."³³ EPA also noted that "...some thermal oxidizers are being used for halogenated wastes, including PFAS precursors. The operating conditions for some thermal oxidizers have the potential to effectively treat PFAS".³⁴ As identified in DoW's 2023 guidance and this analysis, thermal desorption with a thermal oxidizer unit is a currently available technology, receive environmental permits, are an identified option in DoW's and EPA's guidance, and have protections in place to minimize emissions or releases. DoW is therefore identifying thermal desorption units that utilize off-gas collection that is treated by a thermal oxidizer, as a destruction option.

7. Other emerging PFAS destruction technologies with Environmental permits or regulator-acceptance.

DoW's 2023 guidance recognized that other destruction technologies existed or were developing (in addition to HWI and GAC regeneration) and could be considered if the new destruction technologies had an environmental permit or were otherwise approved for PFAS use by the appropriate environmental regulator, and their use is consistent with other criteria in this guidance. EPA's 2024 guidance identifies four emerging technologies (mechanochemical degradation, electrochemical oxidation, gasification and pyrolysis, and supercritical water

³⁰EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 52.

³¹EPA Office of Solid Waste and Emergency Response (Sep 2012). EPA 542-F-12-020. Available online: <https://www.sam.usace.army.mil/Portals/46/docs/library/DDMT/IR%203.23.pdf>

³²EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 51.

³³EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 51.

³⁴EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 51

oxidation (SCWO)) that show promise for PFAS destruction, production of few to no hazardous residuals or byproducts, commercial availability, and cost effectiveness for various impacted matrices.³⁵ To assist PFAS material managers with evaluating whether one of these technologies is suitable, EPA developed the PFAS destruction and disposal technology evaluation framework, which identified priority pieces of information that could provide an evaluation of an emerging technology and its applicability for PFAS disposal or destruction, and focus on innovative technologies that reduce PFAS releases to the environment. EPA has confirmed that users of the framework are not required to fully address each question, rather users should identify the most relevant questions to gain a holistic understanding of the technology's potential for PFAS transformation and/or release to the environment. For example, use of certain air emission methodologies (e.g., USEPA Other Test Method for non-polar volatile PFAS (OTM-50)) may not be relevant for supercritical water oxidation systems because the volume of gaseous effluent emitted is not sufficient to meet the sample collection volumes required by OTM-50. EPA's 2024 guidance acknowledged that they tested the effectiveness of SCWO units from multiple vendors for treating AFFF concentrate. EPA's tests indicated overall destruction efficiencies of greater than 99 percent, although they were not able to monitor all air emissions due to method limitations.³⁶ Over the last eight years, DoW's Environmental Security Technology Certification Program (ESTCP)³⁷ has been conducting rigorous demonstration and validation of multiple PFAS destruction technologies (e.g., SCWO, mechanochemical degradation, hydrothermal alkaline treatment). ESTCP projects collect data comparable to EPA's technology evaluation framework and this data could be useful for DoW decision makers when considering the use of emerging PFAS technologies.

Consistent with EPA's 2024 guidance, DoW's preference is to utilize environmentally permitted technologies which provide additional oversight and emission limitations to safely and effectively destroy or dispose of PFAS-containing materials. Currently, DoW is aware of only a few non-incineration destruction technologies that have environmental permits or regulatory approval to operate at commercial locations (e.g., SCWO units). In addition to SCWO, other technologies are being demonstrated to address PFAS. Therefore, the DoW Components may consider other existing and developing PFAS treatment or destruction technologies that are accepted/permitted by the appropriate State or Federal regulator, on a site-specific basis. When considering other technologies, the DoW Components must follow chain of command procedures to pre-notify OASW(EI&E) of the selected technology on a site-specific basis.

C. On-site hazardous waste storage with environmental permits

EPA's 2020 and 2024 guidance documents recognize that interim storage is not a destruction or disposal method, but asserts that interim storage "may be a short-term option" if the destruction or disposal of PFAS and PFAS-containing materials is "not imperative, on-site storage capacity is readily available, and interim storage has proper controls in place to reduce releases into the environment."³⁸ EPA defines "interim storage" as storage "estimated to be

³⁵EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 119.

³⁶EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 120.

³⁷DoW's Environmental Security Technology Certification Program (ESTCP) funds demonstration and validation of innovative cost-effective environmental technologies and promotes technology implementation by direct insertion and partnering with end users and regulators. <https://serdp-estcp.mil/>

³⁸EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 11.

anywhere from 2 to 5 years.”³⁹ EPA also identifies that interim storage may be less appropriate for high-volumes.⁴⁰ DoW finds that multi-year storage of large quantities of PFAS-containing materials is not a viable option, from either a safety, environmental, logistical, or economic perspective.⁴¹ Thus, in general, DoW assesses that, due to the volume of PFAS-containing waste materials at issue from both its AFFF change out activities and generated from DoW’s nationwide cleanup program, DoW will need to implement actual destruction or disposal solutions for those materials.

DoW is currently conducting cleanup investigations and response actions at over 700 military installations and State Guard facilities.⁴² These investigations and response actions generate PFAS-containing materials (e.g., GAC, soils). While this guidance generally only applies to DoW materials sent off-base to commercial destruction and disposal facilities, if DoW had to plan for, locate, and secure storage of all PFAS-containing materials at applicable DoW Guard facilities, these storage requirements would affect the pace of this necessary cleanup. In addition, the storage would generate its own risks of release to the environment.

DoW is also transitioning to a new firefighting agent for land-based applications and will need to remove existing AFFF. The volume of AFFF that requires disposal is estimated to be over 2 million gallons. DoW does not have the warehouse capacity to properly and safely store this AFFF and associated rinsate at individual bases. DoW is concerned with the risks of release to the environment from storage and believes that secondary containment would be needed to contain releases of PFAS. Storage areas at individual military installations or Guard facilities, where these PFAS-containing materials could potentially be stored if space was available, are not likely to have secondary containment. Building additional storage capacity, to include the necessary contracting actions, would negatively affect the pace of these required cleanup and AFFF replacement activities. DoW believes it does not have the capacity to properly store all PFAS-containing materials at its facilities, and thus disposal or destruction of those materials is imperative.

DoW Components that are storing PFAS-containing materials for over 90 days before the materials are transported to a destruction or disposal facility, shall use onsite hazardous waste storage where available.

V. DOW IMPLEMENTATION OF SECTION 330 OF THE FY 2020 NDAA

The Department is also required to meet Section 330 of the FY 2020 NDAA which requires DoW to ensure that when PFAS-containing materials, including AFFF, are sent to a HWI:

³⁹EPA Interim PFAS Disposal Guidance (Dec 2020). Page 5.

⁴⁰EPA Interim PFAS Disposal Guidance (Apr. 2024). Page 11.

⁴¹EPA. 2024. Final PFAS National Primary Drinking Water Regulation. April 26, 2024. Available online: <<https://www.govinfo.gov/content/pkg/FR-2024-04-26/pdf/2024-07773.pdf>>. Accessed May 3, 2024.

⁴²DoW 2025. PFAS data-cleanup of PFAS. <https://www.acq.osd.mil/eie/eer/ecc/pfas/data/cleanup-pfas.html>

- “(1) all incineration is conducted at a temperature range adequate to break down PFAS chemicals while also ensuring the maximum degree of reduction in emission of PFAS, including elimination of such emissions where achievable;
- (2) all incineration is conducted in accordance with Clean Air Act (42 USC 7401 et seq.), including controlling hydrogen fluoride;
- (3) any materials containing PFAS that are designated for disposal are stored in accordance with the requirement under part 264 of title 40, Code of Federal Regulations; and
- (4) all incineration is conducted at a facility that has been permitted to receive waste regulated under [the Resource Conservation and Recovery Act]⁴³ (42 USC 6921 et seq.).”

Because the first criterion requires a lengthy discussion, this analysis starts by addressing the second, third, and fourth criteria.

The second criterion in section 330 requires that all incineration of PFAS-containing materials is conducted in accordance with CAA requirements. The third criterion in section 330 requires that PFAS-containing materials stored at hazardous waste facilities prior to incineration be stored in accordance with RCRA requirements. The fourth criterion in section 330 requires that incineration is conducted at a RCRA-permitted hazardous waste facility. Based upon the review of these three criteria, if a DoW Component chooses to incinerate PFAS-containing materials in its custody, the DoW Component must send those PFAS-containing materials, including AFFF, only to RCRA- and CAA-permitted HWIs. RCRA-permitted HWIs with CAA Title V permits operate under conditions that represent the maximum commercially available destruction efficiencies for PFAS, including the control of hydrogen fluoride and other PICs. Additionally, RCRA- and CAA-permitted HWI have experience in the proper storage of regulated hazardous wastes and must comply with part 264 of title 40, Code of Federal Regulations, concerning storage of material at their facilities. Therefore, the DoW Components are required to ensure that HWIs utilized for the incineration of PFAS-containing materials, including AFFF, have valid RCRA and CAA operating permits.

The first criterion in section 330 requires that if DoW sends PFAS-containing materials to incinerators, the incinerators utilize a temperature range adequate to break down PFAS while also minimizing emissions of PFAS (i.e., “maximum degree of reduction in emission of PFAS...”). Because the second, third, and fourth criterion in section 330 require incineration at permitted HWIs and because these permitted facilities are required to maintain minimum temperature thresholds, DoW used those minimum thresholds in determining whether it can reasonably conclude that HWIs will achieve the requirements of the first criterion in section 330.

⁴³The Solid Waste Disposal Act of 1965 is commonly referred to as the Resource Conservation and Recovery Act (RCRA), which significantly amended the Solid Waste Disposal Act, in 1976.

A. Relevant RCRA and CAA Permitting Requirements

The regulatory requirements for RCRA- and CAA-permitted HWIs are summarized as follows:

- RCRA-permitted HWIs must follow stringent regulatory requirements and are required by EPA to conduct testing to determine a DRE. The key factors in achieving a high DRE are time in the incinerator (residence time), high temperature, and turbulence (i.e., mixing). The purpose of DRE testing is to demonstrate that virtually all the molecules of a surrogate compound are destroyed in the incinerator. For HWIs, EPA requires a minimum DRE of 99.99 percent. During DRE testing, a surrogate compound is fed into the incinerator that represents classes of compounds that are extremely difficult to destroy. EPA has developed a system of ranking these surrogate compounds, based on their difficulty to destroy.
- After a 99.99 percent DRE is achieved, EPA or the delegated State, issues a CAA Title V permit that includes requirements for operation. This includes a high temperature range and other parameters that are continuously monitored, and if not complied with, the incinerator will stop the flow of materials to the combustion unit automatically and immediately.

While there are several operating conditions specified in a HWI permit, the first criterion in section 330 focuses on a temperature range adequate to break down PFAS. DoW reviewed minimum temperatures specified in nine existing HWI permits to achieve their DRE and found their permits require a minimum temperature in the kiln that range from 1200° Fahrenheit (F) to 1824°F (648°Celsius (C) to 995°C). At these facilities, the kiln is followed by an afterburner/secondary combustion chamber to maximize organic destruction and their permits require a minimum temperature in the afterburner/secondary combustion chamber that ranges from 1488°F to 2026°F (809°C to 1107°C). Based on the studies and information described below, HWIs that operate at temperatures greater than 1100 °C in the afterburner/secondary combustion chamber are adequate to break down detectable PFAS chemicals and significantly minimize PFAS emissions.

B. Existing Data on Destruction Capabilities of Incinerators

EPA's 2024 guidance contains the following findings on the destruction capabilities of HWIs:

- HWIs are designed to optimize temperatures, residence times, turbulence, and other parameters to ensure compliance with organic DRE requirements. Most commercial HWIs use rotary kilns as a primary combustion chamber to facilitate the thermal treatment of containerized wastes (e.g., in drums) and solid wastes such as contaminated soils. The kiln maintains a continuous standing flame fueled by high-heating-value wastes and auxiliary fuels that maintain high temperatures. Typically, solids retention time in the kiln is 0.5 to 1.5 hours, while gas residence time through the kiln is usually around 2 seconds. Kiln flame/solids temperatures range from 650°C to 1,650°C (1,200°F to 3,000°F). The rotary kiln is followed by an afterburner where additional high-heating-

value gaseous and liquid wastes, and auxiliary fuels are added. The afterburner is typically operated at about 1,100°C to 1,370°C (2,000°F to 2,500°F) with a gas residence time from 1 to 3 seconds to maximize organic destruction and minimize the formation of PICs.⁴⁴

Studies and information on PFAS destruction indicate that the temperature ranges used in these types of HWIs are effective in destroying the approximately 50 PFAS that can currently be detected in air emissions through an EPA methodology (e.g., OTM-50):

- In 2021, EPA began conducting pilot-scale PFAS incineration studies using its “Rainbow” furnace, which allows EPA to conduct incineration experiments under controlled conditions.⁴⁵ This research identified fluorocarbon tracer gases (surrogates) that could potentially be used to monitor destruction efficiencies during incineration, and then began experiments. The first publication from these experiments suggests that PFAS can be destroyed when subjected to aggressive thermal environments above 1100°C.
- In 2021, the New York State Department of Environmental Conservation (NYSDEC) announced that it had completed a study to determine if the thermal treatment of PFAS-containing materials at the Norlite facility in Cohoes, New York, resulted in soil and surface water contamination. The Norlite facility is a RCRA- and CAA-permitted hazardous waste combustor that had treated AFFF over a number of years. This NYSDEC study found no clearly discernible pattern of aerial deposition of PFAS that could be traced to Norlite’s operations. Sampling identified low-level detections of PFAS compounds in all soil samples collected at upwind, downwind, and at background locations, consistent with emerging research on the prevalence of PFAS in urban, suburban, and rural environments. Concentrations of PFAS found in soils in the vicinity of the facility were below guidance values NYSDEC developed, indicating that the facility successfully destroyed the PFAS material and did not emit traceable amounts of PFAS during combustion.⁴⁶
- In 2021 and 2022, a commercial RCRA and CAA permitted HWI (Clean Harbors Environmental Services’ in Aragonite, Utah) conducted PFAS-specific studies with the objective to “demonstrate the ability of the high-temperature incineration system at Aragonite to effectively destroy PFAS in the waste feed during normal process operating conditions with a secondary objective of conducting a mass balance of PFAS in process streams and demonstrating high hydrogen fluoride removal”.⁴⁷ In the 2021 study, AFFF was added in high concentrations to a waste feed and sampled at various times throughout the incineration process. A 99.9999

⁴⁴EPA Interim PFAS Disposal Guidance (Apr 2024), page 47.

⁴⁵“Combustion of C₁ and C₂ PFAS: Kinetic Modeling and Experiments,” Krug et al., *Journal of the Air & Waste Management Association*. 2022, 72:3, 256-270. Published Feb. 11, 2022.

⁴⁶*Norlite Environmental Sampling Report*, pp 25-26. New York State Department of Environmental Conservation, March 2021, https://www.dec.ny.gov/docs/materials_minerals_pdf/norlitesamplingfull0321.pdf.

⁴⁷EPA Interim PFAS Disposal Guidance (Apr. 2024) Appendix B, page B-1.

percent DRE was obtained for PFOS, PFOA, Perfluorohexane Sulfonic Acid (PFHxS), and hexafluoropropylene oxide dimer acid (otherwise known as Gen-X) at a temperature range between 1,893°F and 2,008°F (1,034 °C to 1,098 °C) in the kiln, and a range of 2,052°F to 2,110 °F (1122 °C to 1154 °C) in the afterburner.⁴⁸ In the 2022 study, a 99.9999 percent DRE was obtained for PFOS, PFOA, Perfluorohexane Sulfonic Acid (PFHxS), perfluorobutanesulfonate (PFBS), and hexafluoropropylene oxide dimer acid at a temperature range between 1,949°F and 2,219°F (1,065 °C to 1,215 °C) in the kiln, and a range of 2,048°F to 2,075 °F (1,120 °C to 1,135 °C) in the afterburner.⁴⁹ To measure PFAS air emissions, the studies utilized the EPA test method for stack gas sampling of PFAS available at that time (e.g. OTM-45). The studies demonstrate the ability of the high-temperature incineration system to effectively destroy PFAS in the waste feed during normal process operating conditions for the five selected PFAS.⁵⁰ However, the studies did not fully evaluate the formation of PICs due to methodology limitations at the time, i.e., EPA’s OTM-50 methodology was not made available until January 2024.⁵¹ Overall, both tests showed that DREs of 99.999 percent for the five target PFAS were achieved. Of the 49 compounds detected by OTM-45, 34 were non-detect in the stack gas and only 15 were detectable at extremely low concentrations (e.g. insignificant generation of PICs).

- In 2022, a literature review covering 163 published works on thermal treatment of PFAS was published.⁵² This paper suggests that “complete combustion of PFAS will likely be most successful in incinerators that employ a two-stage process. In these, the waste is first fed into the primary combustion chamber where PFAS desorb and partially degrade. The gaseous byproducts are sent to a secondary chamber (the afterburner) that operates in excess air (stoichiometric excess of oxygen) at high temperature (>950°C) and short residence times (1-3 seconds).”⁵³ DoW notes that HWIs employ this two-stage process. This paper also stated that the “general consensus across these lab-scale studies is that even the most stable PFAS (e.g., long-chain sulfonates) desorb at temperatures less than 1000°C, and they are destroyed in the gas phase at temperatures greater than 1000°C.”⁵⁴
- A 2022 performance test report on Chemours Company Fayetteville Works thermal oxidizer summarizes the ability of the thermal oxidizer and 4-stage scrubber system to achieve emission reduction [of 99.99 percent] based on testing

⁴⁸EPA Interim PFAS Disposal Guidance (Apr. 2024) Appendix B, page B-2.

⁴⁹EPA Interim PFAS Disposal Guidance (Apr. 2024) Appendix B, page B-2.

⁵⁰EPA Interim PFAS Disposal Guidance (Apr. 2024) Appendix B, page B-1.

⁵¹EPA Interim PFAS Disposal Guidance (Apr. 2024) page 45.

⁵²“Critical Review of Thermal Decomposition of Per- and Polyfluoroalkyl Substances: Mechanisms and Implications for Thermal Treatment Processes,” Wang et al., *Environ. Sci. Technol.* 2022, 56, 5355-5370. Published April 21, 2022.

⁵³*Id.* at page 5363.

⁵⁴*Id.* at page 5363.

conducted in March 2020 and 2022.⁵⁵ EPA's 2024 guidance provides a high-level review of the report. The test used multiple EPA methods to sample PFAS compounds from polymer and monomer PFAS production process waste gases fed into the thermal oxidizer. The results of the tests showed that the mandated 99.99 percent destruction of targeted PFAS compounds had been achieved. In both testing years the average PFAS DRE was above 99.999 percent with no result being lower than 99.999 percent for the PFAS compounds tested.

- A 2023 publication, described additional work conducted by EPA that built on EPA's 2021 "rainbow furnace" studies and found that for AFFF incineration, PFAS reached non-detect levels (near the detection limit) for PICs as the peak temperature approached 1100°C, and that PFAS destruction efficiencies are high.⁵⁶ This study calculated destruction efficiencies instead of Destruction and Removal Efficiencies because samples were collected before any pollution control devices. OTM-50 was used to quantify 30 PFAS PICs, and results indicated that greater than 99.99 percent destruction efficiencies were achieved with peak temperatures above 1090°C.⁵⁷
- In 2023, research conducted by the University of Notre Dame, investigated ways to improve understanding of PFAS thermal destruction technologies at the laboratory scale in a tube furnace simulating a kiln with use of common alkali and alkaline-earth metal additives to enhance the mineralization of PFOS adsorbed onto GAC. The study found that at 1000°C, PFAS containing solids completely mineralized to hydrofluoric acid (HF) with a minor amount of off gas without the addition of any additives. In the presence of Ca(OH)₂ (hydrated lime) as an additive, mineralization temperatures can be reduced to 800°C with the majority of fluorine being found in the remaining ash, and little production of HF.⁵⁸
- In 2024, researchers from Colorado State University,⁵⁹ published results of their work on quantum machine learning to predict the thermal destruction of PFAS during incineration. The study concluded that PFAS mineralization during hazardous waste incineration requires greater than 950°C and 2 seconds gas residence time for complete mineralization and suggested that thermal destruction of longer carbon chain PFAS found in AFFF may be viable at common afterburner temperatures of $\geq 1000^{\circ}\text{C}$ and 2 seconds gas residence time, given other process variables such as mixing, fuel:oxygen:PFAS

⁵⁵EPA Interim PFAS Disposal Guidance (Apr. 2024) Appendix C.

⁵⁶Shields E., et al (2023). "Pilot Scale Thermal Destruction of Per- and Polyfluoroalkyl Substances in a Legacy Aqueous Film Forming Foam". ACS ES&T Engineering, volume 3, issue 9, pages 1308-1317.

⁵⁷EPA Interim PFAS Disposal Guidance (Apr. 2024) page 45.

⁵⁸Abou-Khalil et al. (2024), "Enhancing the Thermal Mineralization of Perfluorooctanesulfonates on Granular Activated Carbon Using Alkali and Alkaline-Earth Metal Additives". Environ. Sci. Technol., 58, pp. 11162-11174.

⁵⁹Anthony Rappe, Department of Chemistry, Colorado State University. Work conducted in support of SERDP project "ER21-1019: A Quantum Chemical Machine Learning Approach to Prediction of Thermal PFAS Destruction.

ratio, and concentrations of hydrogen donors and fluorine scavenger(s) such as lime are optimized.⁶⁰

- In November 2024, EPA conducted its most comprehensive commercial incinerator PFAS testing at the Clean Harbors Environmental Services (CHES) Aragonite, UT RCRA-permitted hazardous waste incinerator.⁶¹ This testing was a follow-on from two previous comprehensive test programs conducted in 2021 and 2022 at the Aragonite facility (see summary of 2021 and 2022 CHES studies on previous page). The November 2024 testing incorporated most components of the 2021 and 2022 test campaigns, including sampling of liquid waste feed streams (e.g., AFFF concentrate), input chemicals and utilities, process residuals, and exhaust emissions for PFAS target analytes, to support an overall PFAS mass balance. This test program included feeding AFFF concentrate and dynamic spiking of nine PFAS compounds for DRE determination. Tests were conducted under normal operating conditions with mixed solid and liquid waste feeds used to maintain a temperature in the rotary kiln between 950°C and 1196 °C, and over 1100 °C in the afterburner. As during the two previous test programs, EPA’s Other Test Method-45 (OTM-45) was used. Additionally, OTM-50, released by EPA in January 2024, was used in this test for sampling exhaust emissions for 30 volatile fluorinated compounds (VFCs), which could include potential PFAS products of incomplete destruction such as tetrafluoromethane (CF₄) and hexafluoroethane (C₂F₆). The results of the November 2024 PFAS destruction testing demonstrated for the third time the Aragonite RCRA-permitted hazardous waste incineration facility can produce DREs as high as 99.9999 percent for a variety of PFAS. The testing also demonstrated that incineration of PFAS-containing wastes can achieve low to non-detectable concentration of VFCs. DRE values from 99.99 to 99.9999 percent were demonstrated for C₂F₆, supporting the potential use of C₂F₆ as an indicator for PFAS destruction.⁶²
- In 2025, Veolia North America completed a study at its Port Arthur, Texas incinerator. Testing was conducted in July and October of 2024, by a third-party provider. The facility was chosen based on its track record of managing PFAS-containing material, and its ability to maintain an average kiln exit temperature of ~1,660° F (905°C) and reach an average secondary combustion chamber temperature of 2,049 °F (1120 °C) and a residence time of 2.3 seconds. These conditions are conducive to thermal destruction of PFAS compounds with near detection limit concentration for PICs based on previously published pilot and full-scale studies. The testing used the OTM-45 and OTM-50 methods to evaluate PICs in stack air emissions. Although the high detection limit for OTM-50 precluded the quantification of a few thermally stable PICs in the exhaust, nonetheless, Veolia’s results “demonstrated high temperature incineration to be an

⁶⁰Blotevogel, J. et al. 2024. “Thermal Destruction Pathways and Kinetics for *NTf2* and Longer-Chain Bis(perfluoroalkanesulfonyl)imides (Bis-FASIs)”. *Environ.Sci.Technol.Lett.* 11, 1254–1259.

⁶¹EPA 2025. PFAS Destruction by a Hazardous Waste Incinerator: Testing Results. EPA600/R-25/172. September. https://cfpub.epa.gov/si/si_public_file_download.cfm?p_download_id=552166&Lab=CEMM

⁶²EPA 2025. PFAS Destruction by a Hazardous Waste Incinerator: Testing Results. EPA600/R-25/172. September. https://cfpub.epa.gov/si/si_public_file_download.cfm?p_download_id=552166&Lab=CEMM

efficient disposal solution for high concentrations of targeted PFAS, destroying greater than 99% of targeted substances, including up to 99.9999% of PFOS and PFHxS...with few to no PICs detected”⁶³ when temperatures, residence times, and turbulence are optimized. CF₄ was the only PIC detected at the OTM-50 limit of quantitation.

Notwithstanding EPA’s 2024 Guidance general finding that there are uncertainties with PFAS thermal treatment technologies and additional testing was needed with OTM-50 methods, EPA recognized that there is less uncertainty for the permitted facilities that DoW will use for incineration. According to EPA, a subset of permitted HWIs “may operate under conditions more conducive to destroying PFAS and controlling related PICs” relative to thermal treatment units that do not have both RCRA and CAA permits.”⁶⁴ EPA also recognized that permitted HWIs “are designed to optimize temperatures, residence times, turbulence, and other parameters” to “maximize organic destruction and minimize the formation of PICs.”⁶⁵ These controls include pollution control devices which can remove products of combustion.⁶⁶

Updates in research activities have addressed several significant data gaps. Much of this new work has been possible given the 2024 establishment of new air emission sampling and analysis methods for PFAS (i.e., OTM-50) by EPA.⁶⁷ While there are still some uncertainties, controlled pilot-scale studies have empirically demonstrated near-complete mineralization of all PFAS with negligible PIC formation at temperatures ≥ 1000 °C with ≥ 2 seconds residence times.⁶⁸ Additionally, recent testing conducted at the Clean Harbors and Veolia commercial HWI demonstrated that PFAS can be destroyed at high temperatures showing evidence of adequate break down of PFAS and significantly minimized PFAS emissions. Based on the totality of knowledge to date, including that most HWI operate at temperatures greater than minimums set in their operating permits and utilize a two-stage operating process,⁶⁹ the DoW has identified incineration of PFAS-containing materials as a destruction option if the HWI operates at temperatures greater than 1100 °C in the afterburner (secondary combustion chamber) in addition to having RCRA and CAA permits.

⁶³Veolia. 2025. “PFAS Destruction Testing at Veolia’s Port Arthur Incinerator” June. Available online: <https://www.veolianorthamerica.com/media/press-releases/veolia-completes-most-comprehensive-industry-testing-pfas-waste>.

⁶⁴EPA Interim PFAS Disposal Guidance (Apr. 2024), page 2 and 46.

⁶⁵EPA Interim PFAS Disposal Guidance (Apr. 2024), page 47.

⁶⁶EPA Interim PFAS Disposal Guidance (Apr. 2024), pages 46-48.

⁶⁷EPA Interim PFAS Disposal Guidance (Apr. 2024), page 2.

⁶⁸Shields, E. P., Krug, J. D., Roberson, W. R., Jackson, S. R., Smeltz, M. G., Allen, M. R., ... & Linak, W. P. (2023). Pilot-scale thermal destruction of per- and polyfluoroalkyl substances in a legacy aqueous film forming foam. *ACS ES&T Engineering*, 3(9), 1308-1317. Gehrman, H. J., Taylor, P., Aleksandrov, K., Bergdolt, P., Bologna, A., Blye, D., ... & Stapf, D. (2024). Mineralization of fluoropolymers from combustion in a pilot plant under representative european municipal and hazardous waste combustor conditions. *Chemosphere*, 365, 143403.

⁶⁹“Critical Review of Thermal Decomposition of Per- and Polyfluoroalkyl Substances: Mechanisms and Implications for Thermal Treatment Processes,” Wang et al., *Environ. Sci. Technol.* 2022, 56, 5355-5370. Published April 21, 2022

C. DoW's Finding for Section 330

Based on a review of the studies identified above, DoW concludes that RCRA- and CAA-permitted HWIs with a secondary combustion chamber operating temperature above 1100 °C will be adequate to break down detectable PFAS chemicals while also ensuring the significant minimization of PIC formation. Based on the above studies and information that show HWIs utilize this temperature range and other operating parameters to achieve a 99.99 percent DRE, and HWIs are required to have air emission control devices, RCRA and CAA permitted HWIs meet section 330's requirements for an adequate temperature range to break down PFAS that currently can be detected in air emissions and meet emission reduction requirements. Additional research is underway, and DoW will update this guidance as needed to reflect changes as technologies mature, EPA updates its guidance, and additional data, including additional air emission detection methods, becomes available.

SECTION 26 56 20

AIRFIELD AND HELIPORT LIGHTING AND VISUAL NAVIGATION AIDS
02/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2 (2023) National Electrical Safety Code

IEEE C62.11 (2020) Standard for Metal-Oxide Surge Arresters for Alternating Current Power Circuits (>1kV)

IEEE C62.41.1 (2002; R 2008) Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits

IEEE C62.41.2 (2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA WC 71/ICEA S-96-659 (2014; R 2022) Standard for Nonshielded Cables Rated 2001-5000 Volts for use in the Distribution of Electric Energy

NEMA C119.1 (2023) Electric Connectors - Sealed Insulated Underground Connector Systems Rated 600 Volts

NEMA WC 70 (2021) Power Cable Rated 2000 Volts or Less for the Distribution of Electrical Energy

NEMA WC 74/ICEA S-93-639 (2022) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

NFPA 70B (2023) Recommended Practice for Electrical Equipment Maintenance

U.S. DEPARTMENT OF AGRICULTURE (USDA)

REA Bull 1753F-205 (1993) Filled Telephone Cables (PE-39)

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-535-01 (2017; with Change 3, 2021) Visual Air Navigation Facilities

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 70/7460-1 (2016; Rev L; Change 2) Obstruction Marking and Lighting

FAA AC 150/5345-7 (2013; Rev F) Specification for L-824 Underground Electrical Cable for Airport Lighting Circuits

FAA AC 150/5345-10 (2014; Rev H) Specification for Constant Current Regulators Regulator Monitors

FAA AC 150/5345-26 (2021; Rev E) FAA Specification for L-823 Plug and Receptacle, Cable Connectors

FAA AC 150/5345-27 (2021; Rev F) FAA Specification for Wind Cone Assemblies

FAA AC 150/5345-28 (2005; Rev F) Precision Approach Path Indicator (PAPI) Systems

FAA AC 150/5345-42 (2019; Rev J) Specification for Airport Light Bases, Transformer Housings, Junction Boxes and Accessories

FAA AC 150/5345-43 (2019; Rev J) Specification for Obstruction Lighting Equipment

FAA AC 150/5345-44 (2015; Rev K; Errata 2022) Specification for Runway and Taxiway Signs

FAA AC 150/5345-46 (2016; Rev E; Errata 1 2023) Specification for Runway and Taxiway Light Fixtures

FAA AC 150/5345-47 (2005; Rev B) Specification for Series to Series Isolation Transformers for Airport Lighting Systems

FAA AC 150/5345-51 (2005; Rev A) Specification for Discharge-Type Flashing Light Equipment

FAA AC 150/5345-53 (2012; Rev D) Airport Lighting Equipment Certification Program

FAA AC 150/5345-56B	(2011; Rev B) Specification for L-890 Airport Lighting Control and Monitoring System (ALCMS)
FAA E-2519	(1972; Rev A) Types I and II
FAA E-2628	(1979; Rev B) Sequenced Flashing Lighting System, Elevated and Semiflush with Dimming and Monitoring
FAA FO 6850.19	(1978) Frangible Coupling
UNDERWRITERS LABORATORIES (UL)	
UL 6	(2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel
UL 44	(2018; Reprint May 2021) UL Standard for Safety Thermoset-Insulated Wires and Cables
UL 360	(2013; Reprint Apr 2023) UL Standard for Safety Liquid-Tight Flexible Metal Conduit
UL 486A-486B	(2018; Reprint Jul 2023) UL Standard for Safety Wire Connectors
UL 797	(2007; Reprint Apr 2023) UL Standard for Safety Electrical Metallic Tubing -- Steel
UL Electrical Construction	(2012) Electrical Construction Equipment Directory

1.2 SYSTEM DESCRIPTION

Provide airfield and heliport lighting and visual navigation aids as indicated.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Protection plan; G, AE

Training; G

SD-02 Shop Drawings

Lighting and visual navigation aids; G, AE

Composite drawings showing coordination of work of one trade with that of other trades and with structural and architectural

elements of the work. Provide sufficient detail to show overall dimensions of related items, clearances, and relative locations of work in allotted spaces. Indicate where conflicts or clearance problems exist between the various trades.

Wind Cone Indicator Assembly Connection; G, AE

Posted instructions; G, AE

Constant Current Regulator Lift; G, AE

Constant Current Regulator Switchgear Frame; G, AE

Airfield Lighting Control And Monitoring System Components, Complete; G, AE

Airfield Lighting Control And Monitoring System Demo CD Software; G, AE

Airfield Lighting Control And Monitoring System Final CD Software; G, AE

SD-03 Product Data

When equipment or materials are specified to conform to the standards or publications and requirements of AASHTO, ANSI, ASTM, AEIC, ETL, IEEE, IES, NEMA, NFPA, or UL, or to an FAA, FS, or MS, include proof that the items furnished under this section conform to the specified requirements. The label or listing in **UL Electrical Construction** or ETL or the manufacturer's certification or published catalog specification data statement that the items comply with applicable specifications, standards, or publications and with the manufacturer's standards will be acceptable evidence of such compliance. Provide manufacturer prepared certificates when the manufacturer's published data or drawings do not indicate conformance with other requirements of these specifications.

Airfield lighting control and monitoring system (ALCMS) components, complete; G, AE

Approach lighting systems components, complete; G, AE

Precision approach path indicator system; G, AE

Runway edge lights; G, AE

Runway threshold and end lights; G, AE

Taxiway edge lights; G, AE

Guidance signs; G, AE

Runway distance remaining signs; G, AE

Obstruction lighting; G, AE

Light bases, each type; G, AE

Wind direction indicator; G, AE

List of airfield lighting materials and equipment with the
FAA AC 150/5345-53 Appendix C review date.

Isolation transformers; G, AE

Encapsulated isolation transformers; G, AE

Constant current regulators, each size; G, AE

Constant Current Regulator Lift; G, AE

Control cable; G, AE

Frangible couplings; G, AE

Materials and equipment; G, AE

Constant Current Regulators Switchgear Frame; G, AE

Airfield Lighting Control And Monitoring System Components,
Complete; G, AE

SD-06 Test Reports

Visual inspection; G, AE

Photometric testing; G, AE

Discharge-type flashing light equipment; G, AE

PAPIs; G, AE

Progress testing for series lighting circuits; G, AE

Counterpoise system test and inspection; G, AE

Operating test; G, AE

Distribution conductors, 600-volt class; G, AE

Electrical acceptance tests; G, AE

Low-voltage continuity tests; G, AE

High-voltage insulation resistance tests; G, AE

Constant current regulators, each size; G, AE

Airfield Lighting Control And Monitoring System Acceptance Test;
G, AE

SD-07 Certificates

Qualifications of contractor; G, AE

Certified documentation of qualifications, as specified.

Qualifications of photometric tester; G, AE

Special tools; G, AE

List of special tools and test equipment required for installation, maintenance and testing of the products supplied by the Contractor. Items to be listed include, but are not limited to, the following:

4-Jack positioning jig, for in-pavement light bases.; G, AE

Elevated light level; G, AE

Crimping tool; G, AE

Cable penciler

List of parts; G, AE

A list of parts and components for the system by manufacturer's name, part number, nomenclature, and stock level required for maintenance and repair necessary to ensure continued operation with minimal delays.

SD-08 Manufacturer's Instructions

Posted instructions; G

Submit proposed diagrams, instructions, and other sheets prior to posting.

SD-10 Operation and Maintenance Data

Constant current regulators, Data Package 5; G

Constant Current Regulator Switchgear Frame; G, AE

Approach lighting systems components, Data Package 3; G

Maintenance and repair instructions; G

Instructions necessary to check out, troubleshoot, repair, and replace components of the systems, as specified.

Posted operations and maintenance instructions

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

Airfield Lighting Control And Monitoring System Operation And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-built drawings; G

Airfield Lighting Control And Monitoring System As-Built Drawings; G, AE

1.4 MAINTENANCE MATERIAL SUBMITTALS

1.4.1 Spare Parts

Provide spare parts as indicated in this specification as part of this contract. The spare parts must be provided at the end of construction. They must be in addition to any items consumed during construction or in testing.

1.5 QUALITY CONTROL

1.5.1 Regulatory Requirements

In each standard referred to herein, consider the advisory provisions to be mandatory, as though the word "must" has been substituted for "should" wherever it appears. Interpret references in these standards to "authority having jurisdiction," or words of similar meaning, to mean Contracting Officer.

1.5.1.1 Code Compliance

Comply with the requirements and recommendations of **NFPA 70** and **IEEE C2** and local codes where required.

1.5.2 Standard Products

- a. Use only approved equipment listed in **FAA AC 150/5345-53** with addendum for the date of delivery the exception of Air Force threshold lights and Army heliport fixture colors and intensities. Inspect wire and cable for date of manufacture. Materials must be certified and listed as "Approved Airport Lighting Equipment" downloadable from: <http://www.faa.gov/arp/pdf/534553ad.pdf>. Do not use wire and cable manufactured more than one year before delivery to job site.
- b. Provide **materials and equipment** listed by FAA, UL, or ETL, when such equipment is listed or approved. Do not use askarel, tetrachlorethylene and insulating liquids containing polychlorinated biphenyls (PCBs) in equipment.
- c. Material and equipment must be a standard product of a manufacturer regularly engaged in the manufacture of the product and essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening.
- d. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be products of the same manufacturer unless stated in this section.

1.5.3 Qualifications of Contractor

- a. Provide documentation that the aviation lighting equipment contractor and installation electricians are experienced in installing, testing and maintaining aviation lighting systems of a similar complexity. Similar complexity, in increasing complexity, are the following: elevated edge lights and signs, in-pavement lights, and approach light systems. Include a list of government projects and 3 years of experience in constructing similar projects. Include written certification that systems have performed satisfactorily for not less

than 18 months. Provide a list of airfield lighting schools or seminars attended in the last 5 years.

- b. Submit certification containing the names and the qualifications of persons recommended to perform the splicing and termination of medium-voltage cables approved for installation under this contract. Indicate that any person recommended to perform actual splicing and termination has been adequately trained in the proper techniques and has had at least 3 recent years of experience in splicing and terminating the same or similar types of cables approved for installation. Any person recommended by the Contractor may be required to perform a dummy or practice splice and termination, in the presence of the Contracting Officer, before being approved as a qualified installer of medium-voltage power or series lighting cables. If that additional requirement is imposed, provide short sections of the approved types of cables with the approved type of splice and termination kits, and detailed manufacturer's instruction for the proper splicing and termination of the approved cable types.

1.5.4 Qualifications of Photometric Tester

Submit three copies of certification containing the names and the qualifications of persons recommended to perform Field Photometric Testing of Airfield Lighting Fixtures. The general goal of the field photometric testing is to test the airfield lighting fixtures in the field as constructed with respect to [FAA AC 150/5345-46](#). Perform testing utilizing an independent, third-party firm unaffiliated with any lighting manufacturer or construction contractor. This firm must have demonstrated capability for the field measurement of the photometric performance of airfield lighting fixtures in comparison to FAA and UFC standards by having performed work similar to this contract successfully at no less than ten (10) air carrier airports (in the United States) in the past five (5) years.

Submit a list of equipment to be used for the photometric testing with a record of experience of similar projects with references for contact. Additionally, submit a list of equipment used to calibrate the photometric sensors as well as proof of the last date of calibration. The equipment must be calibrated within six months prior to commencement of testing.

1.5.5 Protection Plan

Submit detailed procedures to prevent damage to existing facilities or infrastructures. If damage does occur, the procedures must address repair and replacement of damaged property at the Contractor's expense.

1.5.6 Prevention of Corrosion

1.5.6.1 Metallic Materials

Protect metallic materials against corrosion as specified. Do not use aluminum in contact with earth or concrete. Do not use aluminum conductors.

1.5.6.2 Ferrous Metal Hardware

Ferrous metal hardware must be hot-dip galvanized in accordance with [ASTM A123/A123M](#) and [ASTM A153/A153M](#).

1.6 DELIVERY, STORAGE, AND HANDLING

The Contractor must deliver, store and secure all airfield lighting materials and equipment in accordance with the manufacturers' requirements.

1.7 PROJECT/SITE CONDITIONS

Items furnished under this section must be specifically suitable for the following unusual service conditions:

1.7.1 Altitude

Any equipment must be suitable for operation up to an altitude of 10,000 feet.

1.7.2 Other

Material or equipment to be installed underground; in handholes, manholes, or underground vaults; or in light bases, must be suitable for submerged operation.

1.7.3 Environmental Requirements

The airfield for this project has the following requirements:

Location - KGTB

Altitude - 690 feet above mean sea level.

Maximum Exterior Temperature - 100 degrees F.

Minimum Exterior Temperature - minus 20 degrees F.

Maximum Relative Humidity - 90 percent non-condensing.

Maximum Indoor Temperature - 131 degrees F.

Minimum Indoor Temperature - minus 40 degrees F.

1.7.4 Existing Airfield Lighting Systems

Existing airfield lighting systems must remain in operating condition and interruptions must be held to a minimum. Where interruptions are necessary, they must be scheduled as approved in writing by the Contracting Officer. Prior to the scheduled time for each interruption, all necessary materials and a sufficient labor force must be assembled to permit completing the work within the scheduled time interval. Under no circumstances must any of the existing airfield lighting circuits be left inoperative on usable aircraft pavement without making provisions for suitable temporary connections in the affected area or areas. All airfield lighting circuits covered under this Contract must be replaced in such a manner that they will be operational at dusk each day unless phasing and aircraft pavement shutdowns permit otherwise. The Contractor must submit to the Contracting Officer a plan for outages and maintaining lighting and lighting control.

Submit plans for outages no later than 14-days prior to the scheduled outage to the government for review and approval. Any outages to airfield lighting systems must be approved by Fort Drum, WSAAF Operations.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide airfield lighting and visual navigation aids as indicated on

contract drawings.

2.1.1 Design Requirements

2.2 AIRFIELD LIGHTING CONTROL AND MONITORING SYSTEM (AFLCMS)

The FAA has determined that acceptable airfield lighting control and monitoring systems can be configured in three levels of features. They include personal computer (PC) based systems, programmable logic controller (PLC) based systems, and systems configured with pilot relay panels or basic PLC systems.

2.2.1 Airfield Lighting Control and Monitoring System (ALCMS)

2.2.1.1 Computer Based (PC) Technology

Utilize computer based (PC) technology control systems as indicated on the drawings, compliant with [FAA AC 150/5345-56B](#), L-890 Type. Include touchscreen type control panel. For the essential system PCs provide the following characteristics: 19 inch rack mount form factor, fully enclosed metal housing, redundant hot-swappable solid state software and data storage devices, redundant hot-swappable cooling system, redundant hot-swappable power supplies. Incorporate uninterruptible power supplies (UPS) for primary power to all system components required for normal operation for a duration of fifteen minutes.

2.2.1.2 General Requirements

The ALCMS system shall communicate between major nodes via ethernet and a fiber optic communications network for data transfer between the electrical vault, control tower, and base operations. The computerized ALCMS system shall consist of the following major hardware components:

- a. Touchscreen control station in the tower cab
- b. Touchscreen control station in base operations
- c. Tower computer subsystem consisting of an industrial enclosure, industrial tower computer and communication equipment
- d. Vault computer subsystem consisting of an industrial enclosure, industrial vault computer, communication equipment, laserjet printer and a redundant vault control / monitoring network.

Within the airfield lighting vault shall be a distributed control and monitoring system which operates on a redundant communication network.

The Distributed Control and Monitoring Equipment (DCME) shall be of a distributed nature that shall be installed locally at each controlled element within the vault. The vault industrial computer communicates to each DCME via CAT6a cable.

The system shall monitor the operation of the various lighting systems per [FAA AC 150/5345-10](#)(current edition) requirements.

2.2.1.3 Communication Requirements

The tower, airfield lighting vault, base operations, airfield beacon, and mountain ramp lighting, ~~Runway 15 medium intensity approach light system~~

~~with runway alignment indicator lights~~ shall communicate with the ALCMS as indicated in the contract drawings.

A redundant wireless ethernet link shall be provided between the vault and tower as a backup means of communication. Additional wireless ethernet links shall be provided between the vault and airport beacon. All antennas shall be provided by the ALCMS manufacturer. The wireless network system shall be based on direct line-of-site communication between all radios. A set of spare point to point radios must be provided.

2.2.1.4 Computer Requirements

The vault computer shall be capable of independently carrying out the following functions:

- a. Decode all commands received and transfer them to the corresponding Distributed Control and Monitoring Equipment (DCME) unit for execution.
- b. Interrogate all the DCME units to determine the status of the Constant Current Regulators (CCRs) and other controllable items.
 - (1) Transfer the status of the CCRs and other controllable items to the control tower computer and maintenance center computer.
 - (2) Continuously check for proper operation of all the communication links connected to the computer.
 - (3) Continuously check for proper operation of the vault distributed control and monitoring network.
 - (4) Duplicate the tower control and graphical displays for allowing authorized control from the vaults.
 - (5) The vault shall also duplicate the maintenance center status information.
 - (6) Provide hard copies of real-time and historical information on the status of the airfield lighting systems and other controlled and monitored items.
 - (7) The vault computer application shall not be able to initiate lighting commands unless the control tower authorizes control to Vault.

The Tower computer shall be capable of independently carrying out the following functions:

- a. Receive commands from the Touchscreen control station and transfer lighting control commands to the vault for execution.
- b. Receive the airfield lighting status information from the vault and transfer the status to the Touchscreen display.

2.2.1.5 Distributed Control and Monitoring Equipment

The DCME units shall be installed locally at each device (i.e. CCR) which requires control and/or monitoring within the airfield lighting electrical vault.

- a. Each CCR shall include an integral DCME and each controllable item shall be connected to a DCME.
- b. The DCME shall be a microprocessor-based module that includes all of the communication, control commands, input/output connections and failsafe functionality.
- c. The DCME shall communicate via a redundant (2 independent communication links) communications network.
- d. Connections to the communication network shall be via quick disconnect terminal connectors that can easily be plugged and unplugged from the communication equipment.
- e. The DCME shall communicate back to the Vault computer via either of the networks.
- f. Removal of any DCME units from the vault network shall not affect the operation of the ALCMS system.
- g. The DCME shall be a universal device that can be used on any type of CCR and/or controlled element from any manufacturer.
- h. To minimize spare parts needs, each DCME shall be identical and have interchangeable components.
- i. The DCME unit shall provide optical isolation from all high voltage equipment including the CCR output current, CCR output voltage and CCR input voltage.
- j. All high voltage interfaces to the DCME unit shall be via fiber optic cable.

The DCME shall utilize a redundant vault control network:

- a. The DCME redundant communication network shall use at least two (2) independent communication network cables installed in the electrical vault.
- b. The vault network shall use two (2) CAT6a redundant, daisy-chainable ethernet cables.
- c. The network shall be used to control and monitor all the various controllable elements located within the vault such as CCRs, ATS, and Generator.
- d. Any malfunction in one network shall not affect the operation of the other communications network.
- e. Any malfunction in one of the DCME communication ports shall transfer communication to the remaining port without affecting system functions.

Each externally mounted DCME shall have the following components:

- a. Default Input / Output Board
 - (1) Shall provide eight (6) mechanical latching output points. These control points shall also be self-monitored and provide back-indication to the Vault computer verifying proper execution

of the control command.

(2) Shall provide eight (3) optical-isolated input points.

(3) Shall provide quick-disconnect terminal blocks that can be easily plugged and unplugged from the I/O board.

b. Main Board

(1) Shall provide redundant communication network circuitry.

(2) Shall provide quick-disconnect terminal blocks for redundant communication network connections.

(3) Receives and transmits data to the vault computer.

c. Digital Display

(1) Shall provide visual display of all DCME monitored values on one screen (Power, communications and monitoring).

(2) Brightness Step: display indicating the commanded step of the CCR.

(3) Communication status: display indicating the status of channel A of the redundant communication network.

d. Current and Voltage Monitoring Unit

(1) Shall provide real-time measurements of the output current of the CCR.

(2) Shall provide real-time measurements of the output voltage of the CCR.

Each externally mounted DCME shall have the following operations:

a. Each DCME unit shall have a unique factory set address and a field programmable communication address.

b. Each DCME unit shall have a 7" easily readable, full color, touchscreen display.

c. Each DCME unit shall be able to be fully configured from the front panel display, without the need for a separate PC and/or specialized software.

d. The DCME receives commands via the vault network, executes those commands, and transfers back the status of the element to the vault computer.

e. The DCME shall perform the following functions:

(1) ON/OFF control as required by the controlled element (i.e. generator may only require ON/OFF control).

(2) Perform all failsafe functions.

(3) Communication via both networks to the vault computer.

- (4) Self-diagnostic function to monitor for proper operation.
- (5) Locally store all data and parameters specific to the controlled element.
- f. For maintenance purposes, the DCME shall have an internal ON/OFF switch and shall have a front hinged access door.

Each CCR Integrated DCME shall have the following components:

- a. Monitor Board
 - (1) Shall provide redundant communication network circuitry.
 - (2) Shall provide quick-disconnect terminal blocks for redundant communication network connections.
 - (3) Receives and transmits data to the vault computer.
- b. Digital Display
 - (1) Shall provide visual display of all DCME monitored values on one screen (Power, communications and monitoring).
 - (2) Brightness Step: display indicating the commanded step of the CCR.
 - (3) Communication status: display indicating the status of channel A of the redundant communication network.

Each CCR Integrated DCME shall have the following operations:

- a. Each DCME unit shall have a unique factory set address and a field programmable communication address.
- b. Each DCME unit shall have a 7" easily readable, full color, touchscreen display.
- c. Each DCME unit shall be able to be fully configured from the front panel display, without the need for a separate PC and/or specialized software.
- d. The DCME receives commands via the vault network, executes those commands, and transfers back the status of the element to the vault computer.
- e. The DCME shall perform the following functions:
 - (1) Brightness setting control of the CCRs or ON/OFF control as required by the controlled element (i.e. generator may only require ON/OFF control).
 - (2) Perform all failsafe functions.
 - (3) Communication via both networks to the vault computer.
 - (4) Self-diagnostic function to monitor for proper operation.
 - (5) Locally store all data and parameters specific to the controlled element.

- f. For maintenance purposes, the DCME shall have an internal ON/OFF switch and shall have a front hinged access door.

Each CCR Integrated DCME shall have the following monitoring capabilities:

- a. The DCME unit shall provide full FAA L-827/9 monitoring per FAA AC 150/5345-10 (current edition).
- b. The DCME shall include the monitoring board and provide the following information for each CCR:
 - (1) Loss of input power to the CCR.
 - (2) CCR shutdown by open-circuit / over-current protective devices.
 - (3) Drop of more than 10% in the CCR VA load.
 - (4) Failure of the CCR to deliver the selected output current.
 - (5) Remote / Local status of the CCR.
 - (6) Actual CCR output current.
 - (7) Actual CCR output voltage.
 - (8) Actual CCR output load wattage (W).
 - (9) Actual CCR output load Volts-Amps (VA.)
 - (10) Actual CCR input voltage.
- c. The DCME digital display shall provide local indication of the CCR status including:
 - (1) Remote/Local: display indicating the status of the remote / local switch of the CCR.
 - (2) Primary Power: display indicating the status of the input power to the CCR.
 - (3) Over current: display indicating over current, protective shutdown.
 - (4) Open circuit: display indicating open circuit status.
- d. The DCME shall interface to the regulator control board of the CCR. The regulator control board shall collect current and voltage information and provide this information to the DCME. The regulator control board shall meet the following minimum requirements.
 - (1) Collects analog current and voltage samples at a high sample rate of 50,000 samples/second.
 - (2) Transmits current and voltage samples to the DCME.

The DCME unit shall provide insulation resistance monitoring as an integral component of the DCME unit as follows:

- a. The IRMS is capable of automatically or manually monitoring and

reporting the insulation resistance value of the series circuit cabling (one IRMS per circuit).

- b. The measured resistance shall be displayed locally at the DCME digital display.
- c. The DCME unit shall be capable of reading and recording resistance values from less than 20k Ohms to 2G Ohms.
- d. The DCME shall interface to an external insulation resistance module (IRM). The IRM shall meet the following minimum requirements.
 - (1) Collects insulation resistance samples.
 - (2) Transmits insulation resistance samples to the DCME.
 - (3) Provide digital fiber optical isolation between the DCME and the output of the CCR.
 - (4) Quick disconnect fiber optic connections for interfacing to the DCME.
- e. The IRMS system shall be capable of taking resistance readings on circuits that are energized or de-energized. This shall allow the system to be used as a troubleshooting tool for assisting in locating circuit faults.
- f. The IRMS system shall provide database record keeping that allows for graphical trend analysis of the insulation resistance readings.
- g. The IRMS shall provide configurable insulation resistance warning and alarm limit notification to the system.
- h. The IRMS shall be able to be configured for a minimum of two (2) reading times per day.
- i. The IRMS shall be able to be configured to take readings hourly, daily, weekly or monthly.
- j. All user programmable variables shall be able to be changed at any specified computer within the ALCMS system.
- k. All the IRMS data shall be viewable either as real-time or as historical data at any specified computer location. The IRMS information shall be available at all times and shall not require any special transfer of data between the IRMS system and the control system (since the IRMS is an integral component of the ALCMS).

Each DCME unit shall provide a self-contained latching failsafe feature that shall perform the following functions:

- a. Ensure default operation of the airport lighting, even if the entire airport lighting control system is not functioning.
- b. Display the commands sent by the computer to the CCRs and/or to the other controllable items.
- c. Adaptable to each CCR regardless of internal or external control voltage.

- d. Permits maintenance of portions of the control system, without changing the operational status of the lighting system.

The failsafe mode of each DCME unit shall be "Smart Failsafe" (Latching) mode.

If the CCR was switched ON before the failure, it shall remain ON at the same brightness level.

If the CCR was switched OFF before the failure, it shall remain OFF.

Failsafe shall be able to be bypassed by selecting the CCR locally to any desired brightness level.

2.2.1.6 Tower Touchscreen Display

The Tower Touchscreen display shall control and monitor the airfield lighting system. The display shall show real-time information on the operational status of the airfield lighting systems.

The Touchscreen control stations shall consist of multiple Touchscreen 'pages' each with a specific function. These Touchscreen 'pages' are defined as follows:

- a. Preset: Consists of pre-defined preset buttons used to simplify airfield lighting control commands.
- b. Runway Lights: Consists of runway control touch buttons used to individually control runway circuits. Multiple runway pages may be necessary for airports with several runways.
- c. Taxiway Lights: Consists of taxiway control touch buttons used to individually control taxiway circuits if required.
- d. Utilities: Consists of miscellaneous functions for calibrating the Touchscreen, granting lighting control to other locations, setting the date and time, etc.

All preset and control configurations shall be defined by the airport/owner in conjunction with Air Traffic Control requirements.

The ALCMS manufacturer shall provide preset tables to be used by the airport/owner to define the configuration settings.

2.2.1.7 Overview of Operations

Airfield lighting control commands are entered into the system by touching the corresponding touch button on the Touchscreen video display. When a command is entered, the Touchscreen shall respond by graphically displaying the button as being depressed and change the button color.

The associated circuit graphics shall alternately flash indicating the airfield lighting section that shall be affected when this command is "confirmed".

Once confirmed, the Tower Touchscreen shall register the command, generate a data instruction and transmit the command to the vault computer for implementation. The command is also simultaneously transmitted to the

maintenance computer and all other computers connected to the network.

The tower Touchscreen shall receive confirmation from the vault that the corresponding equipment has responded to the control command and displays the current system status on the Touchscreen display.

In the event that communications is lost between the tower and vault, an alarm is indicated at each computer location.

In the event of a predefined alarm condition, the effected airfield lighting circuit graphic shall flash red and an audible alarm tone shall alert operators to the alarm condition.

2.2.1.8 ALCMS Alarm Functions

a. Touchscreen Audible Alarm

- (1) The audible alarm shall sound at each Touchscreen display when an alarm condition occurs. In addition, the 'ALARM ACK' button shall flash and the associated airfield circuit graphics shall change to red.
- (2) The audible alarm shall stop automatically after three (3) seconds unless the 'ALARM ACK' button is pressed.
- (3) If the alarm is not acknowledged, the audible shall cease for sixty (60) seconds while the 'ALARM ACK' continues to flash. If the 'ALARM ACK' is still not pressed after the sixty (60) seconds, the audible shall sound again for three (3) seconds.
- (4) This sequence shall repeat indefinitely until the alarm is acknowledged.

b. Circuit Alarms

- (1) The ALCMS shall continuously monitor the status of all of the circuits per the monitoring requirements as specified previously.
- (2) If there are any monitoring discrepancies (i.e. incorrect CCR output current, loss of primary power) an alarm shall be generated at the Touchscreen display for the associated circuit.

2.2.1.9 Touchscreen Command Sequences

The Touchscreen control station shall allow the airfield lighting circuits to be controlled individually (i.e. RWY Edge) or as a group based on preset tables (See following section).

Each control command shall require two distinct operator actions in order for the command to initiate any state changes in the airfield lighting. The command sequence shall be as follows:

- a. Select circuit: Operator selects the desired circuit to be changed.
- b. Select intensity: Operator selects the desired brightness step that the circuit is to be changed to.
- c. Graphics flash: The graphics associated with the selected circuit shall begin to flash visually indicating to the operator the airfield

lighting section that is going to be affected by the command.

- d. Confirm/Reject: Operator selects the 'CONFIRM' button to accept the selection and initiate the lighting change. Operator selects the 'REJECT' button to cancel the selections and make another selection.

2.2.1.10 Touchscreen Preset Sequences

The Touchscreen control station shall allow simultaneous airfield lighting circuit changes to be accomplished using preset lighting sequences.

The preset lighting sequences shall be defined by the airport in airfield lighting preset tables.

Each preset lighting change shall be based on the following operator inputs:

- a. Active Runway Selection: Operator selects the runway(s) that shall be active. This is based on runway direction (i.e. "RWY 3")
- b. Day/Night Setting: Operator selects the day/night setting. The day/night setting shall control the intensity of the circuits.
- c. Visibility: Operator selects a single visibility setting that is based upon the current airport visibility.
- d. Confirm/Reject: Operator selects the 'CONFIRM' button to accept the preset selections and initiate the lighting change. Operator selects the 'REJECT' button to cancel the selections and make another preset selection.

Upon confirmation of the preset selections, the intensity of all the circuits associated with the preset condition shall automatically change to match the visibility requirement.

The preset visibility setting of the CCRs is based on FAA document 7110.65 (current version). Presets shall also be coordinated with the COR to properly define airfield lighting operational usage.

Provide a preset sequence for the eleven Mountain Ramp High Mast Lighting in both an on or off condition.

The visibility settings shall include Intensity and Preset Invalid monitoring. This indicates when a preset or intensity setting on the airfield is different than the selected preset intensity.

2.2.1.11 Graphical Airport Pictorial

The ALCMS display screens shall display a graphical pictorial representation of the airport runways, taxiways and other requested airport features.

When there is a change in lighting system status, the appropriate graphical detail shall indicate the status by changing color.

The circuit intensity display colors shall be represented as follows:

- a. Step 5: cyan

- b. Step 4: light green
- c. Step 3: magenta
- d. Step 2: dark green
- e. Step 1: dark blue
- f. Step 0: dark gray

The status monitoring display colors shall be represented follows. This includes ATS monitoring, generator monitoring, communications monitoring and Mountain Ramp High Mast Lighting.

- a. Normal: green
- b. Alarm: red
- c. Off: dark gray

2.2.1.12 Vault Automatic Transfer Switch and Generator Monitoring

The ALCMS shall provide the optically isolated digital inputs to monitor the following feedback points:

- a. Utility Available
- b. Utility On-line
- c. Generator Available
- d. Generator On-line
- e. Generator Alarm

Locating and wiring of the monitoring points within the ATS and generator equipment shall be completed by the contractor in coordination with the equipment manufacturer.

2.2.1.13 Beacon Control

The ALCMS shall provide control of the rotating beacon from the vault ALCMS node via an ethernet radio link.

The ALCMS shall provide one (1) optically isolated, dry-contact output point at the beacon control enclosure. The contact shall be rated 1A at 120Vac.

The ALCMS shall close the output to command the beacon ON and open the output to turn the beacon OFF.

A current sensor shall be provided by the ALCMS manufacturer to provide a contact closure feedback point.

The beacon control and monitoring interface equipment to interface with the existing contactor and equipment shall be provided complete, installed within an equipment enclosure by the ALCMS manufacturer.

Locating and wiring of the output points of the Beacon equipment shall be

completed by the Contractor in coordination with existing conditions.

~~2.2.1.14 MALSR Control~~

~~The ALCMS shall provide control of the RW 15 MALSR from the vault ALCMS node via the communications connection indicated in the drawings.~~

~~The ALCMS shall provide three (3) optically isolated, dry-contact output points within the Airfield Lighting Vault. The contacts shall be rated 1A at 120Vac.~~

~~The ALCMS shall close the appropriate output to command the MALSR to ON Step 1, ON Step 2 or ON Step 3 and open the outputs to turn the MALSR OFF.~~

~~The MALSR control and monitoring interface equipment to interface with the MALSR communications wiring shall be provided complete, installed within an equipment enclosure by the ALCMS manufacturer.~~

~~Locating and wiring of the output points of the equipment shall be completed by the Contractor in coordination with existing conditions.~~

2.2.1.14 Mountain Ramp High Mast Lighting Control

The ALCMS shall provide control of the existing eleven Mountain Ramp High Mast Lighting Poles from the vault ALCMS node via the communications connection indicated in the drawings.

The ALCMS shall provide eleven (11) optically isolated, dry-contact output points within the Airfield Lighting Vault. The contacts shall be rated 1A at 120Vac.

The ALCMS shall close a single output to command a single light pole ON and open the output to turn a single light pole OFF. Provide a preset sequence for the eleven Mountain Ramp High Mast Lighting in both an on or off condition.

The Mountain Ramp High Mast Lighting control and monitoring interface equipment to interface with the Mountain Ramp High Mast Lighting communications wiring shall be provided complete, installed within an equipment enclosure by the ALCMS manufacturer.

Locating and wiring of the output points of the equipment shall be completed by the Contractor in coordination with existing conditions.

2.2.1.15 ALCMS General Requirements and Milestones

The system shall include software, programming, computers, manuals, on-site commissioning, on-site testing, on-site training and any other materials, tools and equipment to provide a fully functional system to the satisfaction of the government.

The following submittals must occur in chronological order:

- a. **Airfield lighting control and monitoring system components, complete.** All system components (separated by node), control architecture, and wiring diagrams with requirements as indicated on the drawings.
- b. **Airfield lighting control and monitoring system demo CD Software.** The ALCMS Manufacturer must produce a demo CD of the planned layout of the

touchscreen that will be used for control of the ALCMS system for review and approval by the government.

- c. **Airfield lighting control and monitoring system final CD Software.** The ALCMS Manufacturer must produce a final CD of the planned layout of the touchscreen that will be used for control of the ALCMS system for review and final approval by the government.
- d. **Airfield lighting control and monitoring system acceptance test.** Following final installation and commissioning of the system, the ALCMS manufacturer must perform a demonstration of the system performance. The demonstration shall include the following: lighting control functions, monitoring functions, alarm functions, print and display functions and further testing with the overall airfield lighting system outlined in Paragraph 3.13.2 Operating Test.
- e. **Airfield lighting control and monitoring system Operation and Maintenance Manuals.** The ALCMS Manufacturer shall provide six spiral-bound hard copies of the operation and maintenance manuals that are hard-covered and suitable for daily operation and maintenance of the system. The manuals shall include the following information: operational overview and system description, graphical User Interface screen operation, system block diagram, detailed external wiring diagrams, detailed input/output terminal diagrams, detailed assembly drawings and wiring diagrams, original Equipment Manufacturer manuals. The ALCMS Manufacturer shall provide six spiral-bound hard copies of the operation and maintenance manuals that are hard-covered and suitable for ATC use. The manuals shall be hard-covered and suitable for daily operation of the system. At a minimum, the manuals shall include the following information: touchscreen operation (graphical user interface), touchscreen maintenance (i.e. calibration).
- f. **Airfield lighting control and monitoring system As-Built Drawings.** The ALCMS Manufacturer shall provide six spiral-bound hard copies of the as-installed drawings in 11" X 17" format after system acceptance with the following information: system block diagram (1-line drawings), system external wiring diagrams, assembly drawings, assembly wiring diagrams.

2.3 APPROACH LIGHTING SYSTEMS

2.3.1 High Intensity Approach Lighting System

Provide approach, centerline, crossbar, threshold bar, side row barrette, centerline barrette, bar lights, sequenced flashing lights, and associated equipment and interconnecting wiring to provide a complete system as indicated on construction drawings.

2.3.1.1 In-Pavement, High-Intensity Steady Burning Approach Lights

Use in-pavement, high-intensity, base-mounted lights in the overrun area and paved areas with traffic. Use fixtures conforming to **FAA AC 150/5345-46** Type L-850E for unidirectional quartz lights with lamps, with filters and transformers as indicated. Mounting must conform to the details indicated. Mount bases level and recess as required by thickness of fixture to provide installation in accordance with manufacturer's instructions.

2.3.1.2 Current Powered Sequenced Flashing Light (SFL) for High Intensity Approach Light System

FAA E-2628. Provide as a complete and integrated part of the approach system including individual power supply units, remote control and monitor units and interconnecting wiring. Provide the system with major components that are the product of a single manufacturer. Install junction boxes as indicated on concrete foundations. Install conduit tapings in the bottom and top of the junction boxes as required to accommodate the incoming and outgoing power and control circuits for the flashing lights. Provide terminal strips in each junction box as indicated for termination and connection of the power and control circuits. Provide signal and monitor cables as recommended by the system manufacturer. Provide current transformers for current powered system. Current powered SFLs and junction boxes must be replaced in-kind, and appropriately coordinated with the existing SFL system per manufacturer requirements. Existing system is provided by Flash Technology.

2.3.2 Glide Slope Indicator System

2.3.2.1 Precision Approach Path Indicator System (PAPI)

FAA AC 150/5345-28 Type L-880(L) LED. Connect the four light units to 6.6 ampere series current circuit (Style B) via appropriate isolation transformers. Provide tilt switches and relays to de-energize all light units when one unit exceeds tilt requirements.

2.4 RUNWAY LIGHTING SYSTEM

Include runway edge lights, runway threshold lights, runway end lights, runway distance remaining signs markers, mounting structures, controls, and the associated equipment and interconnecting wiring to provide complete runway lighting systems as indicated. Use in-pavement light fixtures that withstand a minimum static single wheel load of **50,000 pounds**. Where LED fixtures are used, provide fixtures that are compatible with the associated constant current regulators.

2.4.1 Runway Edge Lights

FAA AC 150/5345-46, Elevated high-intensity (24" height) Type L-862(L) LED with arctic kit (heater) and in-pavement high-intensity Type L-850C(L) LED with arctic kit (heater). Colors: white/white, green/yellow, white/yellow or red/yellow as indicated.

2.4.2 Runway Threshold and End Lights

FAA AC 150/5345-46, Type L-850D(L) LED with arctic kit (heater), in-pavement, high-intensity, unidirectional and bidirectional airfield as indicated. For threshold lights use aviation green filter and for end lights use aviation red filters. Combine these lights in a single bidirectional fixture with the appropriate color filters as indicated.

2.5 TAXIWAY LIGHTING SYSTEMS

Include edge lights, hold position lights, and all associated equipment, power supplies and controls, mounting devices, and interconnecting wiring to provide complete systems. If LED fixtures are used, provide fixtures that are compatible with the associated constant current regulators.

2.5.1 Taxiway Edge Lights

FAA AC 150/5345-46, Type L-861T(L) arctic kit (heater) LED for elevated taxiway edge lights 24 inch height with 20/25-watt, 6.6A lamp and blue LED as indicated.

2.6 AIRFIELD GUIDANCE SIGN SYSTEMS

2.6.1 General

Provide guidance signs compatible with all L-828/L-829 regulators. All guidance signs provided must be Mode 3, 300 mph rated markers.

2.6.2 Photometric Requirements

Guidance signs must meet FAA minimum luminance requirements.

2.6.3 Guidance Signs

FAA AC 150/5345-44, Type L-858Y(L) and L-858L(L) for information/direction and Type L-858R(L) for mandatory signs. Provide signs of the size, style and with the information indicated.

2.6.4 Runway Distance Remaining Signs

FAA AC 150/5345-44, Type L-858B, (L) LED, Size 4, Style 5, with white numerals on a black background as indicated. Provide internally illuminated markers.

2.7 OBSTRUCTION LIGHTING AND MARKING

2.7.1 Obstruction Lights

Mark obstructions on or near the airfield as indicated. Use obstruction marker lights emitting aviation red steady burning light. Use multiple-socket assembly FAA AC 150/5345-43, Type L-810 infrared LED light fixtures. Use double-unit type obstruction marker lights as indicated. Energize the obstruction lights multiple circuits as indicated.

2.8 LIGHT BASES

FAA AC 150/5345-42 Type L-867 or L-868. Use steel bases, Class 1B, Size B, C and D as indicated or as required to accommodate the fixture or device installed. Use Size C where more than one distributed control module is installed in the light base. Where used as pull boxes in (single or multiple) can plazas use L-867D bases. Provide base plates, cover plates, and adapter plates to accommodate various sizes of fixtures.

Furnish each base with internal and external one-hole ground lugs for attaching ground or counterpoise cables.

Furnish each base with a 4 foot braided, #6 AWG equivalent ground strap with one-hole lug compression fittings. Utilize straps made for the purpose of grounding the light fixture to the base can interior grounding lug.

2.9 SNOW PLOW RINGS

Provide cast ductile iron unidirectional and bi-directional snow plow

rings for in-pavement light fixtures, as indicated. The rings must be suitable for the light fixture light beam configuration and L-868 light base cans indicated. Rings must have four stainless steel spring pins and black powder coat finish. Ring must be provided with a silicone rubber O-ring and six ceramic fluoropolymer coated bolts and 2-piece locking washers.

2.10 WIND DIRECTION INDICATOR

FAA AC 150/5345-27, Type L-807L, rigid supporting structure, size 2, 12 feet, Style (IB-lighted). Provide wind cones of the size and color as indicated, including wind cone indicator assembly connection.

2.11 LAMPS AND FILTERS

Provide lamps of the size and type indicated, or as required by fixture manufacturer for each lighting fixture required. Include filters of colors as indicated and conforming to the specification for the light concerned or to the standard referenced.

2.12 ISOLATION TRANSFORMERS

2.12.1 Encapsulated Isolation Transformers

FAA AC 150/5345-47, Type L-830. Provide each transformer with rating as indicated. Insulation Level Primary voltage rating 5000 volts RMS, Secondary 600 V RMS. Operating Temperature range minus 131 degrees F to plus 149 degrees F. Resistant to UV exposure and ozone. Suitable for areas contaminated with oils, aircraft fuels, soil acids, alkalis, and deicing fluids. Compatible with FAA Style 2 and Style 9 connectors.

2.13 SURGE ARRESTERS

Provide surge arresters in accordance with IEEE C62.11, IEEE C62.41.1 and IEEE C62.41.2 as applicable with ratings as indicated. Provide surge arresters on the line side of the constant current regulator (CCR). Provide series circuit surge arresters for locations with a lightning flash density of 8 or more flashes per square kilometer per year.

2.14 SURGE PROTECTIVE DEVICES

As required in Section 26 20 00, INTERIOR DISTRIBUTION SYSTEM.

2.15 CONSTANT CURRENT REGULATORS

2.15.1 Regulators

FAA AC 150/5345-10, Type L-829 with monitoring dry-type system and with current rating of 5.5A and 6.6A. Use regulators of ferroresonant design to reduce EMI, increase efficiency, and provide a power factor not less than 0.90. Provide regulators that operate on 60 Hz, have internal primary switch included, have input voltage of 480 and be controlled by 120 VAC external control voltage. Provide 5, 3 or 1 brightness steps, as indicated. Provide monitors as indicated. Provide regulators with digital power meters. Provide regulators with insulation resistance monitoring systems. Use stackable switchgear style regulators compatible with constant current regulator switchgear frame. Insure that constant current regulators are compatible with signs, LED light fixtures and other connected loads.

See constant current regulator schedule for required switchgear compatible constant current regulators (including spare regulators) to be provided.

2.15.2 Basic Impulse Level (BIL)

Provide 60-kV series circuit BIL except that 4-kW, 7.5-kW and 10-kW regulator series circuits may have a BIL of 25 kV.

2.16 CONSTANT CURRENT REGULATOR SWITCHGEAR FRAME

Provide constant current regulator switchgear frame compatible with constant current regulators. Provide switchgear frame with incoming power bay and S-1 cutout bay as indicated on the drawings.

2.17 CONSTANT CURRENT REGULATOR LIFT

Provide a manually operated constant current regulator lift compatible with the constant current regulator and switchgear construction. Constant current regulator lift must be turned over to the government at completion of the project in a location within the vault as identified by the COR.

2.18 MATERIALS AND HARDWARE

2.18.1 Wire and Cable

Use copper conductors installed in conduit. Do not provide or install wire and cable manufactured more than one year before delivery to the job site.

2.18.1.1 Conductor Sizes

Conductor size conforming to American Wire Gage (AWG) or metric trade size. Use stranded conductors for sizes larger #8 AWG. Unless otherwise indicated #8 AWG and smaller may be solid or stranded.

2.18.1.2 Low Voltage Wire and Cable

- a. UL 44 Type XHHW-2.

2.18.1.3 Wire and Cable for Series Lighting Circuits

- a. FAA AC 150/5345-7, Type L-824 for crosslinked polyethylene Type C 5000-volt cable. Use unshielded cable for series airfield and heliport lighting.

2.18.1.4 Safety (Equipment) Grounding System

Safety (Equipment) Grounding System for constant voltage (parallel) circuits: minimum #6 AWG bare stranded copper, annealed or soft drawn.

2.18.1.5 Sequence Flashing Trigger Circuits

REA Bull 1753F-205 compliant cables.

2.18.1.6 Control Cable

Multiconductor type for 120 V ac control, rated 600 volts, #12 AWG, and conforming to the following unless indicated or specified otherwise.

Insulate each conductor with a thickness of not less than 30 mils and rate for continuous operation at 194 degrees F. Conductors must be color coded. An overall jacket of heavy-duty neoprene rated for direct burial must be included. Cable must conform to NEMA WC 70 for rubber insulation, ANSI/NEMA WC 71/ICEA S-96-659 for cross-linked polyethylene insulation, or NEMA WC 74/ICEA S-93-639 for ethylene-propylene rubber insulation.

2.18.1.7 Cable Tags

Install cable tags for each cable or wire at duct entrances entering or leaving manholes, handholes, and at each terminal within the lighting vault, and in all base cans. Use stainless steel, bronze or copper strip cable tags, approximately 1/16 inch thick or hard plastic 1/8 inch thick suitable for immersion in salt water and impervious to petroleum products. Use sufficient material length for imprinting the legend on one line using raised letters. Permanently mark or stamped with letters not less than 1/4 inch in height as indicated. Two-color laminated plastic is acceptable. When providing plastic tags utilize white colored with markings of black color to provide contrast so that identification can be easily read. Use nylon or stainless steel ties must be of a type that will not deteriorate when exposed to water with a high saline content and to petroleum products.

2.18.1.8 Cable Connectors and Splices

FAA AC 150/5345-26, Item L-823 for connections and splices appropriate for FAA Type L-824 cable.

2.18.2 Conduit, Conduit Fittings, and Boxes

2.18.2.1 Rigid Metal Conduit (RMC) or Electrical Metallic Tubing (EMT) and Fittings

UL 6 and UL 797.

2.18.2.2 Liquid-tight Flexible Metal Conduit (LFMC)

UL 360 liquid-tight flexible metal conduit. See Sections 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION and 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.18.2.3 Outlet Boxes for use with RMC, EMT, of LFMC

See 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

2.18.2.4 Plastic Duct for Concrete Encasement

Provide as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

2.18.2.5 Plastic Conduit for Direct Burial

Provide as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

2.18.2.6 High Density Polyethylene (HDPE) Electrical Conduit for Directional Bore

Provide as specified in Section 33 71 02 UNDERGROUND ELECTRICAL

DISTRIBUTION.

2.18.2.7 Frangible Couplings and Adapters

FAA FO 6850.19 and FAA E-2519. Provide upper section of frangible coupling with one of the following:

- a. Unthreaded for slip-fitter connections.
- b. 2-13/32 inch 16N-1A modified thread for nut and compression ring to secure 2 inch EMT.
- c. 2 inch 11-1/2-N.P.T. (tapered) with 7/32 inch nominal wall thickness to accept rigid conduit coupling.
- d. Frangible Couplings for specialized applications as approved.
- e. Electrical Metallic Tubing UL 797, where indicated for use with frangible couplings and adapters.

2.18.3 Electrical Tape

Provide as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

2.18.4 Ground Rods

As specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

2.18.5 Bolts and Hardware

2.18.5.1 Locking Type Washers

Use locking washers of the two-piece wedge-lock design to prevent damage to the fixture. Do not use split-ring, serrated or star type lock washers.

2.18.5.2 Anti-Seize Compound

Use anti-seize compounds for elevated light fixtures with stainless steel bolts but not for ceramic-metallic/fluorocarbon polymer coated bolts for in-pavement light fixtures. Use as recommended by the fixture manufacturer to provide the required clamping force except as indicated in Part 3 of this specification.

2.18.5.3 Ceramic-Metallic/Fluorocarbon Polymer Coated Bolts

Ceramic-metallic/fluorocarbon polymer coated bolts must be used for in-pavement light fixtures or may be used where recommended by the fixture manufacturer in lieu of using an anti-seize compound.

2.18.5.4 Stainless Steel Bolts for Elevated Fixtures

Use anti-seize lubricating compound.

2.18.6 Manufacturer's Nameplates

Provide on each item of equipment a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be

acceptable.

2.18.7 Field Fabricated Nameplates

Provide field fabricated nameplates as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.18.8 Spare Airfield Lighting Equipment Materials

Contractor must provide the following equipment and materials:

- a. L-862(L) - 9 with optical covers: white:18, red:4, yellow:5
- b. L-850C(L) - 1 with optical reflectors: white:2, red:1, yellow:1
- c. L-850D(L) - 10 with optical reflectors: 10 green, 2 red
- d. L-861T(L) - 10
- e. Isolation Transformers - Same quantity as fixtures above, size per manufacturer recommendations
- f. PAPI LED Light Engine - 1
- g. L-858 LED light engine (bar) - Size 3:2 Size 4:4
- h. Sequenced Flashing Light System - Provide a spare part trunk with parts

Contractor must provide the following airfield lighting and control and monitoring system equipment and materials:

- a. Touchscreen - 1
- b. Distributed Control and Monitoring Equipment (DCMU) Main Board - 3
- c. Current / Voltage Module (CVM) - 3
- d. Insulation Resistance Module (IRM) - 3
- e. Ethernet Fiber Optic Transceiver - 2
- f. Network Interface Card (NIC) - 2
- g. Set of spare point to point radios - 1

2.19 ACCESSORIES

2.19.1 Special Tools

List of special tools and test equipment required for installation, maintenance of testing of the products supplied by the Contractor. Items to be listed include, but are not limited to, the following:

4-Jack Positioning Jig, used to install the light base to ensure correct orientation and leveling of in-pavement fixtures.

Crimping Tool

Cable Penciler

Elevated Light Level

PART 3 EXECUTION

3.1 EXISTING AIRFIELD LIGHTING CIRCUIT LOAD MEASUREMENTS

Conduct a load measurement on each existing airfield lighting circuit where only homerun cabling replacement will effect the circuit as follows. These measurements must include the total load in kw on the circuit at each brightness step under typical operating conditions.

Ensure the circuit is de-energized and provide lockout/tagout procedures prior to connection of measurement tools.

Use a clamp meter and 5kv voltage probe on the output 5kv cabling section of the constant current regulator to the airfield.

- a. Energize the ccr at the lowest step for a minimum of 30 seconds, record both current and voltage measurement.
- b. Energize the ccr at the next subsequent step for a minimum of 30 seconds, record both current and voltage measurement.
- c. Repeat step 2 for each subsequent ccr step until highest step is reached.

Following testing, safely disconnect all testing equipment and restore the circuit back to normal operation. The contractor is responsible for ensuring the airfield lighting circuit and CCR are fully operational and controllable to existing conditions.

Provide data measurement results in table format with the following columns: CCR ID, step (1-5), Voltage Measurement, Current Measurement.

3.2 LIGHT FIXTURE INSTALLATION REQUIREMENTS

3.2.1 Existing Airfield Lighting Systems

Existing airfield lighting systems must remain in operating condition and interruptions must be held to a minimum. Where interruptions are necessary, they must be scheduled as approved in writing by the Contracting Officer. Prior to the scheduled time for each interruptions, all necessary materials and a sufficient labor force must be assembled to permit completing the work within the scheduled time interval. Under no circumstances must any of the existing airfield lighting circuits be left inoperative on usable aircraft pavement without making provisions for suitable temporary connections in the affected area or areas. All airfield lighting circuits covered under this Contract must be replaced in such a manner that they will be operational at dusk each day unless phasing and aircraft pavement shutdowns permit otherwise. The Contractor must submit to the Contracting Officer a plan for outages and maintaining lighting and lighting control.

Submit plans for outages no later than 14-days prior to the scheduled outage to the government for review and approval. Any outages to airfield lighting systems must be approved by Fort Drum, WSAAF Operations.

3.2.2 General Installation Requirements

Conform to IEEE C2, NFPA 70, NFPA 70B, and requirements specified herein. Circuits installed underground must conform to the requirements of Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION, except as required herein. Steel conduits installed underground must be installed and protected from corrosion in conformance with the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Electrical metallic tubing (EMT) must not be installed underground or encased in concrete. Except as covered herein, perform excavation, trenching, and backfilling in accordance with the requirements of Section 31 00 00 EARTHWORK.

3.2.3 Airfield Light Fixture Installation

Use 2-part locking type washers for the installation of all airfield light fixtures. Tighten bolts to washer manufacturers recommended torque based on bolt type used. Use only adjustable torque limiting tools.

Where stainless steel bolts are used for elevated fixture installation use an anti-seize lubricating compound.

Where ceramic-metallic/fluorocarbon polymer coated bolts are used for in-pavement fixture installation do not use anti-seize lubricating compounds.

3.2.4 Light Base Installation

3.2.4.1 Installation in Cored Pavement

Wipe down the sides and bottom of each light base immediately prior to installation. For bored hole installations cover the inside faces of bored hole and bottom and sides of light base with a coating of compatible P-606 sealant that will completely fill the void between concrete and base. Use a jig or holding device when installing each light fixture to ensure positioning to the proper elevation, alignment, level control, and azimuth control. Orient the light fixture with the light beams parallel to the runway or taxiway centerline and facing in the required direction. Outermost edge of fixture must be level with the surrounding pavement. Remove surplus sealant or flexible embedding material. The holding device must remain in place until sealant has reached its initial set. Properly arrange fixture lead wires with respect to their connecting position. Block the wireway entrance into the light recess to retain the sealant material during curing.

3.2.4.2 Installation in Concrete Bed

Where light base can is partially embedded in concrete with jig or holding devices, leave device in place for minimum 24 hours. Let base can set for an additional 48 hours before remaining concrete is placed to the top of the light base.

3.2.5 Frangible Requirements

Install frangible supports, couplings, and adapters as indicated or specified.

3.2.6 Elevated Light Fixtures

Elevated lights must be frangibly mounted, not to exceed 24 inches in

height. For equipment exceeding 14 inches in height, frangibly mount as indicated. Use a 4-jack positioning jig to install the light base to ensure correct orientation and leveling. The individual setting the jig must fully understand reference marks that are provided by the surveyor. Check azimuth by survey before the concrete anchor is placed and again before paving. A factory representative of the light base manufacture must be on-site when the first light base is being installed and verify that the installation crew understands proper azimuth and elevation required for the light base. Do not place the near light base edge closer than two feet from a planned pavement joint. If conflict occurs, immediately notify the Contracting Officer Representative of the conflict.

3.2.6.1 Elevated Light Level

Level elevated light fixture using manufacturers system required for fixture type.

3.2.7 In-Pavement Airfield and Heliport Lights

Remove water, debris, and other foreign substances prior to installing in-pavement light base and light. Use a 4-jack positioning jig, obtained from the L-868 base manufacturer, to install the light base to ensure correct orientation and leveling. The individual setting the jig must fully understand reference marks that are provided by the surveyor. Check azimuth by survey before the concrete anchor is placed and again before paving. A factory representative of the light base manufacture must be on-site when the first light base is being installed and verify that the installation crew understands proper azimuth & elevation required for the light base. Do not place the near light base edge closer than two feet from a planned pavement joint. If conflict occurs, immediately notify the Contracting Officer Representative of the conflict.

3.2.7.1 In-Pavement Light Installation

For in-pavement installations, pavement around the light base must be level with the surrounding pavement; dished or mounded pavement near the light base is not acceptable.

3.2.7.2 Snow Plow Ring Installation

For in-pavement fixtures that require snow plow rings, coordinate the light base can, light fixture and snow plow ring to meet the installation tolerances shown in the paragraph Light Fixture Installation Tolerances. The Contractor may need to provide spacer rings to meet these tolerances. The top edge of the snow plow ring must be slightly higher than the top of the light fixture. Install the silicone rubber O-ring to seal the light base can. Install the ceramic fluoropolymer coated bolts to attach the light fixture and snow plow ring to the light base can.

3.2.8 Light Fixture Installation Tolerances

	IN-PAVEMENT	ELEVATED
ELEVATION (relative to finished pavement surface)	+0 inch, -1/16 inch (fixture edge on low side in snow areas or on high side in non-snow areas)	+1/4 inch, -0 inch
AZMUTH (*) (w/respect to line parallel to RW/TW centerline)	+/- 1/2 degree	+/- 1/2 degree
LEVEL	+/- 1/2 degree	+/- 1/2 degree
STATIONING (in line parallel to RW/TW centerline)	+/- 2 inch	+/- 2 inch
OFFSET (perpendicular to RW/TW centerline)	+/- 1/4 inch	+/- 1/4 inch

(*)For omni-directional fixtures the Azimuth does not apply.

3.2.9 Isolation Transformers

Conform to [FAA AC 150/5345-26](#) for transformer lead connections. Plug transformer secondary connectors directly into a mating connector on the transformer secondary leads. During installation, cover mating surfaces of connectors until connected and clean when plugged together. Refer to Section "Connectors" below for proper installation and mating of L-823 connectors.

3.3 CABLES

3.3.1 Cable Installation

In addition to the requirements of Section [33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION](#), use kit type connectors to splice 5 kV single-conductor series lighting cables. During installation, keep mating surfaces of connectors covered until connected and clean when plugged together. At joint where connectors come together, pull one of the rubber boots over the joint. Cover the joint with two half-lapped layers of tape over the entire joint. Joint must prevent entrapment of air which might subsequently loosen the joint.

3.3.2 Low Voltage Cables

For splices in wires [#8 AWG](#) single conductor cable, use [FAA AC 150/5345-26](#) Type L-823 connectors. Splices below grade or in wet locations must be sealed type conforming to [NEMA C119.1](#) or must be waterproofed by a sealant-filled, thick wall, heat shrinkable, thermosetting tubing or by

pouring a thermosetting resin into a mold that surrounds the joined conductors.

Cable must be rated 600 volts, except that secondaries of isolation transformer to in-pavement lights installed in pavement saw kerf and 48 volt DC control cables may be 300 volts. Other parts of cable systems such as splices and terminations must be rated at not less than 600 volts. Splices in wires #10 AWG and smaller must be made with an insulated, solderless, pressure type connector, conforming to the applicable requirements of UL 486A-486B. Splices in wires #8 AWG single conductor cable must be made with noninsulated, solderless, pressure type connector, conforming to the applicable requirements of UL 486A-486B. They must then be covered with an insulation and jacket material equivalent to the conductor insulation and jacket. Splices below grade or in wet locations must be sealed type conforming to NEMA C119.1 or must be waterproofed by a sealant-filled, thick wall, heat shrinkable, thermosetting tubing or by pouring a thermosetting resin into a mold that surrounds the joined conductors.

3.3.3 Airfield 5kV Series Lighting Cables

3.3.3.1 Connectors

Use kit type connectors to splice 5 kV single-conductor series lighting cables. During installation and prior to covering with earth, cover mating surfaces of connectors until connected and clean when plugged together. At joint where connectors come together, pull one of the rubber boots over the joint. Cover the joint with two half-lapped layers of tape over the entire joint. Joint must prevent entrapment of air which might subsequently loosen the joint.

3.3.3.2 Crimping Tool

Use only splice kit manufacturer's crimping tool on splice connectors for all primary and secondary airfield cable splices. Crimping tool must have an embossed die with gauge marks for gauge of cable being used.

3.3.3.3 Cable Penciler

Airfield cable insulation must only be removed using cabling penciler made specifically for airfield cable.

3.3.4 Cable Markers

Provide cable markers or tags for each cable at duct entrances entering or leaving manholes or handholes and at each termination within the lighting vault. Provide not less than two tags per cable in in each manhole or handhole, one near each duct entrance hole. Immediately after cable installation, permanently attached tags to cables and wires so that they cannot be accidentally detached.

3.3.5 Maximum Allowable Non-Armored Cable Pulling Tension, Using Dynamometer

Cable	Tension LB
2 - 1/C #8 solid	275

Cable	Tension LB
3 - 1/C #8 solid	367
4 - 1/C #8 solid	550
2 - 1/C #6 stranded	420
3 - 1/C #6 stranded	630
4 - 1/C #6 stranded	840
1 - 2/C #8 stranded	305
1 - 3/C #8 stranded	395
1 - 4/C #8 stranded	585
1 - 2/C #6 stranded	455
1 - 3/C #6 stranded	685
1 - 4/C #6 stranded	880
1 - 6/C #12 stranded	315
1 - 12/C #12 stranded	630
1 - 12 pair #19 solid	230
1 - 25 pair #19 solid	541
1 - 50 pair #19 solid	1061
1 - 100 pair #19 solid	2000

3.4 COUNTERPOISE

Install counterpoise above multiple conduits/duct banks for airfield lighting cables, with the intent being to provide a complete cone of protection over the airfield lighting cables. When multiple conduits and/or duct banks for airfield cable are installed in the same trench, the number and location of counterpoise conductors above the conduits must be adequate to provide a cone of protection measured 45 degrees each side of vertical. Install one continuous length of conductor, except where distance exceeds the length usually supplied. Install the counterpoise approximately 6 inches above duct lines as indicated. Where trenches or duct lines intersect, electrically interconnect the counterpoise wires. Connect the counterpoise conductor to a ground rod at every 2,000 feet of cable run, at lighting vault(s) (but not to vault equipment), and at feeder connection to light circuit(s).

The counterpoise must be connected to the exterior one-hole ground lug on fixture bases. Use bolted ground clamps when bases are supplied with ground lug. Bolts and fasteners must be bronze or stainless steel. Torque to manufacturer's recommendation.

3.5 SAFETY (EQUIPMENT) GROUNDING SYSTEM

The purpose of the safety ground is to protect personnel from possible contact with an energized fixture or light base that may result from a shorted power cable or isolation transformer.

Connect the light fixture, whether in-pavement or elevated, to the base interior ground lug by means of a braided ground strap specified in paragraph "Light Bases".

3.6 DUCT LINES

Duct lines as required in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION and as indicated. Ducts must be installed in trench for 24 hours before trench is backfilled to allow ducts to reach final ground temperature.

3.7 MANHOLES AND HANDHOLES

The manholes and handholes as required in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

3.7.1 Handholes Within Unpaved Areas

Handholes within unpaved areas must have their top surface +1 to +2 inches above grade.

3.8 MARKING AND LIGHTING OF AIRWAY OBSTRUCTIONS

Mark and light all towers, poles, buildings of certain shapes and sizes, and other obstructions in accordance with FAA AC 70/7460-1 and as indicated.

3.8.1 Obstruction Marker Lights

Install obstruction marker lights on buildings, and similar structures with 1 inch zinc-coated rigid steel conduit stems using standard tees and elbows, except that lowering devices, when required, must be installed in accordance with equipment manufacturer's recommendations.

3.9 APPROACH LIGHTING SYSTEMS

Install approach lighting system as indicated. Provide nameplates for equipment, controls, devices, and for each lighting circuit.

3.9.1 Alignment of Lights

Align lights in azimuth, with beams axes parallel to the approach lighting system centerline. Aim elevated lights vertically at a point on the glide path with the angular elevation of each light as indicated. In-pavement lights have a preset vertical aiming angle and require alignment in azimuth only.

3.10 WIND DIRECTION INDICATORS

Guy the wind cone direction indicator as indicated on drawings. Energize the wind cone illumination lights and obstruction lights from circuits as indicated.

3.11 CONSTANT CURRENT REGULATORS

Install as indicated in strict accordance with manufacturer's instructions.

3.12 APPLICATION

3.12.1 Exothermic Welding

Utilize only personnel who are experienced in and regularly engaged in this type of work to make these connections. Prior to any installations

in the field, provide documentation that the welding kits, materials and procedures to be used for welded connections are satisfactory. Comply with the manufacturer's recommendations and the following:

- a. Remove all slag from welds.
- b. The light fixture base cans should be provided with internal and external one-hole ground lugs that are coated with hot dipped galvanizing, the same as the rest of the base cans. The external ground lug should be bolted to a separate ground lug that is exothermically welded to a #4 AWG stranded bare copper grounding cable. That is connected to a ground rod for Air Force or Army projects or the counterpoise for Navy projects.
- c. All direct buried welds must be thoroughly coated 6 mil of 3M "Scotchkote", or approved equivalent, or coated with coal tar bitumastic material to prevent surface exposure to corrosive soil or moisture.

3.12.2 Field Fabricated Nameplate Mounting

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.12.3 Equipment for Silicone Sealant

Equipment for silicone sealant must be air-powered pump, components, and hoses as recommended by the sealant manufacturer. Hoses and seals must be lined to prevent moisture penetration and withstand pumping pressures. Equipment must be free of contamination from previously used other type sealant.

3.12.4 Concrete

Concrete used in underground structures, such as manholes, handholes, pull boxes and foundations must have minimum 28-day strength of 4000 psi. Similar structures in areas where freeze/thaw conditions are common, must have minimum 28-day strength of 4500 psi. Concrete used for non-structural items, such as equipment pads, must have minimum 28-day strength of 2000 psi.

3.12.5 Grounding

Ground non-current carrying metallic parts associated with electrical equipment as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

3.13 FIELD QUALITY CONTROL

Notify the Contracting Officer five working days prior to each test. Correct all deficiencies and repeat tests. Field test reports must be written, signed and provided as each circuit or installation item is completed. Include resistance-to-ground and resistance between conductors field tests, and continuity measurements for each circuit. For each series circuit measure the input voltage and output current of the constant current regulator at each intensity. For multiple circuits measure the input and output voltage of the transformer for each intensity setting. Provide report documenting the visual inspection of the lights

operation and installed fixtures or units.

3.13.1 Visual Inspection

Inspection reports must be prepared and provided as each stage of installation is completed. Identify the activity by contract number, location, quantity of material placed, and compliance with requirements.

3.13.2 Operating Test

Upon completion of tests, show by demonstration in service that circuits, control equipment, and lights covered by the contract are in good operating condition. Operate each switch in the control tower lighting panels so that each switch position is engaged at least twice. During this process, observe lights and associated equipment to determine that each switch controls properly corresponding circuit. Provide telephone or radio communication between the operator and the observers. Repeat tests from the alternate control station, from the remote control points, and again from the local control switches on the regulators. Test each lighting circuit by operating the lamps at maximum brightness for not less than 30 minutes. Visually examine at the beginning and at the end of this test to ensure that the correct number of lights are burning at full brightness. Conduct one day and one night operating test for the Contracting Officer.

Provide performance test reports, upon completion and testing of the installed system, in booklet form showing all field tests performed to adjust each component and all field tests performed to provide compliance with the specified performance criteria. For each test, indicate the final position of controls.

3.13.3 Distribution Conductors, 600-Volt Class

Using an instrument which applies a voltage of approximately 500 volts providing a direct reading in resistance, performing testing to verify that no short circuits or accidental grounds exist.

3.13.4 Counterpoise System Test and Inspection

At accessible locations, visually inspect counterpoise system to ensure continuity of counterpoise system. Test continuity of counterpoise system to the vault grounding system in access well closest to the vault.

3.13.5 Progress Testing for Series Lighting Circuits

Conduct a megger test on each section of circuit or progressive combinations of sections as they are installed. Test each section or progressive combination of sections with a megohmmeter providing a voltage of approximately 1000 volts, a direct reading in resistance. Document all results. Eliminate faults found by these tests, and re-test before proceeding with the circuit installation.

3.13.6 Electrical Acceptance Tests

Perform acceptance tests for series and multiple airfield and heliport lighting circuits only on complete and new lighting circuits. Each series and multiple lighting circuit must receive a high voltage insulation test. Check that cable insulation resistance to ground is not less than 50 megohms per FAA-C-1391, Installation and Splicing of Underground Cable.

3.13.7 Low-Voltage Continuity Tests

Test each series circuit for electrical continuity. Eliminate faults indicated by this test and perform retest before proceeding with the high-voltage insulation resistance test.

3.13.8 High-Voltage Insulation Resistance Tests

Subject each new series lighting circuit to a high-voltage insulation resistance test by measurement of the insulation leakage current with a suitable high-voltage test instrument which has a steady, filtered direct current output voltage and limited current. High-voltage tester must include an accurate voltmeter and microammeter for reading voltage applied to the circuit and resultant insulation leakage current. Do not exceed voltage test values specified below.

- a. Test Procedure: Disconnect both leads from regulator output terminals and support so that air gaps of at least 3 inches or as defined by FAA AC 150/5345-7 Table 1 exists between bare conductors and ground. Clean and dry cable sheaths for a distance of 1 foot from ends of cables and exposed insulation at ends of cables. Connect ends of both conductors of the circuit together and to high-voltage terminals of test equipment, and test voltage applied as specified in the following tabulation between conductors and ground for a period of 5 minutes.

Test Voltage, DC	
	First Test on New Circuits
High Intensity Series Lighting Circuits (5,000 volt leads, 500 and 200 watt transformers)	9000
Medium Intensity Series Lighting Circuits (5,000 volt leads, 30/45 watt transformers)	6000
600-Volt Circuits	900

When additions are made to existing circuits, test only new sections in accordance with "First Test on New Circuits" in table above. UNDER NO CIRCUMSTANCE SHALL EXISTING CIRCUIT SECTIONS RECEIVE INSULATION RESISTANCE TESTING.

- b. Leakage Current: Measure and record insulation leakage current for each circuit after a 1 minute application of the test voltage. If leakage current exceeds values specified below, sectionalize the circuit and retest, and repair or replace defective parts. Leakage current limits include allowances for the normal number of connectors and splices for each circuit as follows:

1. Three microamperes for each 1000 feet of cable.

2. Two microamperes for each isolation transformer.
3. If measured value of insulation leakage current exceeds calculated value, sectionalize the circuit and test as specified for each section. Repair or replace defective components until repeated tests indicate an acceptable value of leakage current for the entire circuit.

c. Resistance Values versus Cable Length

An alternate test procedure for circuit validation is to use a megohmmeter. Use 5000V test for new circuits. If the minimum resistance values are not achieved then use the leakage current test indicated above.

Circuit Length in Feet Meters	Minimum Resistance to Ground (Megohms)
10,000	50
10,000-20,000	40
>20,000	30

3.14 PHOTOMETRIC TESTING

3.14.1 Inspection

Provide test reports from an FAA approved, third-party, certification body to insure full compliance with the photometric requirements. Submit for approval for each type of light fixtures to the Contracting Officer prior to final acceptance of the installation. Include certification of compliance with specified requirements, identify deficiencies, and recommend corrective action when appropriate. Type and neatly bind test reports to form a part of the final record. Submit test reports documenting the results of each test not more than 10 days after test is completed. Items to be tested include, but are not limited to, the following:

- a. Runway Edge, End, Threshold Lighting
- b. Discharge-Type Flashing Light Equipment
- c. ALSF-1 In-Pavement Approach Lighting
- d. PAPIs

3.14.2 Test Instrument Calibration

- a. Use a photometric tester that has a calibration program which assures that all applicable test instruments are maintained within rated accuracy.
- b. The accuracy must be directly traceable to the National Institute of Standards and Technology.

- c. Instrument calibration frequency schedule must not exceed 12 months.
- d. Dated calibration labels must be visible on all test equipment.
- e. Calibrating standard must be of higher accuracy than that of the instrument tested.
- f. Up-to-date records which indicate dates and test results of instruments calibrated or tested. For instruments calibrated by the manufacturer on a routine basis, in lieu of third party calibration, include the following:
 - (1) Maintain up-to-date instrument calibration instructions and procedures for each test instrument.
 - (2) Identify the third party/laboratory calibrated instrument to verify that calibrating standard is met.

3.14.3 Pre-Testing Requirements

- a. Verify Constant Current Regulator (CCR) Performance. When the CCR is energized at the highest brightness intensity and at the middle brightness intensity, measure the following electrical characteristics for each CCR powering the light fixtures:
 - (1) Input Current.
 - (2) Input Voltage.
 - (3) Input Power Factor.
 - (4) Input Voltage Harmonic Distortion.
 - (5) Input Current Harmonic Distortion.
 - (6) Output Voltage.
 - (7) Output Current.
 - (8) Output Power Factor.
 - (9) Output Voltage Harmonic Distortion.
 - (10) Output Current Harmonic Distortion.

Analyze the data, and perform any adjustments or calibration required.

- b. Clean Light Fixture. Clean all of the light fixtures to be tested one day prior to commencement of photometric testing. Perform the photometric testing at night. At the end of each night the photometric testing firm must provide a list of the deficient fixtures to the Contractor for corrections. A minimum 10% spare lamps and/or fixtures, transformers, and other accessories required to correct possible deficiencies must be available so that any corrections can be made daily and retested the following night.
- c. Use photometric testing equipment that has an array of photometric sensors capable of taking simultaneous readings along the main photometric beam of the light output in direct correlation with Tables

1, 2, and 3 of the current edition [FAA AC 150/5345-46](#) and the appropriate UFC requirements. The number and angles of measurement as well as sensor location and placement must be in accordance with Paragraphs 4.3.1 and 4.3.3 and the above-mentioned tables of the current edition [FAA AC 150/5345-46](#) which provide specific requirements for photometric testing. Compare the average photometric intensity to the horizontal and vertical requirements detailed in Tables 1, 2, and 3 of the current edition [FAA AC 150/5345-46](#). Include the measurement of each light fixture in both directions. All of the sensor readings for the light fixture being evaluated must be displayed simultaneously for operator and Airport representative review and evaluation. All sensor readings must be recorded automatically through the computer. Hand written data recording will not be accepted. Measure the chromaticity (x, y) or color of 5 percent of the light fixtures to ensure compliance with MIL-C-25050A, "Aviation Colors". Measure the chromaticity along one point in the center of the main photometric ellipse.

- d. The testing firm must submit initial reports daily to the Contractor during the progress of the work so that corrective measures can be taken as required. A list of the deficient fixtures and the photometric output for each fixture must be provided in a field report to the Contractor prior to the testing firm leaving the job-site. Document the final results in a Final Report, with six (6 copies) submitted to the Airport. The final report must present an evaluation of each fixture tested, including proposed or performed corrective measures, such as cleaning or replacement of lenses, re-aiming of fixture, repair or replacement of fixture, for those fixtures that do not meet the performance requirements. Tabulate the photometric test data for each fixture with the following information:

Fixture Number - Provided by Airport.

Light Direction - Direction of light beam (ex. North, South, East, West).

Fixture type - FAA Fixture Type (ex. L-852C).

Lens Color - Color of lens on fixture being tested.

Max CD - Maximum candela output in a point along the main beam.

Avg CD - Average candela output of main beam.

Max/Avg - Ratio of max. candela output over the ave. output.

FAA Min Level - FAA specified output for type of fixture.

Chromaticity - x, y color coordinates of the sample fixture.

3.14.4 Photometric Testing of [Discharge-Type Flashing Light Equipment](#)

To evaluate the equipment performance with respect to the criteria in [FAA AC 150/5345-51](#), paragraph 3.4, perform photometric testing of the discharge-type lights. Conduct tests per FAA-E-1100, Photometric Test Procedures for Condenser Discharge Lights. Include in test results a graph showing the isocandela curve of effective intensity for each brightness setting and oscilloscope photographs or digital image files (e.g., TIF, JPG, BMP) of the discharge pulse shape.

3.14.5 Photometric Testing of [PAPIs](#)

Using photometric testing, evaluate the photometric intensity of the PAPI system and verify that the center of the light beams are aimed at the appropriate vertical and horizontal angle. Provide support to adjust the PAPI while it is flight tested by the government.

- a. Prior to performing the test measure the electrical characteristics of the CCR, if applicable, powering the PAPI or APAPI circuit. Include the measurement of the Volt-Amperes, Current, Voltage, and Harmonic Distortion for each intensity settings on the both the output and input to the CCR.
- b. Sensors. Utilize sensors that are accurate to 5 percent as traceable to the National Institute of Standards and Technology (NIST) secondary standard or approved equal. Sensor must be color corrected to account for the different white and red lens associated with the system.
- c. Measurement Points. Test on a one-degree horizontal interval and one-degree vertical interval inside the photometric ellipse indicated in Figure 3-12, [UFC 3-535-01](#).
- d. Measurement distance from lighting system. Take measurements at a minimum of [10 feet](#) from the edge of the lenses.
- e. Measurement Duration. Each measurement point must last a minimum of 30 seconds. To minimize stray light which may enter the lens, intentensity measurements at each point must be the average measurement over the 30 second time span.
- f. Test Results. Compare measured light intensity values to the minimum values established in Figure 3-12, [UFC 3-535-01](#). The angular difference for successive light units in a light bar must comply with paragraph 3-7.3.1, [UFC 3-535-01](#).

3.14.6 Constant Current Regulators Switchgear Frame

Turn on the fused disconnect power to the bus bar and verify that there is [277 VAC](#) between all phases and both ground and neutral, and ensure there is [480 VAC](#) between any two phases.

3.14.7 Constant Current Regulators

3.14.7.1 Visual Examination

Examine each constant current regulator to ensure that porcelain bushings are not cracked, no shipping damage has occurred, internal and external connections are correct, switches and relays operate freely and are not tied or blocked, fuses, if required, are correct, and oil level of oil-filled regulators is correct. Remove relay panel covers only for this examination; it is not necessary to open the main tank of oil-filled regulators. Accomplish the instructions on the plates attached to the regulators. Replace covers tightly after completing examinations and tests.

3.14.7.2 Electrical Tests

Ensure that supply voltage and input tap correspond. With load disconnected, energize regulator and observe the open circuit protector to ensure that it de-energizes the regulator within 3 seconds. After testing circuits for open connections and grounds and after determining that lamps are good and in place, apply circuit load to the regulator and measure the voltage and current simultaneously on brightness taps. Voltmeter and ammeter must have an accuracy of plus or minus one percent. Record readings and make readings during the day and night in order to obtain the average supply voltage. Output current on each brightness tap must be

within plus or minus 2 percent of the nameplate values after making necessary correction in the supply voltage. Late model regulators have automatic supply voltage correction in lieu of input taps, and output current does not change as supply voltage varies. When output current on tap 5 deviates from nameplate value by more than 2 percent, and regulator is not overloaded, check internal adjustment as described on regulator instruction plate. Since adjustment may be rather delicate, allow a deviation of up to plus or minus 5 percent on taps 1 through 4 before attempting to readjust the regulator.

3.15 FINISHING

Painting required for surfaces not otherwise specified and finish painting of items only primed at the factory must be per manufacturer recommendations.

3.16 CLOSEOUT ACTIVITIES

3.16.1 Demonstration

After completion of installations and the above tests, circuits, control equipment, and lights covered by the contract must be demonstrated to be in acceptable operating condition. Each switch in the control tower lighting panels must be operated so that each switch position is engaged at least twice. During this process, lights and associated equipment must be observed to determine that each switch properly controls the corresponding circuit. Telephone or radio communication must be provided between the operator and the observer. Tests must be repeated from the alternate control station, from the remote control points, and again from the local control switches on the regulators. Each lighting circuit must be tested by operating the lamps at maximum brightness for not less than 30 minutes. At the beginning and at the end of this test the correct number of lights must be observed to be burning at full brightness. One day and one night operating test must be conducted for the Contracting Officer.

3.16.2 Training

Submit requirements of training four weeks before training is scheduled to begin. Submit information describing training to be provided, training aids to be used, samples of training materials, and schedules; instructions necessary to checkout, troubleshoot, repair, and replace components of the systems, including integrated electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting after acceptance of the system.

- a. Provide training on the proper operation and maintenance procedures for the system. Submit a list of **special tools** and test equipment required for maintenance and testing of the products supplied by the Contractor.
- b. Submit six copies of operation for the equipment furnished. One complete set must be furnished prior to performance testing and the remainder must be furnished upon acceptance. Operating manuals must detail the step-by-step procedures required for system startup, operation, and shutdown. Operating manuals must include the manufacturer's name, model number, parts list, and brief description of all equipment and their basic operating features.

- c. Submit six copies of maintenance manuals listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guides. Maintenance manuals must include conduit and equipment layout and simplified wiring and control diagrams of the system as installed.

3.16.3 As-Built Drawings

Submit as-built drawings that provide current factual information including deviations from, and amendments to the drawings and changes in the work, concealed and visible, as instructed. The as-built drawings must show installations with respect to fixed installations not associated with the systems specified herein. Cable and wire must be accurately identified as to direct-burial or in conduit and must locate the connection and routing to and away from bases, housings, and boxes.

3.16.4 Posted Instructions

Submit a typed copy of the proposed posted instructions showing wiring, control diagrams, complete layout and operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation, and procedures for safely starting and stopping the system.

3.17 MAINTENANCE

3.17.1 List of Parts

A list of parts and components for the system by manufacturer's name, part number, nomenclature, and stock level required for maintenance and repair necessary to ensure continued operation with minimal delays.

3.17.2 Maintenance and Repair Instructions

Instructions necessary to check out, troubleshoot, repair, and replace components of the systems, as specified.

3.17.3 Posted Operations and Maintenance Instructions

Submit proposed diagrams, instructions, and other sheets prior to posting.

3.18 SCHEDULES

Refer to Section 01 35 13 SPECIAL PROJECT PROCEDURES for construction outage plan requirements.

-- End of Section --

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SECTION 31 00 00

EARTHWORK
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO T 180 (2017) Standard Method of Test for
Moisture-Density Relations of Soils Using
a 4.54-kg (10-lb) Rammer and a 457-mm
(18-in.) Drop

ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M (2024) Standard Specification for Concrete
Aggregates

ASTM C117 (2017) Standard Test Method for Materials
Finer than 75-um (No. 200) Sieve in
Mineral Aggregates by Washing

ASTM C136/C136M (2019) Standard Test Method for Sieve
Analysis of Fine and Coarse Aggregates

ASTM C150/C150M (2022) Standard Specification for Portland
Cement

ASTM C260/C260M (2010a; R 2016) Standard Specification for
Air-Entraining Admixtures for Concrete

ASTM C618 (2023; E 2023) Standard Specification for
Coal Fly Ash and Raw or Calcined Natural
Pozzolan for Use in Concrete

ASTM C989/C989M (2022) Standard Specification for Slag
Cement for Use in Concrete and Mortars

ASTM D1140 (2017) Standard Test Methods for
Determining the Amount of Material Finer
than 75-µm (No. 200) Sieve in Soils by
Washing

ASTM D1556/D1556M (2015; E 2016) Standard Test Method for
Density and Unit Weight of Soil in Place
by Sand-Cone Method

ASTM D1557 (2012; E 2015) Standard Test Methods for
Laboratory Compaction Characteristics of
Soil Using Modified Effort (56,000

ft-lbf/ft³) (2700 kN-m/m³)

ASTM D2216	(2019) Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D2487	(2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D4253	(2016; E 2019) Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table
ASTM D4254	(2016) Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density
ASTM D4318	(2017; E 2018) Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
ASTM D4829	(2021) Standard Test Method for Expansion Index of Soils
ASTM D4832	(2016; E 2018) Standard Test Method for Preparation and Testing of Controlled Low Strength Material (CLSM) Test Cylinders
ASTM D5268	(2019) Topsoil Used for Landscaping Purposes
ASTM D6023	(2016) Standard Test Method for Density (Unit Weight), Yield, Cement Content, and Air Content (Gravimetric) of Controlled Low-Strength Material (CLSM)
ASTM D6103/D6103M	(2017; E 2021) Standard Test Method for Flow Consistency of Controlled Low Strength Material (CLSM)
ASTM D6938	(2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements
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U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 600/4-79/020	(1983) Methods for Chemical Analysis of
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Water and Wastes

EPA SW-846.3-3

(1999, Third Edition, Update III-A) Test
Methods for Evaluating Solid Waste:
Physical/Chemical Methods

1.2 DEFINITIONS

1.2.1 Structural Fill

Soil material placed to support buildings, walls, pads, and other similar facilities.

1.2.2 Embankment Fill

Soil material placed to construct embankment.

1.2.3 Topsoil

Surface layer of primarily organic soil capable of supporting vegetation growth.

1.2.4 Utility Bedding Material

Fill placed to directly support pipes, conduits, cables, and appurtenant structures. Bedding may also be used to provide a cushion between utilities and bedrock, obstacles, obstructions and other unyielding materials.

1.2.5 Flowable Fill

Fill placed in a plastic or liquid form that flows to near its final placement location with limited assistance and subsequently cures or solidifies to provide a stable or impermeable barrier.

1.2.6 Satisfactory Materials

Satisfactory materials for fill, backfill, and/or any in-situ soils to remain in place comprise any materials classified by ASTM D2487 as GW, GP, GM, GP-GM, GW-GM, GP-GC, GM-GC, SW, SP, SW-SM, SP-SM. Maximum particle size to be no greater than 2 inches in any dimension.

1.2.7 Unsatisfactory Materials

Materials which do not comply with the requirements for satisfactory materials are unsatisfactory. Unsatisfactory materials also include man-made fills; trash; refuse; backfills from previous construction; roots and other organic matter or frozen material. Notify the Contracting Officer when encountering any contaminated materials.

1.2.8 Cohesionless Materials

Cohesionless materials include materials classified in ASTM D2487 as GW, GP, SW, and SP. Materials classified as GM and SM will be identified as cohesionless only when the fines are nonplastic. Perform testing, required for classifying materials, in accordance with ASTM D4318, ASTM C117, ASTM C136/C136M and ASTM D1140.

1.2.9 Cohesive Materials

Cohesive materials include materials classified as GC, SC, ML, CL, MH, and CH. Materials classified as GM and SM will be identified as cohesive only when the fines are plastic. Perform testing, required for classifying materials, in accordance with [ASTM D4318](#), [ASTM C117](#), [ASTM C136/C136M](#) and [ASTM D1140](#).

1.2.10 Hard/Unyielding Materials

Hard/Unyielding materials comprise weathered rock, dense consolidated deposits, or conglomerate materials which are not included in the definition of "rock" with stones greater than 2 [inch](#) in any dimension or as defined by the pipe manufacturer, whichever is smaller. These materials usually require the use of heavy excavation equipment, ripper teeth, or jack hammers for removal.

1.2.11 Unstable Material

Unstable materials are too weak to adequately support the utility pipe, conduit, equipment, or appurtenant structure. Satisfactory material may become unstable due to ineffective drainage, dewatering, becoming frozen, excessive loading.

1.2.12 Expansive Soils

Expansive soils are defined as soils that have an expansion index greater than 20 when tested in accordance with [ASTM D4829](#).

1.2.13 Rock

Solid homogeneous interlocking crystalline material with firmly cemented, laminated, or foliated masses or conglomerate deposits, neither of which can be removed without systematic drilling and blasting, drilling and the use of expansion jacks or feather wedges, or the use of backhoe-mounted pneumatic hole punchers or rock breakers; also large boulders, buried masonry, or concrete other than pavement exceeding [1/2 cubic yard](#) in volume. Removal of hard material will not be considered rock excavation because of intermittent drilling and blasting that is performed merely to increase production.

1.2.14 Capillary Water Barrier

A layer of clean, poorly graded crushed rock, stone, or natural sand or gravel having a high porosity which is placed beneath a building slab with or without a vapor barrier to cut off the capillary flow of pore water to the area immediately below a slab.

1.2.15 Degree of Compaction (Proctor)

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum density obtained by the test procedure presented in [ASTM D1557](#) abbreviated as a percent of laboratory maximum density. Since [ASTM D1557](#) applies only to soils that have 30 percent or less by weight of their particles retained on the [3/4 inch](#) sieve, express the degree of compaction for material having more than 30 percent by weight of their particles retained on the [3/4 inch](#) sieve as a percentage of the maximum density in accordance with [AASHTO T 180](#)-21 paragraph 1.5, Note 1.

1.2.16 Degree of Compaction (Relative Density)

Degree of compaction required for soils with less than 5 percent passing the No. 200 sieve, is expressed as a relative percentage of the maximum index density/dry unit weight and minimum index density/dry unit weight, obtained by the test procedures in accordance with ASTM D4253 and ASTM D4254, respectively, abbreviated as a percent of laboratory relative density.

1.2.17 Borrow

Soil brought to the project site from an external location for the purposes of project construction.

1.2.18 Subgrade

Earth materials directly below foundations and directly below granular base materials in building slab and pavement areas including shoulders.

1.3 CRITERIA FOR BIDDING

Base bids on the following criteria:

- a. Surface elevations are as indicated.
- b. Pipes or other artificial obstructions, except those indicated, will not be encountered.
- d. Ground water was not encountered in any borings drilled a depth of 14 feet below existing surface conditions.
- e. Hard materials and rock will not be encountered in 100 percent of the excavations at 14 feet below existing surface elevations.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Excavation and Trenching Plan; G

Borrow Plan; G

Disposition of Surplus Materials; G

Preconstruction Meeting; G

SD-03 Product Data

Flowable Fill Mix Design; G, AE

SD-06 Test Reports

Material Test Report; G, AE

Borrow Site Testing; G, AE

Geotechnical Evaluation Report; G, AE

1.5 QUALITY CONTROL

1.5.1 Geotechnical Engineer

Provide a Professional Geotechnical Engineer to provide inspection of excavations and soil/groundwater conditions throughout construction. The Geotechnical Engineer is responsible for performing pre-construction and periodic site visits throughout construction to assess site conditions. The Geotechnical Engineer is responsible for preparing and updating the Excavation and Trenching Plan and Dewatering Work Plan as construction progresses to reflect changing conditions and submit an updated plan if necessary. Submit a monthly [Geotechnical Evaluation report](#), informing the Contractor and Contracting Officer of the status of the plan and an accounting of the Contractor's adherence to the plan addressing any present or potential problems. The Contractor is responsible for arranging meetings with the Geotechnical Engineer and Contracting Officer throughout the contract duration.

1.5.2 Qualified Technician

Provide a Qualified Technician to inspect, monitor, sample, and performing field testing. The technician qualifications need to be one of the following: a current National Institute for Certification in Engineering Technologies (NICET) Level II minimum certification in Construction Materials Testing Soils; a Geologist-in-Training with minimum one-year experience; an Engineer-in-Training with minimum one-year experience; a Registered Geologist; or a Professional Engineer.

1.5.3 Lab Validation

Perform testing by a Corps validated commercial testing laboratory or Contractor established testing laboratory meeting the requirements of Section [01 45 00](#) (or similar number) entitled QUALITY CONTROL and approved by the Contracting Officer. Submit testing laboratory validation for the testing to be performed. Do not permit work requiring testing until testing facilities have been inspected, Corps validated and approved by the Contracting Officer.

1.5.4 Preconstruction Meeting

Conduct a preconstruction meeting at the jobsite at least five business days prior to the start of earthwork operations on the project. The [preconstruction meeting](#) is to be arranged by the Contractor and is to follow the written agenda submitted prior to the meeting. The purpose of this meeting is to review the requirements of this specification and the associated plans. The following individuals must be in attendance at this meeting: Contractor's Project Manager and Project Superintendent, earthwork subcontractor's Project Manager and Site Foreman, Contractor's Geotechnical Engineer and Testing Agency, Government Geotechnical Engineer and Civil Engineer, and Government Construction Manager and Engineering Technician.

The minutes of this meeting are to be recorded by the Contractor and published via email within 48 hours to all attendees. The minutes must be re-published within 48 hours via email pending any subsequent comments from the attendees.

PART 2 PRODUCTS

2.1 SOIL MATERIALS

2.1.1 Structural Fill

Materials classified as GW, GP, GM, GC, GW-GM, GW-GC, GP-GM, GP-GC, GC-GM, SW, SP, SM, SW-SM, SP-SM, in accordance with [ASTM D2487](#). Select material type appropriate for the intended purpose.

2.1.2 Embankment Fill

Materials classified as GW, GP, GM, GC, GW-GM, GW-GC, GP-GM, GP-GC, GC-GM, SW, SP, SM, SW-SM, SP-SM, in accordance with [ASTM D2487](#). Select material type appropriate for the intended purpose.

2.1.3 Topsoil

Material suitable for topsoil obtained from offsite areas and excavations is defined as: Natural, friable soil representative of productive, well-drained soils in the area, free of subsoil, stumps, rocks larger than [one inch](#) diameter, brush, weeds, toxic substances, and other material detrimental to plant growth. Topsoil material will be in accordance with [ASTM D5268](#).

2.1.4 Capillary Water Barrier

Conform to [ASTM C33/C33M](#) for fine aggregate grading with a maximum of 3 percent by weight passing [ASTM D1140](#), [No. 200](#) sieve, or coarse aggregate Size 57, 67, or 77.

2.1.5 Utility Bedding Material

Except as specified otherwise in the individual piping section, provide bedding for buried piping in accordance with [ASTM D2321](#). Install bedding for plastic piping to spring line of pipe. Provide geotextile fabric below bedding layer where indicated. Utility bedding material may include the following:

2.1.5.1 Class I

Angular, [0.25 to 1.5 inch](#), graded stone, including a number of fill materials that have regional significance such as coral, slag, cinders, crushed stone, and crushed shells.

2.1.5.2 Class II

Coarse sands and gravels with maximum particle size of [1.5 inch](#), including various graded sands and gravels containing small percentages of fines, generally granular and noncohesive, either wet or dry. Soil Types GW, GP, SW, and SP are included in this class as specified in [ASTM D2487](#).

2.1.5.3 Sand

Clean, coarse-grained sand classified as SP or SP-SM by [ASTM D2487](#) for bedding and backfill.

2.1.5.4 Gravel and Crushed Stone

Clean, coarsely graded natural gravel, crushed stone or a combination thereof identified as having a classification of GW or GP in accordance with [ASTM D2487](#) for bedding and backfill as indicated on the drawings. Do not exceed maximum particle size of [3 inches](#).

2.2 FLOWABLE FILL

Design and submit [flowable fill mix design](#) to consist of Portland cement, fly ash, and/or slag cement and fine aggregate. Include the dry weights of cementitious material(s); quality and gradation of aggregates in the saturated surface-dry weights along with gradation tests; quantities, types, and names of admixtures; and quantity of water per cubic yard. The minimum unconfined compressive strength to be 150 psi at 28 days in accordance with [ASTM D4832](#). The maximum unconfined compressive strength to be 300 psi at 28 days in accordance with [ASTM D4832](#). The aggregates in accordance with [ASTM C33/C33M](#) Fine Aggregates. Air-entrain fill in accordance with [ASTM C260/C260M](#). The air content to be between 8 and 15 percent in accordance with [ASTM D6023](#). The flow to be between [8 and 12 inches](#) in accordance with [ASTM D6103/D6103M](#). Portland cement to be Type I or II in accordance with [ASTM C150/C150M](#). Fly ash to be Class C in accordance with [ASTM C618](#). Provide slag cement in Grade 100 or 120 in accordance with [ASTM C989/C989M](#).

2.3 BURIED WARNING AND IDENTIFICATION MARKERS

Provide polyethylene plastic and metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured specifically for warning and identification of buried utility lines. Provide tape on rolls, [3 inches](#) minimum width, color coded as specified below for the intended utility with warning and identification imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED (intended service) LINE BELOW" or similar wording. Provide permanent color and printing, unaffected by moisture or soil.

Warning Tape Color Codes	
Red	Electric
Yellow	Gas, Oil; Dangerous Materials
Orange	Telephone and Other Communications
Blue	Water Systems
Green	Sewer Systems
White	Steam Systems

Warning Tape Color Codes	
Gray	Compressed Air

2.3.1 Warning Tape for Metallic Piping

Provide acid and alkali-resistant polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.003 inch and a minimum strength of 1500 psi lengthwise, and 1250 psi crosswise, with a maximum 350 percent elongation.

2.3.2 Detectable Warning Tape

Provide polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.004 inch, and a minimum strength of 1500 psi lengthwise and 1250 psi crosswise. Manufacture tape with integral wires, foil backing, or other means of enabling detection by a metal detector when tape is buried up to 3 feet deep. Encase metallic element of the tape in a protective jacket or provide with other means of corrosion protection.

2.3.3 Detection Wire

Insulate a single strand, solid copper detection wire with a minimum of 12 AWG.

2.4 BORROW

Provide borrow materials from sources located outside of Government property meeting the requirements of paragraph STRUCTURAL FILL, EMBANKMENT FILL and TOPSOIL.

PART 3 EXECUTION

3.1 PROTECTION

Perform all work specified in accordance with applicable requirements of the Corps of Engineers publication EM 385-1-1 Safety and Health Requirements Manual. Provide a Geotechnical Engineer to monitor construction activities and to prepare necessary work plans and reports; see paragraph QUALITY CONTROL.

Use equipment of type and size appropriate for the site conditions (soil character and moisture content). Maintenance of exposed subgrades and fills is the responsibility of the Contractor. The Contractor is required to prevent damage by ineffective drainage, dewatering, and heavy loads and equipment by implementing precautionary measures. Repair or replace any defects or damage.

3.1.1 Underground Utilities

Location of the existing utilities indicated is approximate. Physically verify the location and elevation of the existing utilities indicated prior to starting construction. The Contractor is responsible for protecting utilities from damage during construction.

3.1.2 Drainage and Dewatering

Provide for the collection and disposal of surface and subsurface water encountered during construction.

3.1.2.1 Drainage

Provide for the collection and disposal of surface and subsurface water encountered during construction. Construct storm drainage features (ponds/basins) at the earliest stages of site development, and throughout construction grade the construction area to provide positive surface water runoff away from the construction activity and provide temporary ditches, swales, and other drainage features and equipment as required to keep soils from becoming unstable, prevent erosion, or undermining of foundations. Remove unstable material from working platforms for equipment operation and soil support for subsequent construction features and provide new material as specified herein. It is the responsibility of the Contractor to assess the site conditions to employ necessary measures to permit construction to proceed.

3.1.2.2 Dewatering

Control groundwater and stormwater runoff flowing toward or into excavations to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of construction. French drains, sumps, ditches or trenches are not allowed within **3 feet** of the foundation of any structure, except with specific written approval, and after specific contractual provisions for restoration of the foundation area have been made. Perform control measures by the time the excavation reaches the water level in order to maintain the integrity of the in-situ material.

3.1.3 Shoring and Sheet piling

Submit an **Excavation and Trenching Plan** to stabilize features, prevent undermining or unintended horizontal and vertical movement of adjacent structures, and prevent slippage or movement in banks or slopes adjacent to the excavation. Submit drawings and calculations, certified by a registered professional engineer, describing the methods for shoring and sheet piling of excavations. Drawings to include material sizes and types, arrangement of members, and the sequence and method of installation and removal. Calculations are to include data and references used.

3.1.4 Protection of Graded Surfaces

Protect newly backfilled, graded, and topsoiled areas from traffic, erosion, and settlements that may occur. Repair or reestablish damaged grades, elevations, or slopes.

3.2 BORROW

Select borrow material to meet the requirements and conditions of the fill or embankment for which it is to be used. Obtain borrow material from approved private sources. Submit a **Borrow Plan** that includes materials to be excavated, stockpile locations, proposed slopes, drainage, and closure. Unless otherwise provided in the contract, the Contractor is responsible for obtaining the right to procure material, pay royalties and other charges involved, and bear the expense of developing the sources, including rights-of-way for hauling from the owners. Unless specifically

provided, do not obtain borrow within the limits of the project site without prior written approval.

3.2.1 Government Furnished Borrow Area(s)

The Government will not provide a borrow area.

3.2.2 Contractor Furnished Borrow Area(s)

Obtain approved borrow materials from approved offsite sources. If a borrow source is selected that is not a commercial entity from which soil material is directly purchased, submit a Borrow Plan that includes the borrow source location, geotechnical test results showing the fill material meets the Contract requirements, environmental test results in accordance with paragraph ENVIRONMENTAL REQUIREMENTS FOR OFF-SITE SOIL, and any Federal, State, and local permits required for excavation and reclamation of the borrow area.

3.2.3 Environmental Requirements for Off-Site Soil

Test offsite soils brought in for use as backfill for Total Petroleum Hydrocarbons (TPH), Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX) and full Toxicity Characteristic Leaching Procedure (TCLP) including ignitability, corrosivity and reactivity. Backfill may not contain concentrations above appropriate State and EPA criteria, and for hazardous waste characteristics 10 parts per million (ppm) of total petroleum hydrocarbons (TPH) and 10 ppm of the sum of Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX) and pass the TCPL test. Determine TPH concentrations by using EPA 600/4-79/020 Method 418.1. Determine BTEX concentrations by using EPA SW-846.3-3 Method 5030/8020/8260B. Perform TCLP in accordance with EPA SW-846.3-3 Method 1311. Perform hazardous waste characteristic tests for ignitability, corrosivity, and reactivity in accordance with accepted standard methods. Perform PCB testing in accordance with accepted standard methods for sampling and analysis of bulk solid samples. Provide borrow site testing for petroleum hydrocarbons and BTEX from a grab sample of material from the area most likely to be contaminated at the borrow site (as indicated by visual or olfactory evidence), with at least one test from each borrow site. Provide borrow site testing for hazardous waste characteristics (TPH, BTEX and TCLP) from a composite sample of material, collected in accordance with standard soil sampling techniques. Do not bring borrow material to project site until Borrow Plan containing environmental test results has been received and approved by the Contracting Officer.

3.3 SURFACE PREPARATION

3.3.1 Stripping

Strip existing surface soils to a depth of 4 inches below the existing ground surface in all areas within the planned limits of disturbance. All stripped materials not suitable for reuse as topsoil will be wasted in a disposal area on government property specified by Fort Drum within 10 miles of WSAAF. Material stripped within areas specified as "Known ~~/Suspected~~ Areas of PFAS Contamination" the contractor shall manage in accordance with the Special Environmental Requirements of Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS. Screen all stripped soils to remove roots, rocks, and organic material prior to disposal or reuse as topsoil.

Strip suitable soil from the site where excavation or grading is indicated

and stockpile separately from other excavated material. Protect topsoil and keep in segregated piles until needed.

3.3.2 Proof Rolling

Perform proof rolling on exposed subgrade that is unfrozen and free of surface water (wet conditions resulting from rainfall). Notify the Contracting Officer a minimum of three days prior to proof rolling. Perform proof rolling in the presence of the Contracting Officer.

After stripping, excavating, and rough grading to the planned elevation, proof roll the existing subgrade of all building, pavement and embankment locations with four overlapping passes of a 15 ton, pneumatic-tired or smooth drum roller. Operate the roller in a systematic manner to ensure the number of passes over all areas, and at speeds between 2.5 to 3.5 miles per hour. Subgrade materials that exhibit excessive deflection and/or rutting during proof rolling need to be scarified, aerated, and re-compacted to specified density at plus or minus 2 percent of optimum moisture content prior to being considered for remedial action by the Contracting Officer. When proof rolling under buildings, the building subgrade is considered to extend 5 feet beyond the building lines, and make one-half of the passes with the roller in a direction perpendicular to the other passes.

3.3.3 Stockpiling Operations

Place and grade stockpiles of satisfactory as specified by the Contracting Officer. Keep stockpiles in a neat and well drained condition, giving due consideration to drainage at all times. Clear, grub, and seal by rubber-tired equipment, the ground surface at stockpile locations; separately stockpile excavated satisfactory and unsatisfactory materials. Protect stockpiles of satisfactory materials from contamination which may destroy the quality and fitness of the stockpiled material. Do not create stockpiles that could obstruct the flow of any stream, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way. If the Contractor fails to protect the stockpiles, and any material becomes unsatisfactory, remove and replace such material with satisfactory material from approved sources at no additional cost to the Government.

3.4 EXCAVATION

Excavate to contours, elevation, and dimensions indicated. Excavate soil disturbed or weakened by Contractor's operations, and soils softened or made unstable for subsequent construction due to exposure to weather. Use material removed from excavations meeting the specified requirements in the construction of fills, embankments, subgrades, shoulders, bedding (as backfill), and for similar purposes to minimize surplus material and to minimize additional material brought on site. Do not excavate below indicated depths except to remove unstable material as determined by the Geotechnical Engineer and confirmed by the Contracting Officer. Remove and replace excavations below the grades shown with appropriate materials as directed by the Contracting Officer.

If at any time during excavation, including excavation from borrow areas, the Contractor encounters material that may be classified as rock or as hard/unyielding material, uncover such material, and notify the Contracting Officer. Do not proceed with the excavation of this material until the Contracting Officer has classified the materials as common

excavation or rock excavation. Failure on the part of the Contractor to uncover such material, notify the Contracting Officer, and allow sufficient time for classification and delineation of the undisturbed surface of such material will cause the forfeiture of the Contractor's right of claim to any classification or volume of material to be paid for other than that allowed by the Contracting Officer for the areas of work in which such deposits occur.

3.4.1 Ditches, Gutters, and Channel Changes

Finish excavation of ditches, gutters, and channel changes by cutting accurately to the cross sections, grades, and elevations shown. Do not excavate below grades shown. Backfill excessive excavation as directed by the Contracting Officer, with satisfactory, compacted, material or with suitable stone or cobble to grades shown. Dispose excavated material as shown or as directed. Do not allow material to be deposited within 4 feet from edge of a ditch. Maintain excavations free from detrimental quantities of leaves, brush, sticks, trash, and other debris until final acceptance of the work.

3.4.2 Trench Excavation Requirements

Excavate the trench as recommended by the manufacturer of the pipe to be installed. Slope trench walls below the top of the pipe, or make vertical, and of such width as recommended by the manufacturer. Provide vertical trench walls where no manufacturer installation instructions are available. Do not exceed the trench width of 24 inches below the top pipe plus pipe outside diameter (O.D.) for pipes of less than 24 inches inside diameter, and do not exceed 36 inches plus pipe outside diameter for pipe sizes larger than 24 inches inside diameter. Where recommended trench widths are exceeded, provide redesign, stronger pipe, or special installation procedures. The Contractor is responsible for the cost of redesign, stronger pipe, or special installation procedures without any additional cost to the Government.

3.4.2.1 Bottom Preparation

Grade the bottoms of trenches accurately to provide uniform bearing and support for the bottom quadrant of each section of the pipe. Excavate bell holes to the necessary size at each joint or coupling to eliminate point bearing. Remove stones of 3 inch or greater in any dimension, or as recommended by the pipe manufacturer, whichever is smaller, to avoid point bearing.

3.4.2.2 Removal of Unyielding Material

Where unyielding material is encountered in the bottom of the trench, notify the Contracting Officer. Following approval, remove such material 4 inch below the required grade and replaced with suitable materials as provided in paragraph FILLING AND COMPACTION.

3.4.2.3 Removal of Unstable Material

Where unstable material is encountered in the bottom of the trench, remove such material to the depth directed and replace it to the proper grade with suitable material as provided in paragraph FILLING AND COMPACTION. When removal of unstable material is required due to the Contractor's fault or neglect in performing the work, the Contractor is responsible for excavating the resulting material and replacing it without additional cost

to the Government.

3.4.2.4 Excavation for Appurtenances

Provide excavation for manholes, catch-basins, inlets, or similar structures sufficient to leave at least **12 inches** clear between the outer structure surfaces and the face of the excavation or support members.

3.4.2.5 Water Lines

Excavate trenches to a depth that provides a minimum cover of **6 feet** from the existing ground surface, or from the indicated finished grade, whichever is lower, to the top of the pipe.

3.4.3 Underground Utilities

Perform work adjacent to utilities in accordance with procedures outlined by utility owner. Excavation made with power-driven equipment is not permitted within **5 feet** of known utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to utility lines or subsurface construction immediately to the Contracting Officer. Immediately repair damage to utility lines at no additional cost to the Government.

3.4.4 Structural Excavation

Following general excavation and rough grading activities, excavate the entire building footprint area and at least **5 feet** beyond, to **12 inches** below bottom of proposed floor slabs. Scarify the exposed surface to a depth of **6 to 8 inches**, moisture-condition, and compact to at least 95 percent of laboratory maximum density. For new pavement areas including exterior concrete pads, over-excavate to a minimum of **12 inches** below bottom of new pavement/pad base course, scarify, moisture-condition, and compact to at least 95 percent.

Make excavations to the lines, grades, and elevations shown, or as directed. Provide trenches and foundation pits of sufficient size to permit the placement and removal of forms for the full length and width of structure footings and foundations as shown. Clean rock or other hard foundation material of loose debris and cut to a firm, level, stepped, or serrated surface. Remove loose disintegrated rock and thin strata.

Concrete placement is not allowed until footing subgrades have been inspected and approved by the Contracting Officer.

3.4.5 Blasting

Blasting will not be permitted.

3.5 SUBGRADE PREPARATION

3.5.1 General Requirements

Shape subgrade to line, grade, and cross section as indicated. Remove

unsatisfactory and unstable material in surfaces to receive fill or in excavated areas, and replaced with satisfactory materials. Do not place material on surfaces that are muddy, frozen, contain frost, or otherwise containing unstable material. Scarify the surface to a depth of 4 inches prior to placing fill. Step or bench sloped surfaces steeper than 1 vertical to 4 horizontal prior to scarifying. Place 4 inches of loose fill and blend with scarified material. When subgrade is part fill and part excavation or natural ground, scarify to a depth of 8 inches.

3.5.2 Subgrade for Structures, Spread Footings, and Concrete Slabs

Do not excavate below depth shown for structures, spread footings, and concrete slabs. If over excavation occurs, compact disturbed material to 95 percent of ASTM D1557. After final rolling, the surface of the subgrade for buildings and pavements must not show deviations greater than 0.05 foot when tested with a 12-foot straightedge applied both parallel and at right angles to the centerline of the area.

3.5.3 Subgrade for Non-Airfield Pavements

Compact top 12 inches of subgrade for pavements to at least 95 percent of ASTM D1557. After final rolling, the surface of the subgrade for buildings and pavements must not show deviations greater than 0.05 foot when tested with a 12-foot straightedge applied both parallel and at right angles to the centerline of the area.

3.5.4 Subgrade for Airfield Shoulders

Compact the upper 12 inches of subgrade for shoulders to at least 95 percent of ASTM D1557 for the full depth of the shoulder. After final rolling, the surface of the subgrade for airfield shoulder pavements must not show deviations greater than 0.05 foot when tested with a 12-foot straightedge applied both parallel and at right angles to the centerline of the area.

3.5.5 Subgrade for Airfield Pavements

Compact top 12 inches of subgrade, to 100 percent of ASTM D1557. After final rolling, the surface of the subgrade for pavements must not show deviations greater than 0.05 foot when tested with a 12-foot straightedge applied both parallel and at right angles to the centerline of the area. Stabilize the top 6 inches of the existing subgrade by blending in new aggregate base course material in accordance with Section 32 11 20 SUBBASE COURSE FOR RIGID AIRFIELD AND NON-AIRFIELD PAVING.

3.5.6 Subgrade Filter Fabric

Place filter fabric as indicated directly on prepared subgrade free of vegetation, stumps, rocks larger than 2 inch diameter and other debris which may puncture or otherwise damage the fabric. Repair damaged fabric by placing an additional layer of fabric to cover the damaged area a minimum of 3 feet overlap in all directions. Overlap fabric at joints a minimum of 3 feet. Obtain approval of filter fabric installation before placing fill or backfill. Place fill or backfill on fabric in the direction of overlaps and compact as specified herein. Follow manufacturer's recommended installation procedures.

3.6 FILLING AND COMPACTION

Prepare ground surface on which backfill is to be placed and provide compaction requirements for backfill materials in conformance with the applicable portions of paragraphs for SUBGRADE PREPARATION. Do not place material on surfaces that are muddy, frozen, or contain frost. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, or other approved equipment well suited to the soil being compacted. Moisten material as necessary to plus or minus 2 percent of optimum moisture. Fill and backfill to contours, elevations, and dimensions indicated. Compact and test each lift before placing overlaying lift.

3.6.1 Trench Backfill

Backfill trenches to the grade shown. Backfill the trench to 2 feet above the top of pipe prior to performing the required pressure tests. Leave the joints and couplings uncovered during the pressure test.

3.6.1.1 Replacement of Unyielding Material

Replace unyielding material removed from the bottom of the trench with satisfactory material or initial backfill material.

3.6.1.2 Replacement of Unstable Material

Replace unstable material removed from the bottom of the trench or excavation with satisfactory material placed in layers not exceeding 6 inches loose thickness.

3.6.1.3 Bedding and Initial Backfill

Provide bedding of the type and thickness shown. Place initial backfill material and compact it with approved tampers to a height of at least one foot above the utility pipe or conduit. Bring up the backfill evenly on both sides of the pipe for the full length of the pipe. Take care to ensure thorough compaction of the fill under the haunches of the pipe. Except where shown or when specified otherwise in the individual piping section, provide bedding for buried piping in accordance with PART 2 paragraph UTILITY BEDDING MATERIAL. Compact backfill to top of pipe to 95 percent of ASTM D1557. Provide plastic piping with bedding to spring line of pipe.

3.6.1.4 Final Backfill

Do not begin backfill until construction below finish grade has been approved, underground utilities systems have been inspected, tested and approved, forms removed, and the excavation cleaned of trash and debris. Bring backfill to indicated finish grade. Where pipe is coated or wrapped for protection against corrosion, the backfill material up to an elevation 2 feet above sewer lines and one foot above other utility lines need to be free from stones larger than one inch in any dimension. Heavy equipment for spreading and compacting backfill are not to be operated closer to foundation or retaining walls than a distance equal to the height of backfill above the top of footing; compact remaining area in layers not more than 4 inches in compacted thickness with power-driven hand tampers suitable for the material being compacted. Place backfill carefully around pipes or tanks to avoid damage to coatings, wrappings, or tanks. Do not place backfill against foundation walls prior to 7 days after

completion of the walls. As far as practicable, bring backfill up evenly on each side of the wall and sloped to drain away from the wall.

Fill the remainder of the trench, except for special materials for buildings, airfields and pavements with satisfactory material. Place backfill material and compact as follows:

3.6.1.4.1 Buildings, Airfields and Pavements

Place backfill up to the required elevation as specified. Do not permit water flooding or jetting methods of compaction. Compact as specified for Structural Fill in paragraph Structural Fill Placement.

3.6.1.4.2 Turfed or Seeded Areas and Miscellaneous Areas

Deposit backfill in layers of a maximum of 12 inches loose thickness, and compact it to 85 percent maximum density for cohesive soils and 90 percent maximum density for cohesionless soils. Do not permit compaction by water flooding or jetting. Apply this requirement to all other areas not specifically designated above.

3.6.1.5 Heat Distribution System

Free initial backfill material of stones larger than 1/4 inch in any dimension.

3.6.1.6 Electrical Distribution System

Provide a minimum cover of 24 inches from the finished grade to top of conduit, duct line, or concrete encased ductbank, unless otherwise indicated.

3.6.1.7 Buried Tape And Detection Wire

3.6.1.7.1 Buried Warning and Identification Tape

Provide buried utility lines with utility identification tape. Bury tape 12 inches below finished grade; under pavements and slabs, bury tape 6 inches below top of subgrade.

3.6.1.7.2 Buried Detection Wire

Bury detection wire directly above the utility at a distance not to exceed 12 inches above the utility. Terminate the ends of the wire in detector wire access boxes at utility structures, appurtenances, and bends. Detection wires shall have detector wire access boxes installed not exceeding 500-foot spacings. Detector wire access boxes installed in turf areas shall be installed at grade in a 12-inch diameter by 8-inch-thick concrete collar.

3.6.2 Structural Fill Placement

Place fill and backfill beneath and adjacent to structures in successive horizontal layers of loose material not more than 8 inches in depth, or in loose layers not more than 4 inches in depth when using hand-operated compaction equipment. Do not place over wet or frozen materials. Compact to at least 90 percent of laboratory maximum density for cohesive materials or 95 percent of laboratory maximum density for cohesionless materials, except as otherwise specified. Perform compaction in such a

manner as to prevent wedging action or eccentric loading upon or other damage to the structure. Moisture condition fill and backfill material to a moisture content that will readily facilitate obtaining the specified compaction.

3.6.3 Backfill for Appurtenances

After the manhole, catchbasin, inlet, or similar structure has been constructed and the concrete has been allowed to cure for 7 days, place backfill in such a manner that the structure is not be damaged by the shock of falling earth. Deposit the backfill material, compact it as specified for final backfill, and bring up the backfill evenly on all sides of the structure to prevent eccentric loading and excessive stress.

3.6.4 Flowable Fill

Place fill in a manner to completely fill voids in the location indicated. Do not place when atmospheric temperatures are expected to be below 33 degrees F at any time during the 3 day period following placement.

3.6.5 Compaction

3.6.5.1 General Site

Compact underneath areas designated for vegetation and areas outside the 5 foot line of the paved area or structure to 90 percent of ASTM D1557.

3.6.5.2 Adjacent Areas

Compact areas within 5 feet of structures to 95 percent of ASTM D1557.

3.7 EMBANKMENTS

3.7.1 Earth Embankments

Construct earth embankments from satisfactory materials free of organic or frozen material and rocks with any dimension greater than 3 inches. Place the material in successive horizontal layers of loose material not more than 8 inches in depth. Spread each layer uniformly on a soil surface that has been moistened or aerated as necessary and scarified or otherwise broken up so that the fill will bond with the surface on which it is placed. After spreading, plow, disk, or otherwise break up each layer; moisten or aerate as necessary; thoroughly mix; and compact to at least 90 percent laboratory maximum density for cohesive materials or 95 percent laboratory maximum density for cohesionless materials. Backfill and fill material are to be to a moisture content that will readily facilitate obtaining the specified compaction.

Compaction requirements for the upper portion of earth embankments forming subgrade for pavements are identical with those requirements specified in paragraph SUBGRADE PREPARATION. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment.

3.8 FINISHING/FINISH OPERATIONS

During construction, keep embankments and excavations shaped and drained. Maintain ditches and drains along subgrade to drain effectively at all times. Do not disturb the finished subgrade by traffic or other

operation. Protect and maintain the finished subgrade in a satisfactory condition until ballast, subbase, base, or pavement is placed. Do not permit the storage or stockpiling of materials on finished subgrade. Do not lay subbase, base course, ballast, or pavement until the subgrade has been checked and approved, and in no case place subbase, base, surfacing, pavement, or ballast on a muddy, spongy, frozen or otherwise unstable subgrade.

Finish the surface of excavations, embankments, and subgrades to a smooth and compact surface in accordance with the lines, grades, and cross sections or elevations shown. Provide the degree of finish for graded areas within 0.1 foot of the grades and elevations indicated except as indicated for subgrades specified in paragraph SUBGRADE PREPARATION. Finish gutters and ditches in a manner that will result in effective drainage. Finish the surface of areas to be turfed to a smoothness suitable for the application of turfing materials. Repair graded, topsoiled, or backfilled areas prior to acceptance of the work, and re-established grades to the required elevations and slopes.

3.8.1 Capillary Water Barrier

Place a capillary water barrier under concrete floor and area-way slabs grade directly on the subgrade and compact with a minimum of two passes of a hand-operated plate-type vibratory compactor.

3.8.2 Grading Around Structures

Construct areas within 5 feet outside of each building and structure line true-to-grade, shape to drain, and maintain free of trash and debris until final inspection has been completed and the work has been accepted.

3.8.3 Shoulder Construction

Construct shoulders as indicated. Submit advanced notice on shoulder construction for rigid pavements. Construct shoulders immediately after adjacent paving is complete. In the case of rigid pavements, do not construct shoulders until permission of the Contracting Officer has been obtained. Compact the entire shoulder area to at least the percentage of maximum density as specified in paragraph SUBGRADE PREPARATION above, for specific ranges of depth below the surface of the shoulder. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment. Finish shoulder construction in proper sequence in such a manner that adjacent ditches will be drained effectively and that no damage of any kind is done to the adjacent completed pavement. Align the completed shoulders true to grade and shaped to drain in conformity with the cross section shown.

3.8.4 Grading

Finish grades as indicated within one-tenth of one foot. Grade areas to drain water away from structures. Maintain areas free of trash and debris. For existing grades that will remain but which were disturbed by Contractor's operations, grade as directed.

3.8.5 Topsoil and Seed

Provide as specified in Section 32 92 19 SEEDING.

On areas to receive topsoil, prepare the compacted subgrade soil to a 2

inches depth for bonding of topsoil with subsoil. Spread topsoil evenly to a thickness of 6 inch and grade to the elevations and slopes shown. Do not spread topsoil when frozen or excessively wet or dry. Keep topsoil separate from other excavated materials, brush, litter, objectionable weeds, roots, stones larger than 2 inches in diameter, and other materials that would interfere with planting and maintenance operations. Remove from the site any surplus of topsoil from excavations and gradings. Obtain material required for topsoil in excess of that produced by excavation within the grading limits from offsite areas.

3.9 DISPOSITION OF SURPLUS MATERIAL

Remove from Government property all surplus or other soil material not required or not suitable for filling or backfilling, along with brush, refuse, stumps, roots, and timber. Properly disposed of in accordance with all applicable laws and regulations. Prepare plan for [Disposition of Surplus Materials](#) to include permissions document to dispose of nonsalable products.

3.10 TESTING

Perform testing as indicated in Table 1. Submit [Material Test Reports](#) within 7 days of tests being completed.

Material	Location of Material	Test Method	Test Frequency
Fill and Backfill	Any (except structures)	Density - ASTM D6938. When ASTM D6938 is used, check the calibration curves and adjust using only the sand cone method as described in ASTM D1556/D1556M.	One test per 2000 square feet, or fraction thereof, of each lift of fill or backfill areas compacted by other than hand-operated machines. Double testing frequency for areas compacted by hand-operated machines.
Subgrade	Any (except structures)		One test per 2000 square feet, or fraction thereof, of the compacted subgrade for areas compacted by other than hand-operated machines. Double the testing frequency for areas compacted by hand-operated machines. If ASTM D6938 is used, check in-place densities by ASTM D1556/D1556M as follows: One check test per lift for every 10 tests.

Material	Location of Material	Test Method	Test Frequency
Fills and Backfills	Structures	Density - ASTM D6938 When ASTM D6938 is used, check calibration curves and adjust using only the sand cone method as described in ASTM D1556/D1556M	Two random tests of each lift of fill or backfill areas within the building footprint for areas compacted by other than hand-operated machines. Double testing frequency for areas compacted by hand-operated machines. For all other structures, one test per 500 square feet, or fraction thereof, but a minimum of one random test of each lift of fill or backfill within the footprint of the structure. Check in-place densities by ASTM D1556/D1556M as follows: One check test per lift for every 10 tests.

Material	Location of Material	Test Method	Test Frequency
Subgrade	Structures	Density - ASTM D6938 When ASTM D6938 is used, check calibration curves and adjust using only the sand cone method as described in ASTM D1556/D1556M	One test per 500 square feet, or fraction thereof, of the compacted subgrade for areas compacted by other than hand-operated machines. Double the testing frequency for areas compacted by hand-operated machines. Check in-place densities by ASTM D1556/D1556M as follows: One check test per lift for every 10 tests.
Fill and Backfill	Any	Moisture Content - ASTM D2216	Two tests per day for each type of fill and backfill. Sample taken immediately prior to compaction after moisture conditioning.

Material	Location of Material	Test Method	Test Frequency
Fill and Backfill	Any	Moisture Density Relationship - ASTM D1557	One representative test per 500 cubic yards of fill and backfill, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum density. Sample to be taken from stockpile or location of placement.
Fill and Backfill	Any	Relative Density - ASTM D4253 and ASTM D4254	One test per 2000 square feet, or fraction thereof, of each lift of fill or backfill areas compacted by other than hand-operated machines. Double testing frequency for areas compacted by hand-operated machines.

Material	Location of Material	Test Method	Test Frequency
Fill and Backfill	Any	Gradation - ASTM C136/C136M	One representative test per 500 cubic yards of fill and backfill, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum density. Sample to be taken from stockpile or location of placement.
Fill and Backfill	Any	Atterberg Limits - ASTM D4318	One representative test per 500 cubic yards of fill and backfill, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum density. Sample to be taken from stockpile or location of placement.

-- End of Section --

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